OpenText™ Service Manager

Install

Version 9.82

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Install

This section provides instructions on installing Service Manager.

Install on-premises

Install on AWS

Install on Azure

1. Install on-premises

This section provides detailed instructions about planning, preparing, and deploying Service Manager on-premises, and post-installation tasks.

This section includes the following:

Plan

Prepare

Deploy

Post-installation tasks

Uninstall

1.1. Plan

This section provides information required to plan your Service Manager deployment. This section includes the following topics:

Topic	Description
Deployment architecture	Provides details of the Service Manager environment overview and the components. See Deployment architecture.
Sizing	Provides hardware sizing practices. See Sizing.
Support matrix	Provides support matrix information. See Support matrix.

1.1.1. Deployment architecture

You can install Service Manager in a production environment or in non-production environments: development, test, and reporting environments. We recommend that you perform the installation in a development environment and then convert or push the installation to your production environment.

- Production environment: The production environment enables you to deploy your customizations and provide services to your user base. Most production environments run 24 hours a day and 7 days a week, support many simultaneous users, and process large numbers of transactions and requests. In a production environment, you typically install the components of Service Manager on dedicated servers to maximize system performance.
- Development environment: A development environment enables you to evaluate application features and customize your installation prior to deployment in a production environment. In a development environment, you typically install all Service Manager components on one test system with a limited number of users and data.
- Test environment: A test environment is an installation that mirrors your production environment where you can safely test performance, upgrades, and backup and restore procedures. In a test environment, you typically install Service Manager in the same configuration as your production environment.
- Reporting environment: A reporting environment is an installation that mirrors the data from your production environment that you use to generate and view reports. In a reporting environment, you typically install Service

Manager to synchronize data with your production environment but limit the number of users that access the system.

Note

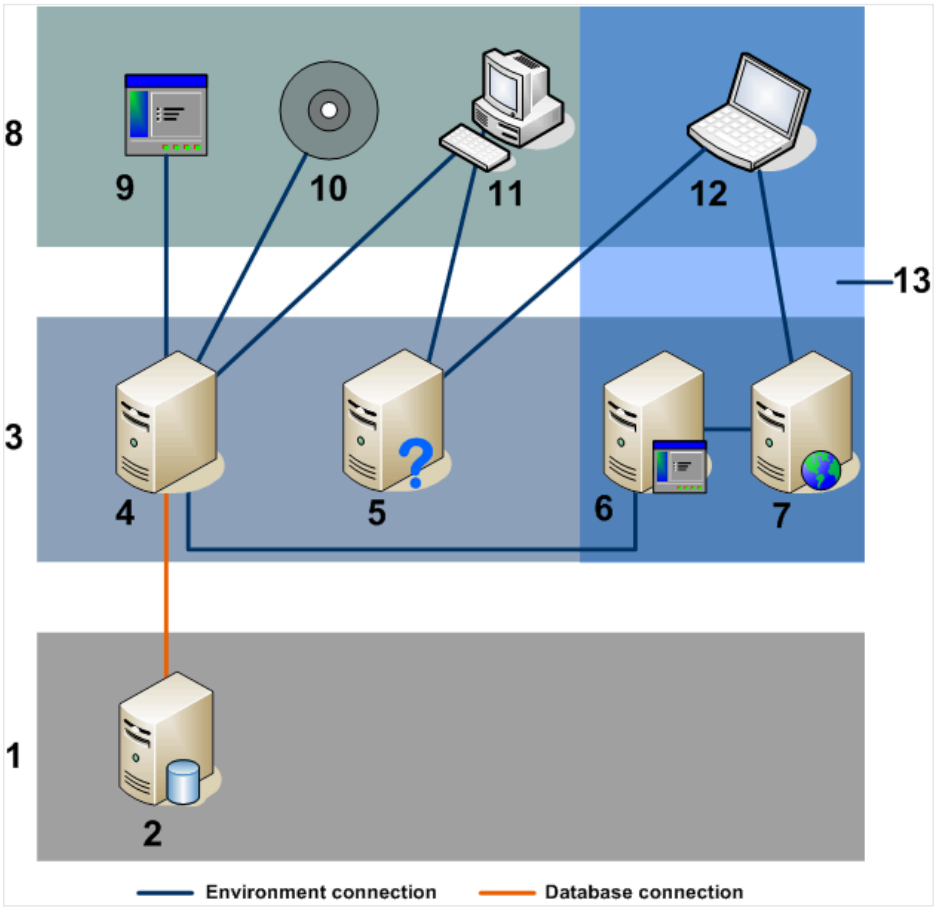
Note For sizing information, see Sizing.

Service Manager environment overview diagram

The Service Manager environment overview diagram describes the components of a Service Manager installation. The Service Manager table below provides a description for each component and lists whether the component is required or optional.

Note

Note This diagram doesn't include the following Service Manager components: Smart Analytics, Service Manager Collaboration, and Identity Manager (IdM). Refer to their specific installation sections for their deployment architectures.



Item	Name	Required?	Description
1	Database Tier	Required	Consists of one or more supported RDBMS servers. Your Service Manager application data must reside in an external RDBMS.
2	RDBMS	Required	A relational database management system for storing Service Manager applications and data. Requires a 1 GB network connection to the Service Manager Server.
3	Server Tier	Required	Contains servers that provide or process data for clients. The server tier includes the Service Manager server, which runs the Service Manager applications and manages the connections between the client and web tiers to the database tier.
4	Service Manager Server	Required	Manages connections between clients and the database tier.
5	Help Server	N/A	This is the documentation server on which the Service Manager Help Center is published.
6	Web Applications Server	Optional	Offers Java applications and content for web clients.
7	Web Server	Optional	Provides HTTP or HTTPS content to web clients.
8	Client Tier	Required	Contains the applications and methods available for connecting to Service Manager.
9	Web Applications	Optional	Applications that can connect to or communicate with Service Manager through a web services API.

10	Other Products	Optional	The suite of applications that can connect to or communicate with the Service Manager server. Temporary licenses are integrated with the Service Manager server. For a complete and up-to-date list of integrations, see the Service Manager Compatibility Matrix on the Software Integration Portal.
11	Windows Client	Required	Intended for Service Manager administrators or implementers. The Windows Client must be installed on a Windows workstation and requires a separate installation on each system that you want to connect to Service Manager. Each installation consumes a license. You must install at least one Windows Client, depending on the number of administrators and power users. The Windows Client enables you to design forms and work with the dbdict, capabilities that aren't available to the Web client user. Requires a 100 MB network connection to Service Manager (SOAP over HTTP or HTTPS).

12	Web Client	Optional	Recommended for end users. A web client enables users to connect to the Service Manager server using a web browser, without needing to install any additional software on the user's system. You can install a web client once and then any number of web users can access Service Manager from a single URL without consuming a license. Service Manager provides the following end user portals: Employee Self-Service (ESS) and SMA Service Portal. You can select one of them according to your needs. A web client can run on non-Windows platforms. **Note** Note There are scaling issues to consider: one web application server can only handle so many concurrent users before you need additional servers to handle the load.

13	Web Tier	Optional	Web application server and web server combinations that enable users to connect via a web browser. Third-party server software that provides the HTTP or HTTPS content to Service Manager web clients. Some web application servers also include built-in or bundled web servers.

Components

A Service Manager installation may include the following components:

- Core components
  - Server
  - Windows Client
    Install at least one Windows Client for the system administrator or group of power users. The Windows Client provides some system administration functionalities that aren't available from the Web Tier client.
  - Web Tier
    The Web Tier client provides several views: Power User, Employee Self-Service (ESS), Accessible, and Self-Service Accessible.
  - Database
  - Applications
- Other components
  - Language Packs: A language pack is required when users need to use a localized version of Service Manager.
  - Smart Analytics
    To enable Knowledge Management search, install Smart Analytics.
  - SMA Service Portal
    SMA Service Portal can be used as a replacement for the ESS user portal. Both of them are a user portal through which users can submit support requests, order catalog items, search knowledge, and complete surveys.
  - Collaboration
    Collaboration is an instant messaging solution combining IT Collaboration and End User Chat.
  - Identity Manager (IdM) service
    The IdM service is required only if you want to enable SAML Single Sign-On for Service Manager.

JRE support

As of version 9.50, Service Manager (SM) adds the support of Open Java Development Kit (OpenJDK) for all SM components that require a JRE or JDK to work. For more information, see JRE support.

1.1.2. Sizing

This section provides hardware sizing practices, existing tuning information, and high availability recommendations to customers who want to implement Service Manager 9.80. All references are based on the benchmark test results and test data in the research & development laboratory (R&D Lab).

The reference configuration data supplied in this document is based on a Service Manager out-of-box (OOB) environment, which includes the Web tier, Mobility, and Smart Analytics running on top of the Service Manager Runtime Environment (RTE).

Individual implementations may consume more resources or require more resources to perform in an acceptable manner.

Failure to test the application with the concurrent user load and transaction rate that is expected at the height of daily system usage and using tailored applications in the test environment may result in an environment that is not large enough to support your requirements.

The recommendations described in this document should be considered the minimum requirements that are needed to run Service Manager effectively.

Terminology

Term	Description
SM	Service Manager.
RTE	Service Manager Runtime Environment.
Concurrent users	Online users who have logged on to the system to perform operations and consume system processing resources. It is different from registered users.
Workload	User activities that are performed against the system.
Throughput	The frequency that end users perform business transactions per hour. For example, 100 orders per hour, 100 requests per hour, and so on.
Test data	Business data populated into database to simulate the customers' data volumes.
Benchmark test	A kind of performance testing method to determine a system's performance quality. In general, it utilizes a load test tool to simulate the concurrent users performing daily operations against the underneath system.
CHO test	CHO stands for Continue Hours Operation, extend the benchmark test to between 12 and 72 hours.

Service Manager sizing questions

The following list of questions is designed to gather the information that is required to make recommendations about the overall system architecture that you need for your Service Manager environment.



Note

Although not all of the information requested here is required to size the environment, it is useful information to obtain for a thorough architecture recommendation.

1. What kind of environment will you use?
    1. An on-premise solution administered by internal IT
    2. An on-premise solution deployed on AWS/Azure cloud infrastructure
    3. Software as a service
2. What is the expected hardware and software environment?
    1. Do you plan to operate in separate development, test, and production environments to ensure quality?
    2. Do you have any ITSCM, disaster recovery, or high availability requirements?
    3. Do you plan to operate in a virtual environment or on physical machines?
3. Do you have existing hardware that you want to reuse?
    1. What operating systems are you using?
    2. How many CPUs are there per computer?

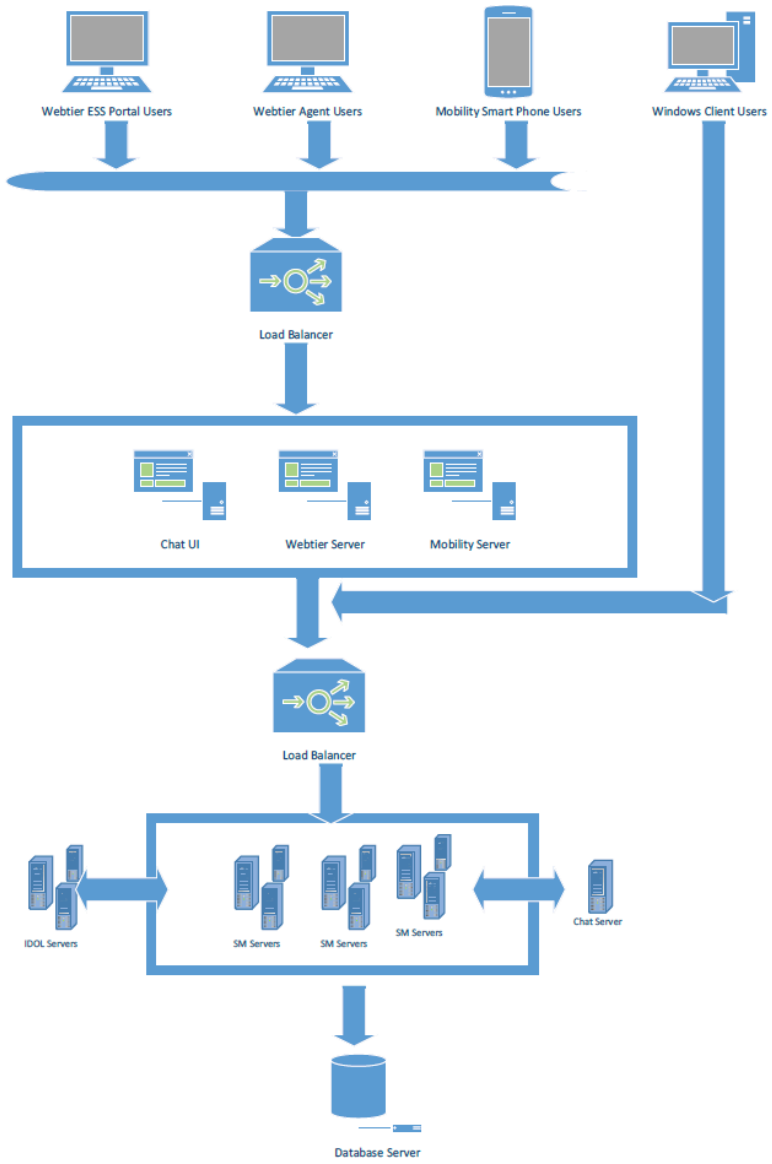
3. How much RAM is available per computer?
4. What is the Storage configuration?
5. Any hardware load balancer existing? Like F5.
6. What relational database management system are you using?

4. What is the average/worst network latency in your environment?
5. Can you provide a Microsoft Visio diagram of your network with minimum latency and bandwidth values (see the example diagram)?
6. Do you plan to use any of the following integrations together with Service Manager? If so, which ones?
    1. Inbound/outbound email, SMTP, POP3, IMAP or EWS
    2. Active Directory (LDAP) integration or single sign-on
    3. Data import of persons or organizations from an HR or other environment
    4. Integration to Universal Configuration Management Database (UCMDB)/CMS
    5. Integration to other **OpenText** Software solutions
    6. Imported configuration items (CIs)
    7. Others
7. Do you have licensing requirements for any of the following?
    1. Service Catalog
    2. Knowledge Management. If yes, what is the expected number of Knowledge Management users (authors, editors, and administrators)?
    3. Additional languages. If yes, which languages do you expect to operate?
    4. IT specialists (Technicians, Administrators, Helpdesk). If yes, what is the overall number of IT specialists? (Technicians, Administrators, Helpdesk)
    5. Named Users. If yes, how many of them should have guaranteed access to Service Manager?
    6. What are your module-level user requirements for Service Manager?
8. How many concurrent users you are expecting to support? Average and peak numbers are expected.
    1. How many users for SM window client?
    2. How many users for SM web tier client?
    3. How many users for SM mobility client?
    4. How many users for SM Collaboration client?
    5. >How many users for Teams Bot client?
    6. How many users for Smart Analytics?
9. What is the geographical breakdown of your user base for each kind of client?
10. What is the expected business transactions throughput in production environment? Average and peak numbers are expected.
    1. How many logins per hour?
    2. How many interactions will be generated per hour?
    3. How many incidents will be generated per hour?
    4. How many requests will be generated per hour?
    5. How many changes will be generated per hour?
    6. How many catalog orders will be generated per hour?
    7. How many smart search requests will be generated per hour?
11. What are your expected data volumes (including attachments) for the following modules?
    1. Service Desk
    2. Incident Management
    3. Change Management
    4. Problem Management
    5. Knowledge Management
    6. Request Management

7. Configuration Management
8. Service Level Management

12. What are your reporting requirements?
    1. An external reporting solution
    2. Data replication into a data warehouse

Service Manager sample deployment



Prepare for sizing

Sizing is the process of determining the underneath hardware resources to support the business requirement targets in production, for both the present and the future. No matter whether you need to build a web farm or not, create a database cluster or adopt new hardware devices such as F5 highly depends on your actual business requirements. High availability or resiliency will produce different hardware configurations for different scalability requirements. Customers should consider their business growth in the future 3 to 5 years, and plan enough hardware resources to accommodate these changes. In general, customers need to keep at least 30% or more capacity for the future growth and refer to the system history to evaluate these growth.

For new customers from scratch, sizing recommendations in this guide should be treated as the minimum reference for deployment. For existing customers who want to upgrade their SM to the latest version, we strongly suggest them to firstly

learn the statistics of the current hardware usage and bottlenecks existing in their production, carefully plan their additional hardware adoption or changes to address existing issues, and then refer to this guide as complementary.

Before diving into the sizing part, you need to understand the key metrics utilized in this guide to provide recommendations. These metrics include concurrent users load, business throughput and backend database data volume size. This guide will provide corresponding values of these metrics for your comparison as well as the related references. For customers who have tremendous different requirements on the metrics, we suggest to implement individual load and stability tests to make sure that no undersized hardware resource are planned based on the minimum hardware reference.

See the following detail steps for each sizing decision:

1. Set expected business needs.
2. Get target deployment size level.
3. Retrieve recommended minimum hardware configuration.
4. Consider redundancy for high availability, or design disaster recovery process to prevent data lost.
5. Refer to **Rules of Thumb** for additional sizing information and tuning practices.

Hardware - size for all-in-one deployment (75 users)

All-in-one deployment is a typical requirement for customers with less than 75 concurrent users' load for demo and test purpose. Aiming to deploy all services on one server with less maintenance efforts, this deployment method has some limitations in high availability and scaling in the future. As a result, we do not recommend an all-in-one deployment for SM in production environment.

At minimum, two servers are recommended to set up all-in-one deployment in production. See the following table:

Usage	CPU / RAM	Deployed Services	OS
SM Clients and Services	8 CPU cores / 16 GB RAM	SM RTE, SM Web Tier, Mobility , SM Collaboration, Smart Analytics	Windows or Linux
SM RDBMS	8 CPU cores / 16 GB RAM	SM database like Oracle, MS SQL Server, PostgreSQL	Windows or Linux

Additional considerations:

- All servers must be in the same subnet, and the network latency (especially between SM Services/SM RDBMS) must be lower than 1ms.
- All the servers can be deployed on either virtual machines or physical machines.
- Set JVM heap size to 1.5 GB as the startup option for each web instance when deploying SM Webtier, Mobility, ChatUI separately.

Set JVM heap size to 3 GB as the startup option for the web instance when deploying SM Webtier, Mobility, ChatUI together.

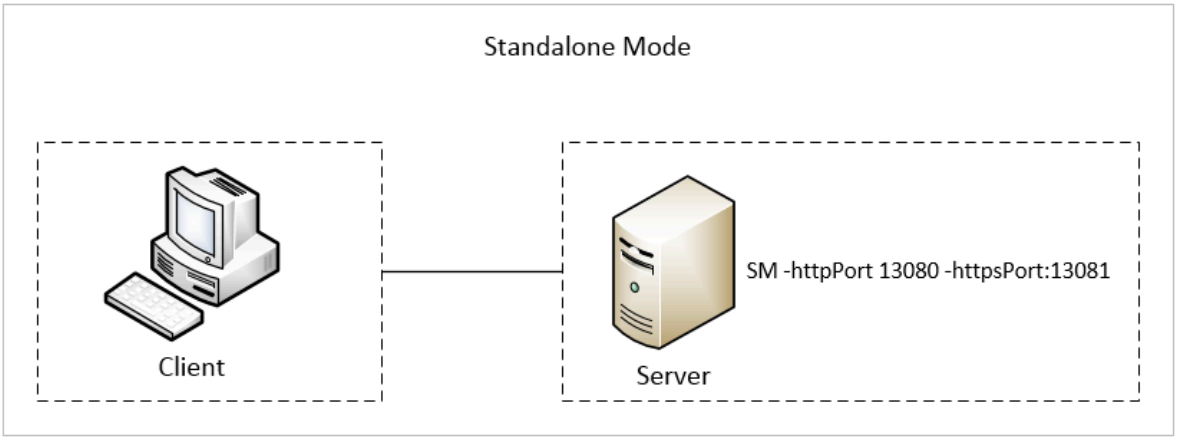
Service Manager server and Web tier deployment sizing

Deployment mode

Service Manager application server and load balancer

Standalone mode

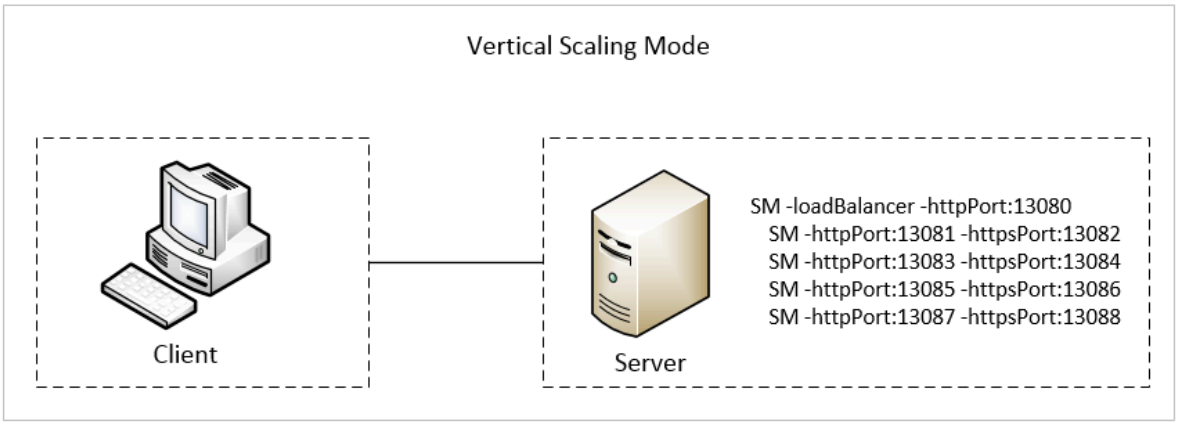
The SM application standalone mode comprises only one SM servlet configured on a SM server that has no software load balancer enabled. This is the simplest deployment mode, you may consider it the base model. SM Windows client/webtier/mobility or any other integration points which need to consume SM application service can be configured to connect to this servlet's open ports directly.



Vertical scaling mode

The SM vertical scaling mode comprises multiple SM servlet container processes running on one SM server that has a software load balancer enabled. This implementation is typically used in an environment with 200-600 concurrent users, where hardware system resources are limited. This implementation supports as many client connections as the Service Manager server has available system resources.

The following diagram demonstrates how SM load balancer deployed with multiple SM servlets on a single server.

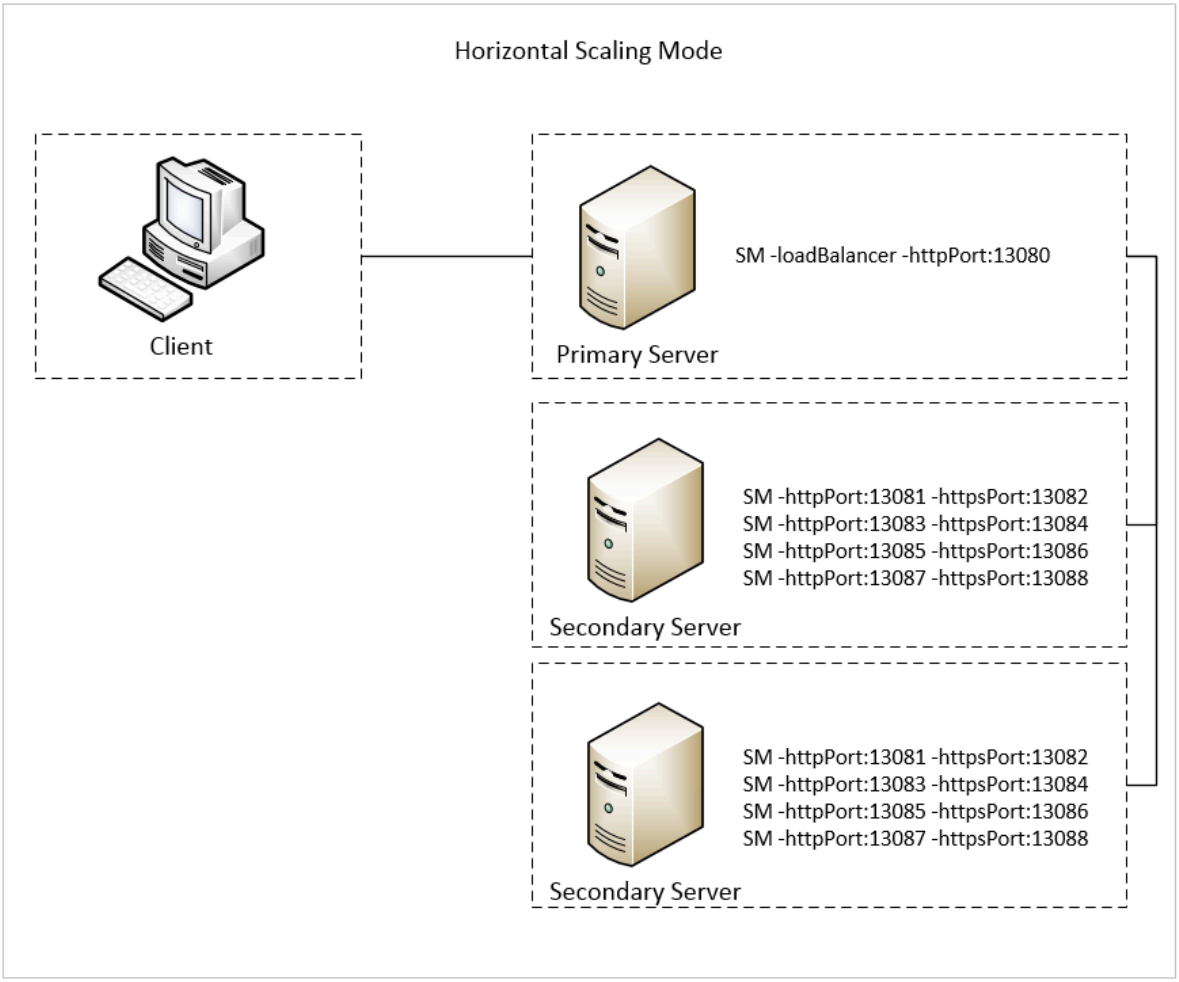


Horizontal scaling mode

The SM horizontal scaling mode comprises several SM servers managed by group parameters. Multiple vertically-scaled Service Manager hosts can be converted to the horizontal scaling mode. Only one software load balancer port needs to be configured. The host that obtains a license file is the primary host, and the other hosts are secondary hosts.

A horizontal scaling implementation is typically used in large environment in which system scalability and resilience is a concern. A horizontal scaling implementation can support as many client connections as the sum of the individual Service Manager instances can support. For example, if each instance can support 250 client connections and there are 6 instances, the system can support 1500 client connections. A horizontal scaling implementation has improved resilience features, such as the ability to stop and start the load balancer process without causing a total system outage and the ability to add a new servlet container process to a virtual group when the system is running.

The following diagram demonstrates how SM load balancer deployed with multiple SM servlets on several servers.



Note

In the horizontal scaling mode, the number of supported concurrent users in Service Manager may reach 10,000. Lab has certified 10,000 concurrent users on Windows server 2019 and Oracle 19c RDBMS.

Service Manager Web tier

Service Manager Web tier is a web application packaged as a WAR that needs to be deployed under web containers such as Apache Tomcat. These containers provide great capabilities on scalability and clustering support. Similar to SM application server, web tier also supports single node deployment, vertical scaling on a big server, and horizontal scaling on clustering of several servers. Refer to the official documents released by the container vendors for the supported load balancers.

Minimum required reference configurations

The following table shows the test data used in lab benchmark and CHO testing. You need to first compare your actual business needs with the data in the table, and then refer to the hardware requirements to plan your deployment sizing.

Index.do

Concurrent users	200	500	1000	2000	3000	10000
Login / hour	90	225	450	900	1350	4500
Interactions / hour	110	275	560	1120	1670	5585
Incidents / hour	55	140	285	575	865	2890
Changes / hour	20	45	95	190	285	960
Problems / hour	20	45	90	190	285	960
Requests / hour	65	155	320	640	1000	3225

Ess.do

Concurrent users	200	500	1000	2000	3000	10000
Login / hour	105	255	515	1030	1545	5160
Order from catalog / hour	75	190	385	765	1155	3855
Submit a request / hour	230	575	1165	2330	3485	11630

Windows server platform

Concurrent users	200	500	1000	2000	3000	10000
Apache	N/A	2 CPU 4GB RAM	2 CPU 4GB RAM	2 CPU 4GB RAM	2 CPU 4GB RAM	2 CPU 4GB RAM
Webtier	2 CPU 4GB RAM	2 CPU 8GB RAM	4 CPU 12GB RAM	8 CPU 16GB RAM	8 CPU 24GB RAM	12 CPU 56GB RAM
SM load balancer	N/A	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM
SM RTE server	4 CPU 16GB RAM	4 CPU 32GB RAM	8 CPU 64GB RAM	12 CPU 128GB RAM	16 CPU 160GB RAM	48 CPU 560GB RAM
SM background	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	4 CPU 8GB RAM	4 CPU 12GB RAM
Oracle 19c	4 CPU 12GB RAM	4 CPU 24GB RAM	6 CPU 48GB RAM	8 CPU 72GB RAM	12 CPU 108GB RAM	32 CPU 192GB RAM

Linux server platform

Concurrent users	200	500	1000	2000	3000
Apache	NA	2 CPU 4GB RAM	2 CPU 4GB RAM	2 CPU 4GB RAM	2 CPU 4GB RAM
Webtier	2 CPU 4GB RAM	2 CPU 8GB RAM	4 CPU 12GB RAM	8 CPU 16GB RAM	8 CPU 24GB RAM
SM load balancer	NA	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM
SM RTE server	4 CPU 20GB RAM	4 CPU 48GB RAM	8 CPU 84GB RAM	12 CPU 144GB RAM	16 CPU 240GB RAM
SM background	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	2 CPU 8GB RAM	4 CPU 8GB RAM
Oracle 19c	4 CPU 12GB RAM	4 CPU 24GB RAM	6 CPU 48GB RAM	8 CPU 72GB RAM	12 CPU 108GB RAM

Note

1. The values provided in the above table have been verified in R&D Lab benchmark testing. If you have different metrics requirements, please adjust hardware size accordingly. See **Service Manager Server (Servlet Container)** for more information.
2. Test data utilized in R&D Lab is 5 million for all kinds of benchmark tests with different concurrent users loads.
3. We recommend that customers who use Service Manager Reports heavily configure another replicated database with the same hardware configuration so as to reduce the performance impact.
4. The test results in the above table reflect the values for Oracle databases. If you use other RDBMS, you can refer to the hardware configurations of Oracle as minimum hardware requirements to plan your performance test.
5. HugePages is configured for Oracle on Linux in R&D Lab.

Service Manager virtualization

To efficiently run all Service Manager components in a virtual environment, make sure there is no resource competition which may affect system stability and performance.

Service Manager Mobility deployment sizing

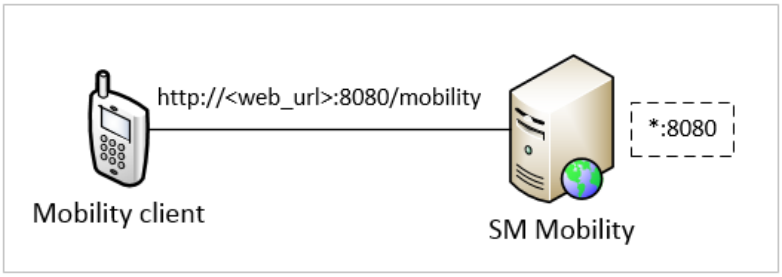
Deployment mode

The Mobility application deployment is based on a specific container. In this guide, we provide Tomcat as an example. Therefore, the deployment's scalability is the same as that of Tomcat containers.

- Standalone mode

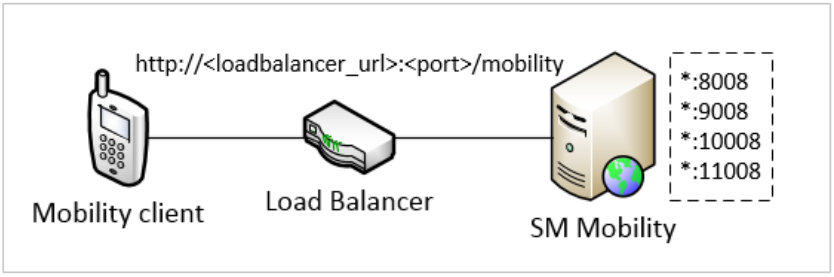
In the standalone mode, only one Mobility server instance is configured. This Mobility server provides services for all incoming requests. To achieve high availability capability, you need to add redundant Mobility servers with same

configurations and adopt load balancer between the clients and the servers.



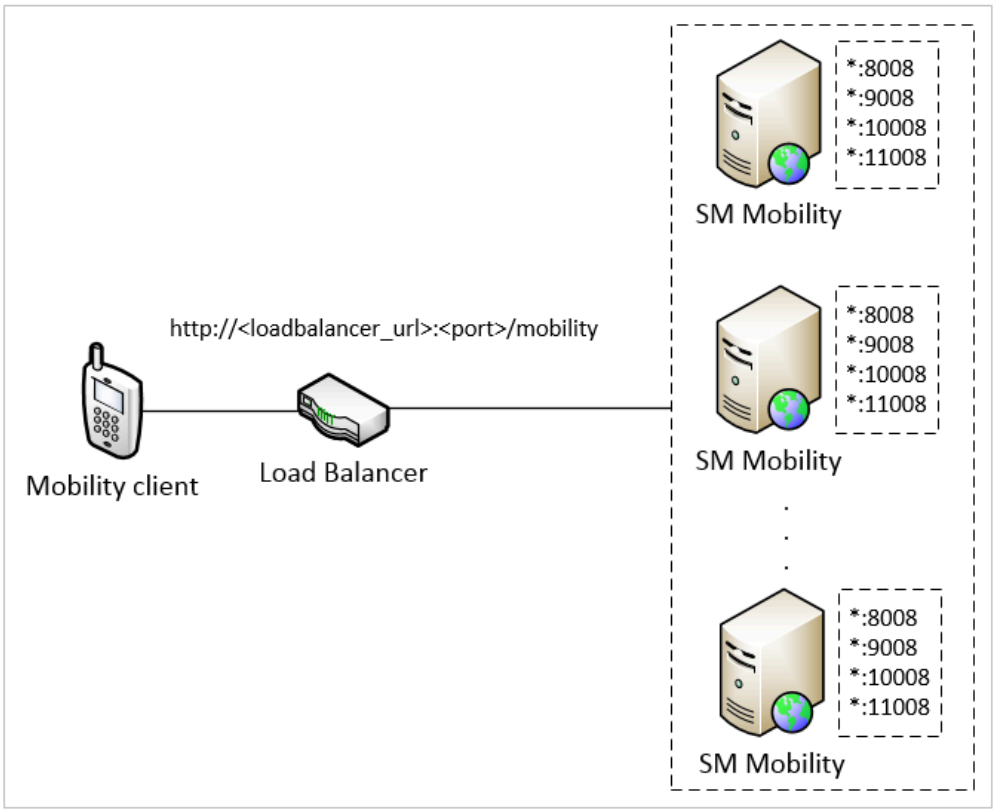
- Vertical scaling mode

To extend Mobility application capabilities, you need to add more instances on the same server and add another load balancer between the clients and the servers. To achieve high availability, you need to add redundant Mobility servers.



- Horizontal scaling mode

If you have a large number of customers, you can set up a Mobility web farm to achieve high availability capabilities in the horizontal scaling mode.



Service Manager Smart Analytics deployment sizing

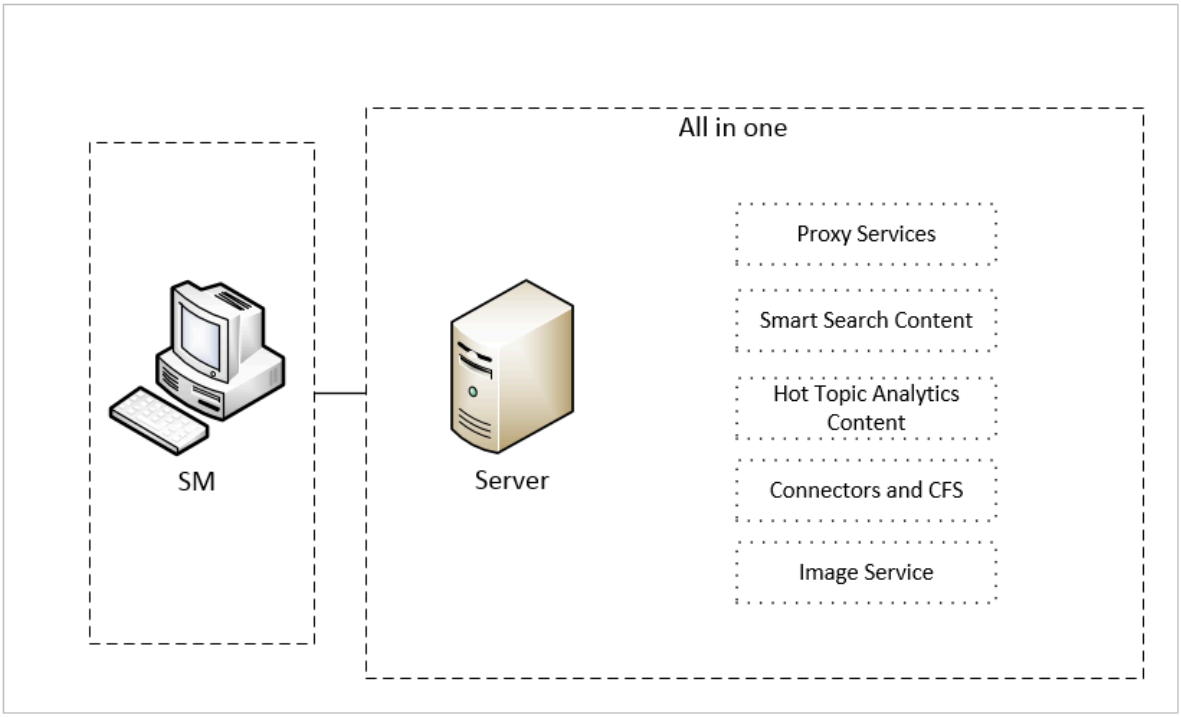
Deployment mode

Service Manager Smart Analytics consists of various components and services to provide different kinds of indexing and searching needs. The main components are as follows:

- Proxy - Includes main proxy, media(image) proxy, and level II proxies such as Smart Search proxy, Hot Topic Analytics proxy.
- Content server - An indexed data storage in IDOL.
- Media(Image) server - A special content server to provide OCR (Optical Character Recognition) capabilities.
- Connectors - IDOL utilizes different kinds of connectors to index and search through external resources such as Sharepoint, Wiki pages, files, and so on.
- CFS - A special service to process attachment of document.

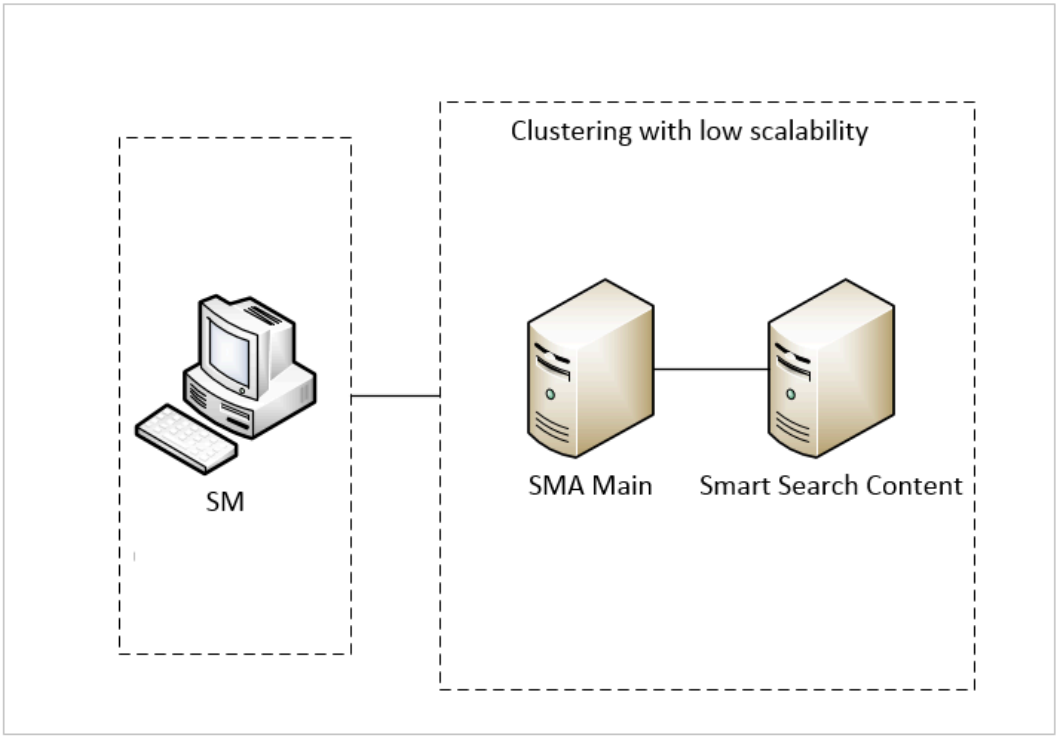
Depending on the data volumes size and the searching service requirements, there are several different topologies available for customers to choose.

- **All-in-one**
  This is the minimum deployment which requires only one server to host proxy services, two Smart Search contents, one HTA content, and media(image) service, connectors, and so on. You must be aware that this deployment has limitations on the high availability support. That is, you cannot avoid single point of failure when the server is down or crashed. Also, we do not suggest this deployment if you have heavy or unpredictable scaling requirements in the future. However, you can still extend capabilities by adding more hardware resources like CPU/RAM/Storage to scale up on the single server.



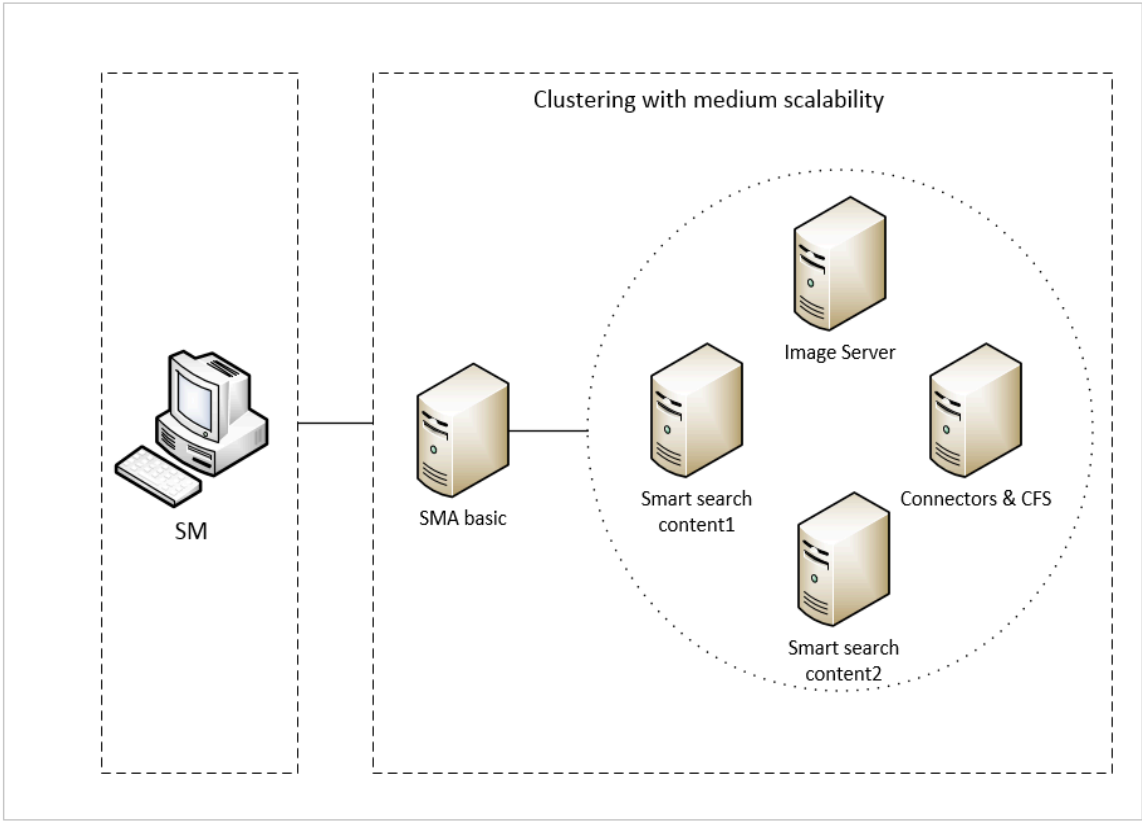
- **Clustering with low scalability**
  This setup is an extended version based on all-in-one deployment. It is for small to medium deployment requirements with limited scalability in the future. It is the basic deployment when you only need 1 to 2 additional content servers besides the all-in-one deployment. Note that the first server need to be configured stronger since it has higher request processing load than the additional content servers do. Similar to all-in-one, you can also extend capabilities by adding more hardware resources such as CPU/RAM/storage to scale up on the single server.
  At minimum, 2 servers are suggested to set up this clustering.
  - 1 x server with all-in-one deployment (SMA Main server)

- 1 x content server for Smart Search



- **Clustering with medium scalability**
  This setup is recommended for customers who has much more requirements on Smart Search services and images than others. It aims to scale out Smart Search content and image servers in the future.
  At minimum, 5 servers are suggested to set up this clustering.
  - 1 x server with proxy, HTA content
  - 2 x content server for Smart Search.
  - 1 x server for media(image).

- 1 x server for connectors and relating CFS service.

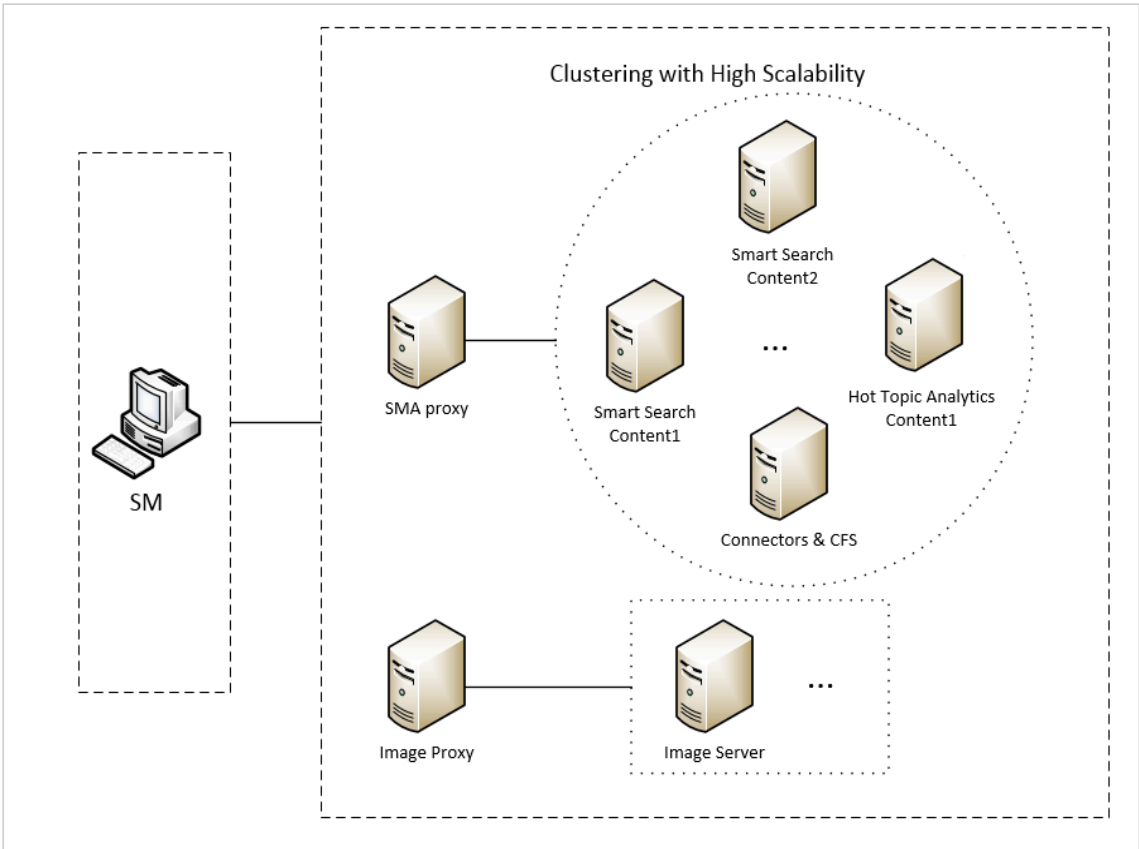


Note:

1. If you have high requirements on HTA, we suggest to split these services from SMA basic server to separated servers. This is more like clustering with high scalability as described below.
2. If you do not have high requirements on image server and connectors, you can even install these services on one computer instead of different computers.

- **Clustering with high scalability**
  This deployment mode considers your large data volumes size, heavy usage on Smart Analytics, and the scaling requirements in the future. We recommend this deployment if you have high scalability requirements on each service. The main logic is to set up a whole clustering but split different service components to different computers. At minimum, 8 servers are suggested to set up this clustering.
  - 1 x server to host proxy services and QMS, including main proxy, level II proxy, and so on.
  - 2 x Smart Search content server.
  - 1 x Hot Topic Analytics content server.
  - 1 x server to host connectors and CFS.
  - 1 x server to host media(image) proxy.

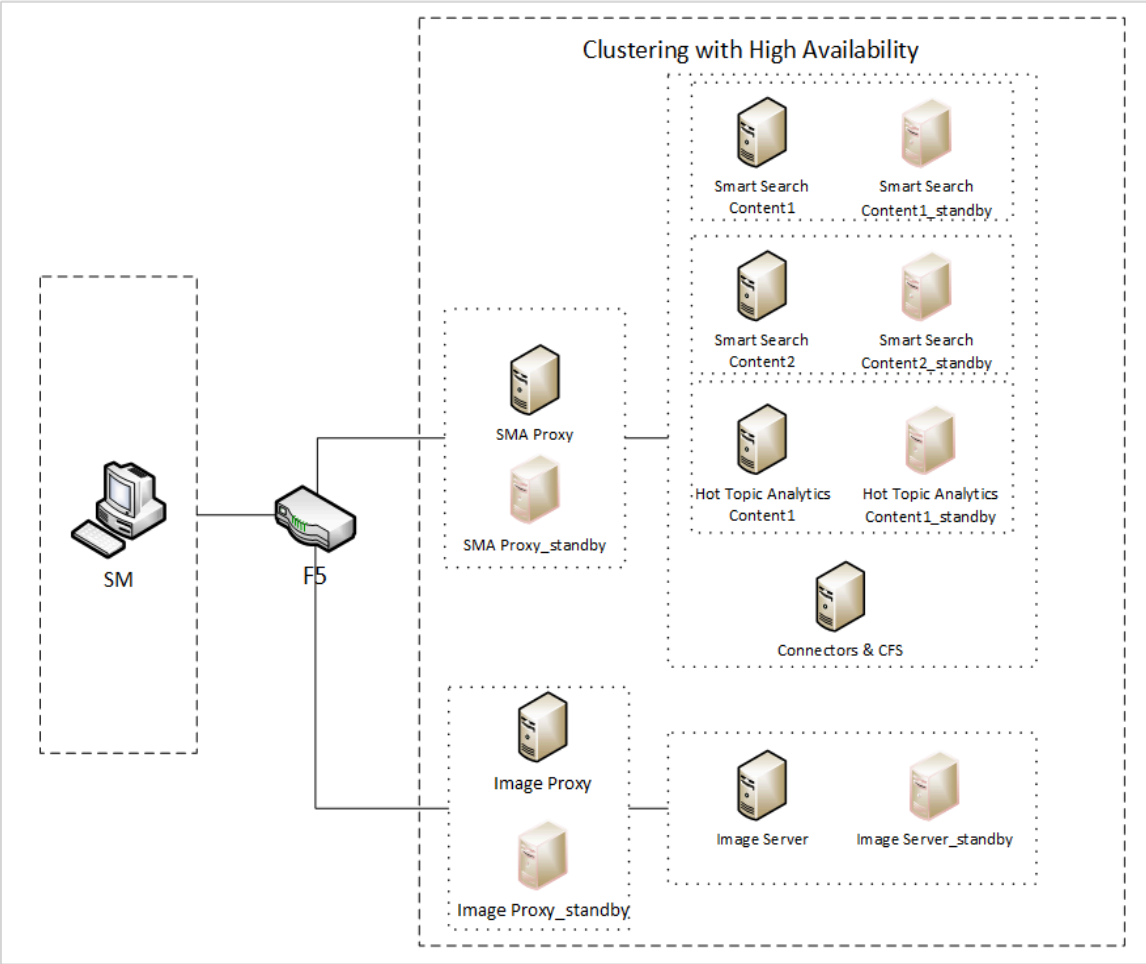
- 1 x server to host media(image) service.



With the above configurations, you can easily extend capabilities by adding more content servers/image servers to clustering to accommodate growing data volume size and concurrency in the future.

- **Clustering with high availability**
  If you have high availability needs, SMA provides another kind of deployment by adding redundant servers on proxy and content server layers.
  At minimum, 15 servers are suggested to setup this clustering.
  - 2 x servers with the same configurations to host proxy services and QMS, including main proxy, level II proxy, and so on.
  - 8 x content servers.
    - 4 x Smart Search content servers with fail-over enabled.
    - 2 x Hot Topic Analytics content servers with fail-over enabled.
  - 1 x server to host connectors and CFS.
  - 2 x servers with the same configurations to host media(image) proxy.

- 2 x servers to host media(image) services.



Minimum required reference configurations

Service Manager Smart Analytics provides plenty of features, such as Smart Search, Smart Ticket and Hot Topic Analytics.The guidelines provided below suppose that you will utilize all these features.

Firstly, you need to consider your expected business needs. There are several factors which need your attention.

- **Indexed data volume size in SMA IDOL**
  There is an upper limit on the data volume for each content server. The maximum size for each content server is 3 million. To achieve a better performance, we still suggest 2 million for each content server especially when you have huge data volume size requirements.
  In this guide, all the recommendations are based on 2 million data size for each content server. For example, if you have 5 to 6 million indexed data requirement, we will suggest 3 content servers for setup.
  From experience, the data volume size needed for Smart Search is much higher than that for Hot Topic Analytics. For customers from scratch, you can plan 10% to 20% data size needs of Smart Search for Hot Topic Analytics
- **Workload**
  There are different kinds of workloads which utilize SMA.
  - Global search
  - Context search
  - Hot Topic Analytics (HTA) search
  - Smart Ticket
  - KM search

From experience, the workloads for global search and context search are much heavier than others. For customers from scratch on Smart Search, you can plan 80% users load on global search, and then plan 20% users load on context search.

Since Smart Search consumes the most workload, the sizing is mainly related to this part. For others, the minimum required references recommended are generally sufficient.

Metrics	Small	Medium	Large	Very Large
Concurrent users load	200	600	1000	2500
Global searches / hour	2400	7200	12000	30000
Context searches / hour	1120	3360	5600	14000
Test Data in R&D Lab	1.5 million	3.5 million	6 million	8 million



Tip

1. If you have different metrics values from the above table and cannot find recommended sizing information, see **Service Manager Smart Analytics**.
2. We suggest you provide a storage instead of local disk for indexed data in IDOL to achieve best performance. Smart Analytics content servers has high requirements for the IO system.

- **Small**
  Windows and Linux
  1 x Smart Analytics: 8 CPU cores, 16GB RAM, 120GB HD
  Deployment: All-in-one. Refer to the **all-in-one** deployment mode.
- **Medium**
  Windows and Linux
  1 x Smart Analytics A: 8 CPU cores, 12GB RAM, 120GB HD
  1 x Smart Analytics B: 6 CPU cores, 8GB RAM, 100GB HD
  Deployment: Refer to the **clustering with low scalability** deployment mode.
- **Large**
  Windows and Linux
  1 x Smart Analytics Proxy: 4 CPU cores, 4GB RAM, 80GB HD
  3 x Smart Search Content: 6 CPU cores, 8GB RAM, 100GB HD
  1 x Hot Topic Analytics Content: 6 CPU cores, 8GB RAM, 100GB HD
  1 x Smart Analytics Connectors and CFS: 6 CPU cores, 8GB RAM, 60GB HD
  1 x Media(Image) Proxy: 4 CPU cores, 4GB RAM, 50GB HD
  1 x Media(Image) Server: 8 CPU cores, 8GB RAM, 50GB HD
  Deployment: Refer to the **clustering with high scalability** deployment mode. If you want to implement high availability, add redundant servers for each content and proxy and refer to the **clustering with high availability** deployment mode.

Note

1. The above configuration is based on the assumption that customers will have high business requirements on all SMA services. For customers who only have partial requirements on certain services, please adjust hardware resources accordingly.
2. For example, if you have high requirements on Smart Search only, you can refer to the **clustering with medium scalability** deployment mode and reduce the necessary computers. The minimum required reference configurations are as follows:
   1 x Smart Analytics Proxy, HTA: 8 CPU cores, 8GB RAM, 100GB HD
   3 x Smart Search Content: 6 CPU cores, 8GB RAM, 100GB HD
   1 x Smart Analytics Connectors and CFS: 6 CPU cores, 8GB RAM, 60GB HD
   1 x Media(Image) Server: 8 CPU cores, 8GB RAM, 50GB HD

- **Very large**
  Windows and Linux
  1 x Smart Analytics Proxy: 4 CPU cores, 4GB RAM, 80GB HD
  1 x Media(Image) Proxy: 4 CPU cores, 4GB RAM, 50GB HD
  4 x Smart Search Content: 8 CPU cores, 8GB RAM, 100GB HD
  1 x Hot Topic Analytics Content: 8 CPU cores, 8GB RAM, 100GB HD
  1 x Smart Analytics Connectors and CFS: 8 CPU cores, 8GB RAM, 60GB HD
  1 x Media(Image) Server: 8 CPU cores, 8GB RAM, 50GB HD

Deployment: Refer to the **clustering with high scalability** deployment mode. If you want to implement high availability, add redundant servers for each content and proxy and then refer to the **clustering with high availability deployment** mode.

Note

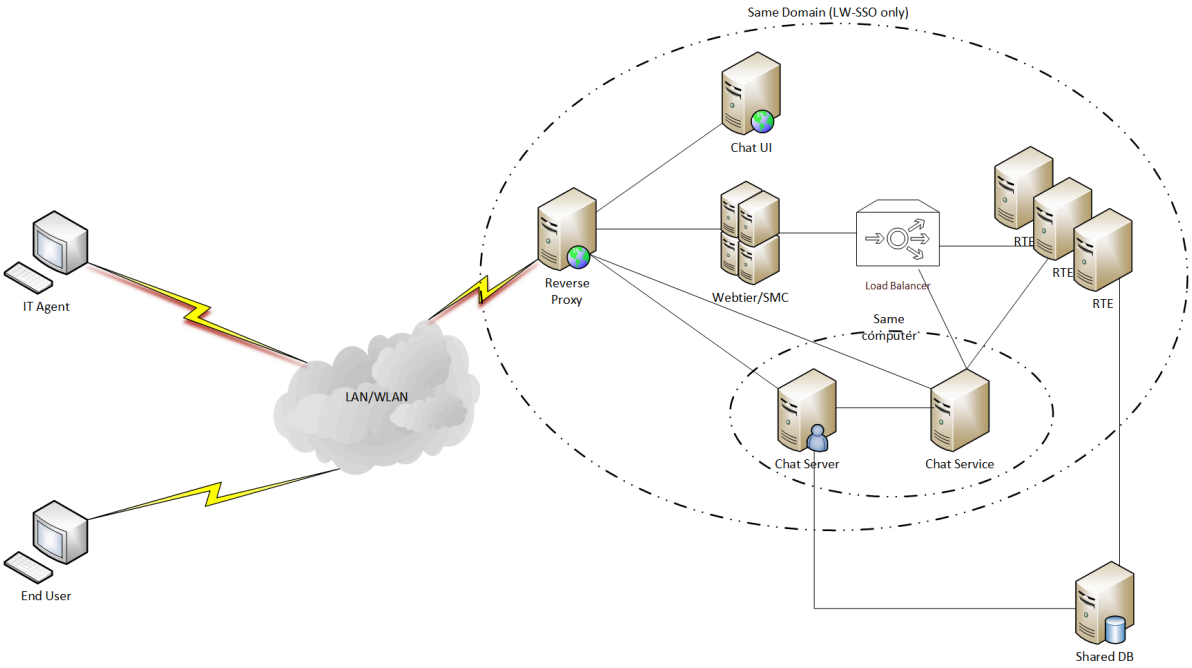
1. The above configuration is based on the assumption that customers will have high business requirements on all SMA services. For customers who only have partial requirements on certain services, please adjust hardware resources accordingly.
2. For example, if you have high requirements on Smart Search only, you can refer to the **clustering with medium scalability** deployment mode and reduce the necessary computers. The minimum required reference configurations are as follows:
   1 x Smart Analytics Proxy, HTA: 8 CPU cores, 8GB RAM, 100GB HD
   3 x Smart Search Content: 6 CPU cores, 8GB RAM, 100GB HD
   1 x Smart Analytics Connectors and CFS: 6 CPU cores, 8GB RAM, 60GB HD
   1 x Media(Image) Server: 8 CPU cores, 8GB RAM, 50GB HD

- For deployment requirements which exceed the above sizes, we recommend that you adopt the **clustering with medium scalability** or **clustering with high scalability** deployment mode and scale out by adding more content servers with the following configuration. See **Service Manager Smart Analytics** for more information.
  Windows and Linux
  1 x Smart Analytics Proxy: 4 CPU cores, 4GB RAM, 80GB HD
  1 x Media(Image) Proxy: 4 CPU cores, 4GB RAM, 50GB HD
  1 x Smart Analytics Connectors and CFS: 8 CPU cores, 8GB RAM, 60GB HD
  Content Server (for each): 8 CPU cores, 8GB RAM, 100GB HD
  Media(Image) Server (for each): 8 CPU cores, 8GB RAM, 50GB HD

Service Manager Collaboration deployment sizing

Deployment mode

Example SMC deployment diagram



Minimum required reference configurations

- Chat Server and Chat Service configuration
  Windows
  8 CPU cores, 16GB RAM
- We recommend that you install the Chat Server and the Chat Service on a dedicated server for better performance.
- JRE 11 (64-bit) is recommended for Chat Server and Chat Service.
- The performance results for Chat Server and Chat Service with Oracle JDK and Zulu OpenJDK are same.
- Windows (64-bit) is recommended for Chat Server and Chat Service.
- In Apache HTTP Server, update the parameters as follows:
  `ThreadLimit 6000`

  `ThreadsPerChild 6000`
- You need to configure the Chat Server database schema together with the Service Manager Database schema. In most cases, the Chat Server database schema requires 300MB of free disk space for every 1 million message history records.

Service Manager Teams integration deployment sizing

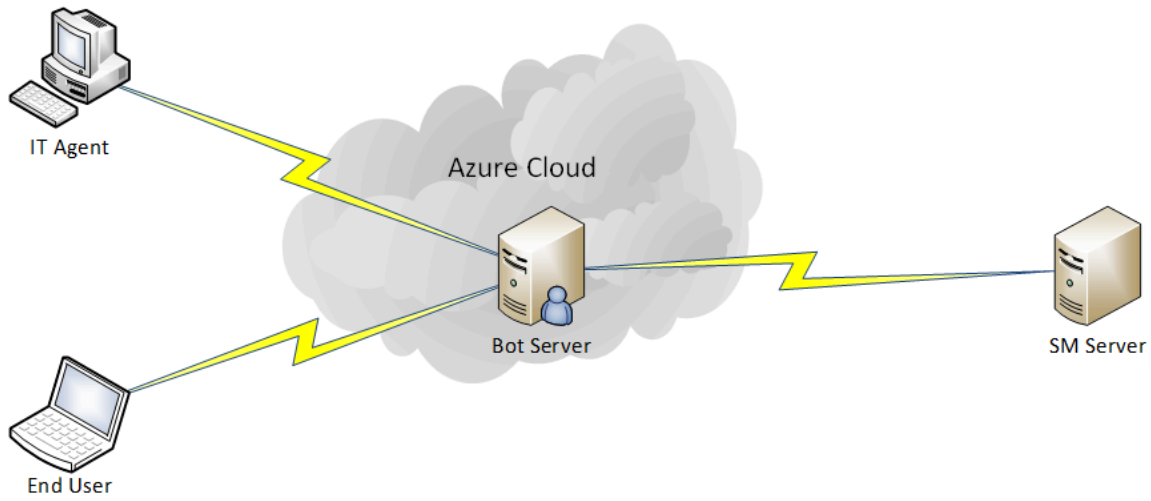
War room experience

In the integration for war room experience, SM server calls Microsoft Graph API through Azure application. Therefore, you don't need to set up a dedicated SM server for war room experience.

Note that the onboarding of war room experience will increase the workload on the Background Scheduler servlet. So when you set up the integration for war room experience, be sure to evaluate your existing resources and allocate some additional resources if required.

Agent Bot and Virtual Agent

Deployment mode



In addition to the existing SM server(s) for other integration points, You need to further set up dedicated SM server(s) for Teams Bot integration.

For the dedicated SM server(s) for Teams Bot, both standalone mode and scaling mode are supported.

- If the dedicated SM server is deployed in the standalone mode, you can deploy one servlet in the dedicated SM server to provide service for all incoming requests for Teams Bot integration.
- If the dedicated SM servers are deployed in the scaling mode, you can use an F5 hardware load balancer. Note that SM software load balancer isn't supported.

Reference configurations

Teams Bot client contains both Agent Bot and Virtual Agent users.
For the hardware requirement of the dedicated SM Server, you may compare your actual business needs with the reference data below:

Metrics	Small	Medium	Large
Concurrent Users	150	300	600
SM RTE Server	2 CPU / 8GB RAM	4 CPU / 12GB RAM	8 CPU / 24GB RAM

Session timeout configuration

Set session timeout for dao thread to 30 seconds:
`dao_sessiontimeout:30`

Azure bot server

Teams bot is deployed to the Azure cloud. The bot resource in Azure can be scaled up as described below:

In the Azure portal, go to bot **Scale up (App Service plan)**, and then configure the bot server resource according to your business requirements.

Service Manager Application Lifecycle Management (ALM) integration deployment sizing

In addition to the existing SM server(s) for other integration points, you need to further set up dedicated SM server(s) for ALM integration. Refer to the following table for the hardware requirement of the dedicated SM server(s):

Metrics	Small, Medium, or Large
SM RTE Server	2 CPU / 8GB RAM

Service Manager Jira Software (Jira) integration deployment sizing

In addition to the existing SM server(s) for other integration points, you need to further set up dedicated SM server(s) for Jira integration. Alternatively, you may add additional 2CPU and 8GB RAM to the main SM server and run the Jira Integration in a dedicated servlet.

Refer to the following table for the hardware requirement of the dedicated SM server(s):

Metrics	Small, Medium, or Large
SM RTE Server	2 CPU / 8GB RAM

You can refer to the *Minimum required reference configurations* table in the **Service Manager Web tier** section for the number of records per hour and concurrent users. For the number of threads in SMIS of the dedicated SM Server, you may determine the number of records for Jira synchronization and then compare your actual business needs with the reference data below.

Records per hour	Number of threads
Records/hour < 2000	1
2000 < Records/hour < 4000	2
4000 < Records/hour < 6000	3
6000 < Records/hour < 8000	4

Service Manager Smart Email deployment sizing

Deployment mode

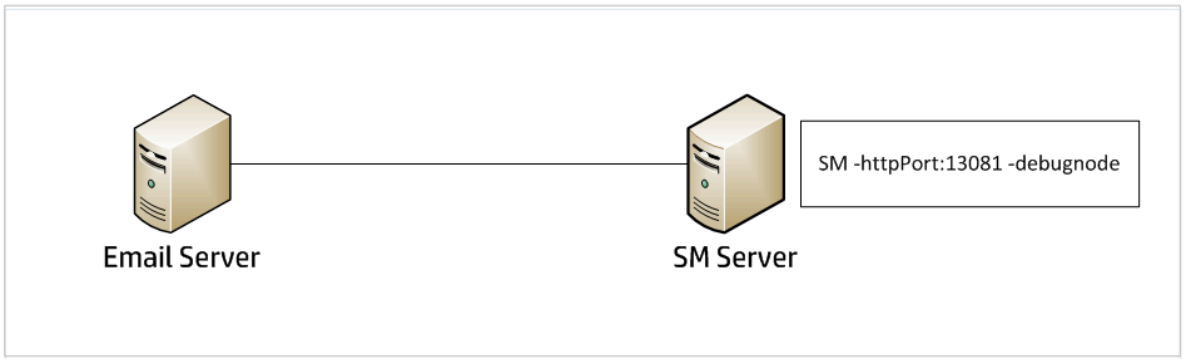
You need to configure the SM Smart Email servlets as a dedicated server and set the Smart Email servlets as debug nodes to isolate the SM load balance nodes.

We recommend that you start Smart Email process on a dedicated SM server to avoid performance impact and use a debug node to isolate all the SM nodes which are be used by Smart Email. A debug node can start 5 to 10 Smart Email Schedulers.

- **Standalone**

In the standalone mode, you need to configure only one SM Smart Email debug node to process all email tasks. To achieve high availability capability, you can add more debug nodes of the same configurations.

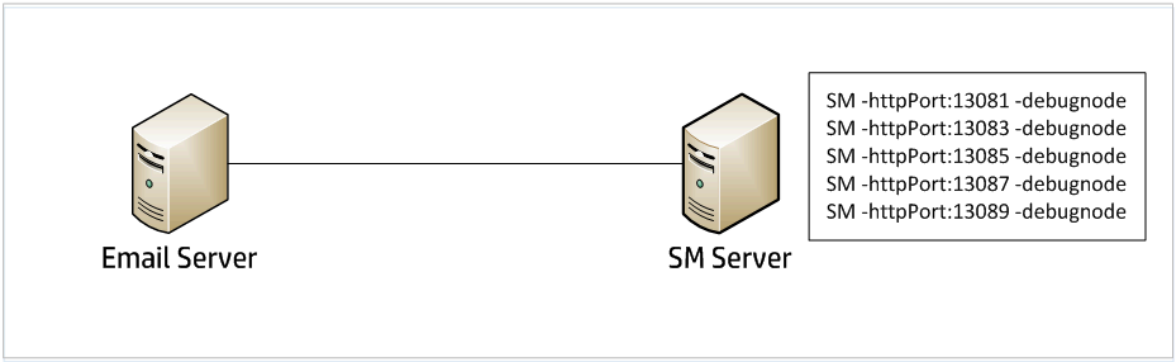
The email process rate is about 0.9 emails/minute in this mode. This value does not include reading emails from the network.



- **Vertical scaling**

You can extend Smart Email capabilities by adding more SM debug nodes to the same server and start other Smart Email Scheduler in new SM debug nodes.

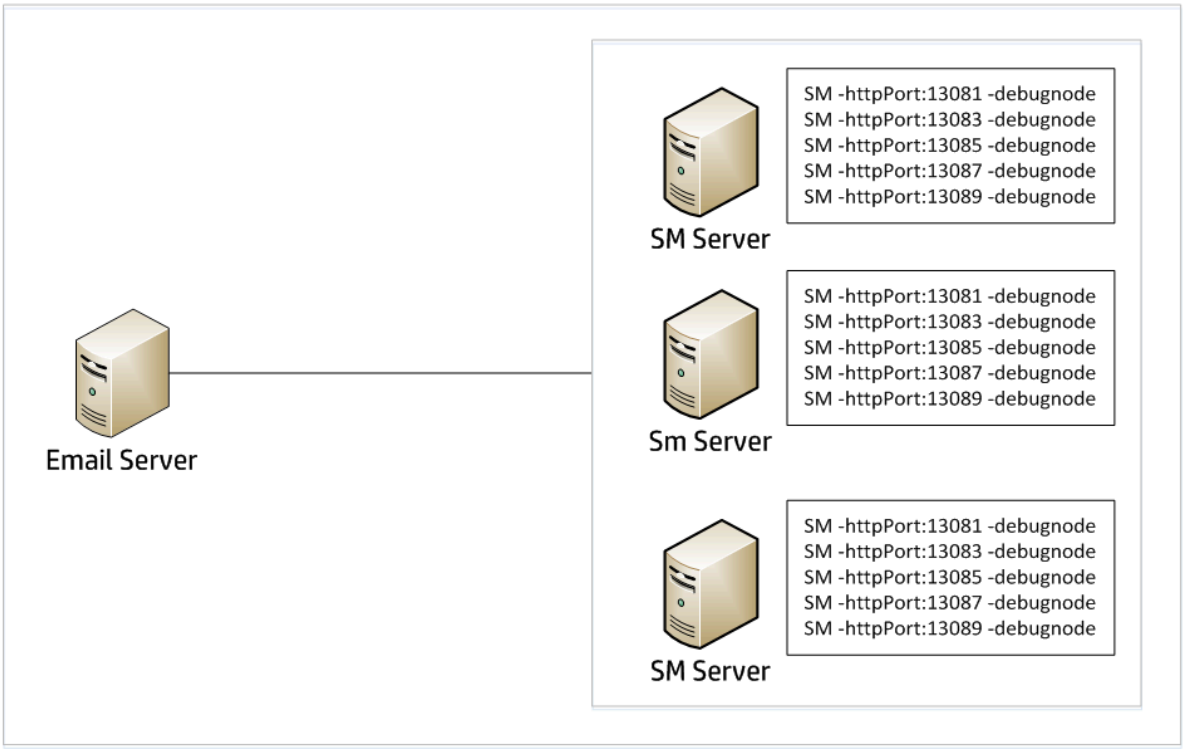
In this mode, you can configure 5 debug nodes at maximum and the email process rate is about 45 emails/minute. This value does not include reading emails from the network.



- **Horizontal scaling**

This deployment mode provides high availability capabilities. For large customers (>45 emails/minute), you can choose to set up a dedicated SM Server web farm for Smart Email.

In this mode, you can configure 5 debug nodes at maximum in each SM Server.



Minimum required reference configurations

Metrics	Small	Medium	>Large
Processed Emails / Hour	54	2700	> 2700

- **Small**
  Windows and Linux
  1 x SM Smart Email Server: 4 CPU Cores, 16GB RAM
  Deployment: standalone mode. 1 SM debug node is configured.
- **Medium**
  Windows and Linux

1 x SM Smart Email Server: 8 CPU Cores, 48GB RAM
Deployment: vertical scaling mode. 5 debug nodes can be configured at maximum.

- **Large**
  If you have higher requirements on Smart Email, you can extend by adding more dedicated servers based on the following configuration. 5 debug nodes can be configured at maximum in each SM Server.
  Windows and Linux
  SM Smart Email Server (for each): 8 CPU Cores, 48GB RAM
  Deployment: horizontal scaling mode. 5 debug nodes can be configured at maximum.

Service Manager IdM deployment sizing

Deployment mode

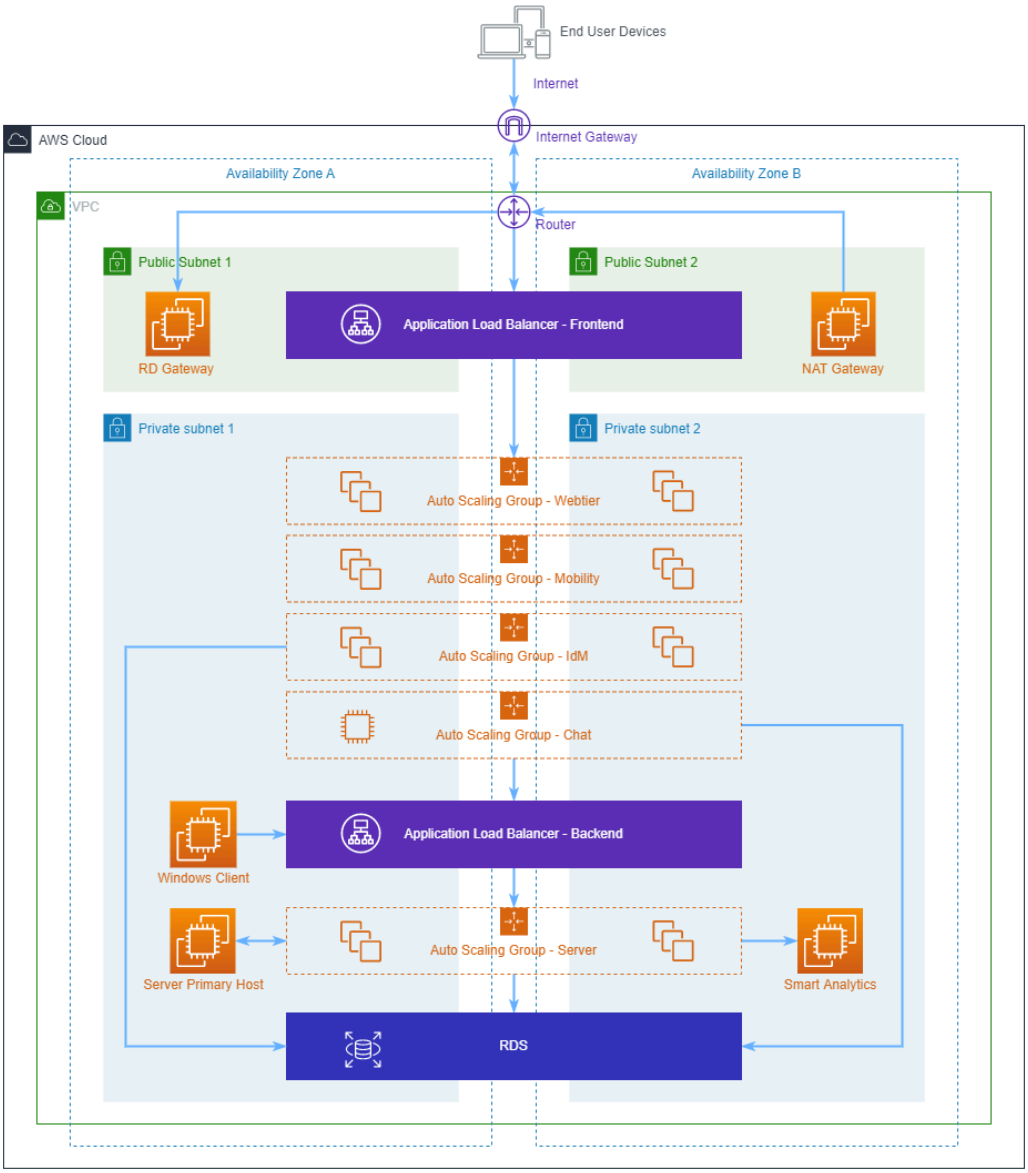
Service Manager (SM) leverages **OpenText** Identity Manager (IdM) to support Single Sign-On (SSO) using SAML 2.0 or OpenID Connect. IdM server supports single node deployment. Refer to the heap size for SM Web-tier as the same Java heap setting (`-Xms1536m -Xmx1536m`) for your deployment.

Minimum required reference configurations

- Windows:
  1 x IdM Server 4 CPU cores, 4GB RAM

Service Manager on AWS deployment sizing

Deployment mode



Reference configuration for 2000 concurrent users

Web tier

- EC2 instance type: 2 vCPUs, 4 GB RAM
- ASG desired capacity: 8

Server primary host

- EC2 instance type: 2 vCPUs, 8 GiB RAM
- EC2 instance count: 1

Server secondary hosts

- EC2 instance type: 4 vCPUs, 32GiB RAM
- ASG desired capacity: 5

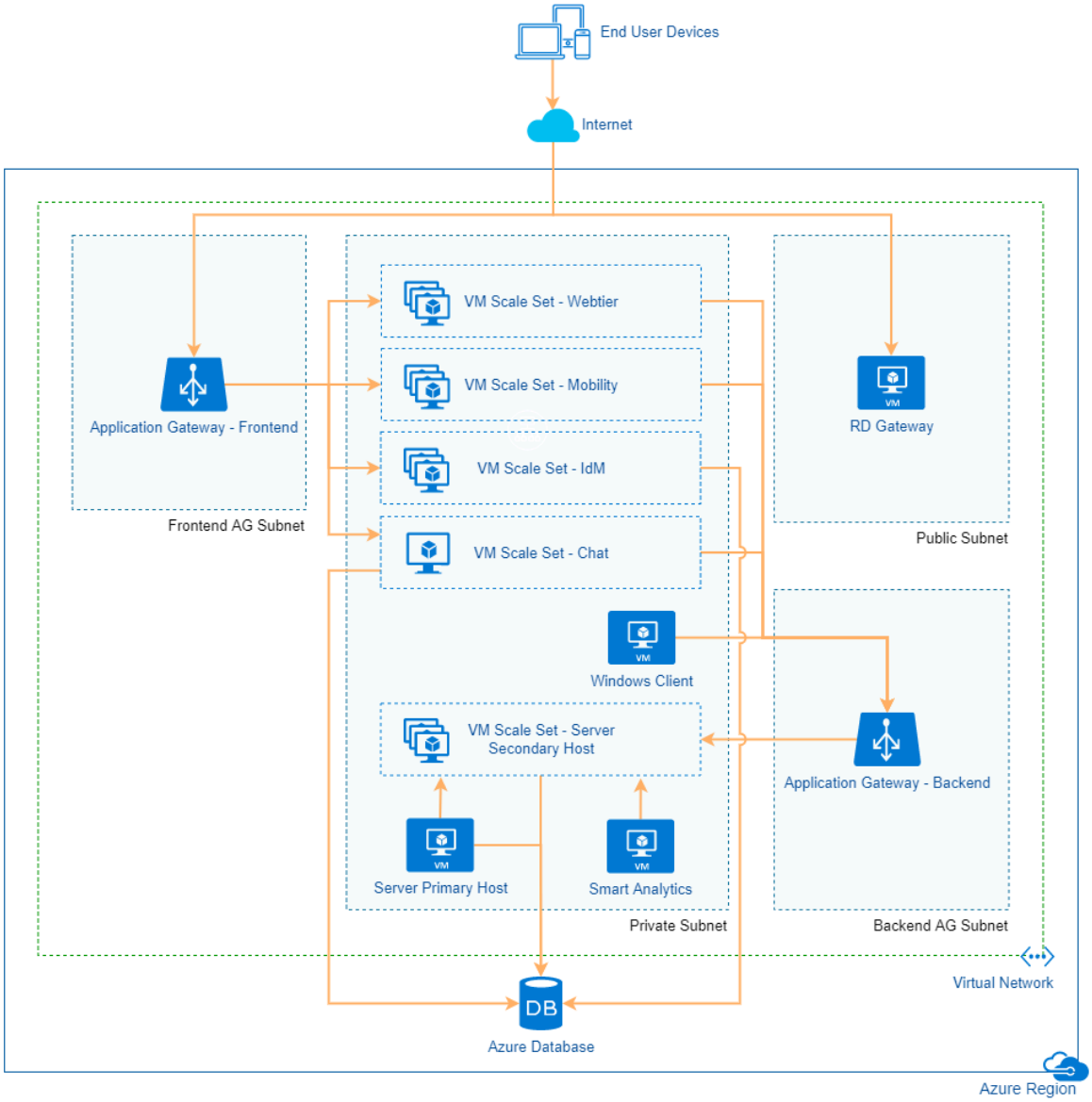
RDS Database

- Engine type: PostgreSQL
- Version: PostgreSQL 13

- DB instance size: 8 vCPUs, 48 GiB RAM

Service Manager on Azure deployment sizing

Deployment mode



Reference configuration for 2000 concurrent users

Webtier

- Size: 2 vcpus, 4 GiB memory
- Virtual machine scale sets: 7

Server primary host

- Size: 2 vcpus, 8 GiB memory
- Virtual machine count: 1

Server secondary hosts

- Size: 4 vcpus, 32 GiB memory
- Virtual machine scale sets: 6

Database

- Engine type: SQL Server
- Version: SQL Server 2019
- DB instance size: 16 vCPUs, 64 GiB RAM

Database Performance Tips

- To get lower network latency between RTE and DB, it is recommend to use AZURE "SQL virtual machines" to create SQL server within virtual machine, which has the same private subnet with SM server autoscaling set.
- To get lower network latency between RTE and DB, it is recommend to create AZURE "Proximity placement group" and place SM server autoscaling set and DB virtual machine in this "Proximity placement group".
- To get higher performance, it is recommend to set "Accelerated networking: Enabled" for SM server autoscaling set and DB virtual machine.

Rules of thumb

Service Manager server (servlet container)

For an out-of-box environment (limited to 75 concurrent users), a servlet container requires approximately 8GB of RAM, which is distributed in the following ways:

- Java virtual machine requirement, the Java heap size is 2 GB.
- User session memory usage:
  - For windows platform, the memory usage is 40 MB - 55 MB per user session.
  - For Linux platform, the memory usage is 55MB - 70MB per user session.
- Shared_memory setting value is equals to or greater than 512 MB (Counted only once for all servlet containers on a single computer)

Since the number of concurrent users assigned to one servlet container can be much more than 75 in 64-bit platform, the memory consumed will also be larger than 8 GB accordingly. To conclude a proper number of concurrent users assigned to one servlet container and the memory sizing, we recommend that you start with 150 threads (users) per process (servlet container). Then, run a performance test and tune the number based on the test results (process memory usage and transaction response time) , up to a maximum of 375 threads (if you have deployed SMA Service Portal, then the maximum threads number is 250, that is, threadsperprocess:250 and dao_threadsperprocess:250). For the planning of Java heap size, use the following logic:

- 4 GB if the number of user sessions is larger than 250, else
- 2 GB within 250 user sessions.

Note

1. Increases in user session memory usage may be caused by large global lists, global variables that are not garbage collected, or multiple open application threads. These activities have a direct effect on the number of user threads that each servlet container can support.
2. The network latency between Service Manager and the database server has a significant impact on the overall system performance. For example, every 5ms latency between the Service Manager applications server and the database server will double the login time. Therefore, we highly recommend that the network latency between the Service Manager applications server and the database server is less than 1ms.
3. If you are using horizontal scaling mode, we highly recommend that the network latency between each Service Manager application server is less than 1ms.
4. Most Linux operating systems provide ways to limit and control the usage of system resources such as threads, files, and network connections on a per-process and per-user basis. These limits prevent single users from using too many system resources. Sometimes, these limits have low default values that can cause a number of issues if there is a high number of concurrent users. We recommend that you use the ulimit -a command to check the resource limitation in the operating system, and ensure that the resources are sufficient for Service Manager servlets. We recommend that the value set by the ulimit -d command is over 3GB or "unlimited" mode.

Service Manager Web tier and Mobility

The Service Manager web tier configuration that was used to determine the sizing recommendations is based on an Apache Tomcat web application server with a number of processes(from 1 to 10 per different concurrent users load) running 1.5 GB of heap size each
(-Xms1536m -Xmx1536m), and an Apache web server to perform load balance the Java virtual machines (by using the Apache software load balancer). For Mobility, it shares similar configurations as web tier. To support additional users and higher loads, add computers that have additional Tomcat Java virtual machines to create a web farm as required.

Customers can also configure hardware load balancer instead of software load balancer to work with tomcat clustering. Details please refer to Service Manager Help server documents.

Note

In an initial setup, we recommend 150 to 350 concurrent users for each Tomcat instance.

Service Manager software load balancer

The Service Manager software load balancer should be located on a separate computer and should always be sized as a small computer because it performs no other function than redirecting connections to an available servlet. This computer also must run the same operating system as the Service Manager applications servers.

Because this is the one component in a configuration that is a single point of failure, it should be replicated and placed in a clustered environment for high availability. Failure of this component will only affect users who are attempting to establish an initial connection into the environment (additionally, the users will be affected only until the load balancer is restarted or fails over). Any currently active users that are connected to the environment will be unaffected by the loss of the load balancer.

Load balanced computers and servlet computers should use the server sizing that is described in this document. The number of nodes depends on the size of the computer and on the total number of concurrent production users.

Service Manager RDBMS server

The RDBMS server sizing that is specified above represents the configurations that were used during the benchmarking runs on the OOB Service Manager system. You should plan the storage requirements of the actual servers that you will use in a production environment based on expected data volumes, including attachments. The CPU and memory requirements for the selected database should be based upon the recommendations of the database vendor.

It is recommended that you install the RDBMS server on a separate physical machine, and not share the database across other services or products.

HugePages is recommended to be configured for Oracle and PostgreSQL databases.

When the total number of concurrent users exceeds 7000 (or 2000 if you deployed SMA Service Portal) and you are using an Oracle database, it is suggested that you set the `_ash_size_` parameter to a larger value in your Oracle database.

Service Manager virtualized environment (VMware)

Service Manager can be deployed in both physical and virtualized environments. Provided that the sizing guidelines are followed, there is no significant performance impact.

The Service Manager applications server, the Service Manager software load balancer, and the web tier can be deployed in a virtualized environment if you follow the sizing that is described in **Service Manager virtualization**. To deploy a virtualized environment, we recommend that you make the following configurations:

- Upgrade to vSphere7 or a later version.
- Assign dedicated resources (such as vCPU, memory, and disk I/O) to a guest operating system that acts as a Service Manager server.
- Deploy all Service Manager servers in the same virtualization resource pool.

Service Manager Collaboration

The Chat Service support about 400 concurrent users.

We recommend you to run the Chat server/service on 64-bit Java. The default JVM option is as follows:

```
-XX:ThreadStackSize=1024 -Xms512m -Xmx1536m
```

Service Manager Smart Analytics

Service Manager Smart Analytics server

Hardware size

If you cannot find recommended minimum configuration reference in the previous sizing section, follow the steps below to evaluate the target SMA content configurations:

1. **Determine the target deployment mode**
   Clustering with medium scalability is recommended for customers from scratch since most of them have much more requirements on Smart Search and images OCR processing than other services. With this deployment mode, you do not have to provide very big single server, and the load can be distributed to separate computers for better performance. Also, you can extend capabilities in the future.
   Note: For additional content server computers, we suggest to configure one content per computer.
2. **Determine the number of content servers**
   From the performance perspectives, the recommended volume size for each content server is 2 million for customers from scratch. The maximum volume size for each content server is 3 million.
   For example, if you plan 5 million indexed data in total for Smart Search, you need to configure 3 content servers. You can follow the same logic for HTA content.
3. **Determine the number of CPU cores**
   The number of CPU cores is highly dependent on the throughput produced to SMA IDOL services. For global search, HTA search and SMSP search, the recommended value is 4000 searches per core per hour based on the R&D Lab benchmark. For context search, the recommended value is 580 searches per core per hour.
   For example, if you plan 15000 global searches per hour, you need to configure at least 4 CPU cores. If you plan 4000 context searches per hour, configure at least 7 CPU cores. And the total needed CPU cores is 11 CPU cores to support both global searches and context searches.
   For proxy servers, we recommend 4 CPU cores for the clustering with high scalability deployment, and 8 CPU cores for the clustering with medium scalability deployment.
   For connectors and CFS, we recommend 6 ~ 8 CPU cores.
4. **Determine the number of RAM**
   For each content server, we recommend 6 to 8GB RAM.
   For proxy servers, we recommend 4GB RAM for the clustering with high scalability deployment, and 8GB RAM for

the clustering with medium scalability deployment.
For connectors and CFS, we recommend 6 to 8GB RAM.

5. **Determine the IO subsystem.**
   We recommend you to configure storage instead of local disk for better performance. IDOL has provided a good utility tool named PerfTool to evaluate your storage IO capabilities. Please consult the R&D team for more information about this tool. The minimum IO speed is 120 MB/s, while the recommended is 280 MB/s for read/write which is verified in R&D Lab benchmark and this value is captured from PerfTool. If you do not have supported storage, configure a separate disk for each content server at least.

Network

- We suggest that you locate the Service Manager server, IDOL server, and image server under the same switch in order to achieve the best performance.
- We suggest that you ensure a network latency that is less than 1ms between the Service Manager server and the IDOL server, between the Service Manager server and the image server, and between the IDOL server and the image server.
- On Windows-based platforms, your computer may run out of available TCP ports if your IDOL service receives a large number of actions. If your system does not have enough ports, you can edit the following Windows registry parameters:
  `TCPTimedWaitDelay`

  `MaxUserPort`
  These parameters are located in the HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\ directory. For more information about these registry parameters, refer to the Windows documentation.

Operating system

- We recommend that you install the IDOL server on a Linux system for better stability and maintainability.

Indexing and training

- Indexing: It takes about 40 minutes to index 100,000 interaction records and 6.5 hours to index 1,000,000 interaction records. The growth of the time according to the interaction count is linear.
- Training: It takes about 16 hours to train 2,000,000 records in more than 3400 affected services.

Service Manager Smart Analytics media(image) server

- We recommend that you install the media(image) server on a dedicated server to achieve better performance.
- We recommend to use the distributed media(image) server mode if you process more than 1.82 OCR records (with the average size of 200 KB) per second. This mode can improve the speed in which OCR records are processed.
- Distributed media(image) servers do not support large images (over 1MB) well in environments that have a throughput greater than 3.33 OCR records per second.
- Smart Analytics does not have a significant impact on the run time performance of Service Manager. But, as a result of implementing Smart Analytics, end users and operators can submit tickets very quickly. Therefore, the following points must be considered for optimal performance:
  - The overall throughput and load on the system will increase due to rapid ticket submission. Please continue monitoring the system resources to prevent any bottlenecks.
  - The media(image) server will save all the requested content on a local server. Disk space usage on this server may grow quickly.

Deployment we recommend the following best practices for the best performance:

- Although the database can be deployed on the same computer as the Service Manager server, we strongly suggest that you deploy the database on a separate computer. However, the server and database should in the same subnet and have a latency lower than 1ms.
- You can implement small environment deployments on virtual machines or on physical machines, as long as you provide sufficient hardware resources.
- We strongly suggest that you use a solid state drive (SSD) instead of a hard disk drive (HDD). An SSD significantly increases the speed of disk I/O operations.

- You can deploy all web applications, including the Service Manager web tier, Service Manager Mobility, , on the same Tomcat node or separate Tomcat nodes. For high availability (HA), we suggest that you use a separated deployment.
- If you deploy all web applications on same Tomcat node, we recommend the following startup options in the Java virtual machine instance:

```
-Xms2048m -Xmx2048m
```

- If you deploy the web applications on separate Tomcat nodes, we recommend the following startup options in the Java virtual machine instance:

```
-Xms1536m -Xmx1536m
```

- For high availability (HA) and fail-over capabilities in the web applications layer, we recommend that you deploy the web applications on two Tomcat nodes, and configure an Apache server in front of the nodes as a software load balancer to avoid a single point of failure.

For high availability (HA) and fail-over capabilities in the Service Manager applications server, we recommend that you use a vertical scaling model in which two servlet container processes run on one Service Manager server that has a software load balancer enabled.

1.1.3. Support matrix

This section provides information about the supported hardware and software that you must have to successfully install and run Service Manager (SM).

Java support

To align with the latest Java technology and keep data integrity with third-party vendors, customers on this version of Service Manager should use Java 17.

This applies to the SM components such as chat server, chat Service, and web elements such as web client and mobility client running on Tomcat application server.

Zulu OpenJDK 17

In 2018 Oracle announced changes to the Oracle Java lifecycle and roadmap detailed here. To accommodate the needs of all our customers, we're moving toward providing all products in a form compatible with an Open JDK environment (where Java is used). The plan is to adopt the following approach for Service Manager:

- Stop redistributing Oracle JRE/JDK and generally adopt a Zulu Open JDK distribution policy
- Continue to support customer-supplied Oracle JDK

By following the overall approach, Service Manager bundles Zulu OpenJDK JRE (64-bit) to Windows server, Windows client, and Linux server.

Service Manager server

The Service Manager server communicates with the database using the appropriate database client software. The appropriate database client software must be installed and configured on the Service Manager server. The database should reside on a different server, which may use a different operating system. The company doesn't make compatibility statements about the operating systems supported by the database. Database vendors are responsible for indicating supported server platforms. A 64-bit JRE 17 (Zulu OpenJDK 17) is provided in the Service Manager installation for Windows and Linux servers.

Server platforms

The Service Manager server supports the following 64-bit operating systems.

Operating system	Version
Windows	Windows Server 2022, 2025
Linux	Red Hat Enterprise Linux **8.x, 9.x** and **10.x** Oracle Enterprise Linux **8.x** and **9.x** Novell SUSE Linux Enterprise Server **15** and **16** Rocky Linux **9.x** and **10.x**

Databases

The Service Manager server supports the following back-end databases. For better performance and stronger security, we recommend that you use the Enterprise edition of the databases.

RDBMS	Version
PostgreSQL	15, 16, 17, 18
Oracle 1	Oracle 19c, 21c
SQL Server	SQL Server 2019, 2022, 2025

1 When you upgrade the Oracle database client to 19c, make sure that the following settings are correct:

- After upgrade, the following two lines in sm.ini are the same as before:
  Windows:

  ```
  sqldictionary:oracle10
  [oracle10]
  ```

  Linux:

  ```
  sqldictionary:oracle
  [oracle]
  ```

- Make sure that the RDBMS driver setting for the **sqllibrary** parameter in sm.ini is as follows:
  - Windows - sqoracle.oci10.DLL
  - Linux - libsqoracle19.so
- Sevice Manager works with Pluggable Database (PDB) inside Oracle Container Database (CDB) by connecting to PDB with SID or Service Name.

Virtualization support

The Service Manager server supports the following virtualization platforms.

Virtualization platform	Notes
VMWare vSphere 7.x, 8.x	vMotion is supported.
KVM	The kernel version must be 4.18 or later.

Note

VMHA is transparent to Service Manager. A client reconnection is required after a fail-over.

Note

VMWare's Snapshot features should be used with caution. In addition, there are some known issues and recommendations in vMotion that need user attention. For details, see the Service Manager vMotion Test Report white paper.

Case sensitivity

Service Manager supports the default case-sensitivity for all the RDBMS platforms listed, including the case-insensitive collations in Microsoft SQL Server. In addition to these defaults, Service Manager supports case-insensitive mode in the Oracle database and the PostgreSQL database.

Oracle Real Application Cluster and Transparent Application Failover

Oracle Transparent Application Failover (TAF) is a feature that allows database clients to reconnect to surviving nodes in an Oracle Real Application Cluster (RAC) in the event of a failure of an instance. All supported Service Manager server versions currently perform similar session recovery operations within our own application. When detecting a connection failure, Service Manager will attempt to reestablish the connection, set up necessary session properties, and then attempt to repeat the failed transactions. Service Manager will continue to retry the connection for 1 minute. If the database is within an Oracle RAC configuration, this should allow time for failover and reconnection to another available instance. We fully support Oracle RAC configurations and will honor this re-connect strategy. For more information, see *Transparent Technology and Virtualization Support*. Since similar functionality is already available in Service Manager, the product has not been modified to run in an Oracle TAF configuration.

Caution

Using Service Manager in combination with Oracle TAF may actually cause connectivity issues in the database. Do not run Service Manager in an Oracle TAF configuration.

Support of SQL Server AlwaysOn

Service Manager supports SQL Server AlwaysOn, which is a high-availability solution for SQL Server databases.

Caution

To support this functionality, Microsoft ODBC Driver 17 for SQL Server (or a later version) must be installed and used to create a system DSN for the Service Manager database.

For more information, see Support of SQL Server AlwaysOn.

Service Manager clients

This section provides support matrix information for the Service Manager clients.

Note

No new features are being added to the Windows (Eclipse) client. We recommend that Service Manager administrators deploy other Service Manager clients (web client, Mobility client, or SMA-SM Service Portal) instead of the Windows (Eclipse) client for end users.

Note

Viewing Service Manager forms with either the web or the Windows client requires a minimum screen resolution of 1280x800 and a maximum display aspect ratio of 16:9.

Web tier

The Service Manager web tier supports the following configurations.

Application server	• **Apache Tomcat 10.1.x 3** • **IBM WebSphere Application Server Liberty 24.0.0.12 4** • **JBoss Web Server (JWS) 6.0**
Web server 1	Apache HTTP Server 2.4 IIS (bundled with Windows Server 2022, 2025)
Web client 2	Chromium-based Edge (the latest version) Firefox (the latest version) Chrome (the latest version)

1 The web server must be compatible with the web tier application server.

2 We recommend that you use Windows desktop operating system to access the Service Manager web client.

3 As of Tomcat 10.1.x, the out-of-box Tomcat `server.xml` sets the value of `maxParameterCount` to 1000, which overwrites the default value of 10000. Because some modules in Service Manager, such as Data Policy, include more than 1000 fields, this change will result in unexpected data loss. If you deploy Tomcat 10.1.x and above versions, you need to change the `server.xml` settings, either by removing the `maxParameterCount` parameter or updating the value of the `maxParameterCount` parameter to 10000.

4 JAVA SE 17 requires IBM WebSphere Application Server Liberty fix pack 23.0.0.12 or later. For more information, see Updating the Liberty Java runtime environment or software development kit.

Screen reader support

The Service Manager web tier is compatible with screen reader software used by blind and visually impaired users on supported Windows operating systems.

To enable screen reader accessibility, you must have the supported screen reader software installed on your local client.

Supported screen reader software

Service Management supports the following screen reader software on supported Windows operating systems:

- JAWS Screen Reader software version 2025

Supported browsers

- Chromium-based Edge (the latest version)

Mobility client

The Service Manager mobility client supports the following configurations.

Application server	• Apache Tomcat 10.1.x • IBM WebSphere Application Server Liberty 24.0.0.12 • JBoss Web Server (JWS) 6.0
Handset	The Service Manager Mobility client supports handsets that are running the following operating system versions and their built-in browsers. 1 iOS 16.x, 17.x, 18.x: Safari Android 11.x, 13.x, 14.x, 15.x: Chrome, Android browser

1 Other third-party web browsers have not been certified and are therefore not supported. To access and use the Service Manager Mobility client, your phone must have a touch screen. The performance of the Service Manager Mobility client is highly dependent on the performance of the handsets that you use.

Windows client

The Service Manager Windows client supports Windows 11 Pro and Windows 10.

Note

For the HTML Editor to work correctly in the Windows client, the client machine must have a version of Edge installed that's supported for the web client.

Note

Virtualization options, such as Citrix, are considered transparent technologies. See *Transparent Technology and Virtualization Support* below for more information.

Smart Analytics

Before you install Smart Analytics, make sure that your Service Manager Applications version and your Service Manager platform version meet the version requirements listed below.

Smart Analytics Installer version	Service Manager Applications version	Service Manager platform version
9.82	9.80 or later	9.80 or later

The Service Manager Smart Analytics supports the following 64-bit operating systems.

Operating system	Version
Windows	Windows Server **2022, 2025**
Linux	Red Hat Enterprise Linux **8.x, 9.x** and **10.x** Oracle Enterprise Linux **8.x** and **9.x** Novell SUSE Linux Enterprise Server **15** and **16** Rocky Linux **9.x** and **10.x**

Service Manager Collaboration

The Service Manager Collaboration chat server (Openfire) supports the following configurations.

Database	**The same as Service Manager server databases.**
Operating systems	Windows Server **2022, 2025** Red Hat Enterprise Linux **8.x, 9.x** and **10.x** Oracle Enterprise Linux **8.x** and **9.x** Novell SUSE Linux Enterprise Server **15** and **16** Rocky Linux **9.x** and **10.x**
Web server	Apache HTTP Server 2.4 IIS (bundled with Windows Server 2022, 2025)

Identity Manager (IdM) service

The Service Manager Identity Manager (IdM) service supports the following configurations.

Web application server	**For idm version before 1.40.1: Tomcat 10.x** **For idm version on 1.40.1 or above: Tomcat 10**
Identity provider (IdP)	• Identity Providers (IdPs) for SAML: ○ Microsoft Active Directory Federation Services (ADFS) 3.0 1 ○ Azure Active Directory ○ Oracle Identity and Access Management 12 or higher • Identity Providers (IdPs) for OAuth: ○ Keycloak 21.0.0 or higher
Database	Oracle 19c, 21c SQL Server 2019, 2022, 2025 PostgreSQL 15, 16, 17, 18

1 You must configure ADFS to authenticate users stored in an LDAP server.

Hardware load balancers

F5 hardware load balancer is the certified load balancer for Service Manager. For more information, see Hardware load balancers.

Compatibility

This section provides compatibility information about Service Manager components.

Client/server compatibility

Service Manager server and clients (including the web client, Mobility client, and Windows client) must be at the same minor level. In other words, the combinations of client and server at different minor levels aren't supported. For example, using an SM 9.7x client together with the SM 9.8x server or using an SM 9.8x client together with an SM 9.7x server is unsupported. In addition, we strongly recommend that you use Service Manger server together with the clients that are from the same release. For example, use the Service Manager 9.82 server together with the Service Manager 9.82 clients.

Platform/application compatibility

The Service Manager client/server version must be no earlier than the applications version. If you use the 9.82 applications, you must use a client/server version of 9.80 or greater. If you use the 9.80 client/server, you can use the 9.80, 9.72, 9.71, or 9.70 applications.

Service Manager Collaboration

Service Manager Collaboration (SMC) requires the same applications version as its own. SMC 9.82 requires Service Manager applications version 9.82.

SMA-SM Service Portal compatibility

We recommend that you download and install the latest version of SMA-SM SP that supports your version of Service Manager. For compatibility information and installation documentation of SMA-SM SP, see https://portal.microfocus.com/s/article/KM000001627.

Search engine compatibility

Service Manager 9.82 supports the IDOL Search Engine.

> **Note**
>
> The IDOL Search Engine is also called Smart Search as a feature in Service Manager Smart Analytics. If you enable Smart Analytics, the IDOL Search Engine is also enabled.

Compatibility with other software products

Service Manager supports many portfolio integrations of the company, as well as those of any third parties. These integrations are identified in the integration support matrix. To view the integration support matrix, visit INTEGRATION CENTRAL.

AWS and Azure support matrix

Service Manager supports AWS and Azure deployments with the following items certified.

Items	Applicable to	Supported version
Operating system	Server, web tier, Mobility, Smart Analytics, Service Manager Collaboration	Windows Server 2022, 2025 Red Hat Enterprise Linux 8.x and 9.x
Database	Server, Service Manager Collaboration	Oracle 19c, 21c SQL Server 2019, 2022, 2025 PostgresSQL 15, 16, 17, 18

> **Note**
>
> Service Manager supports Azure AD as the LDAP server.

SCAuto SDK Support Matrix

SCAuto SDK 9.82 supports the following operating systems with Service Manager version 9.82.

Operating system	Version
Windows	Windows Server 2022, 2025
Linux 1	Red Hat Enterprise Linux 8.x, 9.x and 10.x Oracle Enterprise Linux 8.x and 9.x Novell SUSE Linux Enterprise Server 15 and 16 Rocky Linux 9.x and 10.x

1 All UNIX operating systems are deprecated from SCAuto SDK 9.70.

Languages, localization, and internationalization

The Service Manager 9.82 server supports all Service Manager Application languages, localization, and internationalization versions. Service Manager supports Unicode (UTF-8) on the server and client. Unicode is a worldwide standard compatible with ISO 10646 (www.iso.org). UTF-8 is part of the Unicode standard, which enables you to encode text in practically any script and language. It also supports a comprehensive set of mathematical and technical symbols that simplify scientific information exchange. Service Manager 9.82 supports UTF-8 as an encoding method for new or existing ASCII and multi-byte characters. For more information about the languages and character sets that are supported by UTF-8, visit the www.unicode.org website.

Service Manager 9.82 approaches languages, localization, and internationalization as follows:

- Language packs provide a translated UI for Service Manager.
- Service Manager 9.82 language packs are available for the following languages: Arabic, French, German, Italian, Japanese, Simplified Chinese, Spanish, Brazilian Portuguese, Korean, and Turkish.
  Service Portal supports Swedish in addition to all the languages that are listed above.
  The Mobility client supports all the languages that are listed above, except for Arabic (the right-to-left display language).
- Service Manager accepts and displays data for any language that's supported by UTF-8, regardless of the language pack installed. Furthermore, no translation is required for this feature to apply. For example, a French Service Manager system can accept and display German. A Japanese system can accept and display Spanish. Note that appropriate SQL database data types or code pages are required.

Transparent technology and virtualization support

In recent years, a number of “transparent” hardware and software technologies and virtualization solutions have become increasingly prevalent. For example, the hardware load balancers using transparent technology, and the virtualization solutions such as Citrix, Microsoft Cluster Software, and VMware. These solutions operate in the technology layers adjacent to the operating systems or, in some cases, as extensions of the operating systems. Similarly, database solutions offer transparent components as supported elements.

The company supports Service Manager running on operating systems and databases on particular platforms as described in the matrix above, not specific hardware and software configurations. The company will support Service Manager customers who run software products of the company on supported operating systems and databases, irrespective of whether they are running transparent or virtualization solutions in their environment. The company doesn't support these transparent or virtualization technologies directly. Since the providers of these technologies support a set of certified operating systems and hardware, the customer and the providers of these technologies will be responsible for any interactions or issues that arise at the hardware or operating system layer as a result of their use.

The company will not require customers to re-create and troubleshoot every issue in a non-transparent environment; however, the company does reserve the right to request that its customers diagnose certain issues in a native certified operating system environment without transparent technology. The company will only make this request when there is reason to believe that the environment is a contributing factor to the reported issue.

While Service Manager is expected to function properly with these transparent technologies in place, there may be performance implications, which can invalidate the company’s typical sizing and recommendations. Analysis must be

performed within the context of the specific application to be hosted in a virtual environment to minimize potential resource overload, which can have a significant impact on performance and scalability, particularly under peak load.

Underlying technology version policy

Third-party components, such as databases and operating systems, are supported at the minor level unless a different minimum level is specified. For example, Oracle 11.2 is supported at the minimum release of Oracle 11.2.0. Future releases of the same minor release (second numeral) are expected to be supported, unless a conflict specific to that release arises. For example, you can expect Oracle 11 to be supported on 11.2.0.3, 11.2.0.4, etc. Refer to the support matrix provided by the vendor for restrictions and other considerations.

It's not our policy to recertify a released product against a new version of a vendor product, unless the current version of our product will be supported well past the end of obtainable or extended support of the associated vendor product, and there is not a viable extension to the support of that product. We also, as a rule, don't recertify on minor releases (for example, Oracle 11.0, then 11.2, then 11.2g, and so on); we only list the latest version of the vendor product that we actually certified at the time of our product release.

Obsolescence plans

To learn the obsolescence plans for the previously released versions of Service Manager and the company's products that integrate with Service Manager, go to Product Support Lifecycle.

1.2. Prepare

This section provides information required to prepare your Service Manager deployment. This section includes the following topics:

Topic	Description
Download the Service Manager installation packages	Provides a detailed list of packages shipped with the release. See Download the Service Manager installation packages.
Prepare your RDBMS	Provides details of how to prepare the databases. See Prepare your RDBMS.

1.2.1. Download the Service Manager installation packages

This SM release includes the **SM982.zip** file that you can download from the Software Licenses and Downloads website. It consists of the following installation packages.



Note

"xxxx" represents the build number of the corresponding package.

Package	Description
sm.9.82.xxxx_Windows_Server.zip	Server package for Windows
sm.9.82.xxxx_Linux.tar.xz	Server package for Linux
sm.9.82.xxxx_UP.zip	Application Upgrade Utility
sm.9.82.xxxx_Web_Tier.zip	Web tier
sm.9.82.xxxx_Windows_Client.zip	Windows client
sm.9.82.xxxx_Mobility.zip	Mobility client
sm.9.82.xxxx_TeamsIntegration.zip	Teams integration
sm9.82.xxxx_SmartAnalytics_Linux.zip sm9.82.xxxx_SmartAnalytics_Windows.zip	Smart Analytics for Linux and Windows
chat-server-9.82.xxxxxxx.zip chat-ui-9.82.xxxxxxx.war chat-msvc-9.82.xxxxxxx.zip	Collaboration Chat Server, Chat UI, and Chat Service
idm-x.x.x.zip	Identity Manager (IdM) service, IdM admin console, and IdM data protection tool
sm9.82.xxxx_apm_all.zip	Application Performance Management (APM) package
sm9.82.xxxx_l10n.zip	Contains language packs

1.2.2. Prepare your RDBMS

The following recommendations assume the implementation of conventional database tuning and performance measures. Actual results may vary on a system-by-system basis, based on the tuning expertise available and hardware and software selections. These recommendations are intended only as a guide and should not be implemented on a production system without extensive testing.

A fully qualified database administrator should assist with this preparation.

> **Note**
>
> The Service Manager server needs to be within reasonable proximity to the RDBMS. Utilizing a RDBMS for use with Service Manager Server over a WAN link is not recommended and will have a negative impact on overall product performance.

Follow the instructions in this section to prepare your Relational Database Management System (RDBMS) prior to installing the Service Manager data.

Prepare a SQL Server database

To prepare a SQL Server database, follow these steps:

1. Meet general space requirements:
   Place all Service Manager data in one or more dedicated table spaces within a single SQL Server instance. These table spaces must contain Service Manager data only. Multiple instances consume more system resources than a single-instance solution.
   Allocate at least 1 GB of data space for a test system. The amount of space necessary for a production system depends on the amount of data that you need to store and your specific implementation.
2. Set the sorting of characters for localized systems and set the case sensitivity of the database.
   Service Manager supports both case-sensitive and case-insensitive Microsoft SQL server. To use Service Manager in case-insensitive mode, you must select a case-insensitive collation on the SQL Server before installing Service Manager. You can specify the desired case-sensitivity for sort order during the creation of the database. Set the SQL Server database to the desired collation when you create it. Service Manager automatically detects the settings.
   - To run in case-sensitive mode, pick a collation that ends with _BIN, such as Latin1_General_BIN.
   - To run in case-insensitive mode, pick a collation that ends with _CI_AS, such as Latin1_General_CI_AS.

   MS SQL Server collation support
   Service Manager supports two types of collations :
   - Those ending in some form of _BIN (binary)
   - Those ending in some form of _CI_AS (case-insensitive, accent sensitive)

   _BIN is used for case-sensitive collation, meaning both, comparisons (WHERE NAME="ADMINISTRATOR") and sorting (ORDER BY NAME) are case-sensitive. Collations ending in _CS_AS claim to be case-sensitive, but that only applies to comparisons. For sorting, they use a "dictionary order," which is essentially case-insensitive.
   _CI_AS is the supported case-insensitive collation. It uses case-insensitive comparisons (WHERE NAME="AdMinIstrator") and case-insensitive sorting. Do not use _CI_AI since Service Manager has no concept of accent insensitivity.
   The sort order becomes important in the case of a combined SQL/IR search. In that case, IR has to match its own results with the results returned by SQL, and for that it assumes binary or case-insensitive sorted order.
   Note that such queries might take place even if you don't know about them. For example in the Service Catalog, when ordering, a user enters an IR query, but the RAD appends a long non-IR query to it. This results in a mixed query.
   The other aspect of the collation is the code page. The code page defines the code page of VARCHAR columns, NVARCHAR are always in UTF-16. "General" refers to the American default Windows code page, Win 1252.

Service Manager supports the following SQL Server code pages:

Code page	Description
874	MS cp874 Thai
932	MS cp932 Japanese
936	MS cp936 Simplified Chinese
949	MS cp949 Korean
950	MS cp950 Traditional Chinese
1250	MS cp1250 Central European
1251	MS cp1251 Russian, Bulgarian, Serbian
1252	MS cp1252 Latin/Western European
1253	MS cp1253 modern Greek
1254	MS cp1254 Turkish
1255	MS cp1255 Hebrew
1256	MS cp1256 Arabic
1257	MS cp1257 Estonian, Latvian, Lithuanian
1258	MS cp12578 Vietnamese

You can generate the list of all the code pages supported by Service Manager with the command, `sm -reportlanguages`. Only those code pages that start with "mswin" are used in the context of SQL Server.
If you are using an unsupported code page, you will see this warning in the log file: `W SQL Server uses a code page <nnn> unsupported by Service Manager which retrieved codepage <nnnn>`.



Note

- MSSQL Server 2019 is now UTF-8 compliant with collation suffix as _UTF8. However, Service Manager does not support such collation.
- If you have selected the **Use Unicode Data Type** option when configuring the Service Manager server, unicode sorting is used instead of collation sorting. For more information about the **Use Unicode Data Type** option, see Install the Service Manager Server.

3. Create server connections. Every Service Manager thread, foreground, or background, requires a connection to your RDBMS. Service Manager background processors require 17 connections to run. Make sure that you allocate enough connections for all of your users. For additional information, refer to your RDBMS vendor documentation.
4. Create a login ID and password for Service Manager to use to connect to your database.
   Create a login ID and password for Service Manager to use to connect to your SQL Server database.
   The login must have CREATE/ALTER/DROP TABLE authority for the target database. The CREATE/ALTER/DROP TABLE authority is only required during the installation and creation of new Service Manager tables, and only if you allow Service Manager to issue the DDL to create tables and indexes. When Service Manager connects to your database using the login ID, tables are created in the default table space defined for that login ID.

Caution

You must grant the DB account of the "sqllogin" SM parameter the privilege to create Stored Procedures and User Defined Functions.

5. (Optional) If you use IR Expert, Full-Text Search is an optional component of the Microsoft SQL Server Database Engine. If you did not select Full-Text Search when you installed SQL Server, run SQL Server Setup again to add it. The installation process does not require a shutdown of your SQL Server Database Engine.

Note

Note: For more information, see IR Expert solution comparison and https://docs.microsoft.com/en-us/sql/relational-databases/search/full-text-search?view=sql-server-2017.

6. Set up a connection to Service Manager:
   Service Manager connects to the database through your RDBMS client. To set up the connection between the Service Manager application server and your RDBMS, know the name of the database and the login ID and password required to connect to the database server that you created above. The Service Manager initialization file, sm.ini, must be present in the Service Manager server RUN directory.
   Follow these connectivity rules when you set up the connection to your SQL Server:
   - Make sure that the database name you enter in the configuration tool corresponds to the ODBC Driver system data source.
   - Configure the ODBC Driver data source as a System DSN. Set it up to use the following settings:
     - SQL server authentication
     - ANSI quoted identifiers
     - ANSI nulls, paddings, and warnings
   - Do not create the system DSN by using a 32-bit ODBC administrator for SQL Server driver.
     It is recommended that you use 64-bit Microsoft ODBC Driver for SQL Server as the ODBC DSN.

   Starting from Service Manager 9.72, you can connect to Microsoft SQL Server 2019 or Microsoft SQL Server 2022 installed on Linux platform.
   Follow the instructions in this article to set up the ODBC driver on "RHEL and Oralce Linux" and "SLES".
   Run the following command to create the ODBC configuration file:
   `touch smodbc.ini`
   Edit the file. For example:

```
[DSN name]
Description=Service Manager 9.81 Connection
# Use the same driver name defined in /etc/odbcinst.ini
Driver=ODBC Driver 18 for SQL Server
Server=serveraddress.net
Database=DSN name
```

   Run the following command to add the configurations into `~/.odbc.ini`:
   `odbcinst -i -s -h -f smodbc.ini`
   Run the following command to check if the `odbc.ini` file is updated successfully:
   `cat ~/.odbc.ini`

Note

Service Manager supports SQL Server AlwaysOn for SQL Server. AlwaysOn is a high availability solution for SQL Server databases. To support SQL Server AlwaysOn, Service Manager Server introduces the sqlalwayson:1 parameter and the SQMSSQLAO.ODBC.DLL file. To use this functionality, you must install Microsoft ODBC Driver 17 for SQL Server (or a later version) and configure a System DSN with this ODBC driver. For more information, see Support of SQL Server AlwaysOn.

7. Set time zones for reporting:
   If you plan to report on Service Manager data using your RDBMS tools, set the sqltz parameter in the sm.ini file. This is an optional parameter that the Service Manager server uses to control the storing of date and time values in the RDBMS. This parameter defines the time zone to use as a base for all date and time values. The time zone is specified as the name of the time zone record in the Service Manager tzfile table. The default time zone is Greenwich/Universal (GMT). For more information about using the sqltz parameter, see the System Configuration Parameters topic in the Service Manager Help.

   Caution

   If you use different time zone settings, the dates contained in reports made by your RDBMS utility may be inaccurate.

8. Set the Truncate Log On Checkpoint option for the target database on your SQL server. During initial system load, Service Manager places a high insert transaction load on your SQL server. Set the Truncate Log On Checkpoint option to prevent the transaction log from growing too large.
9. Secure connections with TLS. See Connect to SQL Server with strict encryption and Connect to SQL Server 2022 with strict encryption for more information.

Prepare an Oracle database

Oracle Transparent Application Failover (TAF) enables database clients to switch the connection to surviving nodes in an Oracle Real Application Cluster (RAC) without re-establishing the database connection or setting up necessary session properties in the event of a failure of an instance.

All supported Service Manager server versions perform similar session recovery operations within the application. When Service Manager detects a connection failure, it attempts to reestablish the connection, set up necessary session properties, and repeat the failed transactions. Service Manager retries the connection for one minute. If the database is in an Oracle RAC configuration, this should allow time for failover and reconnection to another available instance.

Since similar failover functionality is available within Service Manager, the product has not been modified to run in an Oracle TAF configuration.



Caution

Using Service Manager in combination with Oracle TAF could cause connectivity issues. Do not run Service Manager in an Oracle TAF configuration.

Do not change the `NLS_LENGTH_SEMANTICS` parameter value from `Byte` to `CHAR` in **Oracle** databases. For details, see `NLS_LENGTH_SEMANTICS` in Oracle Database Reference.

To prepare an Oracle database, follow these steps:

1. Meet general space requirements:
   Place all Service Manager data in a dedicated table space within a single Oracle instance. This table space must contain Service Manager data only. Multiple instances consume more system resources than a single-instance solution.
   Allocate at least 1 GB of data space for a test system. The amount of space necessary for a production system depends on the amount of data that you need to store and your specific implementation.
2. Set the sorting of characters for localized systems.

   Note

   Setting the sorting of characters in Service Manager is dependent on the settings defined within the back-end database or client connections to these databases. Please check with your company's Database Administrator when setting up your Service Manager database to ensure you have the correct settings for the native language(s) that will be used within Service Manager.

Database vendors provide different options for these settings and continue to add/modify settings for languages around the world. Refer to the technical documentation provided by the database vendor for more information and recommendations on the correct localization settings. For additional information, search the vendor documentation for the following terms.
- Globalization Support
- Oracle client installation globalization

3. Create server connections. Every Service Manager thread, foreground, or background, requires a connection to your RDBMS. Service Manager background processors require 17 connections to run. Make sure that you allocate enough connections for all of your users. For additional information, refer to your RDBMS vendor documentation.
4. Create a login ID and password for Service Manager to use to connect to your database.
   Create a login ID and password for Service Manager to use to connect to your Oracle server. When you sign in to Service Manager, it creates a table in the default table space defined for that login ID. The login ID must have the following privileges:
   - Connect
   - Create, Alter, Drop a table
   - Create, Alter, Drop an index
   - Select on v_$parameter
   - Alter Session Privileges

   You can provide these privileges to an Oracle user by using the following oracle statements:

```
create user <smadmin> identified by <smadmin> default
tablespace <users> quota unlimited on <users>;
grant connect, resource to <smadmin>;
grant select on v_$parameter to <smadmin>;
```



Note

- CREATE/ALTER/DROP TABLE privileges are only required during installation and creation of new Service Manager tables if you allow Service Manager to issue the DDL to create tables and indexes.
- You must grant the DB account of the "sqllogin" SM parameter the privilege to create Stored Procedures and User Defined Functions.
- For IR Expert, the following privileges are required:
  `grant execute on ctxsys.ctx_ddl to <smadmin>;`

  `grant create job to <smadmin>;`
- The following privileges are required for sqldebug:
  `grant select on sys.col$ to <smadmin>;`

  `grant select on sys.obj$ to <smadmin>;`

  `grant select on sys.icol$ to <smadmin>;`

  `grant select on sys.ind$ to <smadmin>;`

5. Set the case sensitivity of the database.
   Service Manager supports case-insensitivity for Oracle. See the appropriate task below for instructions on setting case-sensitivity.

   Set case-sensitivity on a new Oracle database

   1. While creating a new Oracle instance, choose **All Initialization Parameters** and modify the parameters.
   2. Set the case-sensitivity for the NLS_SORT and NLS_COMP parameters:
      - For a case-sensitive database, set NLS_SORT and NLS_COMP to **BINARY**.
      - For a case-insensitive database, set NLS_SORT to **BINARY**_CI and set NLS_COMP to **LINGUISTIC**.
   3. In Service Manager, run the following SQL statement to verify that the parameters are in effect:
      `select parameter, value from nls_instance_parameters;`
   4. Make sure that the values for NLS_SORT and NLS_COMP match your selection in Step 2.

   Set case-sensitivity at Oracle database session level

   1. Create a user which will be used for SM server to connect with Oracle database.
   2. Log in with the user created in step 1, and then create a Logon trigger.
      The following example configures Base Type as "database", so as to set the Oracle instance to case-insensitive.

      ```
      create or replace trigger AFTER_LOGIN_BINARY_CI
      after logon on database
      begin
        execute immediate 'alter session set NLS_SORT=''BINARY_CI'' NLS_COMP=''linguistic''';
      end;
      /
      ```

   3. Check the sm.log file in the <SM_install_location>\Server\logs directory. The following entry in the log file indicates that the Oracle instance is set to case-insensitive and that you connected to it successfully:
      `RTE I Oracle server settings for language, territory and character set:`
      `AMERICAN_AMERICA.AL32UTF8 (AL16UTF16)`
      `RTE I OCI Client settings for language, territory and character set:`
      `AMERICAN_AMERICA.AL32UTF8 (UTF16)`
      `....`
      `RTE I Oracle instance setting for NLS_SORT is set to BINARY_CI`
      `RTE I Oracle instance setting for NLS_COMP is set to LINGUISTIC`
      `...`
      `RTE I Oracle session is set up in CASE INSENSITIVE mode`

   Change an existing case-sensitive Oracle database with Service Manager data to be case-insensitive

   1. Back up all Service Manager data.x
   2. Run the following command:
      `sm -system_unload -system_directory:<path to where you want to store the Service Manager data unload files>`
   3. Sign in to Oracle as a sys admin user, and issue the following statements to set NLS_SORT to **BINARY_CI** and NLS_COMP to **LINGUISTIC**:
      `alter system set NLS_SORT=BINARY_CI SCOPE=SPFILE;`
      `alter system set NLS_COMP=LINGUISTIC SCOPE=SPFILE;`
      `create pfile from spfile;`
   4. Shut down and restart the Oracle instance.

      Note
      If you are running Oracle on parallel servers, be sure to replicate the parameter file to all instances of Oracle.

   5. Drop all Service Manager tables.

6. In Service Manager, run the following SQL statement to verify that the parameters are in effect:
   `select parameter, value from nls_instance_parameters;`
7. Make sure that the values for NLS_SORT and NLS_COMP match your selection in Step b.
8. Remove groupname, sqldebug, and any other debugging parameters from the sm.ini file, and then run the following command:
   `sm -system_load -system_directory:<path to the Service Manager data unload files>`
9. Validate the case-insensitive unique indexes.
   During the initial system load, Service Manager creates a set of case-insensitive indexes for each table, based on the keys in that table. When the sqldebug:1 parameter is in the sm.ini file, Service Manager logs these indexes the first time it reads a table. You can review the settings created for each table by viewing the sm.log file.
   The case-insensitive unique indexes should be as Oracle function-based indexes where the Column Expression is `NLSSORT("<field name>",'nls_sort=BINARY_CI')`.
   Follow these steps to verify that the case-insensitive unique indexes are created with the correct column expression:
   1. Set sqldebug:1 in the sm.ini file, located in the <SM_install_location>\Server\RUN directory.
   2. Start Service Manager.
   3. Check the sm.log file in the <SM_install_location>\Server\logs directory. The following entry in the log file indicates that the Oracle instance is set to case-insensitive and that you connected to it successfully:
      `RTE I Oracle server settings for language, territory and character set:`
      `AMERICAN_AMERICA.AL32UTF8 (AL16UTF16)`
      `RTE I OCI Client settings for language, territory and character set:`
      `AMERICAN_AMERICA.AL32UTF8 (UTF16)`
      `....`
      `RTE I Oracle instance setting for NLS_SORT is set to BINARY_CI`
      `RTE I Oracle instance setting for NLS_COMP is set to LINGUISTIC`
      `...`
      `RTE I Oracle session is set up in CASE INSENSITIVE mode`
      The following information in the log file indicates that the Dbdict table has an index, DBDICTM1C989DE64, with a key called "NAME", which is case-insensitive:

```
RTE D Table Name: DBDICTM1

RTE D Schema Name                  Index Name
RTE D ------------------------------ ------------------------------
RTE D SMDB                         DBDICTM1C989DE64
RTE D ------------------------------ ------------------------------
```

(Lines continue)

```
Type Column Name                   Column Expression
---- ------------------------------ ----------------------------------------
U    SYS_NC00003$                  NLSSORT("NAME",'nls_sort=''BINARY_CI''')
---- ------------------------------ ----------------------------------------
```

6. If you use IR Expert, you need to grant the following additional privileges for $SM_DB_USER:

> **Note**
> **Note:** For more information, see https://docs.oracle.com/en/database/oracle/oracle-database/19/ccapp/getting-started-with-oracle-text.html and IR Expert solution comparison.

```
GRANT EXECUTE ON CTXSYS.CTX_DDL TO $SM_DB_USER;
GRANT SELECT ON DBA_SCHEDULER_JOBS TO $SM_DB_USER;
GRANT CREATE JOB TO $SM_DB_USER;
```

7. Set up a connection to Service Manager:
   Service Manager connects to the database through your RDBMS client. To set up the connection between the Service Manager application server and your RDBMS, know the name of the database and the login ID and password required to connect to the database server that you created above. The Service Manager initialization file, sm.ini, must be present in the Service Manager server RUN directory.
   To set up connectivity to your Oracle database, follow these steps:
   1. Install the Oracle client on your Service Manager server machine.

      Note
      Service Manager requires a 64-bit Oracle Client library, which can be found in the Oracle 64-bit Client installation directory or Oracle 64-bit Instant Client directory. Visit the Oracle site to download the proper Oracle client for the platform where you plan to run the Service Manager server.

   2. Configure a connection to the Oracle server in the tnsnames.ora file. Define the database name.
      - On UNIX systems, the tnsnames.ora file is located in $ORACLE_HOME/network/admin or can be specified using the TNS_ADMIN environment variable.
      - On Windows systems the tnsnames.ora file is located in the Oracle Home [%ORACLE_HOME%/network/admin] directory.
   3. For Oracle 11.1 or above only, disable ADR tracing by turning off the diag_adr_enabled parameter in the sqlnet.ora file: diag_adr_enabled=OFF. ADR tracing is enabled by default.

      Note
      An Oracle bug ("Multi Threaded OCI Client Dumps Core After Reconnecting To Database") is triggered when ADR tracing is enabled. Once this bug is triggered, the Service Manager server will crash every time when setting up a connection to the Oracle server.

   4. Specify the name of the Oracle database connection in the sqldb parameter in the sm.ini file. You can do this after you install the Service Manager Server by running the Configuration tool and specifying the database name you provided in the tnsnames.ora file.
8. Set time zones for reporting:
   If you plan to report on Service Manager data using your RDBMS tools, set the sqltz parameter in the sm.ini file. This is an optional parameter that the Service Manager server uses to control the storing of date and time values in the RDBMS. This parameter defines the time zone to use as a base for all date and time values. The time zone is specified as the name of the time zone record in the Service Manager tzfile table. The default time zone is Greenwich/Universal (GMT). For more information about using the sqltz parameter, see the System Configuration Parameters topic in the Service Manager Help.

   Caution
   If you use different time zone settings, the dates contained in reports made by your RDBMS utility may be inaccurate.

9. Set Oracle table spaces. Most tables on an Oracle server hold less than 50 KB of data. Service Manager sets the initial storage space size when creating the SQL tables. When manually creating a new Oracle instance, follow these guidelines:
   - Create the database with a block size of 8 KB or a multiple thereof.
   - Create a separate table space for the Service Manager data, and make this the default table space for the Service Manager user.
   - Set the TEMPORARY table space for the Service Manager user to an appropriate temporary table space.
10. Set your UNIX environment variable for Oracle:
    1. Find the path to your Oracle client's 64-bit shared libraries.

2. Set the UNIX environment variable as shown in the following examples. In these examples, the path to the Oracle client shared libraries is set relative to the Oracle environment variable $ORACLE_HOME.
   - C shell: `setenv LD_LIBRARY_PATH $LD_LIBRARY_PATH:$ORACLE_HOME/lib32`
   - Korn shell: `export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:$ORACLE_HOME/lib32`

   Note
   For Linux, you must set the LD_LIBRARY_PATH variable.

11. Create your Oracle database with a UTF-8 code page. All data passed from Service Manager to the Oracle client is encoded in UTF-8; using a UTF-8-based Oracle database reduces the overhead of converting data and prevents the loss of special characters.
12. Secure connections with TLS. See Configuring Transport Layer Security Authentication for more information.
    As Oracle 19c does not support TLSv1.3, we recommend setting up Oracle 19c with TLSv1.2.

Prepare a PostgreSQL database

To prepare a PostgreSQL database, follow these steps:

1. Meet general space requirements:
   Place all Service Manager data in one or more dedicated table spaces within a single PostgresSQL Server instance. These table spaces must contain Service Manager data only. Multiple instances consume more system resources than a single-instance solution.
   Allocate at least 1 GB of data space for a test system. The amount of space necessary for a production system depends on the amount of data that you need to store and your specific implementation.
2. Configure your PostgreSQL to use the UTF-8 encoding.
3. Secure connections with TLS.
   Service Manager Server supports secure connections with TLS. We recommend setting up PostgreSQL Server with TLSv1.2 and above enabled.

   Important
   By selecting to have the communication not via TLSv1.2 or TLSv1.3, you are disabling or bypassing security features, thereby exposing the system to increased security risks. By using this option, you understand and agree to assume all associated risks and hold the company harmless for the same.

   Important
   In case TLSv1.2 or TLSv1.3 communication to the PostgreSQL server is changed from the configuration above, you may need to add relevant protection measures to protect against risks associated with the settings which were changed, which is not provided by the company. By not implementing relevant protection measures you may be exposing the system to increased security risks. You understand and agree to assume all associated risks and hold the company harmless for the same. It remains at all times the Customer's sole responsibility to assess its own regulatory and business requirements. The company doesn't represent or warrant that its products comply with any specific legal or regulatory standards applicable to Customers in conducting Customer's business.

4. Set database schema.
   Service Manager supports to use schema. If no schema is specified during Service Manager configuration, the default schema "public" is used. To create a schema, you must be a database superuser or have the CREATE privilege.
5. Set the sorting of characters for localized systems
   Service Manager supports both case-sensitive and case-insensitive PostgreSQL Server database.
   To use Service Manager in case-insensitive mode, you must select a case-insensitive collation on the PostgreSQL Server before installing Service Manager, and create Extension "citext" for the PostgreSQL Server database schema.

> **Note**
> Make sure that the contrib package is installed on each PostgreSQL database server.

6. Create server connections. Every Service Manager thread, foreground, or background, requires a connection to your RDBMS. Service Manager background processors require 17 connections to run. Make sure that you allocate enough connections for all of your users. For additional information, refer to your RDBMS vendor documentation.
   Make sure that Service Manager Server can connect to the database server by adding the Service Manager server hosts to `<installation dir>\data\pg_hba.conf`.
7. Create a login ID and password for Service Manager to use to connect to your database.
   Create a login ID and password for Service Manager to use to connect to your PostgreSQL Server database.
   The login must have CREATE/ALTER/DROP TABLE authority for the target database.

   > **Note**
   > You must grant the DB account of the `sqllogin` SM parameter the privilege to create Stored Procedures and User Defined Functions.

8. (Optional) If you use IR Expert, when the PostgreSQL database owner is not the super user "postgres", the IR stop list might not be applied correctly and you might get unexpected search results. To resolve this issue, follow the workaround provided in IR solution comparison - Limitation 2.

   > **Note**
   > **Note:** For more information, see https://www.postgresql.org/docs/10/textsearch.htm and IR Expert solution comparison.

9. Set up a connection to Service Manager.
   To set up the connection between the Service Manager application server and your RDBMS, know the name of the database and the login ID and password required to connect to the database server and PostgreSQL database schema that you created above. The Service Manager initialization file, sm.ini, must be present in the Service Manager server RUN directory.
   To set up connectivity to your PostgreSQL database, you must specify the PostgreSQL database connection information in the sm.ini file. You can do this after you install the Service Manager Server by running the Configuration tool and specifying the following database connection information:
   - SQL DB Name
   - SQL User and Password
   - PostgreSQL Schema
   - Case Insensitive

   For more information, see the detailed steps in Install the Service Manager Server.

1.3. Deploy

This section provides detailed instructions about how to deploy Service Manager on-premises.

For new installations, the following order is recommended:

1. Prepare an RDBMS. For details, see Prepare your RDBMS.
2. Install the Service Manager 9.80 Server, and load the applications and demo data. For details, see Install the Service Manager Server.
3. Install the Service Manager 9.81 Windows Client and verify you can connect from the Client to the SM Server. For details, see Install the Service Manager Windows Client.
4. Install the Service Manager 9.81 Web Tier and verify you can connect from the Web Tier client to the SM Server. For details, see Install the Service Manager Web Tier.
5. Install the Service Manager 9.81 Server patch.
6. Install the Service Manager 9.80 language packs. For details, see Install the Service Manager 9.80 language packs.

> **Note**
> If you have already earlier language packs installed, you must upgrade your applications and language packs by running the Upgrade Utility. The Upgrade Utility will upgrade both the applications and language packs at the same time.

7. Upgrade the applications and language packs to 9.81 by using the Upgrade Utility.
8. Install other components as needed, including Smart Analytics, Mobile Applications Client, SMA Service Portal, and Collaboration. For detailed steps, click the appropriate links under the **Deploy** section.

1.3.1. Install the Service Manager Server

This section provides instructions on how to install and configure the Service Manager Server.

Note: Install Service Manager in a development environment and then convert or push the installation to your production environment.

Generate and install your Service Manager licenses

The Service Manager installer automatically copies AutoPass libraries as part of the server installation. AutoPass validates your license and determines what product features are enabled. Follow the steps in this section to obtain your permanent licenses.

1. Go to the Software Licenses and Downloads website.
2. Sign in to **OpenText Passport**.
3. Follow the instructions provided on the website to obtain license keys for your product.
4. Save the license key file to your system. Install license key(s) directly from a license key file. Do not manually transcribe and edit them from the activation certificate. Copy each license key file to the appropriate target system. As part of the process of obtaining a perpetual AutoPass license, a .dat file or several .dat files were sent to the email address that you provided. These files contain the licensing data required to use the applicable Service Manager modules. After you receive these files, follow these steps to move them to the Service Manager Server:
    1. Rename the .dat file to a .txt file so that you can open it with a text editor. For example, rename J8888X1624204.dat to J8888X1624204.txt.
    2. Create a text file named **LicFile.txt** and place it in the *<Service Manager server installation path>*/RUN/ directory.
    3. Copy the license data from the license .txt file that you created in step **1** and paste it in the LicFile.txt file that you created.

    > **Note**
    > Create the LicFile.txt file only once. If you request any additional licenses, append those licenses to the end of this file.

    > **Tip**
    > You can run the **sm -updatelicense:"<file path>/NewLicFile.txt"** command to load a new license .txt file into the existing one.

    > **Note**
    > Make sure that the LicFile.txt file is not read-only. When you upgrade Service Manager from an earlier version, the first time when you start the Service Manager Server, the Server might need to change the content of the license file to the XML format if the AutoPass jar files were updated. If the license file is read-only, the Service Manager Server will fail to start.

5. If you plan to run your system in a horizontally-scaled environment, copy LicFile.txt to the *<Service Manager server installation path>*\RUN\ folder on all hosts running in the horizontal group. In addition, provide the **grouplicenseip**

parameter for each host. The **grouplicenseip** value should match the IP address that you provided when obtaining the license key.

> **Note**
> If you have horizontal scaling implementation setup as a high availability failover cluster, you must request a permanent Service Manager license for both the virtual and the physical IP addresses of the host. Otherwise, the Service Manager Server will fail to start.

Temporary licenses

You can use the following command line option to install a 90-day temporary license to evaluate, test, or develop your Service Manager system: **sm -instantOn**.

You can install a temporary license once per system. The license is valid for 90 days. Within this 90-day period, you must obtain a perpetual license key or a trial evaluation extension to continue using the product. During the last 10 days of the evaluation period, every user who signs in to the system will see a license expiration warning message. To remove the warning message, a system administrator must follow the steps above to obtain perpetual license files.

> **Note**
> If you have not purchased all modules and want to consider adding modules for review during the Instant-On process, contact your account manager.

Next, make sure you meet the Service Manager server installation requirements.

Meet the Service Manager Server requirements

Before you install the Service Manager Server, you should meet the following requirements.

For information about the supported operating systems and databases, refer to the *Service Manager* Support matrix.

Meet the requirements for a Windows system

1. Make sure that you have a compatible operating system with the current updates.

Platform	Server OS
x86-64 / Microsoft Windows	Windows Server 2025

2. The Service Manager Server requires one of the following database resources.

RDBMS	RDBMS server needed	RDBMS client needed
Oracle	Oracle database 19c	64-bit Oracle database client
MS SQL	MS SQL Server 2019, 2022	64-bit Microsoft ODBC Driver 17 for SQL Server
PostgreSQL	PostgreSQL Server 15, 16, 17	n/a

3. Have 8 GB RAM minimum.
4. Create a local administrator account on the Windows server.
5. Have 1 GB free disk space for Service Manager server application.
6. Have a valid TCP/IP port that's greater than 1024.
7. Specify the appropriate number of processes. The system starts one process for each sm command line in the sm.cfg file. By default, each process is limited to 250 concurrent user sessions. The system assigns each user session or background process a dedicated thread.

> **Note**
> If you start background processes by using the sm system start command in the sm.cfg file, then the sm processes own the background process threads. If you start the background processes from a user session inside Service Manager, then the thread controller process that owns the user session also owns the background process threads.

8. Specify the amount of shared memory that you want the system to allocate. You can specify the amount of shared memory the system allocates by using the shared_memory parameter in the **sm.ini** file. By default, the share memory size is 512MB.
9. For an Oracle database, update your system's PATH variable to include the path to the 64-bit versions of your RDBMS client. Refer to your operating system documentation for information on setting variables.
   For a Microsoft SQL Server database, configure a 64-bit ODBC DSN. It is recommended to use Microsoft ODBC Driver 17 for SQL Server as the ODBC DSN.

Meet the requirements for a Linux system

1. Make sure that you have a compatible operating system with the current updates.

Platform	Server OS
Linux 64-bit	○ Red Hat Enterprise Linux 8.8, 8.10, 9.0, 9.2, 9.4, 9.5 ○ Oracle Enterprise Linux 8.9, 8.10, 9.3, 9.4, 9.5 ○ Novell SUSE Linux Enterprise Server 15sp5, 15sp6 ○ Rocky Linux 9.3, 9.4

2. The Service Manager Server requires one of the following database resources.

RDBMS	RDBMS server needed	RDBMS client needed
Oracle	Oracle database 19c	64-bit Oracle database client
MS SQL	MS SQL Server 2019, 2022	64-bit Microsoft ODBC Driver 17 for SQL Server
PostgreSQL	PostgreSQL Server 15, 16, 17	n/a

3. Have 8 GB RAM minimum.
4. Have 1 GB free disk space for Service Manager server application.
5. Have a valid TCP/IP port that's greater than 1024.
6. Specify the appropriate number of processes. The system starts one process for each sm command line in the sm.cfg file. By default, each process is limited to 250 concurrent user sessions. The system assigns each user session or background process a dedicated thread.

> **Note**
> If you start background processes by using the sm system start command in the sm.cfg file, then the sm processes own the background process threads. If you start the background processes from a user session inside Service Manager, then the thread controller process that owns the user session also owns the background process threads.

7. Create a separate user ID that owns Service Manager. Service Manager will not run from a root account. You only need root access to modify the system kernel.

8. Run the installation script from an ANSI terminal to avoid rendering errors.
9. For an Oracle database, update your system's LD_LIBRARY_PATH variable to include the path to the 64-bit versions of your Oracle instant client. Refer to your operating system documentation for information on setting variables. For a Microsoft SQL Server database, configure a 64-bit ODBC DSN. It is recommended to use Microsoft ODBC Driver 17 for SQL Server as the ODBC DSN.
10. Adjust kernel resources. To set proper OS parameters, copy the `<Service Manager server installation path>/RUN/97-sm-sysctl.conf` file into /etc/sysctl.d directory. Follow the instructions in the file to apply the settings.
11. Make sure that the user running the installation script has permission to create new directories in the chosen installation path.

Next, install the Service Manager Server.

Install the Service Manager 9.80 Server

Important: This section is intended for new customers. If you are upgrading from an existing installation, see Upgrade the Service Manager Server.

Install the Service Manager 9.80 Server (Windows)

To install the Service Manager Server on a Windows system, follow these steps:

1. Sign in to the Windows server as a user with local administrator privileges.
2. Extract the server package file into the appropriate drive of the server.
3. Navigate to the **Installation\Server** folder, and double-click **setupserver-9.80.exe**.
4. Click **Next** to read and accept the licensing agreement.
5. Select the **I accept the agreement** option. The **Next** button becomes active.
6. Click **Next** to accept the default installation folder: **C:\Program Files\OpenText\Service Manager 9.80\Server**, or click **Browse** to select a different installation location then click **Next** to continue.

   Caution
   Do not install the server over existing versions of Service Manager. Install in a new folder. The server can't start if installed in a folder with a non-compliant folder name.

   Note
   If you have two different versions of the Service Manager Server (for example, the 9.70 server and 9.80 server) installed on the same server host, you will not be able to start both of them with the same `system` parameter in **sm.ini** - your attempt to start the second server will fail with an error in the server log that indicates the program that created the shared memory is incompatible with the current program. This is because the two servers are different binaries, which can't be attached to the same shared memory.
   If you want to start the second server successfully, you can set a different `system` parameter in **sm.ini**.
7. Click **Install** to start the installation process.
8. When the installation is complete, click **Finish**.
   Note: If you select the **Run the Server Configuration Utility?** option, the server configuration tool will automatically open when you click **Finish**.
9. Follow the steps in the next section to configure the server.

Install the Service Manager 9.80 Server (Linux)

Note:

- The following convention identifies variables that may change depending on your particular installation: `<variable>`. When you see a variable in brackets during the installation, replace the variable with information specific to your system. Do not type the brackets (< >) as part of the command.
- Before you install the Service Manager Server, make sure you have installed the Perl language by running the following command:

`yum install perl`

- You need to install the required libraries by running the following command as "root":
  `yum -y install libnsl`
  You may need to update /etc/yum.repos.d with the working links for the required package if necessary.

To install the Service Manager Server, follow these steps:

1. Sign in to the Linux server.
2. Extract the server package file into the appropriate drive of the server.
3. Navigate to **Installation/Server** directory.
4. Run the server installation script:
   ./setupLinuxX64-9.80.bin
5. Accept the license agreement.
6. At the installation script prompt, type the absolute installation directory where you want to install Service Manager, or accept the default one. Follow these rules:
   - Do not install the server over an existing version of Service Manager. Install in a new folder.

     Note
     If you have two different versions of the Service Manager Server (for example, the 9.70 server and 9.80 server) installed on the same server host, you will not be able to start both of them at the same time - your attempt to start the second server will fail with an error in the server log that indicates the program that created the shared memory is incompatible with the current program. This is because the two server versions are different binaries, which can't be attached to the same shared memory.

   - The installer will ask you to confirm if the specified installation directory is correct.
7. Exit the installer after the server installation is completed.

Install the Service Manager 9.81 Server patch

Note
Before you start, be sure to set up ODBC driver if you use the Microsoft SQL Server database on Linux. See Prepare a SQL Server database.

After you have installed the Service Manager 9.80 Server, you are ready to apply the Service Manager 9.81 Server patch. The server patch provides a script that enables you to apply the patch in an automated way. For detailed steps, see the *Upgrade from version 9.80* section in Upgrade the Service Manager Server.

Install the rng-tools package

If your Service Manager Server is installed on a Linux system, you need to ensure that the the rngd service is enabled in the system.

The **rng-tools** package can increase the entropy collected into the `/dev/random` file and improve the performance of Service Manager.

If you haven't installed the **rng-tools** package, run the following command to install:

- For RedHat or Oracle Linux:
  `sudo yum install rng-tools`
- For SUSE:
  `sudo zypper install rng-tools`

After you install the **rng-tools** package, run the `systemctl start rngd` command to start the rngd service.

Run the Service Manager Server configuration utility

Once you have installed the Service Manager 9.80 Server, run the configuration utility for the Service Manager Server as follows.

You can customize your server installation by modifying the Service Manager initialization file (sm.ini).

You can define the processes that the system starts automatically and the system's startup parameters from the Service Manager configuration file (sm.cfg), which determines how the system starts when started from a service.

When you install the server for the first time, the default settings are configured for the sample database. While this tool is intended to configure new or test implementations running the out-of-box sample data, you can use it whenever you want to change the settings in your system's sm.ini file.



Caution

The configuration utility overwrites your current sm.ini settings. You should back up your system's sm.ini file prior to running the configuration utility to prevent any accidental data loss or loss of service.

Configure the Service Manager Server (Windows)

To configure the Service Manager server, follow these steps:

1. Run the **configure.bat** script, which is located in the <*Service Manager server installation path*> directory. Click **Next** and specify the configurations:
    1. Specify the listener ports:



Note

For a new server installation, the configuration wizard doesn't display the **Enable HTTPS Port** and **HTTPS Port** options, because SSL is disabled by default (the sm.ini file contains the **sslconnector:0** parameter). If you have already set **sslconnector:1** in the sm.ini file and then rerun the configuration utility, the configuration wizard displays two HTTPS options so that you can reconfigure the HTTPS settings:

- Enable HTTPS Port: Select this option to enable an HTTPS port.
- HTTPS Port: The communication port on which you want Service Manager to listen to secure client connection requests.

    - HTTP Port (system): The communications port on which you want Service Manager to listen to client connection requests. The default port is 13080.
    2. Select your deployment environment
        - Demo: Load demo data and set all demo users' password as expired.
        - Production: Load configuration data; Purge all demo data except for the falcon account. Set falcon account's password as expired.
          Note: It is recommended to set a password here. Once you set this password, you need to enter the password before you reset the password for the expired users.
    3. Specify the database type and connection information:
        - Database Type: The database that you want to use to store your data.
        - SQL Database Name:
          For Microsoft SQL Server, this is the ODBC DSN name.
          For Oracle Database, this is the Network Service Name in tnsnames.ora.
          For PostgreSQL Database, this includes database host, port, and database name. The format is as follows: `host=<db_server_host> port=<db_port> dbname=<db_name>`
        - SQL User: The user that Service Manager should connect to your database with. **Note**: For PostgreSQL, make sure the SQL User, db owner, and schema owner are the same user.
        - SQL Password: The password for the user that Service Manager should use to connect to your database.

- Use Unicode Data Type: Use this option if you need to support multiple languages that aren't supported by one collation. Otherwise, use the right collation and don't select this option. For information about how to select SQL Server collations, refer to the SQL Server documentation. This option is for SQL Server only.

  Caution
  Be aware that this operation is not reversible.

- PG Schema: Specify the schema created for the Service Manager database. If you leave it empty, the default "public" schema is used, which is not recommended. This option is for PostgreSQL only.
- Case Insensitive: By default, Service Manager doesn't support case-insensitive searching when using PostgreSQL. Select this option if you want to use case-insensitive searching. Because this option is set when initializing the database, if you need to change this option, you must load the SM data again after the change. This option is for PostgreSQL only.

4. Click **Verify Connection** to confirm that Service Manager can connect to the database.
5. If you have configured and connected your existing database according to the steps above, don't upload the sample data and skip this step.
   If you have configured a new database and want to use the SM demonstration data for testing, upload the sample data. Uploading the demonstration data also uploads the out-of-box applications.

   Note
   If you select **Demo** environment in step 2, before your system goes live, it is mandatory to purge the out-of-box demonstration data by running the **PurgeOutofBoxData** unload script. For details, see knowledge article KM718390.

   Caution
   Do NOT upload the data into an existing production system. Instead, you need to perform an applications upgrade. For more information, see *Service Manager Applications Upgrade* .

   Note
   If you selected the Unicode data type for SQL Server, the applications and demonstration data are uploaded as Unicode. Additionally, to indicate that your SQL Server database uses Unicode, the read-only **Use Unicode Data Type** flag is turned on in the **sqlserver** record of the **sqldbinfo** table.

2. In a text editor, open the sm.ini file and add the following parameters:
   - *jsaccessfilereadregex*
   - *jsaccessfilewriteregex*
   - *jsaccesscmdregex*

   Important
   These parameters specify access restrictions from JavaScript code for file access and command execution. Configuration of these parameters is required so that users with Process Designer tailoring rights aren't able to use the available Service Manager JavaScript APIs to directly access the underlying host file system and command line.
   For more information about how to configure these parameters, see Startup parameters: JavaScript access.

Configure the Service Manager Server (Linux)

To configure the Service Manager server, follow these steps:

1. From a console, run the server configuration script, **configure**, which is located in the *<Service Manager installation path>* directory:

   ```
   ./configure -consolemode
   ```

   1. Specify the listener ports:
      - HTTP Port (system): The communications port on which you want Service Manager to listen to client connection requests. The default port is 13080.
      - Enable HTTPS Port: Select this option to enable an HTTPS port.
      - HTTPS Port: The communication port on which you want Service Manager to listen to secure client connection requests.

      Note

      For a new server installation, the configuration wizard doesn't display the **Enable HTTPS Port** and **HTTPS Port** options, because SSL is disabled by default (the sm.ini file contains the **sslconnector:0** parameter). If you have already set **sslconnector:1** in the sm.ini file and then rerun the configuration utility, the configuration wizard displays these two HTTPS options so that you can reconfigure the HTTPS settings.

   2. Select your deployment environment:
      - Demo: Load demo data and set all demo users' password as expired.
      - Production: Load configuration data; Purge all demo data except for the falcon account. Set falcon account's password as expired.
        Note: It is recommended to set a password here. Once you set this password, you need to enter the password before you reset the password for the expired users.
   3. Specify the database type and connection information:
      - Database Type: The database that you want to use to store your data.
      - SQL Database Name:
        For Microsoft SQL Server, this is the ODBC DSN name.
        For Oracle Database, this is the Network Service Name in tnsnames.ora.
        For PostgreSQL Database, this includes database host, port, and database name. The format is as follows: `host=<db_server_host> port=<db_port> dbname=<db_name>`
      - SQL User: The user that Service Manager should connect to your database with.
      - SQL Password: The password for the user that Service Manager should use to connect to your database.
      - PG Schema: The schema created for the Service Manager database. If you leave it empty, the default "public" schema is used, which is not recommended. This option is for PostgreSQL only.
      - Case Insensitive: By default, Service Manager doesn't support case-insensitive searching when using PostgreSQL. Enable this option if you want to use case-insensitive searching. Because this option is set when initializing the database, if you need to change this option, you must load the SM data again after the change. This option is for PostgreSQL only.
   4. If you have configured and connected your existing database according to the steps above, don't upload the sample data and skip this step.
      If you have configured a new database and want to use the SM demonstration data for testing, upload the sample data. Uploading the demonstration data also uploads the out-of-box applications.

> **Note**
> If you select **Demo** environment in step 2, before your system goes live, you need to purge the out-of-box demonstration data by running the **PurgeOutofBoxData** unload script. For details, see knowledge article KM718390.

> **Caution**
> Do NOT upload the data into an existing production system. Instead, you need to perform an applications upgrade. For more information, see Upgrade Service Manager.

2. In a text editor, open the sm.ini file and add the following parameters:
   - *jsaccessfilereadregex*
   - *jsaccessfilewriteregex*
   - *jsaccesscmdregex*

> **Important**
> These parameters specify access restrictions from JavaScript code for file access and command execution. Configuration of these parameters is required so that users with Process Designer tailoring rights aren't able to use the available Service Manager JavaScript APIs to directly access the underlying host file system and command line.
> For more information about how to configure these parameters, see Startup parameters: JavaScript access.

Start the Service Manager Server

Once the Service Manager Server is installed and is successfully connected to the database, you are ready to start it.

Start the Service Manager Server on Windows

Start the Service Manager server so that users can connect with client sessions. You can start the server from Windows services or from the Windows command prompt.

> **Caution**
> Before proceeding to the next step, check that the SM_pdf.txt file exists in this folder. Do not delete or modify this file; otherwise the server will fail to start. Additionally, if you have copied the HPSM7_pdf.txt or HPSM_pdf.txt file from the RUN folder of a previous server installation, you are recommended to delete it.

Start the server from Windows services

1. From the Windows Control Panel, click **Administrative Tools** > **Services** to open the Services applet.
2. In the Services list, select the **Service Manager 9.80 Server** service.
3. Click **Start** to start the service.
   Windows displays a message that the Service Manager service is starting. After several seconds, the service starts and displays Started in the Status field.

Start the server from the Windows command prompt

1. Open the Windows command prompt on the Service Manager server. Click **Start** > **All Programs** > **Accessories** > **Command Prompt**.
2. Change directories to the RUN folder of your Service Manager installation. For example, `C:\Program Files\OpenText\Service Manager 9.80\Server\RUN`.
3. Type the following command:
   sm -httpPort:13080 -httpsPort:13443

> **Tip**
> You can omit the httpsPort parameter if SSL is disabled in your Service Manager environment.

> **Note**
> You can omit the httpPort and httpsPort parameters if you provide them in the Service Manager initialization file (sm.ini).

4. Press **Enter**.

> **Caution**
> Leave the command prompt open while the Service Manager server runs. Closing the command prompt window stops Service Manager immediately without cleaning up any processes or releasing any record locks.

Start the Service Manager Server on Linux

Start the Service Manager server so that users can connect with client sessions.

1. Run the following command:
   export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:<sm RUN folder>:.
2. Run the following command to go to the RUN directory of your Service Manager:
   cd <install path>/RUN

> **Caution**
> Before proceeding to the next step, check that the SM_pdf.txt file exists in this folder. Do not delete or modify this file; otherwise the server will fail to start. Additionally, if you have copied the HPSM7_pdf.txt or HPSM_pdf.txt file from the RUN folder of a previous server installation, you are recommended to delete it.

3. Type the following command, and then press Enter.
   ./smstart

Now, the Service Manager server is started. Next, you must install either the Windows Client or the Web Tier to get a user interface for accessing the server. See Install the Service Manager Windows Client and Install the Service Manager web tier.

> **Tip**
> The Service Manager server supports several different implementation options to manage large numbers of client connections to the server. For information on how to manage your client connections, see Configuring installation and setup options.

Apply the unload file (SM9.6x compatible)

If you want to use your existing business data instead of the sample data for testing, you must apply the **64bitCompatible_SM970_SM960.unl** unload file to Service Manager. You must use Unload Manager to apply this unload file. See Load an unload file.

If you have uploaded the SM demonstration data when configuring the Service Manager server, there is no need to apply this unload file.

Check and update your custom JavaScript code

> **Important**
> The embedded Mozilla SpiderMonkey JavaScript engine was upgraded to a higher version in the 9.70 server. In order to avoid any potential problems, you need to check your existing scripts and pay attentions when you add new scripts. See JavaScript tailoring guideline.

Regenerate the IR index

As of version 9.62, Service Manager introduces the new IR Expert solution by leveraging the native Full-Text Searching feature in DBMS. For more information, refer to IR Expert.

If you want to use your existing business data instead of the sample data for testing and you are using the IR Expert feature, you need to run the **sm -upgradeir** command to regenerate the IR indexes.

1. Stop the Service Manager server.
2. Go to the RUN directory.
3. Execute the following command to run the IR upgrade utility: **sm -upgradeir**

The IR upgrade utility scans all tables, and regenerate IR indexes on DBMS for the tables with the IR keys defined.

If you have uploaded the SM demonstration data when configuring the server, or if you don't use the IR Expert feature, there is no need to regenerate the IR indexes.

1.3.2. Install the Service Manager web tier

The Service Manager web tier enables users to access the Service Manager server by using a web browser. It can run on both Windows and Linux platforms. One web application server can handle certain numbers of concurrent users and you may need to consider scaling issues and add additional servers to handle the load.

Follow the instructions in this section to install the Service Manager web tier.

Meet the browser and architecture requirements for the web tier

Install the web tier by deploying the webtier-x.xx.war to your web application server. Web application servers may also require you to install the Java Runtime Environment (JRE) or the Java SE Development Kit (JDK). The web tier requires JRE/JDK 17. See Support matrix.

1. Enable the following browser settings:
   - Cookies
   - Java
   - JavaScript
   - Pop-ups. You can add the Service Manager server URL to the pop-up exception list.
2. The Service Manager Web tier uses SSL encryption between the web browser and web application server by default. You must provide a valid web application server certificate to use the following SSL features:
   - Encrypt all communication with the web application server
   - Protect against complex SSL-related attacks
   - Authenticate that the web application server is a valid host

   Note

   If you don't need extra SSL configuration and want to demonstrate the web tier functionality only, you can disable the `secureLogin` parameter from the web tier configuration file (<web tier>\WEB-INF\webtier.properties or <web tier>\WEB-INF\web.xml)

3. Determine the web architecture that you need to support your web tier. A Service Manager web tier requires at least one web application server to run. Depending on the features and scale of your web tier, it may also require a dedicated production web server and additional web application servers. If you use any of the implementation options list below, you need to install and configure a dedicated production web server. If you're not running any of the configurations listed below, then you can run your web tier from a single web application server:
   - A trusted sign-on implementation: You want web client users to sign in to Service Manager without entering a username and password.
     A trusted sign-on implementation requires a web server to accept the pre-authenticated HTTP header information from your authentication software (such as SiteMinder or Integrated Windows Authentication). You must install and configure the authentication software separately. See your web server documentation for information about the HTML headers that your web server expects from your authentication software. For

additional information, see Example: Configuring Single Sign-on with third party authentication on the web tier.

- A load balanced implementation: You want to distribute web client connections among multiple web application servers.
  A load-balanced implementation uses a web server to route connection requests to two or more web application servers. You must configure the web server to identify the web application servers (also known as workers) that are available to accept web client requests. For some web server and web application server combinations, you may need to install additional connection software. For example, to route requests to Tomcat web application servers using the Apache web server, you must install the proper connector. See your web server and web application server documentation for information about routing HTTP requests to available worker web application servers.
- A scaled implementation to support a large number of concurrent users: You want to support 300 or more concurrent web client connections.
  A scaled implementation uses the load balanced implementation described above to support a large number of concurrent web client users. As a general rule, we recommend starting one worker web application server for every 300 concurrent web client connections you want your web tier to support.

Deploy the web tier

The Service Manager web tier contains a JakartaEE-compliant web application that must run on a web application server. Before you proceed, you must have a supported web application server installed. For the supported versions of different web application servers, see Support matrix.

Once your web application server is ready for use, deploy the web tier on your web application server. Each web application server has its own method of deploying web applications. See your web application server documentation for specific instructions on deploying a web application. This section provides sample implementation instructions.



Tip

If you are upgrading from an earlier version of the web tier, see Upgrade the Web Tier.

Enable HttpOnly and Secure session cookies in your web application server



Caution

Be sure to set up your Web Tier with only the minimum required functionality. Enabling more functionality in web application servers (extra languages or scripts such as VB, PHP, CGI, and so on), increases the risk of security breaches. We recommend that you follow well-known best practices for a secure application server.

It is recommended to enable HttpOnly and Secure cookies in your web application server to help prevent malicious JavaScript injection and make the browser (or other HTTP clients) only send cookies over SSL connections. For more information, see KM02233778.

Deploy the web tier

The following table provides a summary of the deployment method required for each supported web application server.

Web application server

Web application server	Deployment method
Apache Tomcat	Copy the webtier-x.xx.war file to the <Tomcat_HOME>\webapps directory and start the web application server. For detailed steps, see *Example deployment on Tomcat on Windows* below.
IBM WebSphere Application Server Liberty	Deploy the webtier-x.xx.war file in Liberty. For general procedures, see *Example deployment in WebSphere Application Server Liberty* below.
JBoss Web Server (JWS)	Copy the webtier-x.xx.war file to the <JWS_HOME>\tomcat\webapps directory and start the web application server. For detailed steps, refer to the configurations in *Example deployment on Tomcat on Windows* below.

Web tier log files

Tomcat:

The default log file is sm.log. The location of this file varies depending on how you start the web application server.

Linux: <TOMCAT_HOME>/bin/
Windows (start by Windows Services): <TOMCAT_HOME>/
Windows (start by startup.bat or catalina.bat): <TOMCAT_HOME>/bin/

You can change the default log file and location in log4j2.properties, which is located in <web app install dir>\webtier-x.xx\WEB-INF.

Note: On Linux, you need to set the **write** permission to the location for the users who are authorized to start and stop the web application server.

WebSphere Application Server Liberty:

The log file is ${wlp.user.dir}/servers/serverName/logs/console.log. To display the log information in sm.log, you can add the following configuration to ${wlp.user.dir}/servers/serverName/server.xml.

```
<library id="webtierLogRef">
        <folder dir="${server.config.dir}\apps\expanded\webtier.war\WEB-INF"/>
</library>

<application type="war" id="webtier" location="webtier.war" name="webtier">
        <classloader commonLibraryRef="webtierLogRef" />
</application>
```

JBoss Web Server:

The default log file is sm.log. The location of this file varies depending on how you start the web application server.

Linux: <JWS_HOME>/tomcat/bin/
Windows (start by Windows Services): <JWS_HOME>/tomcat/
Windows (start by startup.bat or catalina.bat): <JWS_HOME>/tomcat/bin/

You can change the default log file and location in log4j2.properties, which is located in <web app install dir>\webtier-x.xx\WEB-INF.

Note: On Linux, you need to set the **write** permission to the location for the users who are authorized to start and stop the web application server.

Example deployment on Tomcat on Windows

This example describes the steps to deploy the webtier-x.xx.war file on Tomcat 10.1.x.

1. Sign in to the server as a user with local administrator privileges.
2. Stop Tomcat.
3. Set the appropriate heap size for Tomcat.
   If you use the Java 17 default GC (G1), set the heap size to 1.5G (1536M). If you prefer to use a smaller heap size 1G (1024M), you need to use the Parallel GC.
   1. If you start Tomcat by running the batch file (startup.bat), follow these steps:
      1. Open the <TOMCAT_HOME>/bin/setenv.bat file (create it if it does not exist) with a text editor.
      2. Add a line for JAVA_OPTS according to the GC used.
         Java 17 default GC:
         `set "JAVA_OPTS=%JAVA_OPTS% -Xms1536m -Xmx1536m"`
         Parallel GC:
         `set "JAVA_OPTS=%JAVA_OPTS% -Xms1024m -Xmx1024m -XX:+UseParallelGC"`
   2. If you start Tomcat by using the Tomcat Windows service, follow these steps:
      1. Start tomcat10.1w.exe.
      2. Switch to the Java tab.
      3. Set the memory pool according to the GC used.
         `Initial memory pool: 1536 (1024)`
         `Maximum memory pool: 1536 (1024)`
      4. (Parallel GC only) Add the following new line to the Java Options list.
         `-XX:+UseParallelGC`
4. Set the value of `maxParameterCount` to 10000 in the <TOMCAT_HOME>\conf\server.xml file.
5. Use similar steps as in step 3 to add `-Djdk.xml.xpathTotalOpLimit=20000` to the java options of the Tomcat configuration.
6. From the web tier package, copy or save the webtier-x.xx.war file to the <TOMCAT_HOME>\webapps directory. Extract the webtier-x.xx.war file to the webtier-x.xx sub-folder.
7. Set the customize-folder parameter in <webtier>\WEB-INF\web.xml if you want to customize branding or to share the web tier configuration (webtier.properties) among multiple web application sever instances on the same host.
8. Edit the <web tier>\WEB-INF\webtier.properties file or edit the <customize-folder>\config\webtier.properties file if you have configured the customize-folder parameter. The settings you define in this file determine the client preferences for all web clients. View the Service Manager online help for a complete list and more detailed explanation of each parameter.
   1. Open the webtier.properties file in a text editor.
   2. Set the secureLogin, sslPort, serverHost, and serverPort parameters.

Parameter

Parameter	Default value	Description
secureLogin	true	Controls the encryption of network communication between the web application server and the web browser. Set it to false if you don't use Secure Sockets Layer (SSL) connections to the web server. **Note:** To use secure login, you must enable SSL on your web application server. For details, refer to your web application server documentation.
sslPort	8443	This parameter is needed only when secureLogin is set to true. Set it to the SSL port of the web application server.
serverHost	localhost	Specifies the name of the Service Manager host server.
serverPort	13080	Specifies the communications port number to which the Service Manager server listens.

3. Set other common parameters as desired. The table below lists the commonly set parameters and their default values.

Parameter

Parameter	Default value	Description
cacerts	WEB-INF/cacerts	Lists the path to the CA certificates required for SSL support.
compress_soap	true	Specifies if you want to use data compression between web clients and the web tier.
helpServerHost		Specifies the name of the Help Server. See Web parameter: helpServerHost.
helpServerPort		Specifies the communications port number to which the Help Server listens. See Web parameter: helpServerPort.
helpServerContext	help	Specify the context path of the help server home page URL. See Web parameter: helpServerContext.
showHelp	false	Enabling this parameter causes Web clients to display the **Online Help** and **Shortcut List** options after users click the **Help** (question mark) button. The **Online Help** option allows users to access the Help Server you define in the web.xml file. If this parameter is set to **false**, clicking the **Help** button will directly bring up the web client keyboard shortcut list.
refreshMessages	true	Determines whether the browser checks for new messages from the application server.
refreshMessagesInterval	15000	Determines how often (in milliseconds) the browser checks for new messages from the application server.
ssl	false	Enables the web client to encrypt communications using the server's demonstration certificate.
viewactivenotes	false	Determines whether you see a pop-up message when the server sends a message.

4. If there is a hardware load balancer exists on top of the web tier, you must set the webtierFullUrl parameter in the webtier.properties file.

   > **Note**
   > The AWS ELB passes the original host in the Host header, and the Azure Application Gateway passes the original host in the `X-Original-Host` header. The web tier can get the original host from these headers, so it is unnecessary to set the *webtierFullUrl* parameter when you deploy the web tier behind the AWS ELB or the Azure AG.

9. Save and close webtier.properties.
10. Start the Tomcat server.

Example deployment in WebSphere Application Server Liberty

This example provides the general procedures to deploy the webtier-x.xx.war file in WebSphere Application Server Liberty.

1. Deploy the webtier-x.xx.war file in Liberty. For detailed steps, see Deploying applications in Liberty.
   Note: In the server.xml file, you need to do the following:
   - Enable the pages-3.1 feature.
   - Add the following configuration line to the server.xml file. For more information, refer to Configuration for HTTP session management.

   ```
   <httpSession idLength="128" useContextRootAsCookiePath="true"/>
   ```

2. Continue to follow *step 6* to *step 8* as described in *Example deployment on Tomcat on Windows* above.

Access Service Manager by using a web client

For detailed instructions, see Sign in to the web client.

1.3.3. Install the Service Manager Windows Client

Follow these instructions to install the Service Manager Windows Client.

Meet the Service Manager Windows Client requirements

Install a Windows Client

Define a connection from the Windows Client to the Server

Connect the Windows Client to the Service Manager Server

Customize images used by the Windows Client

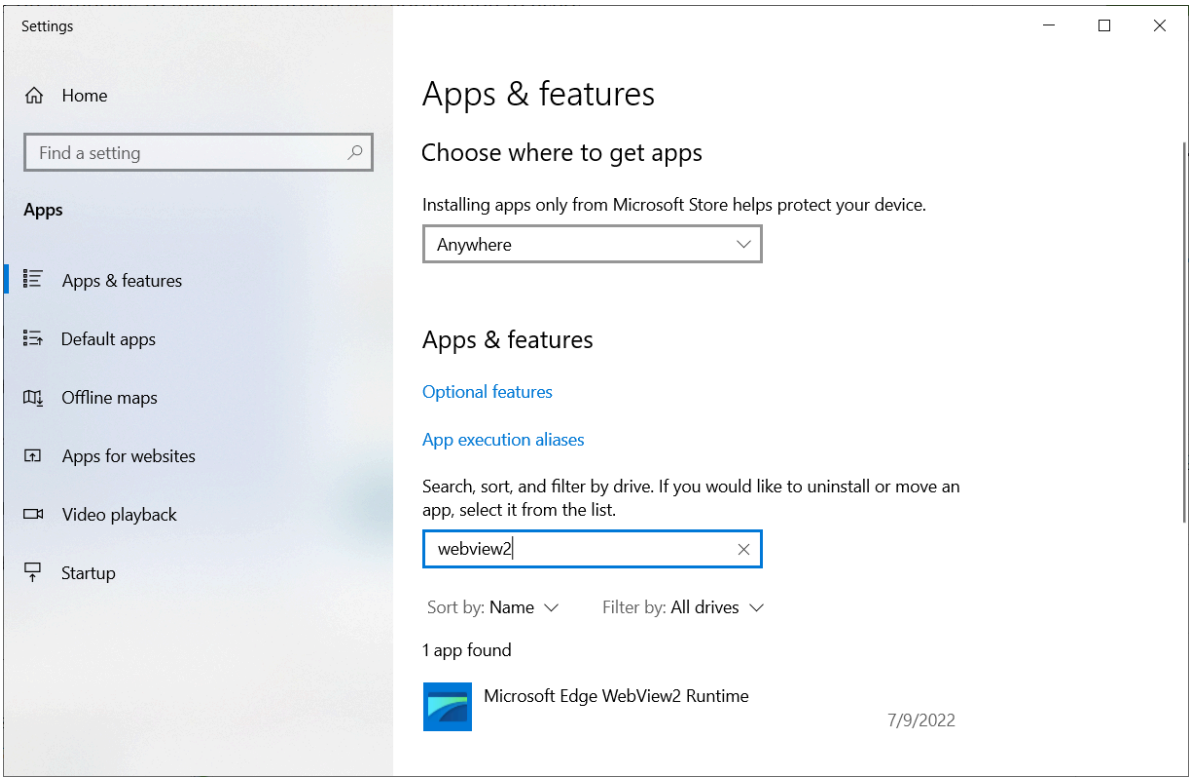
Customize the Windows Client

1.3.3.1. Meet the Service Manager Windows Client requirements

Install at least one Windows client for the system administrator or group of power users. You must install the Windows Client on a Windows workstation. You can support clients running on other operating systems by installing the Service Manager web tier.

You must have local administrator privileges to install the Windows Client. By default, the client workspace and configuration information is stored in the **C:\Users\<*username*>\Service Manager** directory.

The Service Manager Windows Client now uses the SWT Edge browser to render the HTML Editor, the HTML Viewer, and the Help Contents. As a prerequisite, you need to download and install the Microsoft Edge WebView2 Runtime on your workstation. To check whether the Microsoft Edge WebView2 Runtime is installed or not, search it in the **Apps & features** settings. See the following screenshot as an example:



Important

Important

- The Windows Client installer also installs the Client Configuration Utility.
- You can't upgrade previous Service Manager clients to the current version. Instead, simply install the current version of the Service Manager Windows Client in a new folder. We recommend that you remove previous Service Manager Clients.

For more information about Windows client hardware sizing practices, see Sizing.

1.3.3.2. Install a Windows client

The Service Manager Windows Client is mainly for Service Manager administrators and implementers. We recommend that you use the web client for end users.

Note

Note The Windows Client installer also installs the Client Configuration Utility in the ***<Service Manager installation path>*\Client\ClientConfiguration** directory. To run the utility, double-click the **confutil.bat** file in this directory.

You can install a Service Manager Windows client (including the Client Configuration Utility) locally or share a local Windows client on a network where multiple users run from the shared client installation. See the following sections to view instructions for installing the Windows Client on a local machine or share a local Windows client on a network.

Install a local Windows Client and Client Configuration Utility

You can install a local Windows Client by either running the Installer or perform the silent installation.

Run the Installer to install a local Windows Client

Follow these steps:

1. Make sure that the workspace and configuration folders are writeable.
2. Sign in to the Windows server as a user with local administrator privileges.
3. Extract the Windows client installation package into the appropriate drive of the server.
4. Run **setupclient.exe** to start the Installer. The installation wizard starts.
5. Select the language for the installer and click **OK**. The Service Manager Client Installer opens.
6. Click **Next** to read the licensing agreement.
7. Select **I accept the terms of the License Agreement**. The **Next** button becomes active.
8. Click **Next** and the Choose Installation Folder page opens.
9. The default installation folder is `C:\Program Files\OpenText\Service Manager x.xx\Client`. Click **Choose** to navigate to a different location if you want to choose a different path.
10. The feature selection page opens, on which the **Service Manager Client Administrative Tools** option selected by default. This option will install the client administrative tools, such as **Console** and **RAD Debugger**. If you don't want to install these administrative tools, deselect this option.
11. Click **Next** to prepare the installation process. The Summary page opens.
12. Click **Install** to begin copying the installation files. The Service Manager Client Installer displays summary information when the installation is complete.
13. Click **Finish** to exit.

Perform silent installation to install a local Windows Client

Follow these steps:

1. Follow *step 1* to *step 3* as described in *Run the Installer to install a local Windows Client*.
2. Manually create a response file (For example, setupclient.prop) and its content may resemble the following:

```
# Replay feature output
# ---------------------
# This file was built by the Replay feature of InstallAnywhere.
# It contains variables that were set by Panels, Consoles or Custom Code.

#Choose Install Folder
#---------------------
USER_INSTALL_DIR=<SM client installation dir>

#Select the features you would like to install
#---------------------------------------------
USER_INPUT_ADMIN_TOOL=\"Service Manager Client Administrative Tools\"
USER_INPUT_ADMIN_TOOL_1=Service Manager Client Administrative Tools
USER_INPUT_ADMIN_TOOL_BOOLEAN_1=1

#Install
#-------
-fileOverwrite_<SM client installation dir>\\_uninstall\\uninstaller.lax=Yes
-fileOverwrite_<SM client installation dir>\\_uninstall\\resource\\iawin32.dll=Yes
-fileOverwrite_<SM client installation dir>\\_uninstall\\resource\\win64_32_x64.exe=Yes
-fileOverwrite_<SM client installation dir>\\_uninstall\\resource\\remove.exe=Yes
-fileOverwrite_<SM client installation dir>\\_uninstall\\resource\\invoker.exe=Yes
-fileOverwrite_<SM client installation dir>\\ServiceManager.exe=Yes
```

   For example, the response file's content may resemble the following if you hope to install the Windows client in the `C:\SM\Client` directory.

```
# Replay feature output
# --------------------
# This file was built by the Replay feature of InstallAnywhere.
# It contains variables that were set by Panels, Consoles or Custom Code.

#Choose Install Folder
#---------------------
USER_INSTALL_DIR=C:\\SM\\Client

#Select the features you would like to install
#---------------------------------------------
USER_INPUT_ADMIN_TOOL=\"Service Manager Client Administrative Tools\"
USER_INPUT_ADMIN_TOOL_1=Service Manager Client Administrative Tools
USER_INPUT_ADMIN_TOOL_BOOLEAN_1=1

#Install
#-------
-fileOverwrite_C\:\\SM\\Client\\_uninstall\\uninstaller.lax=Yes
-fileOverwrite_C\:\\SM\\Client\\_uninstall\\resource\\iawin32.dll=Yes
-fileOverwrite_C\:\\SM\\Client\\_uninstall\\resource\\win64_32_x64.exe=Yes
-fileOverwrite_C\:\\SM\\Client\\_uninstall\\resource\\remove.exe=Yes
-fileOverwrite_C\:\\SM\\Client\\_uninstall\\resource\\invoker.exe=Yes
-fileOverwrite_C\:\\SM\\Client\\ServiceManager.exe=Yes
```

3. Run the following command to perform silent installation with the response file:
   setupclient.exe -i silent -f <*Response file name*>
   For example:
   setupclient.exe -i silent -f setupclient.prop
   When the silent installation process starts, two setupclient.exe processes appear in the Windows Task Manager. The silent installation process may take some time. Wait until there are 14 files and folders in the target directory. When the silent installation is completed, the two setupclient.exe processes terminate and disappear from the Windows Task Manager.

> **Note**
> Note To keep the console waiting till the silent installation is completed, you can save the above command in a batch file such as install.bat, and then run the **call install.bat** command.

Share the Windows Client on a network

> **Caution**
> Caution If you share a Windows Client on a network share, you run the risk of running out of user licenses or network system resources to run the client.

Follow these steps:

1. Refer to *Run the Installer to install a local Windows Client* to install a local Windows client first.
2. Create a Windows network share to the folder where you installed the Service Manager client and grant users access to the network share. For example, \\my_server\Service Manager Client.
3. Sign in to the computer system of each user who will use the shared client and map the network share to a drive letter on the local system. For example, Drive letter: F: Mapped to: \\my_server\Service Manager Client.
4. Create a Windows shortcut to the ServiceManager.exe file on the network share. For example, F:\ServiceManager.exe.
   By default, all users share common client settings. If you want each user to have individual local client settings, modify the target properties of the Windows shortcut to add the following information after the executable name: `-`

`data %USERPROFILE%\Service Manager\`

The `-data` parameter enables you to specify a path where you want to store client settings. The example path above places a Service Manager folder in the Document and Settings folder of the currently logged in user.

> **Note**
>
> Note If your path name includes spaces, enclose the path in double quotation marks. For example, `F:\ServiceManager.exe -data "%USERPROFILE%\Service Manager\workspace"`.

The Windows Client is now installed. Next, you need to define a connection from the Windows Client to the Service Manager Server.

1.3.3.3. Define a connection from the Windows Client to the Server

The first time you access the Windows Client, the Connections window opens, enabling you to define a connection to a Service Manager host server. You can add and save multiple connection settings from the Connections window.

You must set your connection to an active server process. These are the default settings:

- Client listener on ports 13080 (http) and 13443 (https/SSL) for HTTP clients, including Windows, web, and SOAP API
- Special listener on port 12690 for ServiceCenter Automation (SCAuto)

> **Note**
>
> Note
>
> - The login account (User name/Password) that you want to define for a new client connection must already exist in Service Manager.
> - Service Manager provides an out-of-the-box login account with System Administrator privileges: **System.Admin** (with a blank password). You can use this login account the first time you connect to a Service Manager Server. Disable this account or change its password after creating accounts for all your users. For information about how to create user accounts, see the Service Manager help.
> - If the client computer already has a previous version of Service Manager Client installed, all the existing connection settings are still available in the new client version. However, some client settings in certain old versions may prevent Service Manager Client from running. In this case, you must remove the `<user_home>\ServiceManager` folder on the client computer (where `<user_home>` is the user profile folder) and restart Service Manager Client. This will remove all client settings and the user has to reconfigure the client.

1. From the Windows **Start** menu, click **All Programs > OpenText > Service Manager 9.xx > Service Manager Client**. The Connections window opens.
2. Click **New**. The Connections window displays a new node in the Connections pane.
3. Type or select the connection parameters:

Parameter	Default option	Description
Name	New_configuration	The name of this configuration
User name	User name of the Windows user currently signed in.	The name that you use to sign in to the Server
Password	blank	The password that you use to sign in to the Server
Remember my password	False	An option to tell the system whether to store your password
Automatically log in	False	An option to sign in automatically when you start the Service Manager client
Server host name	localhost	The name of the server that hosts the Service Manager service
Server port number	13080	The port number that your computer uses to connect with the server
Language	blank	The language to use for this session (can differ from the language set on the computer)
Connection identified by a color	blank	An option to change the background color of your connection

4. Click **Advanced** if you want to set the optional advanced connection options:
   - **Connect to External Load Balancer**: An option that enables the Windows Client to connect to the server through an external hardware load balancer.
   - **Compress SOAP Messages**: An option that compresses SOAP messages using GNU zip (gzip) encoding to reduce the amount of data transmittal to and from the server.
   - **Use SSL Encryption**: An option that uses a Secure Socket Layer (SSL) encryption tool to protect your data when transmitting it over the network.

     Caution

     Caution You must define a valid CA certificates file to enable SSL encryption. The client installation includes a sample CA certificate file `cacerts`. The `cacerts` file exists in the `<Service Manager installation path>\Client\ plugins\com.hp.ov.sm.client.common_x.xx.xxxx` folder.

   - **Trace SOAP Traffic**: An option that logs SOAP messages for debugging.
5. Click **OK** to add the advanced features.
6. Click **Apply** to add the connection.
7. To add additional connections, continue to follow *steps 2* to *step 6*.

1.3.3.4. Connect the Windows Client to the Service Manager Server

You can connect to multiple servers from one Service Manager client. Each connection opens in its own window.

1. From the Windows **Start** menu, click **All Programs > OpenText > Service Manager x.xx > Service Manager Client**. The Connection window opens.
2. Double-click a connection or click a connection and then click **Connect**.

1.3.3.5. Customize images used by the Windows Client

You can customize the images that the Windows Client uses by providing alternate versions of the images from a local folder or web server virtual directory. The following guidelines and considerations apply to customized images:

- All custom images must retain their original file name.
- All custom images must retain their original relative path from the icons/obj16 folder.
- You only need to save customized images in the branded/obj16 folder. If the Service Manager client does find updated images in the branded/obj16 folder it uses the default images in the icons/obj16 folder.
- Providing custom images from a web server allows you to automatically update images without having to reinstall the Windows Client.

To prepare custom images from a local folder

You can use the following steps to prepare custom images with the repackaged client.

Note: This image customization method increases the amount of hard disk space required to install the Service Manager Windows Client as the custom images are installed in addition to the default images.

1. Copy the images from the Service Manager client into a temporary folder.
   The client images are located in the following folder: **C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\ com.hp.ov.sm.client.eclipse.user_x.xx\src\resources\icons\obj16** .
2. Edit the images that you want to customize in the temporary folder.
3. Delete any images that you don't customize from the temporary folder.
4. Run the Client Configuration Utility and select the local images option.

   Tip

   Tip To run the Client Configuration Utility, double-click the **confutil.bat** file in the ***<Service Manager installation path>*\Client\ClientConfiguration** directory.

   The Client Configuration Utility creates the following new folder in the Service Manager client installation: **C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\ com.hp.ov.sm.client.eclipse.user_x.xx\src\resources\icons\branded\obj 16** .
5. Copy your custom images to the branded\obj16 folder. You can copy the custom images to the local folder while the Client Configuration Utility is open.
6. Repackage the client as a .zip file or another distribution format.

To prepare custom images from a web server virtual directory

You can use the following steps to prepare custom images from a central web server. This image customization method doesn't increase the amount of hard disk required to install the Service Manager Windows Client. In addition, any changes you make to images on a web server are automatically applied to the Windows Client.

1. Create a virtual directory on your web server to store the custom images.
2. Copy the images from the Service Manager client into a temporary folder. The Service Manager client images are located in the following folder: **C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\ com.hp.ov.sm.client.eclipse.user_x.xx\src\resources\icons\obj16** .
3. Edit the images that you want to customize in the temporary folder.
4. Delete any images that you don't customize from the temporary folder.
5. Copy your custom images to the virtual directory on your web server.
6. Run the Client Configuration Utility and select the web server virtual directory option. The Client Configuration Utility configures the Service Manager client to point to the URL of your web server virtual directory.

   Tip

   Tip To run the Client Configuration Utility, double-click the **confutil.bat** file in the ***<Service Manager installation path>*\Client\ClientConfiguration** directory.

1.3.3.6. Customize the Windows Client

The Service Manager Client Configuration Utility enables you to customize a Windows installation for deploying to end users. This utility is included in the Windows Client installation. For its installation requirements, see Meet the Service Manager Windows Client requirements.

> **Note**
>
> Note The Client Configuration Utility can't push customization changes to previously installed Windows Clients. To change existing installations of the Windows Client, you uninstall the existing client and reinstall using the customized files you create. The Client Configuration Utility only picks up changes made directly from the Windows Client interface or within the utility itself. The Client Configuration Utility can't pick up changes made directly to Windows Client initialization files.

The Service Manager Client Configuration Utility changes the following Windows Client settings:

- Splash screen image that Service Manager displays when users open the Windows Client.
- Name of a provider listed for Service Manager.
- Name of the Service Manager application. For example, Service Manager.
- Location of application images and icons.
- Location of the Help Server where the Windows Client can access documentation.
- Changes that an administrator saves within the Windows Client interface prior to running the Client Configuration Utility, including default login options, connection dialog box configuration options to display, default connection configuration settings, and Help Server configuration options.

Known issues

The Client Configuration Utility has the following known issues:

- You must customize images before you run the Client Configuration Utility. The Client Configuration Utility enables you to change the location of images but not to edit them directly.
- If you deploy a repackaged Windows Client that has a predefined connection over SSL, the Windows Client may display the connection error message `No trusted certificate found.` This error indicates that you are installing the customized client in a different path than the original client used. You can restore the client connection by providing the correct path to the CA certificate file in the client Preferences dialog box.

To customize the Windows Client, follow these steps:

1. Run the Windows command prompt as an administrator. Browse to the *<Service Manager installation path>*\Client\ClientConfiguration directory and then run confutil.bat. The Windows Client Configuration Utility opens.
2. Click **Next**. The Specify Service Manager Directory page opens.
3. Type or select the path to an existing installation of the Service Manager Windows Client, and then click **Next**. The Change Startup Splash Image page opens.
4. Click **Skip** if you want to use the default splash screen image, or type or select the path to the splash screen image that you want to use.
   The default splash screen image, splash.bmp, is located in the following folder: **`C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\ com.hp.ov.sm.client.eclipse.user_x.xx\src\resources\icons\obj16`** .
5. Follow these guidelines when you edit the splash screen image:
   - The image must retain its original file name.
   - The image must be in the Windows bitmap (.bmp) file format.
   - The image should be approximately 500 wide by 600 pixels high. The Client Configuration Utility crops larger images to this size.
6. Click **Next**. The Replace Provider and Application Strings page opens.
7. Click **Skip** to use the default application strings or type the text strings that you want to use for the following items:
   - Provider: Type the company name that you want to display in the Windows Client interface. The default name is Hewlett-Packard Enterprise Development Company, L.P.

- Application: Type the application name that you want to display in the Windows Client interface. The default name is Service Manager.

8. Click **Next**. The Customize Where Service Manager Application Images are Located page opens.
9. Click **No customization** or **Skip** to use the default images. Or select the path to your customized images:
   - Locally: The Client Configuration Utility creates a \branded\obj16 folder where you can place customized images to override the default Windows Client images.
   - Remotely: Type the URL where the Windows Client can access customized images.
10. Click **Next** to continue. The Customize Default Login Options page opens.
11. Click **Skip** if you don't want to create a default connection. Or select whether to display the following options on the Connections dialog box on your customized client:
    - Show the "Remember my password" option: Enabled by default. When disabled, the client's Connections dialog box will not display the **Remember my password** checkbox option.
    - Show the server parameters: Enabled by default. When disabled, the client's Connections dialog box will not display the "Use Login/ Password" and "Use Trusted Sign-on" radio buttons, or the "Server host name" and "Server port number". Also, the New and Delete buttons will be disabled.
    - Show the "Advanced" options page: Enabled by default. When disabled, the client's Connections dialog box will not display the "Advanced" notebook tab. Additionally, the "Trace SOAP Traffic" feature will not be available.
12. Provide the following information about the default connection that you want to create:
    - Hostname: Type the network name or IP address of the Service Manager server that you want the Windows Client to connect to.
    - Port Number: Type the communications port on which the Service Manager server listens to client connection requests. The default communications port is 13080.
    - Compress Messages: Select true to enable message compression between the Windows Client and the Service Manager server. Select false to have messages remain uncompressed.
    - Use SSL Connection: Select true to enable an SSL connection between the Windows Client and the Service Manager server. Select false to use a standard connection.
    - CA Certificate Path: Type or select the local path to the CA certificate used by your SSL connection. Leave this entry blank if you don't use an SSL connection.
      Note: You can find a sample CA certificate file cacerts in the following path: **C:\Program Files\OpenText\Service Manager x.xx\Client\plugins\com.hp.commons_x.xx** .
13. Click **Next**. The Use and Configure Help Server page opens.
14. Click **Skip** if you don't want to provide online help from a Help Server. Or select the **Use Central Help Server** option to establish a connection to a Help Server. Type the following Help Server information:
    - Help Server Host: Type the network name of the Service Manager Help Server to which you want the Windows Client to connect.
    - Help Server Port: Type the communications port on which the Service Manager Help Server listens to client connection requests. The default communications port is 80.
15. Click **Next**. The Client Configuration Utility page opens.
16. Click **Exit**. The client is now configured with your customizations.
17. Zip the contents of the Service Manager x.xx directories to prepare and deploy the customized configuration of the Service Manager x.xx client to other users. The following two directories are necessary for a proper deployment:
    - **C:\Program Files\OpenText\Service Manager x.xx**
    - **%HOMEPATH%\Service Manager**
18. Make sure that end users have the installation DLLs in their windows\system32 directory in order for the deployed client to work properly. Add the following path to their system environment variables: `<Service Manager Home>\plugins\com.hp.ov.sm.client.thirdparty_x.xx\lib;`

1.3.4. Install language packs

For new installations, the language pack installation order is as follows:

1. Install the Service Manager 9.80 Server and load the applications and demonstration data. For details, see Install the Service Manager Server.
2. Install the Service Manager 9.80 language packs. For details, see Install the Service Manager 9.80 language packs.
3. Upgrade the applications to version 9.80. For details, see Upgrade the applications. The Service Manager 9.80 Upgrade Utility upgrades not only your applications but also your language packs.

1.3.4.1. Install the Service Manager 9.80 language packs

The Service Manager 9.80 release includes a language pack package, which provides language packs that must be installed on top of the Service Manager 9.80 applications.



Important

Important If you are upgrading from a version earlier than 9.80, you need to run the Service Manager Upgrade Utility to upgrade your language packs instead of installing the language packs. The Upgrade Utility not only upgrades your applications to the current version, but also upgrade your language packs. For details, see Upgrade the applications.



Note

Note The officially supported languages of Service Manager can meet most localization requirements. However, if you need to localize Service Manager into an unofficially supported language, see Localize Service Manager into unsupported languages.

Language pack installation prerequisites



Important

Important If you are using a SQL Server database, make sure your database supports the languages that you want to install. When configuring a SQL Server database connection for the Service Manager Server, use the **Use Unicode Data Type** option if you need to support multiple languages that aren't supported by one collation, otherwise use the right collation and don't select this option (see Prepare your RDBMS). For information about how to select SQL Server collations, refer to the SQL Server documentation.

Before you install Service Manager 9.80 language packs, you must perform the following tasks:

1. Install the following components from the Service Manager installation media, if you have not done so already:
   - Service Manager 9.80 server
   - Service Manager 9.80 applications
2. Back up your Service Manager application data.

Perform the following steps if you want to customize your Windows client.

1. Install the Client Configuration Utility from the English installation media.
   This task is needed only if you want to make customizations to your Windows clients such as changing the splash screen, adding custom images, adding company branding, saving default settings and connections, and configuring connections to a help server or update site.
2. Open the Client Configuration Utility and customize any additional settings you want the multilanguage Windows client to have.
3. Deploy the customized and localized Windows client installer to your development environment.

Installing the language pack

You must install the language pack on the same system where you installed the Service Manager server. You can install multiple languages on a Service Manager server. This allows users to select the language in which they want Service Manager to display the application forms.

While the server displays Service Manager application forms in the selected language, the operating system language of the client determines the language in which the client menus are displayed. For example, a Windows client running on an English operating system will display menus in English even though the application forms might be in Japanese.

You can install the Service Manager server language pack on either a Windows or Linux system.

Windows installation requirements

- Compatible Windows operating system (See the Support matrix)
  - Install the most current Windows updates
  - Install the Windows language pack
- Existing Service Manager server installation (in English)
  - Back up your application data
- 90 MB disk space for each language pack you install

Linux installation requirements

- Compatible Linux platform (See the Support matrix)
  - Install the most current Linux platform updates
  - Install the Linux language pack
- Existing Service Manager server installation (in English)
  - Back up your application data
- 210 MB disk space per language installed

Service Manager language pack setup

Extract the Service Manager Language Pack installation package into an appropriate drive of the server, and then follow the steps below to install the Service Manager Server language packs.



Note

Note Make sure that you use the same account that you used for the Service Manager Server installation.

Installation for Windows system

1. Run the Windows language pack installer (setup.exe) to start the installation for Windows.
2. Click **Next** to read and accept the license agreement. Select the **I accept the terms of the License Agreement** option if you accept the terms. The **Next** button becomes active.
3. Click **Next** to set the destination folder. The Destination Folder page opens.

   Note

   Note The language pack must be installed in the directory where the Service Manager server is installed.

4. Click **Next** to select one or more language packs you want to install, and then click **Next** to prepare the installation process. The Pre-installation Summary page opens.
5. Click **Install** to start the installation of the selected language packs. You can stop the installation by clicking **Cancel**.

   Note

   Note The setup wizard automatically uploads language data to your database. Depending on your system performance, this process may take twenty minutes or more.

   The Install Complete page opens when the language data is successfully uploaded.
6. Click **Done** to exit the Setup wizard.
7. Restart your Tomcat server. The installed language is available upon your next login.

Installation for Linux

Run the Linux language pack installer (setup_*xx*.bin) to start the installation for Linux.

You can install the language pack for Linux by using the command interface or the GUI interface. Follow the corresponding instructions in the installer to install the language pack step by step.

> **Note**
>
> Note It may take twenty minutes or more to load the language pack data, depending on your system performance.

1.3.5. Install SMA Service Portal

Important: Our company has chosen SMA Service Portal (SMA SP) as the strategic focus for further Service Portal development, and Service Manager Service Portal (SMSP) will no longer evolve functionally. You need to adopt SMA SP as a replacement for SMSP.

SMA SP is included in the Service Management Automation (SMA) suite. The SMA suite is delivered on the basis of ITOM Container Deployment Foundation (CDF). CDF runs on many environments by design. Deploy your IT Operations Management suite on bare metal or a virtual environment, or even in cloud. CDF uses a native cloud toolset, including Docker technologies and Kubernetes from the Cloud Native Computing Foundation (CNCF), which allows the management of container-based applications at scale.

SMA is an analytics-driven service management solution for managing the entire service portfolio and lifecycle within and beyond IT. To enable existing customers of Service Manager to take advantage of the containerized services offered by SMA and keep their existing Service Manager implementations, SMA supports mixed mode deployment.

SMA SP is normally released in a quarterly cadence. You need to install the latest version of SMA-SM that supports your version of Service Manager. For more information about the compatibility between Service Manager and SMA Service Portal, see Compatibility between Service Manager and SMA-SM Service Portal. For more information about how to install the SMA Service Portal, see the Install SMA Service Portal section in the SMA-SM documentation.

1.3.6. Install Service Manager Mobile Applications

The field engineers are typically tasked to work with business users at a customer site, and they spend most of their time away from the office to resolve tickets. When Change Approvers are away from the office, they need a way to quickly approve or deny changes, so that pending work orders can be assigned. Service Manager Mobile Applications connect your company's people and information by providing your team access to the Change Management, Incident Management, Service Desk, and Request applications through the use of smartphones. When these IT operators are away from the office, they can receive and view work assignments and perform the tasks below by using the Mobile Applications:

- Approve and deny change requests.
- Approve and deny interaction.
- Reassign incidents/changes to the proper support group.
- Search the knowledge base.
- Open a support ticket.
- Continue working on-site with customers.

These quick responses improve business metrics, as service level objectives are met.

Before you start

As the System Administrator, you must:

- Have already installed Service Manager on a web accessible location.
- Your web application server meets the requirements that are described in the Service Manager compatibility matrix. For details, refer to the Support matrix.
- Configure the JVM memory settings on the web application server as below:
  - Minimal setting: `-Xms:1024m, -Xmx:1024m, -XX:PermSize=256m`

    - Recommended setting: `-Xms:1500m, -Xmx:1500m`
- Set `JAVA_HOME` to the location where the JDK was installed.
- Review the browser requirements for smartphones.
- Set up your Mobile Applications preferences.
- Configure Service Manager.

Edit the configuration file in the war archive

To edit web.properties in webapp-x.xx.xxxx.war, follow these steps:

1. Open webapp-x.xx.xxxx.war in an archive management program.
2. Extract the web.properties file from the WEB-INF directory to your local system, and then open this file in a text editor.
3. Set the Service Manager server and port parameters as follow to add your server connection information.

Required parameters for Service Manager Mobile Applications client connection

Parameter	Default value	Description
endpoint	http://localhost:13080/SM/ui	Change `localhost:13080` to your Service Manager server host name and port number.
resourceRoot	None	This parameter is commented-out by default. You need to update this value only when the *.js image files and the *.css files are stored on another server. For example, if you moved the static files to another Apache server, change this value to `resourceRoot=http://<host name of another Apache server>:8080/mobile/statics/`.



Tip

- Optionally, you can also define the number of records that appear in every page of the List View by configuring the value in the *recordlistcount* parameter.
- If FIPS (Federal Information Processing Standards) mode is enabled on the Service Manager side, you need to enable FIPS mode in the Mobility client. For details, see the following topics:
    - Configure LW-SSO in the Mobility Client for FIPS mode
    - Configure FIPS mode in the Mobility Client

4. Save the file and add the updated file back to the WEB-INF directory of the webapp-x.xx.xxxx.war archive.

Install Service Manager Mobile Applications

The Service Manager Mobile Applications contains a JakartaEE-compliant web application that runs on your web application server. Each web application server has its own method of deploying web applications. See your web application server documentation for specific instructions on deploying a web application.

The following table provides a summary of deployment methods required.

Web application server	Deployment method
Apache Tomcat	Copy the webapp-x.xx.xxxx.war file to the <Tomcat>\webapps folder and start the web application server.
IBM WebSphere Application Server Liberty	Deploy the webapp-x.xx.xxxx.war file in Liberty. For detailed steps, see Deploying applications in Liberty. **Note:** In the server.xml file, you need to enable the pages-3.1 feature.
JBoss Web Server (JWS)	Copy the webapp-x.xx.xxxx.war file to the <JWS_HOME>\tomcat\webapps folder and start the web application server.

Note

If you are planning to upgrade both platform and applications to version from Service Manager 9.34 or later 9.3x versions, the Mobile Applications will be upgraded to the latest version which supports both power user view and self-service user view.

Mobility log files

Tomcat

The default log file is mobility.log. The location of this file varies depending on how you start the web application server.
Linux: <TOMCAT_HOME>/bin/
Windows (start by Windows Services): <TOMCAT_HOME>/
Windows (start by startup.bat or catalina.bat): <TOMCAT_HOME>/bin/

Note: On Linux, you need to set the **write** permission to the location for the users who are authorized to start and stop the web application server.

WebSphere Application Server Liberty

The log file is ${wlp.user.dir}/servers/serverName/logs/console.log. To display the log information in mobility.log, you can add the following configuration to ${wlp.user.dir}/servers/serverName/server.xml.

```
<library id="mobileLogRef">
    <folder dir="${server.config.dir}\apps\expanded\webapp.war\WEB-INF"/>
</library>

<application type="war" id="webapp " location="webapp.war" name="webapp">
    <classloader commonLibraryRef="mobileLogRef " />
</application>
```

JBoss Web Server

The default log file is mobility.log. The location of this file varies depending on how you start the web application server.
Linux: <JWS_HOME>/tomcat/bin/
Windows (start by Windows Services): <JWS_HOME>/tomcat/
Windows (start by startup.bat or catalina.bat): <JWS_HOME>/tomcat/bin/

Note: On Linux, you need to set the **write** permission to the location for the users who are authorized to start and stop the web application server.

Launch Service Manager Mobile Applications on your smartphone

Before launching Service Manager Mobile Applications on your smartphone, make sure:

1. Your touchscreen smartphone meets the requirements that are described in the Support Matrix.
2. Your smartphone browser conforms to the following configurations:
   - Cookies are enabled.
   - JavaScript is enabled.
   - Pop-ups are enabled. At a minimum, add the Service Manager Mobile Applications host name to the pop-up exception list.
3. You have a valid Service Manager username and password.

To launch Service Manager Mobile Applications on your smartphone, follow these steps:

1. In the web browser on your smartphone, type the Service Manager Mobile Applications web URL.
   Power user view: **http://<*servername*>:<*portnumber*>/<*appname*>/std/** (/std/ can be omitted)
   Self-service user view: **http://<*servername*>:<*portnumber*>/<*appname*>/ess/** (/ess/ can't be omitted)
   The Login page opens.
2. On the Login page, type your user name and password. The user name is auto-filled in the login screen the next time you log in.
3. Tap the **Language** field and scroll down to select your preferred language from the list, and then tap **Done**. The language defaults to your settings on Service Manager.

   Note
   - Service Manager Mobile Applications doesn't support right-to-left displayed languages such as Arabic and Hebrew.
   - Available languages are retrieved from the Service Manager server. Only when your Service Manager server has a language pack installed, the corresponding language appears in the language selection list.
   - If you clear your browser cookies from the smartphone, you need to type your user name and select the preferred language again the next time you log in.

4. Tap **Login** to log on to Mobile Applications.

   Note
   Service Manager Mobile Applications supports multi-tenancy mode.

Service manager no longer support desktop browsers. If you're accessing the application from a desktop browser, use the DevTools Toggle device toolbar or another mobile simulator.

1.3.7. Install Smart Analytics

Built on Service Manager (SM) and using an OEM-licensed version of IDOL, SM Smart Analytics heralds the debut of the "Big Data" edition of Service Manager.

Smart Analytics offers the following capabilities: Smart Ticket, Hot Topic Analytics, and Smart Search. For more information, see Smart Analytics overview.

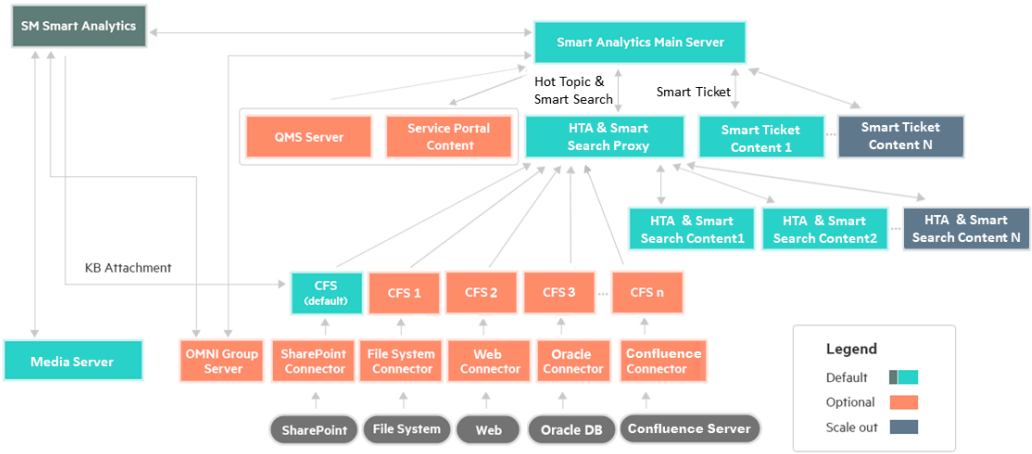
If you are upgrading your Smart Analytics from an earlier version, see Upgrade Smart Analytics. For information about how to configure and administer Service Manager Smart Analytics, see Smart Analytics Administration.

Note
Note

Before you install Smart Analytics, make sure that your Service Manager Applications version is no earlier than the Smart Analytics version.

1.3.7.1. Installation overview

Smart Analytics is powered by IDOL. You need to prepare servers to deploy Smart Analytics. Check the information of Smart Analytics architecture as displayed in the following diagram before installation.



Smart Analytics deployment scenario

This section describes Smart Analytics installation options based on different deployment scenarios.

Small to medium enterprises with data volume less than 6 million records

For small to medium enterprises with data volume smaller than 6 million records, Service Manager provides four templates to fit different needs:

- Quick Install
  If you don't need search file systems, web sites or SharePoint, you can choose **Quick Install** to deploy only necessary components for Smart Analytics on one server. When performing quick install, you can only configure the server ports, while service ports and index ports are automatically configured based on the server ports that you specify.

  Tip
  Tip You can install corresponding connectors using the **Customize** template if you want to search external data sources in the future.

- Basic for Smart Analytics
  If you don't need search file systems, web sites or SharePoint, but need configure all ports, you can choose the **Basic for Smart Analytics** template in Advanced Install to deploy only necessary components for Smart Analytics on one server. When using **Basic for Smart Analytics**, you can manually configure the server ports, service ports and index ports.

  Tip
  Tip You can install corresponding connectors using the **Customize** template if you want to search external data sources in the future.

- All in One
  If you would like to search file systems, web sites or SharePoint, and install all components on one server, you can choose the **All in One** template in Advanced Install. In this scenario, all Smart Analytics components will be installed. When using **All in one**, you can manually configure the server ports, service ports and index ports.

Large enterprises with data volume more than 6 million records

For large enterprises with data volume larger than 6 million records, it is recommended that you install Smart Analytics on multiple servers using the **Customize** template in Advanced Install. For more information, see Example: Deploying Smart Analytics on multiple servers

Smart Analytics installer components

The following table lists the components available in Quick Installation and Advanced Installation respectively:

Installation type	Components included
Quick installation	• Smart Analytics Main Server • Main Content Server (content server for Smart Ticket) • Smart Search Proxy Server (proxy server for Hot Topic Analytics and Smart Search) • Two Smart Search Content Servers (content servers for Hot Topic Analytics and Smart Search) • Default Connector Framework Server (for attachment index for Smart Search) • Media server for OCR
Advanced installation	• All components included in Quick Installation • Distributed Content Server • Distributed Media Server • CFS server • SharePoint Connector • Web Connector • File System Connector • Oracle Connector • Confluence Connector • OMNI Group Server

Here is the description of the components for the Smart Analytics main server.

Component	Description
DIH (Distributed Index Handler)	Allows you to efficiently split and index extremely large quantities of data into multiple Smart Analytics content servers, in order to create a completely scalable solution that delivers high performance and high availability. It provides a flexible way of transparently batching, routing, and categorizing the indexing of internal and external content into Smart Analytics main server.
DAH (Distributed Action Handler)	Distributes actions to multiple child components of Smart Analytics main server. It enables you to scale your system in a linear manner, increasing the speed that runs actions and saving processing time.
Community	A Smart Analytics main server component that manages users and communities.
Agentstore	A Smart Analytics main server component that stores agents and categories.
Category	A Smart Analytics main server component that manages categorization and clustering
View	A Smart Analytics main server component that converts files in a repository to HTML formats for viewing in a Web browser.

Here is the description of the Smart Analytics connectors and protocol required for the installation.

Component	Description
OMNI Group Server	Communicates with SharePoint connector and LDAP to retrieve access permissions for your users. In this way, the access permissions can be applied to documents in the Smart Analytics main server. **Note** Note LDAP - Lightweight Directory Access Protocol. A protocol that applications can use to retrieve information from a server. LDAP is used for directory services (such as corporate e-mail and telephone directories), and user authentication.
SharePoint Connector	Retrieves information from a Microsoft SharePoint repository, through the SharePoint web services. The connector can also retrieve information from an instance of SharePoint Online.
File System Connector	Retrieves various document types from file system so that the documents are available in Smart Search.
Web Connector	A powerful tool for retrieving documents from a web site. The Web Connector uses spiders to find web pages and to process the web pages for content and links to other web sites. Web Connector can retrieve various document types, including web documents, Word, Excel, and PDF files.
Oracle Connector	Retrieves Knowledge Management documents from an Oracle database server so that the documents are available in Smart Search.
Confluence Connector	Retrieves information from a Confluence server so that the documents are available in Smart Search.

Note

Note After you install a connector, you must configure the parameters for this connector from the corresponding .cfg file before you start the service. For a configuration example, see Configure connectors.

Default configuration for server ports

Server name	Default port number
SM Smart Analytics Main Server port	9000
Main Content Server (content server for Smart Ticket) port	10010
Smart Search Proxy Server (proxy server for HTA & and Smart Search) port	20010
First Smart Search Content Server (content server for HTA & Smart Search) port	30010
Second Smart Search Content Server (content server for HTA & Smart Search) port	30020
Media Server port	18000
CFS Server port	7000
OMNI Group Server port	5057
SharePoint Connector port	36000
Web Connector port	5678
File System Connector port	1234
Oracle Connector port	5555
Confluence Connector port	7128
Query Manipulation Server port	16000
Portal Content Server port	10020

To check ports occupied by Smart Analytics, see Get ports occupied by Smart Analytics.

Note

Note For SM Smart Analytics Main Server and Smart Search Proxy Server, it is recommended that you reserve the next 99 consecutive ports following the port you configured for the server. For example, if you specify the port number of Smart Analytics Main Server as 9000, then you need to reserve all the port numbers from 9000 to 9099.

1.3.7.2. System Requirements

This section lists the hardware requirements and the supported operating systems for the Smart Analytics servers.

Hardware requirements

Quick installation

- 64 GB RAM (8 GB minimum)
- 8 Core (a minimum of 4 dedicated CPU - XEON 3 GHz or above)
- 500 GB disk

Advanced installation

About hardware requirements for typical install scenarios, please see table below.

Installation template	Hardware requirements
Basic for Smart Analytics	• 64 GB RAM (8 GB minimum) • 8 Core (a minimum of 4 dedicated CPU - XEON 3 GHz or above) • 500 GB disk
All in one	• 64 GB RAM (8 GB minimum) • 8 Core (a minimum of 4 dedicated CPU - XEON 3 GHz or above) • 700 GB disk
Basic for Service Portal	• 64 GB RAM (8 GB minimum) • 8 Core (a minimum of 4 dedicated CPU - XEON 3 GHz or above) • 600 GB disk
Distributed IDOL content server	• 4 GB RAM • A minimum of 2 dedicated CPU - XEON 3 GHz or above • 100 GB disk
Distributed media server	• 4 GB RAM • A minimum of 2 dedicated CPU - XEON 3 GHz or above • 100 GB disk
Customized scenarios	For the Smart Analytics proxy server (including DIH, DAH, community, category, agentstore, view, and one second-level DIH and DAH), one Hot Topic Analytics content, and two sets of Smart Search content: • 64 GB RAM (8 GB minimum) • 8 Core (a minimum of 4 dedicated CPU - XEON 3 GHz or above) • 500 GB disk For each content or media server: • 4 GB RAM • A minimum of 2 dedicated CPU - XEON 3 GHz or above • 100 GB disk For each connector and CFS and OMNI Group server: • 2 GB RAM • CPU - XEON 3 GHz or above • 20 GB disk

Note

Note For more information, see Sizing.

Supported operating systems

For the information about supported operating systems, see Support matrix.

1.3.7.3. Install Smart Analytics on Windows

Before you install Smart Analytics, make sure that your servers meet the system requirements as specified in System requirements.

Caution

Caution If you are re-installing content servers for a different Smart Analytics deployment architecture, don't manually add or remove Smart Search content servers in the IDOLServer.cfg file when the installation is complete. Otherwise, you would encounter data loss when doing full index to index data.

To properly add or remove Smart Search content servers, see the following sections:

Add a content server for Smart Search

Remove a content server for Smart Search

Pre-install considerations

There are a number of key factors you should consider before deploying Smart Analytics:

- **Operating system** Smart Analytics supports Windows family and Linux operating system. It is recommended that all servers used to utilize the same OS for ease of management.
- **Storage I/O performance**: Ideally, each SM Smart Analytics Component should have access to its own disk or partition, independently capable of 120 MB/s bandwidth and 180–200 IOPS (bandwidth being the more important metric).
- **Storage capacity**: Due to bandwidth bottlenecks of conventional HDDs, it is recommended that the size of the partitions utilized for each SM Smart Analytics Component should not exceed 300 GB.
- **Network** To properly interface with data repositories, Smart Analytics requires high-speed network access to the data repositories. In practice, this is accomplished by placing connectors physically "close" to the original data sources.

Install

To install Smart Analytics on a Windows-based system, follow these steps:

1. Obtain the Smart Analytics installer for Windows.
2. Unpack the .zip file and then double-click the setup application (`setupSmartAnalyticsWindowsX64.exe`).
3. The SM SmartAnalytics Setup wizard opens. Read the introduction, and then click **Next**.
4. Read the License Agreement. To continue the installation, select **I accept the terms of the License Agreement**, and then click **Next**.
5. Select **New Installation**, and then click **Next**.
6. Choose an installation folder, and then click **Next**. The default installation folder is **`C:\Program Files\OpenText\Service Manager x.xx\SmartAnalytics`** .
7. Select **Quick Install** or **Advanced Install** as your installation type, and then click **Next**.
8. Continue with the corresponding installation steps.
   Quick Install
   Quick Install deploys the minimum required components to perform Smart Analytics on Service Manager internal data only. It sets default configurations, and no extra data source connectors are installed.
   To perform quick installation of Smart Analytics on Windows, continue with these steps:
   1. Select **Quick Install** as the installation type, and then click **Next**.
   2. Specify the IP address of your Service Manager Server, and then click **Next**.

You need to specify the IP addresses (or hostnames) of the Service Manager servers that are permitted to send administrative and query actions to the Smart Analytics servers. Use commas to separate multiple addresses (don't use a space before or after a comma).

> **Note**
>
> Note
>
> - Service Manager Server uses load balance, therefore, all slave Service Manager addresses should be specified in this step.
> - Please use a valid FQDN or IP address for the server address. Do not use the localhost or 127.0.0.1.

3. Follow the configuration steps to configure ports for these servers. Click **Next** after each step.
   - SM Smart Analytics Main Server
   - Main Content Server (content server for Smart Ticket)
   - Smart Search Proxy Server (proxy server for HTA & and Smart Search)
   - Smart Search Content Server 1 (content server 1 for HTA & Smart Search)
   - Smart Search Content Server 2 (content server 2 for HTA & Smart Search)
   - Media Server
   - CFS Server
4. Check the pre-installation summary. If you want to change your configuration, click **Previous**. Otherwise, click **Install** to start the installation.
5. Wait for the installation to complete

Advanced Install

Advanced Install provides multiple installation templates, each including a group of server components and connectors. You can deploy components by selecting different installation templates for different scenarios.
To perform advanced installation of Smart Analytics on Windows, continue with these steps:

1. Select **Advanced Install** as the installation type, and then click **Next**.
2. Specify the IP address of your Service Manager Server.
   You need to specify the IP addresses of the Service Manager servers that are permitted to send administrative and query actions to the Smart Analytics servers. Use commas to separate multiple addresses (don't use a space before or after a comma).

   > **Note**
   >
   > Note
   >
   > - Service Manager Server uses load balance, therefore, all slave Service Manager addresses should be specified in this step.
   > - Please use a valid FQDN or IP address for the server address. Do not use the localhost or 127.0.0.1.

3. Choose an installation template, and then click **Next**. To customize your installation, select the **Customize** template to choose from the available components.
   Service Manager provides six out-of-box installation templates.

Installation template	Components included	Description
Basic for Smart Analytics	▪ Proxy Server Components (including DIH, DAH, community, category, agentstore, view, one second-level DIH and DAH, one Main Content Server, and two Smart Search Content Servers) ▪ Media server ▪ CFS Server	Deploys the minimum required components to perform Smart Analytics on Service Manager internal data only. No extra data source connectors are installed.
All in one	▪ Proxy server components (including DIH, DAH, community, category, agentstore, view, one second-level DIH and DAH, one Main Content Server, and two Smart Search Content Servers) ▪ Media Server ▪ CFS Server ▪ OMNI Group Server ▪ SharePoint Connector ▪ Web Connector ▪ File System Connector ▪ Oracle Connector ▪ Confluence Connector	Deploys all components in one physical machine. In this way, you can use the full Smart Analytics functionality on both Service Manager without a large volume of data.
Distributed IDOL content server	Content Server	Installs remote or additional content servers to handle more data.
Distributed Media server	▪ Media Server ▪ Media Proxy Server	By default, one Media server is included in the basic installation to handle OCR process. If there are too many image processing requests, you can deploy a distributed Media server.
Customize	No pre-set components	Install any component based on your customized requirements.  **Note** Note Do not select **Portal Content** and **Query Manipulation Server**.



Note

Note If you plan to install the content servers and the proxy server on different machines, it is recommended that you install the content servers first. Because without the content server installed and started, the proxy server can't start and the OOB data can't be imported. For detailed steps of installing Smart Analytics components on different machines, see Example: Deploying Smart Analytics on multiple servers

4. Specify the IP addresses of the remote machines on which you have installed or will install Smart Analytics components such as content servers, CFS servers, and connectors, and then click **Next**.
5. Follow the configuration steps to configure the components you selected. Click **Next** after each configuration step.
   If you did not select Main Content, Smart Search Content 1, or Smart Search Content2 in the installation template as described in step c, the installer displays configuration pages for you to specify the IPs and ports of the remote servers on which you have installed or will install these contents servers.

If you don't specify IPs and ports for remote content servers on the configuration pages, or you want to change your settings after installation, you need to manually edit the IPs and Ports in the following configuration files and corresponding sections:

File	Configurations
<*Smart Analytics Installation*>/IDOL/IDOLServer.cfg	[Service] ..... ServiceStatusClients=...,<Remote Main Content Server IP> ServiceControlClients=...,<Remote Main Content Server IP> ..... [AdminRole] ..... Clients=<Remote Main Content Server IP> ..... [QueryRole] ..... Clients=<Remote Main Content Server IP> ..... [IndexRole] ..... Clients=<Remote Main Content Server IP> ..... [IDOLServer1] ..... Host=<Remote Main Content Server IP> Port=<Remote Main Content Server Port> .....

<Smart Analytics Installation>/level2proxy/IDOLServer.cfg	[Service] ..... ServiceStatusClients=.....,<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ServiceControlClients=.....,<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ..... [AdminRole] ..... Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ..... [QueryRole] ..... Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ..... [IndexRole] ..... Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ..... [DistributionIDOLServers] Number=2 [IDOLServer<0>] Host=<First Remote Smart Search Content Server IP> Port=<First Remote Smart Search Content Server Port> ..... [IDOLServer<1>] Host=<Second Remote Smart Search Content Server IP> Port=<Second Remote Smart Search Content Server Port>

Some useful information for configuration:

- **Configure Smart Search Proxy Server** > **Replicas**: Specifies the number of identical copies of each document to index. This is configured in the level2proxy\IDOLServer.cfg file.
  The default value is set to **0**, which means that there is only one copy of each document for Smart Search. If you set the number as 1, it means that there will be two mirrored copies of the document in your Smart Search content servers.
  In the Smart Search proxy level, content is distributed between virtual nodes, which the DIH assigns to its child servers. When you configure replicas, DIH copies the documents in a particular virtual node to two or more child servers. This method ensures there are two mirrored copies of the document in your system without you having to set up specifically mirrored child servers.

> **Note**
> Note The number of copies (that is, the value of Replicas plus one) must be no more than the number of child servers. If you create more copies than existing Smart Search child servers, DIH doesn't start.

- **Configure SharePoint Connector** > **SharePoint URL Type**:
  - To retrieve all content databases and site collections, set the value of this parameter to **WebApplication**. Set the value of the SharepointUrl parameter to the URL of the web application. You can't use this value with SharePoint Online.
  - To retrieve only one site collection, set the value of this parameter to **SiteCollection**. Set the value of the SharepointUrl parameter as the URL of the site collection.
- **Configure LDAP Repository** (for OMNI Group server):
  - LDAP User Base: the base DN or top level of the directory tree you want to search for users. This indicates where in the LDAP directory tree you want to begin the search.
  - LDAPGroupBase: the base DN or top level of the directory tree you want to search for groups. This indicates where in the LDAP directory tree you want to begin the search.
  - (Optional) UserFilter: the query you want to use to retrieve the set of users you want to store in the group server. For example, `UserFilter=(objectClass=hpeEmployee)`.
  - (Optional) GroupFilter: you can use GroupFilter to set the filter that is passed to the LDAP server to request a list of groups. For example, `GroupFilter=(objectClass=hpeGroup)`.

6. Check the pre-installation summary. If you want to change your configuration, click **Previous**. Otherwise, click **Install** to start the installation.
7. Wait for the installation to complete.

9. The Start Service page opens when the installation is complete. If you want to start the services now, select **Yes** and then click **Next**. Otherwise, select **No** and then click **Next**

> **Note**
> Note We recommend you start the services manually after you configure the .cfg files for each connector.

10. If you selected **Yes** in step 8, the Import OOB Data page opens. If you want to import the out-of-box data now, select **Yes** and then click **Next**. Otherwise, select **No** and then click **Next**.

> **Note**
> Note Importing the out-of-box data will erase the previous data for Hot Topic Analytics.
> You can also import the out-of-box data at any time after the installation, by running the *<Smart Analytics Installation>*/OOBData/oobdata.cmd file.

11. Wait a few minutes for the installer to finish starting the services and importing the OOB data, and then click **Done**.

> **Note**
> Note
> - Make sure that all the required components are started. If not, you need to start the corresponding component manually.
> - Under some circumstances, users are unable to connect to the main server in Service Manager while the status of SM Smart Analytics Main Server shows **Started** in the Services snap-in. You can check `<Smart Analytics Installation>/IDOL/logs/application.log` to see if agentstore was started. If not, restart the SM Smart Analytics Main Server service to troubleshoot the problem.

Post-install actions

After installation, you can modify the following configuration files to change the previous settings:

- Smart Analytics main server: <*Smart Analytics Installation*>/IDOL/IDOLServer.cfg
- Media Server: <*Smart Analytics Installation*>/MediaServer#/MediaServer#.cfg
- Media Proxy Server: <*Smart Analytics Installation*>/MediaProxyServer/dah.cfg
  This is only required if you have installed multiple Media Servers in the distributed mode.
- Content server: <*Smart Analytics Installation*>/Content#/Content#.cfg
- CFS server: <*Smart Analytics Installation*>/CFS/CFS.cfg
- SharePoint connector: <*Smart Analytics Installation*>/SharepointRemoteConnector/SharepointRemoteConnector.cfg
- OMNI group server: <*Smart Analytics Installation*>/OmniGroupServer/OmniGroupServer.cfg
- Web Connector: <*Smart Analytics Installation*>/WebConnector/WebConnector.cfg
- File system connector: <*Smart Analytics Installation*>/FileSystemConnector/filesystemconnector.cfg
- Oracle connector: <*Smart Analytics Installation*>/OracleConnector/OracleConnector.cfg
- Confluence connector: <*Smart Analytics Installation*>/ConfluenceRESTConnector/ConfluenceRESTConnector.cfg
- Query Manipulation Server: <*Smart Analytics Installation*>/QMS/QMS.cfg

Note

Note If you installed any connectors, you may need to edit their configuration files manually. For more information, see Configure connectors.

Restart the corresponding components after you modify the related configuration files.

Tip

Tip If you want to uninstall SM Smart Analytics, see Uninstall Smart Analytics.

1.3.7.4. Install Smart Analytics on Linux

Before you install Smart Analytics, make sure that your servers meet the system requirements as specified in System requirements.

Caution

Caution If you are re-installing content servers for a different Smart Analytics deployment architecture, don't manually add or remove Smart Search content servers in the IDOLServer.cfg file when the installation is complete. Otherwise, you would encounter data loss when doing full index to index data.

To properly add or remove Smart Search content servers, see the following sections:

Add a content server for Smart Search

Remove a content server for Smart Search

Pre-install considerations

There are a number of key factors you should consider before deploying Smart Analytics:

- **Operating system** Smart Analytics supports Windows family and Linux operating system. It is recommended that all servers used utilize the same OS for ease of management.
- **Storage I/O performance**: Ideally, each IDOL engine should have access to its own disk or partition, independently capable of 120 MB/s bandwidth and 180–200 IOPS (bandwidth being the more important metric).
- **Storage capacity**: Due to bandwidth bottlenecks of conventional HDDs, it is recommended that the size of the partitions utilized for the IDOL services should not exceed 300 GB.
- **Network** To properly interface with data repositories, Smart Analytics requires high-speed network access to the data repositories. In practice, this is accomplished by placing connectors physically “close” to the original data

sources.

Install

To install or re-install the SM Smart Analytics servers on a Linux-based system, follow these steps:

1. Obtain the SM Smart Analytics installer for Linux .
2. Unpack the .zip file and then execute the binary file (`setupSmartAnalyticsLinuxX64.bin`) from the command line on the Linux server.
3. Read the introduction, and then press **Enter**.
4. Read the License Agreement, and then press **Enter** repeatedly until you see **DO YOU ACCEPT THE TERMS OF THIS LICENSE AGREEMENTS? (Y/N)**.
5. Type **Y**, and then press **Enter**.
6. Select **New Installation**, and then press **Enter**.
7. Specify an installation folder, and then press **Enter**. The default installation folder is `/opt/MicroFocus/ServiceManager9.xx/SmartAnalytics` .
8. Select **Quick Install** or **Advanced Install** as your installation type, and then press **Enter**.
9. Continue with the corresponding installation steps.

   Quick Install

   Quick Install deploys the minimum required components to perform Smart Analytics on Service Manager internal data only. It sets default configurations, and no extra data source connectors are installed.

   To perform quick installation of Smart Analytics on Linux, continue with these steps:

   1. Specify the IP address of your Service Manager Server, and then press **Enter**.
      You need to specify the IP addresses of the Service Manager servers that are permitted to send administrative and query actions to the Smart Analytics servers. Use commas to separate multiple addresses (don't use a space before or after a comma).

      Note

      Note

      - Service Manager Server uses load balance, therefore, all slave Service Manager addresses should be specified in this step.
      - Please use a valid FQDN or IP address for the server address. Do not use the localhost or 127.0.0.1.

   2. Follow the configuration steps to configure ports for these servers. Press **Enter** after each step.
      - SM Smart Analytics Main Server
      - Main Content Server (content server for Smart Ticket)
      - Smart Search Proxy Server (proxy server for HTA & and Smart Search)
      - Smart Search Content Server 1 (content server 1 for HTA & Smart Search)
      - Smart Search Content Server 2 (content server 2 for HTA & Smart Search)
      - Media Server
      - CFS Server
   3. Check the pre-installation summary, and then press **Enter**.
   4. Wait for the installation to complete.

   Advanced Install

   Advanced Install provides multiple installation templates, each including a group of server components and connectors. You can deploy components by selecting different installation templates for different scenarios.

   To perform advanced installation of Smart Analytics on Linux, continue with these steps:

   1. Specify the IP address of your Service Manager Server.
      You need to specify the IP addresses of the Service Manager servers that are permitted to send administrative and query actions to the Smart Analytics servers. Use commas to separate multiple addresses (don't use a space before or after a comma).

Note

Note

- Service Manager Server uses load balance, therefore, all slave Service Manager addresses should be specified in this step.
- Please use a valid FQDN or IP address for the server address. Do not use the localhost or 127.0.0.1.

2. Choose an installation template, and then press **Enter**. To customize your installation, select the **Customize** template to choose from the available components.
   Service Manager provides six out-of-box installation templates.

Installation template	Components included	Description
Basic for Smart Analytics	▪ Proxy Server Components (including DIH, DAH, community, category, agentstore, view, one second-level DIH and DAH, one Main Content Server, and two Smart Search Content Servers) ▪ Media Server ▪ CFS Server	Deploys the minimum required components to perform Smart Analytics on Service Manager internal data only. No extra data source connectors are installed.
All in one	▪ Proxy Server Components (including DIH, DAH, community, category, agentstore, view, one second-level DIH and DAH, one Main Content Server, and two Smart Search Content Servers) ▪ Media Server ▪ CFS Server ▪ OMNI Group Server ▪ Web Connector ▪ File System Connector ▪ Oracle Connector ▪ Confluence Connector	Deploys all components in one physical machine. In this way, you can use the full Smart Analytics functionality without a large volume of data.
Distributed IDOL content server	Content Server	Installs remote or additional content servers to handle more data.
Distributed Media server	▪ Media Server ▪ Media Proxy Server	By default, one Media server is included in the basic installation to handle OCR process. If there are too many Media processing requests, you can deploy a distributed Media server.

Customize	No pre-set components	Install any component based on your customized requirements. **Note** Note Do not select **Portal Content** and **Query Manipulation Server** unless you plan to run Service Portal. Do not select **IDOL content server** at the same time when you have selected **Proxy Server Components**. If you plan to install distributed IDOL content servers on the same machine where you want to deploy the Proxy Server Components, install Proxy Server Components and distributed IDOL content servers separately by performing the install process at different times.

Note

Note If you plan to install the content servers and the proxy server on different machines, it is recommended that you install the content servers first. Because without content server installed and started, the proxy server can't start and the OOB data can't be imported. For detailed steps of installing Smart Analytics components on different machines, see Example: Deploying Smart Analytics on multiple servers.

3. Specify the IP address of the remote machine on which you have installed or will install Smart Analytics components, and then press **Enter**.
4. Follow the configuration steps to configure the components you selected. Press **Enter** after each configuration step.
   If you did not select Main Content, Smart Search Content 1, or Smart Search Content 2 in the installation template as described in *step c*, the installer displays configuration pages for you to specify the IPs and ports of the remote servers on which you have installed or will install these contents servers.

If you don't specify IPs and ports for remote content servers on the configuration pages, or you want to change your settings after installation, you need to manually edit the IPs and Ports in the following configuration files and corresponding sections:

File	Configurations
<*Smart Analytics Installation*>/IDOL/IDOLServer.cfg	[Service] ..... ServiceStatusClients=...,<Remote Main Content Server IP> ServiceControlClients=...,<Remote Main Content Server IP> ..... [AdminRole] ..... Clients=<Remote Main Content Server IP> ..... [QueryRole] ..... Clients=<Remote Main Content Server IP> ..... [IndexRole] ..... Clients=<Remote Main Content Server IP> ..... [IDOLServer1] ..... Host=<Remote Main Content Server IP> Port=<Remote Main Content Server Port> .....

<*Smart Analytics Installation*>/level2proxy/IDOLServer.cfg	[Service] …… ServiceStatusClients=……,<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> ServiceControlClients=……,<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> …… [AdminRole] …… Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> …… [QueryRole] …… Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> …… [IndexRole] …… Clients=<First Remote Smart Search Content Server IP>,<Second Remote Smart Search Content Server IP> …… [DistributionIDOLServers] Number=2 [IDOLServer<0>] Host=<First Remote Smart Search Content Server IP> Port=<First Remote Smart Search Content Server Port> …… [IDOLServer<1>] Host=<Second Remote Smart Search Content Server IP> Port=<Second Remote Smart Search Content Server Port>

Some useful information for configuration:
Configure Smart Search Proxy Server > **Replicas**: Specifies the number of identical copies of each document to index. This is configured in the level2proxy\IDOLServer.cfg file.
The default value is set to **0**, which means that there is only one copy of each document for Smart Search. If you set the number as 1, it means that there will be two mirrored copies of the document in your Smart Search content servers.
In the Smart Search proxy level, content is distributed between virtual nodes, which the DIH assigns to its child servers. When you configure replicas, DIH copies the documents in a particular virtual node to two or more child servers. This method ensures there are two mirrored copies of the document in your system without you having to set up specifically mirrored child servers.

> **Note**
> Note The number of copies (that is, the value of Replicas plus one) must be no more than the number of child servers. If you create more copies than existing Smart Search child servers, DIH doesn't start.

5. Check the pre-installation summary, and then press **Enter**.
6. Wait for the installation to complete.

10. Press **Enter** to exit the installer.

Post-install actions

1. Run the following command to import the out-of-box data before you start any components of Smart Analytics:
   [INSTALL_DIR]/OOBData/ImportOOBData.sh
2. Start the components of SM Smart Analytics.
   To start all the components, run the following command:
   [INSTALL_DIR]/scripts/StartALL.sh

   > **Note**
   > Note
   > 1. This script starts all the components that you installed one by one.
   > 2. By running this command, you also start components like connectors, if they are installed.

   > **Tip**
   > Tip If you want to stop all these components, run the following command:
   > **[INSTALL_DIR]/scripts/StopALL.sh**

   Start the components with separate commands
   To start a single component that you installed, run the corresponding command as follows:
   - To start a content server, run the following command:
     [INSTALL_DIR]/scripts/StartContent[x].sh

     > **Note**
     > Note Replace [x] with the number of your content server, for example, `StartContent1`.

     > **Tip**
     > Tip If you want to stop a content server, run the following command:
     > **[INSTALL_DIR]/scripts/StopContent[x].sh**

   - To start both the main proxy server and the Smart Search proxy server, run the following command:
     [INSTALL_DIR]/scripts/StartIDOL.sh

     > **Note**
     > Note Before you start the main proxy server, make sure all the content servers have started.

     > **Tip**
     > Tip If you want to stop the main proxy server, run the following command:
     > **[INSTALL_DIR]/scripts/StopIDOL.sh**

   - To start a Connector Framework Server (CFS), run the following command:
     [INSTALL_DIR]/scripts/StartCFS.sh

> **Tip**
> Tip If you want to stop a Connector Framework Server (CFS), run the following command:
> **[INSTALL_DIR]/scripts/StopCFS.sh**

- To start a Media Server, run the following command:
  [INSTALL_DIR]/scripts/StartMediaServer[x].sh

> **Note**
> Note Replace [x] with the number of your Media server; for example, `StartMediaServer1`.

> **Tip**
> Tip If you want to stop a Media Server, run the following command:
> **[INSTALL_DIR]/scripts/StopMediaServer[x].sh**

- To start a Media Proxy Server, run the following command:
  [INSTALL_DIR]/scripts/StartMediaDAH.sh

> **Tip**
> Tip If you want to stop a Media proxy server, run the following command:
> **[INSTALL_DIR]/scripts/StopMediaDAH.sh**

- To start a Web Connector, run the following command:
  [INSTALL_DIR]/scripts/StartWebConnector.sh

> **Tip**
> Tip If you want to stop a Web Connector, run the following command:
> **[INSTALL_DIR]/scripts/StopWebConnector.sh**

- To start a file system connector, run the following command:
  [INSTALL_DIR]/scripts/StartFileSystemConnector.sh

> **Tip**
> Tip If you want to stop a file system connector, run the following command:
> **[INSTALL_DIR]/scripts/StopFileSystemConnector.sh**

- To start an Oracle connector, run the following command:
  [INSTALL_DIR]/scripts/StartOracleConnector.sh

> **Tip**
> Tip If you want to stop an Oracle connector, run the following command:
> **[INSTALL_DIR]/scripts/StopOracleConnector.sh**

- To start a Query Manipulation Server, run the following command:
  [INSTALL_DIR]/scripts/StartQMS.sh

> **Tip**
> Tip If you want to stop a Query Manipulation Server, run the following command:
> **[INSTALL_DIR]/scripts/StopQMS.sh**

3. After installation, you can modify the following configuration files to change the previous settings:
   - Smart Analytics main server: <*Smart Analytics Installation*>/IDOL/IDOLServer.cfg

- Media Server: <*Smart Analytics Installation*>/MediaServer#/MediaServer#.cfg
- Media Proxy Server: <*Smart Analytics Installation*>/MediaProxyServer/dah.cfg
  This is only required if you have installed multiple Media Servers in the distributed mode.
- Content server: <*Smart Analytics Installation*>/Content#/Content#.cfg
- CFS server: <*Smart Analytics Installation*>/CFS/CFS.cfg
- Web Connector: <*Smart Analytics Installation*>/WebConnector/WebConnector.cfg
- File system connector: <*Smart Analytics Installation*>/FileSystemConnector/filesystemconnector.cfg
- Oracle connector: <*Smart Analytics Installation*>/OracleConnector/OracleConnector.cfg
- Confluence connector: <*Smart Analytics Installation*>/ConfluenceRESTConnector/ConfluenceRESTConnector.cfg
- Query Manipulation Server: <*Smart Analytics Installation*>/QMS/QMS.cfg



Note

Note If you installed any connectors, you may need to edit their configuration files manually. For more information, see Configure connectors.

Restart the corresponding components after you modify the related configuration files.



Tip

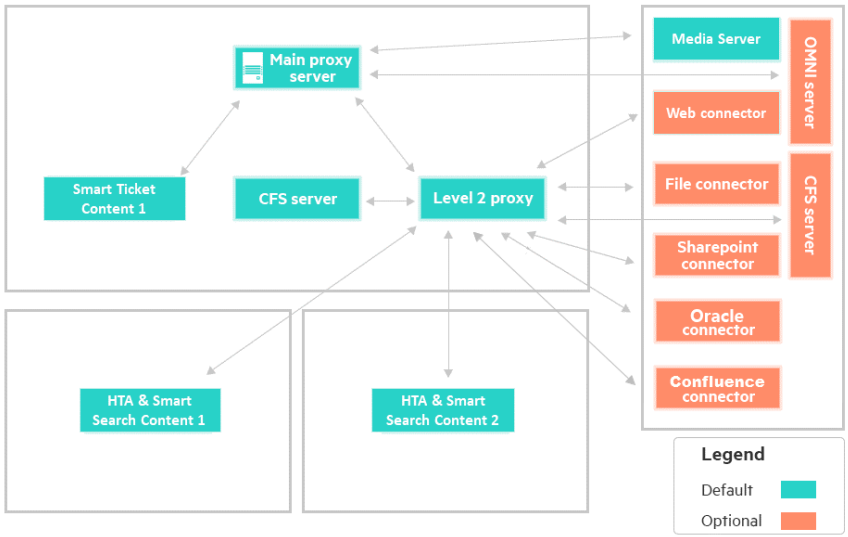
Tip If you want to uninstall SM Smart Analytics, see Uninstall Smart Analytics.

1.3.7.5. Example: Deploy Smart Analytics on multiple servers

This section provides an installation example for typical large enterprise environment. The example describes how to deploy Smart Analytics on four physical servers with an architecture as follows:

- Server 1 (Smart Search Content 1, default port number 30010)
- Server 2 (Smart Search Content 2, default port number 30020)
- Server 3 (Main Server)
  - Main proxy server (default port number 9000)
  - Level 2 proxy server (default port number 20010)
  - Main content server (content server for Smart Ticket) (default port number 10010)
  - CFS server (default port number 7000)
- Server 4 (Media server and other optional components)
  - Media server (default port number 18000)
  - CFS server (default port number 7000)
  - OMNI group server (default port number 5057)
  - File system connector (default port number 1234)
  - Web connector (default port number 5678)
  - Sharepoint connector (default port number 36000)
  - Oracle connector (default port number 5555)
  - Confluence connector (default port number 7128)

The following diagram provides a graphical illustration of the architecture in this example:



To deploy Smart Analytics on four physical servers, follow these steps:

1. On the first machine, install the first Smart Search Content server.
    1. Run the Smart Analytics installer.
    2. Follow the on-screen instructions until you see the Select Installation Type page. Select **Advanced Install**.
    3. Specify SM Server IP.
    4. On the Choose Distributed Components page, select **Distributed Content Server** from the **Install Template** drop-down list.
    5. Specify Smart Analytics main server IP, which is the IP address of the machine where you plan to install the main server.
    6. Specify the number of content servers to 1.
    7. Specify the content server ports.
    8. Confirm the Pre-Installation Summary and start the installation.
2. On the second machine, repeat the steps above to install the second Smart Search Content server.
3. On the third machine, install the main server.
    1. Run the Smart Analytics installer.
    2. Follow the on-screen instructions until you see the Select Installation Type page. Select **Advanced Install**.
    3. Specify SM Server IP.
    4. On the Choose Distributed Components page, select **Customize** from the **Install Template** drop-down list.
    5. Select **Main Content** under **Proxy Server Components**.
    6. Select **CFS Server**.
    7. Specify remote server IPs; separate IPs with commas.
    8. Follow the on-screen instructions to continue the configuration steps.
    9. Confirm the Pre-Installation Summary and start the installation.
4. On the fourth machine, install the rest components.
    1. Run the Smart Analytics installer.
    2. Follow the on-screen instructions until you see the Select Installation Type page. Select **Advanced Install**.
    3. Specify SM Server IP.
    4. On the Choose Distributed Components page, select **Customize** from the **Install Template** drop-down list.

5. Select the following components:
    - Media server
    - CFS server
    - OMNI group server
    - File connector
    - Web connector
    - Sharepoint connector
    - Oracle connector
    - Confluence connector
6. Specify the Smart Analytics main server IP, which is the IP address of the machine where you have installed the main server.
7. Follow the on-screen instructions to continue the configuration steps.
8. Confirm the Pre-Installation Summary and start the installation.

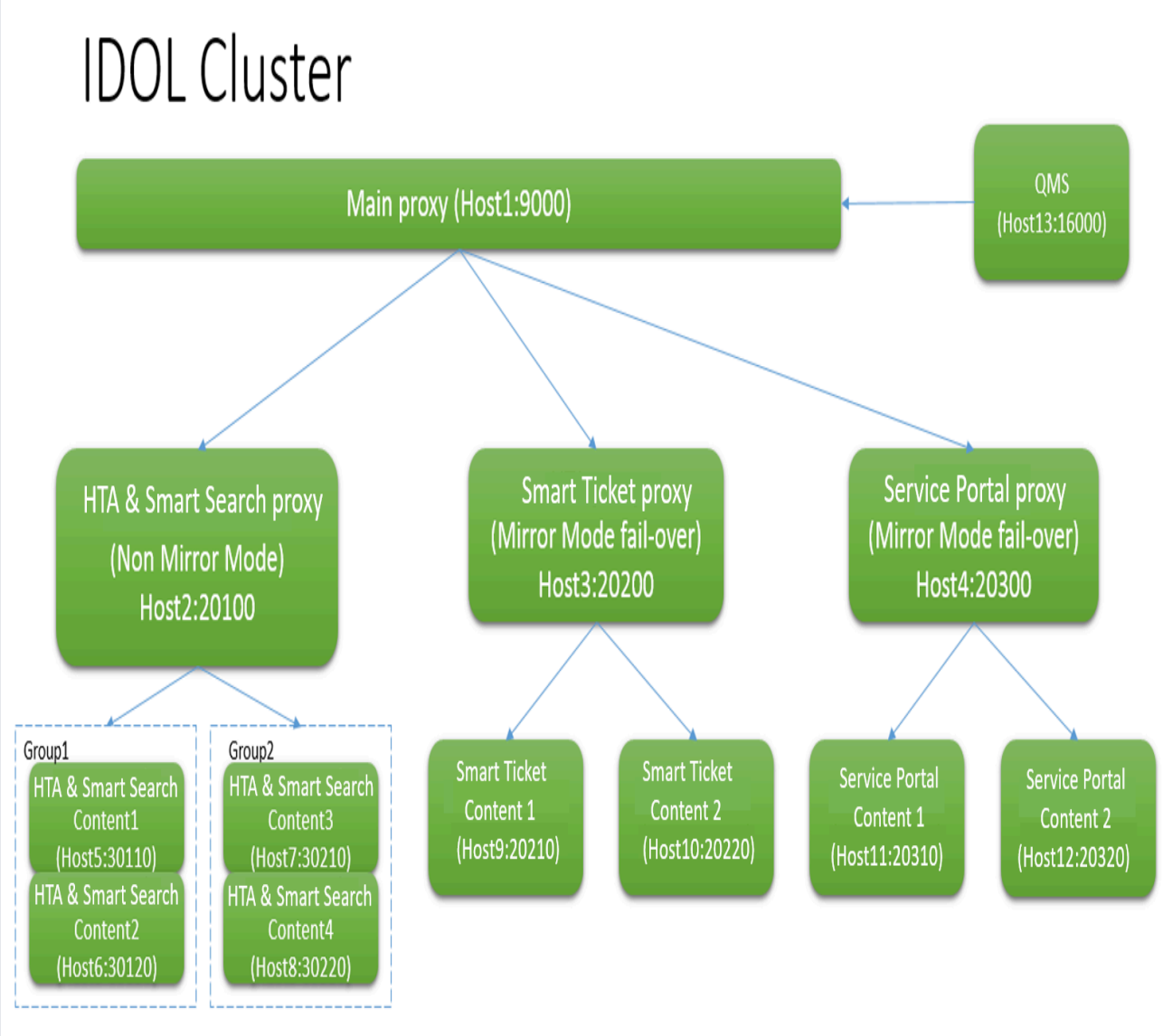
For more information about configuring Smart Analytics, see Configure Smart Analytics in Service Manager.

1.3.7.6. Configure Smart Analytics for high availability

This section provides instructions for you to configure Service Manager Smart Analytics for high availability.

Overview

To reduce potential down time, it is recommended that you configure Smart Analytics for high availability. The following diagram is an example architecture of Smart Analytics configured for high availability:



In this diagram:

- Service Portal content2 is the failover content of Service Portal Content1 and vice versa.
- Smart Search content2 is the fail-over content of Smart Search content1 and vice versa.
- Smart Ticket content2 is the fail-over content of Smart Ticket content1 and vice versa.

Note

Note

- Smart Search data is distributed among groups, while content servers in a group are mirrored to each other as fail over backup. For better index and query performance, it is suggested to store no more than 3 million records for each content server. You can adjust the number of groups accordingly to fit actual scenarios. For example, if Smart Search would have 9 million records, at least 3 groups should be used for Smart Search, which means 6 content servers in total used as Smart Search content servers.
- You can configure high availability for all contents or contents of a specific component only, for example, Smart Search contents.
- The ports are provided only for reference. You can change the ports by modifying the corresponding configuration files.
- You can locate the proxy and QMS servers on the same machine or different machines. However, it is recommended that the rest content servers be located on different machines. For example, Host 1, 2, 3, 4 and 13 in the diagram can be the same machine, while Host 5-12 different ones.

Configuration steps

To configure Smart Analytics for high availability as depicted in the diagram, follow these steps:

1. Install and configure the proxies for high availability as needed.
   In the diagram, Smart Analytics uses four proxies to organize its contents: one main proxy and three level 2 proxies under the main proxy.
   Install and configure main proxy
   1. Install the main proxy by running the Smart Analytics installer.
   2. Select **Advanced Install**.
   3. From the Install Template drop-down list, select **Customize**.
   4. Select the **Proxy Server Components** checkbox with the other checkboxes unchecked.
   5. Follow the on-screen instructions to finish the install.
   6. Once the installation is complete, modify the .\IDOL\IDOLServer.cfg file so that main proxy locates its three level 2 proxy servers, as shown below:

      ```
      Main proxy setting:
      ……
      [DistributionIDOLServers]
      Number=3
      [IDOLServer0]
      Name=SmartSearch
      Host=Host2
      Port=20100
      DistributeByFieldsValues=GlobalSearch
      [IDOLServer1]
      Name=Content1
      Host=Host3
      Port=20200
      DistributeByFieldsValues=CONTENT1
      [IDOLServer2]
      Name=Content-Propel
      Host=Host4
      Port=20300
      DistributeByFieldsValues=PROPEL
      ……
      ```

   Install and configure the three Level 2 proxies: (Smart search proxy, Smart Ticket proxy, and Service Portal proxy)

1. From the Main proxy installation location, copy the level2proxy folder to each of the three locations where you plan to set up the level 2 proxies. The three level 2 proxies can use the same or different hosts.

   Note

   Note Copy the “level2proxy” folder to two other locations if you plan to use the original “level2proxy” as one of the three level 2 proxies.

2. Modify the corresponding configuration files of the level 2 proxies, as shown below:
   - Smart search proxy:
     Modify the < Smart Search Proxy directory>\IDOLServer.cfg file so that the proxy locates its four content servers and arranges them as two groups with non-mirror mode, as shown below

```
Smart Search Proxy setting:
……
[DistributionSettings]
mirrormode=false
//DistributeOnBatch=true //To send documents in batches
//DIH settings:
DistributeByReference=true
UseConsistentHashing=true
//DAH setting:
SimpleCombinatorMode=true
DistributionMethod=0
[ConsistentHashing]
//number of virtual nodes
VirtualNodes=4096
//The number of copies (that is, the value of Replicas plus one) must be smaller than
number of DistributionIDOLServers.
Replicas=0
//This section is equivalent to the [engines] section in the DAH and DIH standalone
configuration
[DistributionIDOLServers]
Number=2
[IDOLServer0]
Host=Host5,Host6
Port=30110,30120
……
[IDOLServer1]
Host=Host7,Host8
Port=30210,30220
……
```

- Smart Ticket proxy:
  Modify the <*Smart Ticke proxy directory*>*IDOLServer.cfg* file so that the proxy locates its two content server as mirror mode, as shown below:

```
Smart Ticket proxy setting:
……
[DistributionSettings]
mirrormode=true
//DistributeOnBatch=true //To send documents in batches
//DIH settings:
//DistributeByReference=true
//UseConsistentHashing=true
//DAH setting:
//SimpleCombinatorMode=true
DistributionMethod=0

//[ConsistentHashing]
//number of virtual nodes
//VirtualNodes=4096
//The number of copies (that is, the value of Replicas plus one) must be smaller than
number of DistributionIDOLServers.
//Replicas=0
//This section is equivalent to the [engines] section in the DAH and DIH standalone
configuration
[DistributionIDOLServers]
Number=2
[IDOLServer0]
Host=Host9
Port=20210
……
[IDOLServer1]
Host=Host10
Port=20220
……
```

- Service Portal proxy:
  Modify the <*Service Portal proxy directory*>*IDOLServer.cfg file* so that the proxy locates its two content server as mirror mode, as shown below:

```
Propel proxy setting:
……
[DistributionSettings]
mirrormode=true
//DistributeOnBatch=true //To send documents in batches
//DIH settings:
//DistributeByReference=true
//UseConsistentHashing=true
//DAH setting:
//SimpleCombinatorMode=true
DistributionMethod=0

//[ConsistentHashing]
//number of virtual nodes
//VirtualNodes=4096
//The number of copies (that is, the value of Replicas plus one) must be smaller than
number of DistributionIDOLServers.
//Replicas=0
//This section is equivalent to the [engines] section in the DAH and DIH standalone
configuration
[DistributionIDOLServers]
Number=2
[IDOLServer0]
Host=Host11
Port=20310
……
[IDOLServer1]
Host=Host12
Port=20320
……
```

2. Install and configure failover content servers by using the Smart Analytics installer or copying existing ones from the Smart Analytics installation, as described below:
   Install four content servers for Smart Search on four machines by performing either of the following:
   - On each of four machines, run the Smart Analytics installer and select **Advanced Install**. After that, from the Install Template drop-down list, select **Distributed Content Server**, and then select the **Content Server** checkbox.
   - From the Smart Analytics installation directory, copy the Content2/Content3, modules and langfiles folders, as well as the synonym.txt file to each of the four machines. Then, re-modify the configuration files for all the four content servers by correcting the ports and the related file location paths of modules, langfiles and synonym.txt.

   Install two content servers for Smart Ticket on two machines by performing either of the following:
   - On each of four machines, run the Smart Analytics installer and select Advanced Install. After that, from the Install Template drop-down list, select Distributed Content Server, and then select the Content Server checkbox.
   - From the Smart Analytics installation directory, copy the Content1, modules and langfiles folders, as well as the synonym.txt file to each of the two machines. Then, re-modify the configuration files for both content servers by correcting the ports and the related file location paths of modules, langfiles and synonym.txt.

1.3.7.7. Enable Smart Analytics in Service Manager

User Role: Administrator

To enable Smart Analytics in Service Manager and set up connections, follow these steps:

> **Note**
>
> Note If you want to set up an SSL connection, see Configure TSL/SSL for two-way authentication .

1. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
2. Click the **Enable Smart Analytics** button to enable Smart Analytics.
   After you click this button, a message is displayed to state that once you migrate to IDOL, you can't use SOLR as the search engine any more and you have to log out and re-log in to Service Manager before Smart Analytics is applied.
3. Click **Yes** to migrate to IDOL. Your account logs out automatically and you need to re-log in to Service Manager.
4. From the System Navigator, click **System Administration** > **Ongoing Maintenance** > **Smart Analytics** > **Configuration**.
5. Enter the address and port for Smart Analytics server, and then click **Test Connection**.
6. Enter the address and port for the default CFS server, and then click **Test Connection**. This default CFS server is used for Service Manager attachment index.
7. Enter the address and port for the Media Server, and then click **Test Connection**.
8. Click **Save**.
   Once you click **Save**, all the Service Manager Integration Suite tasks for the Smart Analytics integration are automatically started.

1.3.8. Install Service Manager Collaboration

Service Manager Collaboration (SMC) is an instant messaging solution combining IT Collaboration and End User Chat. IT Collaboration enables Service Manager IT operators to collaborate in real-time (or anytime) when handling an Interaction, Incident, Incident Task, Request, Request Task, Problem, Problem Task, Change, or Change Task. End User Chat builds a communication channel between the Employee Self-Service (ESS) portal users and the Service Manager Service Desk agents so as to increase IT efficiency, improve customer satisfaction and reduce IT support costs.

Follow these instructions to install Service Manager Collaboration.

Service Manager Collaboration deployment scenario

Prerequisite tasks

Task 1: Install and configure Java for the chat server

Task 2: Deploy and configure the chat server

Task 3: Deploy and configure the chat service

Task 4: Deploy the End User Chat UI

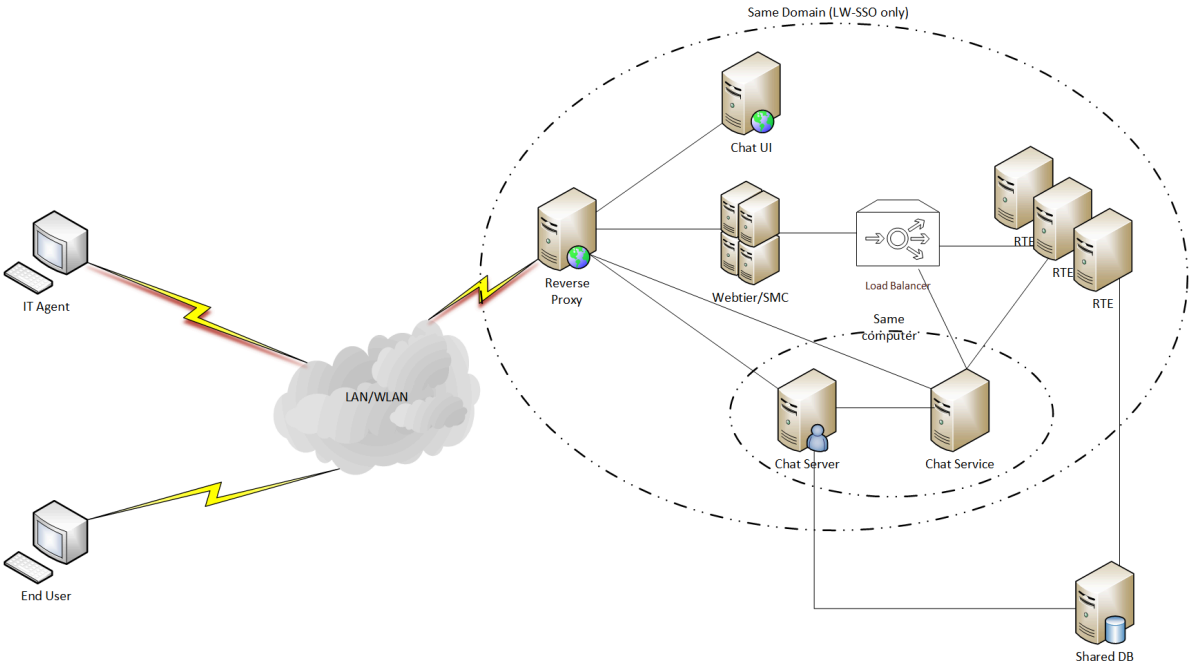
Task 5: Configure reverse proxy for the chat components

Task 6: Select a portal for End User Chat

Task 7: Enable Service Manager Collaboration

1.3.8.1. Service Manager Collaboration deployment scenario

The following diagram illustrates a typical Service Manager Collaboration (SMC) deployment scenario.



Service Manager Collaboration cooperates with the following essential factors and Service Manager components:

- **Chat server**:
  The chat server is based on Openfire, which is a Real Time Collaboration (RTC) server licensed under the Open Source Apache License. SMC communicates with Openfire by using the widely adopted open protocol Extensible Messaging and Presence Protocol (XMPP) to approach all the operations, such as setting up conversations, posting messages, presence management, notification management and so on. The chat server also links Service Manager and other messaging applications.
- **Reverse proxy**:
  The chat server must be deployed behind a reverse proxy such as the Apache HTTP server, Internet Information Services (IIS) or the F5 load balancer to protect the sensitive information, such as the IP addresses, ports, and so on.
- **Service Manager web tier**:
  SMC is embedded in Service Manager web tier to access the chat server through reverse proxy for instant messaging.
- **Chat service**:
  Service Manager chat service provides both the RESTful web service and the virtual agent chat service to the chat clients. It also connects the chat server to the Service Manager server (RTE) and manages chat requests between an end user, a virtual agent and a live agent.
- **Chat UI**:
  The chat UI is an End User Chat window which enables a Service Manager web tier ESS user or a Service Portal user to chat with either a virtual agent or a live agent.

The following topics contain tasks of detailed steps and guide you through the SMC configuration from scratch.

Important notes

1. The tasks are described using examples, which you may need to adjust according to your actual environment. See the following table for a list of sample host names:

Component	Sample host name
Service Manager server (RTE)	smserver.yourdomain.net
Service Manager web tier	smwebtier.yourdomain.net
Database server	dbserver.yourdomain.net
Chat server	chatserver.yourdomain.net
Reverse Proxy server	proxyserver.yourdomain.net
Chat UI server	chatuiserver.yourdomain.net

2. The tasks marked as HTTP Only are required only if you are configuring HTTP. Similarly, the tasks marked as HTTPS Only are required only if you are configuring HTTPS. All other tasks are shared by HTTP and HTTPS.
3. Tomcat is used as the application server for the Service Manager web tier and chat UI.
4. The chat server, chat service, reverse proxy server, Tomcat, and the Service Manager server share the same domain name: **yourdomain.net**, but are installed on different hosts in order to reflect real-world scalable deployment.
5. When configuring SSL, the chat server, chat service, proxy server, Tomcat, and the Service Manager server share the same root CA certificate.

1.3.8.2. Prerequisite tasks

You may skip some or all of these steps in this section depending on the actual configuration in your environment.

Task 1: Install the Service Manager server and web tier

Install the Service Manager Server

Install the Service Manager Web Tier

In this example, the context root for the webtier is **webtier**.

Note

Note

After the Service Manager web tier is installed, open %TOMCAT_HOME%/conf/server.xml and take a note of the port of the AJP 1.3 connector. The default one is 8009.

For example:

```
<!-- Define an AJP 1.3 Connector on port 8009 -->
    <Connector protocol="AJP/1.3"
               secret="MyT0mcAt"
               tomcatAuthentication="false"
               address="0.0.0.0"
               port="8009"
               redirectPort="8443" />
```

Task 2: Configure authentication on the Service Manager server and web tier

Perform this task on smserver.yourdomain.net and smwebtier.yourdomain.net.

For Service Manager Collaboration (SMC) to function, you need to configure Lightweight Single Sign-On (LW-SSO) or Identity Manager (IdM) on the Service Manager server, Service Manager web tier, chat server, and chat service so that Service Manager users can use SMC without signing in to the Service Manager server separately.

LW-SSO

Follow these steps:

1. Configure LW-SSO on the Service Manager server.
2. Configure LW-SSO on the Service Manager web tier.

IdM

Follow these steps:

1. Install Identity Manager (IdM).
2. Configure IdM authentication on the Service Manager server.
3. Configure IdM authentication on the Service Manager web tier.

Task 3 (HTTPS Only): Configure SSL in the Service Manager server and web tier

Perform this task on **smserver**.yourdomain.net and **smwebtier**.yourdomain.net.

Configure SSL in the Service Manager server and web tier by following the instructions as described in Configure two-way SSL.

If you don't require Service Manager clients to present signed certificates to connect to the server and also be on the list of trusted clients, follow the instructions as described in Configure one-way SSL.

Task 4: Test authentication on the Service Manager web tier

Perform this task on any computer that has a supported browser. In this task, you will verify if your Service Manager configuration works.

Follow these steps:

1. Access your web tier with the Tomcat server's FQDN. In this example, access http://webtier.yourdomain.net:8080/webtier/index.do in your web browser to display the Service Manager login screen.
2. Sign in to Service Manager as a system administrator. The system displays the administrator's To Do Queue.

Task 5: Configure reverse proxy for the Service Manager web tier

This task is performed on proxyserver.yourdomain.net and currently only the following three types of reverse proxies are certified.

> **Note**
>
> Note
> If you have already configured Trusted sign-on (TSO) for the Service Manager system with a web server (such as Apache HTTP server or Internet Information Services), you don't need to install a new one and set the location of the reverse proxy for web tier. However, you still need the other steps to prepare the reverse proxy for the chat components.

Apache HTTP server

For detailed instructions, see the **Apache HTTP server** section in Configure a web server to redirect requests to the web tier.

Internet Information Services (IIS)

For detailed instructions, see the **Windows Internet Information Services (IIS)** section in Configure a web server to redirect requests to the web tier.

F5

Follow these steps:

1. Create one pool and set properties and members as listed in the below table.

Pool name	Active health monitor	Member	Description
smc-webtier	tcp	smwebtier.yourdomain.net:8080 or smwebtier.yourdomain.net:8443 (HTTPS)	For web tier

2. Create a new iRule **SMC** and configure it as follows:

```
when HTTP_REQUEST {
if { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/webtier" } {
  pool smc-webiter
  return
}
}
```

3. Create a new virtual server to use the SMC iRule.
4. Save your changes.

Task 6: Test Service Manager with the reverse proxy

Perform this task on any computer that has a supported browser.

On any computer that has a supported browser, access http(s)://proxyserver.yourdomain.net/webtier/index.do, and then sign in to Service Manager as a system administrator. The system displays the administrator's To Do Queue.



Caution

Caution

From now on, you must use the fully qualified domain name (FQDN) of the reverse proxy server to access the Service Manager web tier.

Now all the prerequisites are complete and you can go on to the next steps.

Next

The Service Manager Collaboration deployment involves the following tasks. You must complete the prerequisite tasks before proceeding with deployment. To verify this, access Service Manager with the reverse proxy. For example:

`http(s)://proxyserver.yourdomain.net/webtier/index.do`

Note: The SMA suite includes Service Manager Collaboration. When you are installing SMA-SM, you install Collaboration as well. You can skip task 1 to task 4 if you are working with SMA-SM.

- Task 1: Install and configure Java for the chat server
- Task 2: Deploy and configure the chat server
- Task 3: Deploy and configure the chat service
- Task 4: Deploy the End User Chat UI
- Task 5: Configure reverse proxy for the chat components
- Task 6: Select a portal for End User Chat
- Task 7: Enable Service Manager Collaboration

1.3.8.3. Task 1: Install and configure Java for the chat server

This task is performed on chatserver.yourdomain.net.

Follow these steps:

1. Install Oracle JDK 17 or Zulu OpenJDK 17.
2. Add the `JAVA_HOME` environment variable.

For example, add a new System variable `JAVA_HOME=C:\Program Files\Java\jdk17.0.x_xx\` in the Windows Environment Variables dialog.
For Linux, you can add this environment variable in a later step.

1.3.8.4. Task 2: Deploy and configure the chat server

This task is performed on chatserver.yourdomain.net and any computer that has a supported browser. The company provides a pre-configured version of Openfire as the Service Manager Collaboration chat server, which is easy to set up and administer and offers rock-solid security and performance. In this task, you will install the chat server and go through configuration steps for it.

Note

- The chat server can be deployed on the same OS as the Service Manager server.
- The chat server shares the Service Manager database. You may want to back up the Service Manager database before beginning this task.
- If you are upgrading the chat server from an earlier version to 9.80, skip this task and refer to Upgrade Service Manager Collaboration.

Deploy the chat server

Follow these steps:

1. Save the chat server installer sm9.80.*xxxx*_ChatServer.zip from Service Manager installation package to your chat server.
2. Extract it to any directory. For example, C:\chat-server-9.80.0 or /opt/chat-server-9.80.0. This directory is referred to as %CHAT_SERVER_HOME% in the sections below.

Configure the chat server

Windows

1. Open the Windows command prompt and change the directories to **%CHAT_SERVER_HOME%**\bin.
2. Run the **install-service.bat** command to install the Windows service for the chat server.
3. Go to Windows Services, and verify that the new **SmChatServer** service is installed.

Tip

- You can run the **nssm edit SmChatServer** command to edit the corresponding configurations after the Windows service is installed.
- To remove this Windows service, run the **nssm remove SmChatServer** command.

Linux

1. Open the %CHAT_SERVER_HOME%/bin/openfire-sysconfig file with a text editor.
2. Uncomment the lines for `JAVA_HOME`, `OPENFIRE_HOME` and `OPENFIRE_USER`. Set `OPENFIRE_HOME` to the root directory of the chat server, set `OPENFIRE_USER` to the user with enough rights and you want to run chat server, and then set `JAVA_HOME` to your installed JDK or JRE. For example:

```
OPENFIRE_HOME="/opt/chat-server-9.80.0"
# If there is a different user you would like to run openfire as,
# change the following line.
# OPENFIRE_USER="daemon"
JAVA_HOME=/usr/lib/jvm/java-17.0.9_linux-x64
```

Note

The `OPENFIRE_USER` must have write access to the % OPENFIRE_HOME%/logs directory.

3. Copy the file to the /etc/sysconfig/ directory and rename it to openfire. You need to manually create this directory if it doesn't exist.

Start/stop the chat server

Windows

Follow these steps to start/stop the chat server from the Windows command prompt:

1. Open the Windows command prompt and change directories to %CHAT_SERVER_HOME%\bin.
2. Run the **openfire.bat** command.
3. To stop the chat server, press Ctrl+C in the command window.

Follow these steps to start/stop the chat server from Windows services:

1. Go to Windows Services, select the **SmChatServer** service and click **Start Service** to start the service.
2. To stop the chat server, go to Windows Services, and click **Stop Service** to stop the SmChatServer service.

Linux

> **Note**
>
> You may need to add the execute permission to the file before execution. For example, run the **chmod +x** or **chmod 755** command.

Follow these steps to start/stop the chat server as a foreground process:

1. Launch a Linux terminal to connect to the server.
2. Navigate to the %CHAT_SERVER_HOME%/bin directory.
3. Run the **./openfire.sh** command to start the chat server.
4. To stop the chat server, press Ctrl+C in the terminal.

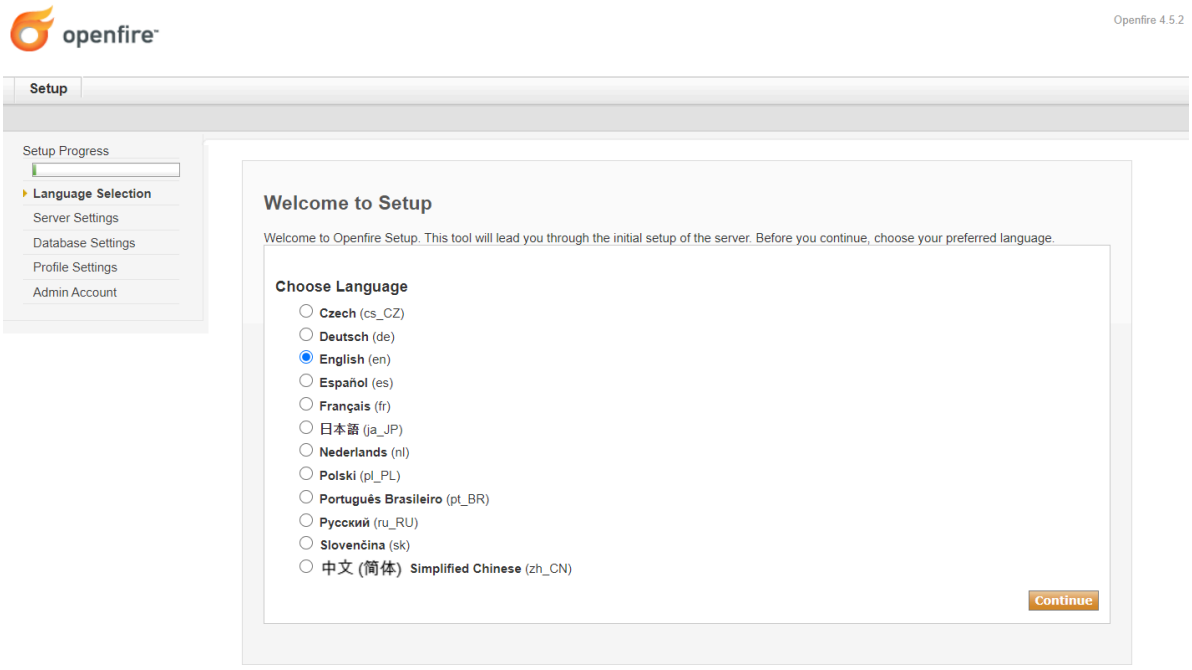
Follow these steps to start/stop the chat server as a background process:

1. Launch a Linux terminal to connect to the server.
2. Navigate to the %CHAT_SERVER_HOME%/bin directory.
3. Run the **./openfire start** command to start the chat server.
4. To stop the chat server, run the **./openfire stop** command.

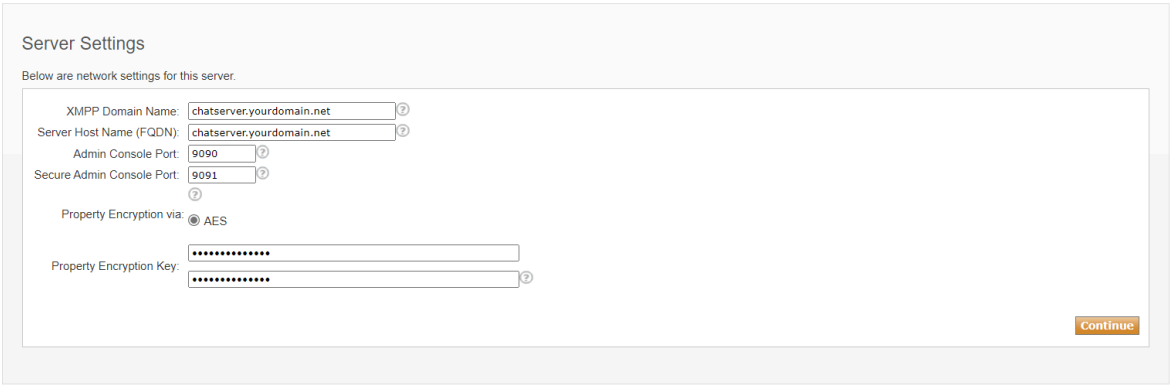
Configure Openfire

Follow these steps:

1. (Optional) If you are working with an Oracle database, download and copy the JDBC driver (for example, ojdbc8.jar) to the %CHAT_SERVER_HOME%\lib directory.
2. Start the chat server.
3. Access http://chatserver.yourdomain.net:9090/setup/index.jsp from the web browser. The Openfire Setup: Welcome to Setup screen is displayed.



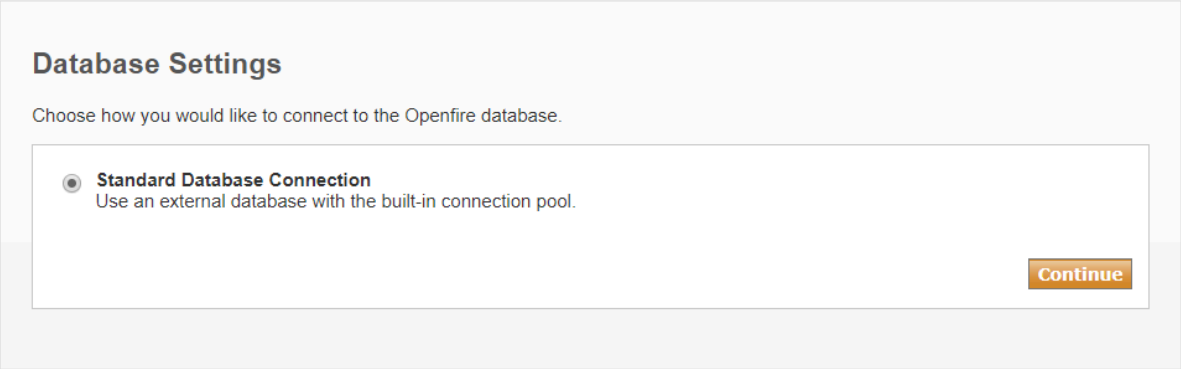
4. Select **English** (or any other language you prefer) and click **Continue**.
5. On the Server Settings page, set the FQDN (for example, chatserver.yourdomain.net) to the Domain field. Fill in other fields and click **Continue**:



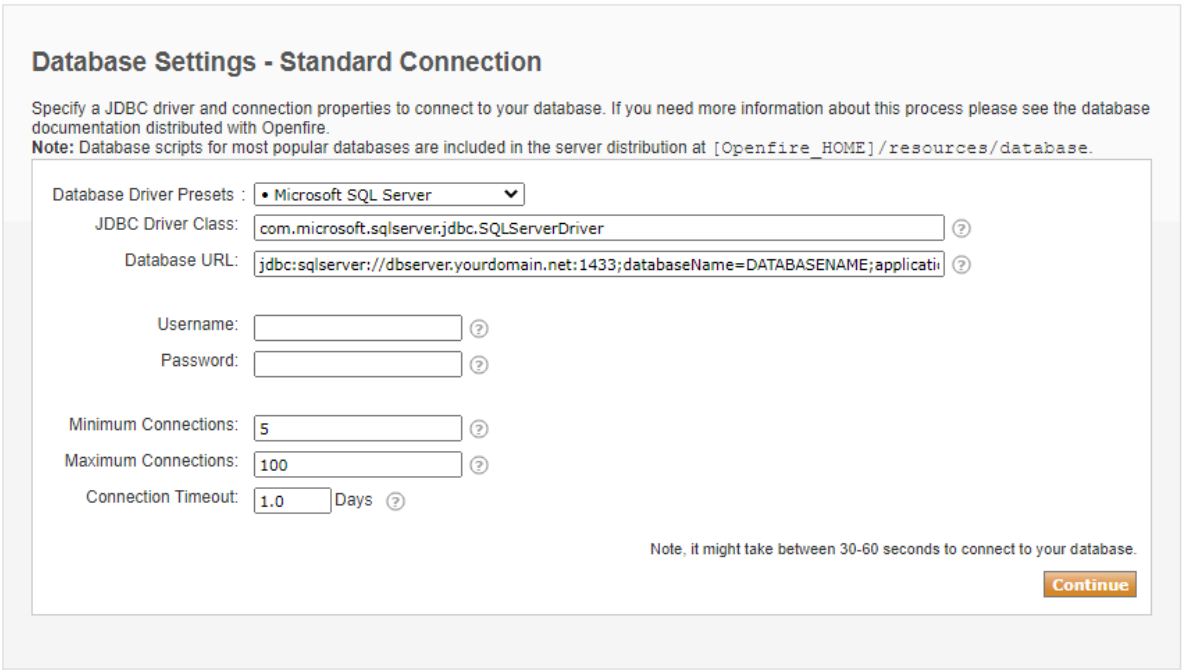
Server settings

Parameter	Value in this task	Description
Domain	chatserver.yourdomain.net	The XMPP domain of this instance of Openfire. Two different XMPP domains can't communicate with each other.
Admin Console Port	9090	The port used for unsecured Admin Console access. The default value is 9090. Leave this port to its default value if you don't need to open an HTTP port.
Secure Admin Console Port	9091	The HTTPS port used for secured Openfire Admin Console access. The default value is 9091.
Property Encryption via	AES	The encryption algorithm used by the Openfire server to prevent sensitive data from being exposed. The default option is AES.
Property Encryption Key	smdemotraining	Specify the AES encryption key. This field is mandatory. You can specify any value in the first field, and then type this value again in the second field.

6. Click **Continue** on the Database Settings screen.



7. On the **Database Settings - Standard Connection** page, specify the database details so that Openfire can connect to your Service Manager database and create the related tables there.


Database Settings - Standard Connection
Specify a JDBC driver and connection properties to connect to your database. If you need more information about this process please see the database documentation distributed with Openfire.
Note: Database scripts for most popular databases are included in the server distribution at [Openfire_HOME]/resources/database.
Database Driver Presets :
• Microsoft SQL Server
JDBC Driver Class:
com.microsoft.sqlserver.jdbc.SQLServerDriver
Database URL:
jdbc:sqlserver://dbserver.yourdomain.net:1433;databaseName=DATABASENAME;applicati
Username:
Password:
Minimum Connections:
5
Maximum Connections:
100
Connection Timeout:
1.0
Days
Note, it might take between 30-60 seconds to connect to your database.
Continue

Database Settings - Standard Connection

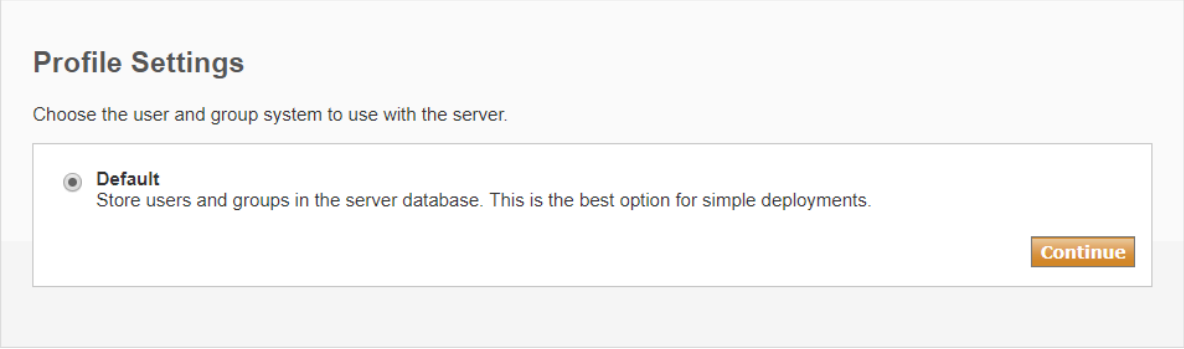
Parameter	Value in this example	Description
Database Driver Presets	Microsoft SQL Server	Select the database type of Service Manager. You can select SQL server, Microsoft SQL Server (legacy), Oracle, or PostgreSQL.
JDBC Driver Class	com.microsoft.sqlserver.jdbc.SQLServerDriver Do not modify the default value	Value in this field is populated automatically after the database type is selected.

Database URL	jdbc:sqlserver://dbserver.yourdomain.net :1433; databaseName=smdb; applicationName=Openfire	Value in this field is populated automatically after the database type is selected. You should update the values in `[]` with the actual values of your Service Manager database. ○ The default Oracle database URL is `jdbc:oracle:thin:@[host-name]:1521:[SID]`. You need to download and copy the JDBC driver (for example, ojdbc8.jar) to the **%CHAT_SERVER_HOME%**\lib directory first. If you are working with Oracle 19c in FIPS mode, your database password must be at least 14 bytes (112 bits). ○ The default PostgreSQL server database URL is `jdbc:postgresql://[host-name]/[database-name]`. If you use a private schema, specify the schema name after the database URL: `jdbc:postgresql://[host-name]/[database-name]?currentSchema=[private-schema]` If you have enabled SSL on the PostgreSQL server, refer to Enable SSL for PostgreSQL for detailed configuration. If you are working with PostgreSQL in FIPS mode, your database password must be at least 14 bytes (112 bits). ○ The default Microsoft SQL server database URL is `jdbc:sqlserver://[host-name:port];databaseName=[database-name];applicationName=Openfire`. If you have multiple database instances on an SQL server, refer to the Named and Multiple SQL Server Instances section on the Building the Connection

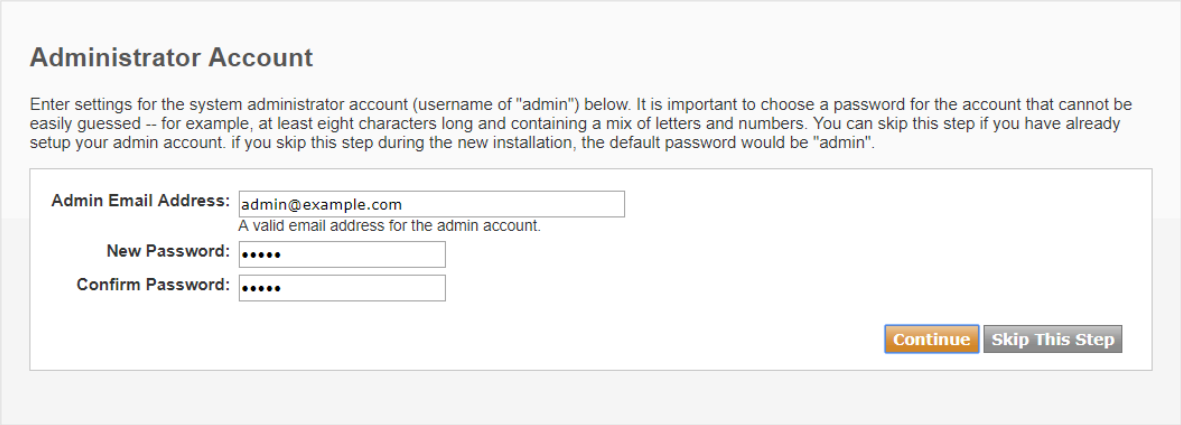
		URL web page for more information about the database URL configuration. ○ The default Microsoft SQL server (legacy) database URL is `jdbc:jtds:sqlserver://[host-name]/[database-name];appName=Openfire`. You MUST use this legacy driver for Microsoft SQL Server in FIPS mode.
Username	*<Your Service Manager database user name>*	Specify the user name to sign in to the Service Manager database.
Password	*<Your Service Manager database password>*	Specify the password to sign in to the Service Manager database. We suggest that you use a strong password.
Minimum Connections	5	Specify the minimum number of database connections the connection pool should maintain. The default value is 5.
Maximum Connections	100	Specify the maximum number of database connections the connection pool should maintain. The default value is 100.
Connection Timeout	1.0	Specify the time (in days) before connections in the connection pool are recycled. The default value is 1.0.

Note
Service Manager Collaboration uses the Service Manager database and inserts a number of Openfire tables into the database.

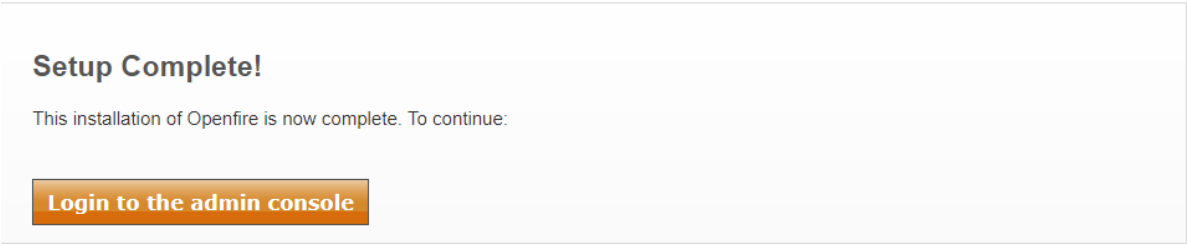
8. After the Profile Settings screen is displayed, check your Service Manager database for the Openfire tables, each table name is prefixed with "of". Then click **Continue**.



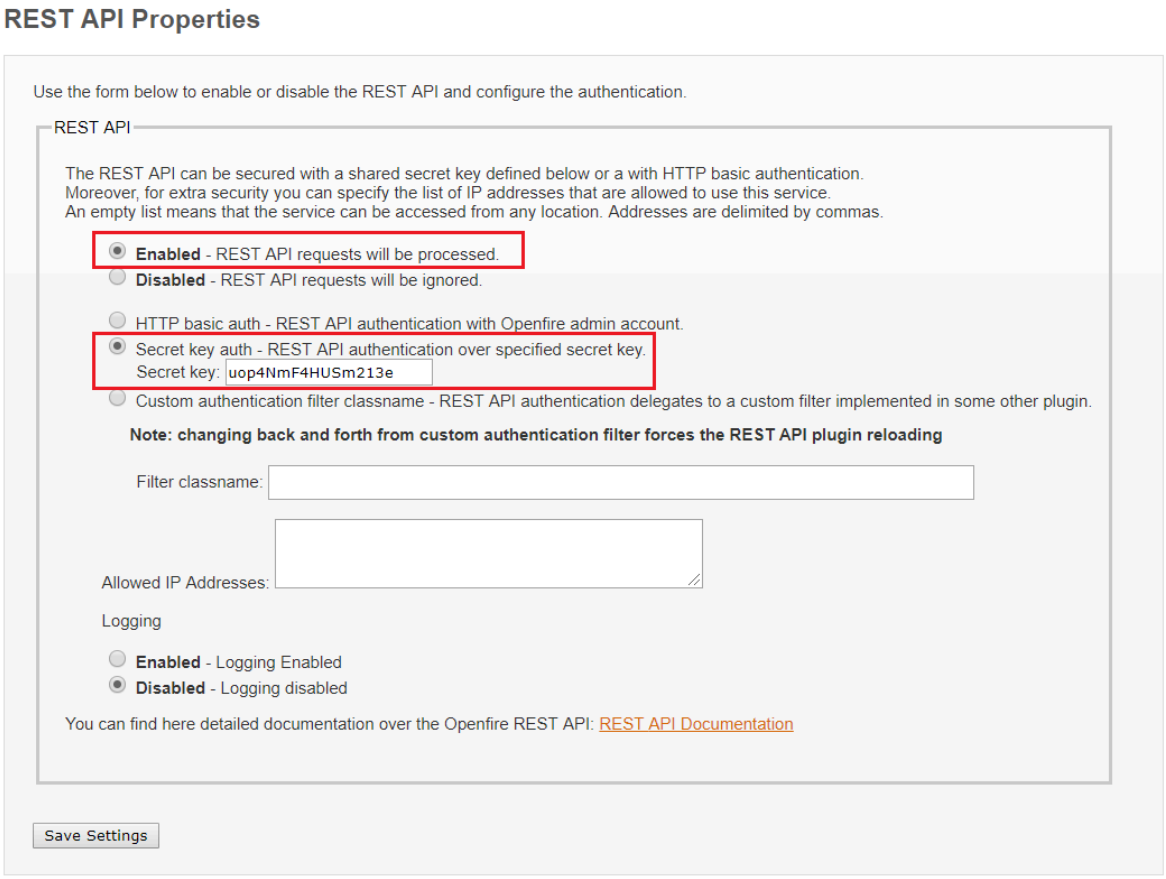
9. Create the Email address and password for your Openfire system administrator account (username of "admin") on the Administrator Account screen. Later you will sign in to Openfire as administrator with this account. Click **Continue** to finish the Openfire installation.



10. Your Openfire setup is complete now.



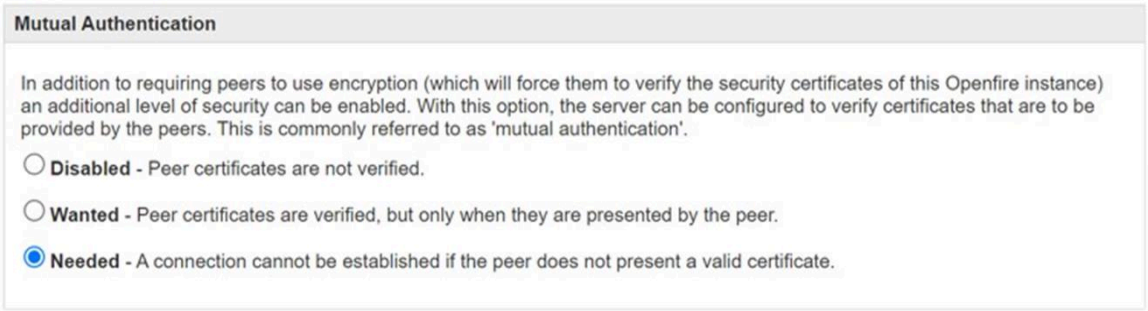
11. Click the **Login to the admin console** button to sign in to your Openfire Administration Console with the account information you specified in step 9.
12. Click the User/Groups tab and create an Openfire user. For example:
    User name: publishadmin
    Password: admin123
13. To enable the virtual agent, click **Server** > **Server Settings** > **REST API**. Enable REST API and take a note of the secret key string then click **Save Settings**.



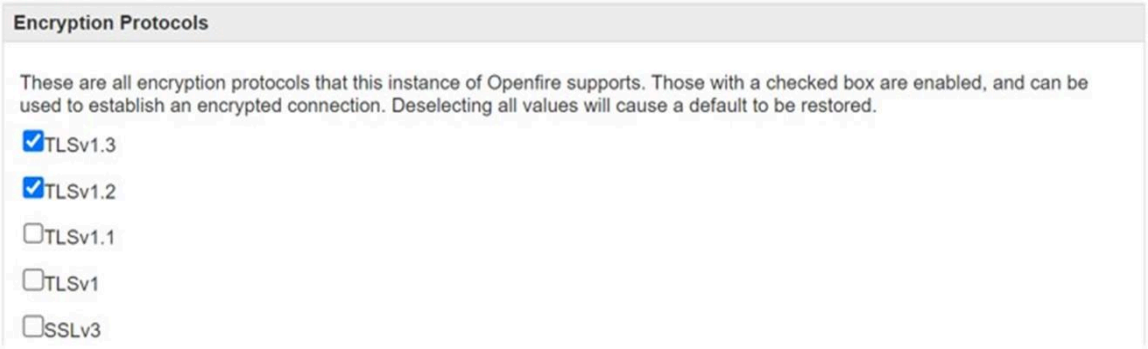
14. (HTTPS Only) Configure TLS Client Connections Settings.

> **Note**
> We highly recommend enabling **Mutual Authentication** on port 5223 to allow connection from the chat service only.

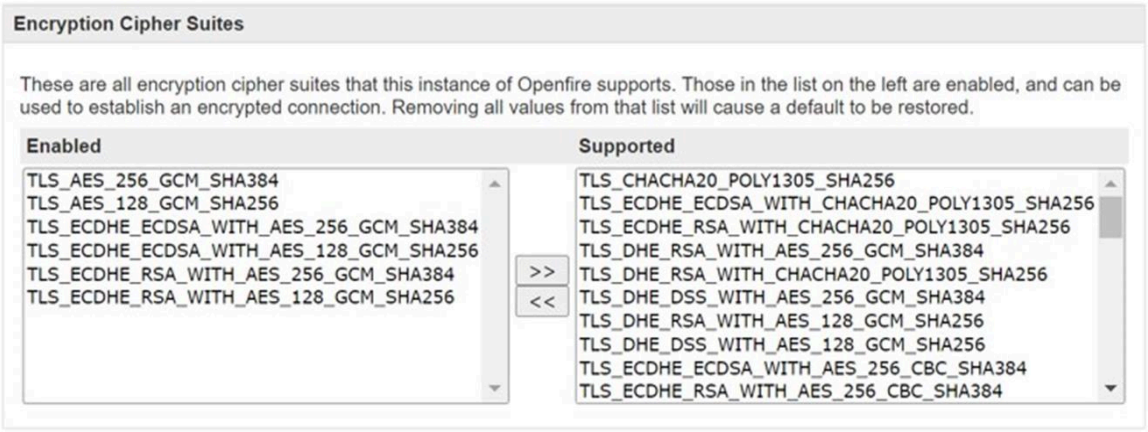
1. Sign in to your Openfire Administration Console.
2. Go to **Server Settings** > **Client Connections**.
3. In the **Encrypted (legacy-mode) connections** section, activate the **Advanced Configuration...** link. Note that if you have applied 9.80.p1 or later patches, then you need to activate the **Advanced Configuration...** link in the **Encrypted (Direct TLS) connections** section.
4. Update the configuration as follows:
   Set Mutual Authentication to **Needed**.



Enable TLSv1.3 and TLSv1.2, and disable other protocols.



Enable strong cipher suites only.



5. Click the **Save Settings** button.

15. (HTTPS Only) Configure the SSL settings by following the instructions described in Chat server.
    Note: If you are configuring FIPS mode, refer to Configure FIPS mode in the Chat Server.
16. Restart the chat server.

Configure authentication on the chat server

The chat server supports the LW-SSO or the IdM authentication, but not both at the same time. You need to use the IdM authentication on the chat server if you have configured the IdM authentication in the web tier. Otherwise, use LW-SSO instead.

LW-SSO

Follow these steps to configure LW-SSO authentication on the chat server:

1. Navigate to the %CHAT_SERVER_HOME%\conf folder.
2. Open lwssofmconf.xml with a text editor (You can choose to make a backup of it before making any changes).
3. Set the `domain` parameter, the `initString` parameter, and other `crypto` parameters to the same values with what you configured in the web tier's lwssofmconf.xml file.
4. Save your changes and close the file.
5. Restart the chat server.

IdM authentication

To enable Identity Manager (IdM) authentication on the chat server, see Configure IdM authentication in Collaboration for detailed instructions.

1.3.8.5. Task 3: Deploy and configure the chat service

This task is performed on chatserver.yourdomain.net.

Deploy the chat service

Follow these steps:

1. Save the chat service installer sm9.80.*xxxx*_ChatMSVC.zip from Service Manager installation package to the same server where your chat server is deployed.
2. Extract sm9.80.*xxxx*_ChatMSVC.zip to any directory. For example, C:\chat-msvc-9.80.0. This directory is referred to as %CHAT_SERVICE_HOME% in the sections below.

Configure the chat service

Shared steps

1. (HTTPS Only) Configure the SSL settings by following the instructions described in Chat service.
2. Open the chat service configuration file app.properties with a text editor.
3. Update the related parameters:

Parameter	Value in this example	Description
daoFactory.serviceEndPoint	http://smserver.yourdomain.net:13080 or https://smserver.yourdomain.net:13443	The Service Manager server's URL.
daoFactory.username	falcon	An SM operator with the "system administrator" privilege and the RESTful API capability.
daoFactory.password		The password of the SM operator.
pubSubServiceConfig.userName	publishadmin	The user name created in Task 2: Deploy and configure the chat server > Configure Openfire > step 12.
pubSubServiceConfig.password	admin123	The user password created in Task 2: Deploy and configure the chat server > Configure Openfire > step 12.
chatServerConfig.domain	chatserver.yourdomain.net	The FQDN or IP address of the chat server.
chatServerConfig.host	chatserver.yourdomain.net	The FQDN or IP address of the chat server.
chatServerConfig.restApiSecretKey	spEKHXM9KnQkaX1J	The secret key string you recorded in Task 2: Deploy and configure the chat server > Configure Openfire > step 13.

4. Save your changes and close this file.

Windows

1. Open the Windows command prompt and change the directory to %CHAT_SERVICE_HOME%\bin.
2. Run the **install-service.bat** command to install the Windows service for the chat service.
3. Go to Windows Services, and verify that the new Service **SmChatService** service has been installed.

Tip

- You can run the **nssm edit SmChatService** command to edit the corresponding configurations after the Windows service is installed.
- To remove this Windows service, run the **nssm remove SmChatService** command.

We recommend you to use 64-bit Java in Windows, because 32-bit Java may have potential performance limitations. However, if you are working with 32-bit Java, follow these steps so that the Windows service **SmChatService** can be started successfully:

1. Open %CHAT_SERVICE_HOME%\bin\startup.bat with a text editor.
2. Update the memory setting to `SET CHAT_SVC_JVM_OPTIONS=-XX:ThreadStackSize=256 -Xms512m -Xmx1024m`.
3. Save your changes and close this file.
4. Open %CHAT_SERVICE_HOME%\bin\ install-service.bat with a text editor.
5. Update the `ThreadStackSize` memory setting to `ThreadStackSize=256 -Xms512m -Xmx1024m`.
6. Save your changes and close this file.

Linux

1. Open the file /etc/sysconfig/openfire with a text editor.
   In Linux, the Chat service shares the same system configuration file as the chat server.
2. Uncomment the line for `CHATMSVC_HOME` and then update it with the correct path. For example: `CHATMSVC_HOME="/opt/chat-msvc-9.80.0"`

Start/stop the chat service

Windows

Follow these steps to start/stop the chat service from the Windows command prompt:

1. Open the Windows command prompt and change the directory to %CHAT_SERVICE_HOME%\bin.
2. Run the **startup.bat** command.
3. To stop the chat service, press Ctrl+C in the command window.

Follow these steps to start/stop the chat service from Windows services:

1. Go to Windows Services, select the **SmChatService** service and then click **Start Service** to start the service.
2. To stop the chat server, go to Windows Services, and then click **Stop Service** to stop the SmChatService service.

Linux

Note

You may need to add the execute permission to the file before execution. For example, run the **chmod +x** or **chmod 755** command.

Follow these steps to start/stop the chat service as a foreground process:

1. Open a Linux terminal to connect to the server.
2. Change directories to **%CHAT_SERVICE_HOME%**/bin.
3. Run the **./startup.sh** command to start.
4. To stop the chat service, press Ctrl+C in the terminal.

Follow these steps to start/stop the chat service as a background process:

1. Open a Linux terminal to connect to the server.
2. Change directories to **%CHAT_SERVICE_HOME%**/bin.
3. Run the **./chatmsvc start** command to start.
4. To stop the chat server, run the **./chatmsvc stop** command.

Configure authentication on the chat service

The chat service supports the LW-SSO or the IdM authentication, but not both at the same time. You need to use the IdM authentication on the chat service if you have configured the IdM authentication in the web tier. Otherwise, use LW-SSO instead.

LW-SSO

Follow these steps to configure LW-SSO authentication on the chat service:

1. Navigate to the %CHAT_SERVICE_HOME%\conf folder.
2. Open lwssofmconf.xml with a text editor.
3. Set the `domain` parameter, the `initString` parameter, and other `crypto` parameters to the same values with what you configured in the webtier's lwssofmconf.xml file.
4. Save your changes and close the file.
5. Start the chat service.

IdM authentication

To enable Identity Manager (IdM) authentication on the chat service, see Configure IdM authentication in Collaboration for detailed instructions.

Perform health checks

To perform a health check for the connection to the Chat Server and the Service Manager server, visit http://chatserver.yourdomain.net:8089/healthcheck?pretty=true or https://chatserver.yourdomain.net:8449/healthcheck?pretty=true.

See the following example for a successful response:

Successful response:

```
{
 "ChatServerRESTConnection" : {
  "healthy" : true
 },
 "ChatServerXMPPConnection" : {
  "healthy" : true
 },
 "SMServerRESTConnection" : {
  "healthy" : true
 },
 "deadlocks" : {
  "healthy" : true
 }
}
```

For another example, if the daoFactory.userName is not set correctly, the response message of the health check will show that the dependent data service is unavailable:

```
{
  "ChatServerRESTConnection" : {
  "healthy" : true
  },
  "ChatServerXMPPConnection" : {
    "healthy" : true
  },
  "SMServerRESTConnection" : {
    "healthy" : false,
    "message" : "The dependent data service at https://smserver.yourdomain.net:13443 is unavailable. "
  },
  "deadlocks" : {
    "healthy" : true
  }
}
```

In the above examples,

- The `ChatServerRESTConnection` checks the Representational State Transfer (REST) connection from the chat service to the chat server.
- The `ChatServerXMPPConnection` checks the Extensible Messaging and Presence Protocol (XMPP) connection from the chat service to the chat server.
- The `SMServerRESTConnection` checks the Representational State Transfer (REST) connection from the chat service to the Service Manager server.

1.3.8.6. Task 4: Deploy the End User Chat UI

Perform this task on `chatuiserver.yourdomain.net`. You should have Tomcat installed on this server.

The End User Chat UI consists of the End User Chat window and the End User Chat button. Follow these steps to deploy the End User Chat UI:

1. Save the End User Chat UI installer `sm9.80.xxxx_ChatUI.zip` from Service Manager installation package to your computer.
2. Unzip the zip file and then rename the `WAR` file to `chatui.war`, and then unzip the `WAR` file.
3. Open the `chatui/conf/env.js` file with a text editor.
4. Update the `chatServiceUrl` value, which must be the real URL that you can access through the reverse proxy. You will configure the reverse proxy later.
   For example, `chatServiceUrl: 'http(s)://proxyserver.yourdomain.net/chatservice'`.
5. Save your changes and close this file.
6. Copy the `chatui` folder to the Tomcat `webapps` folder to deploy it.
7. Start Tomcat.



Note

Note

The chat UI contains static resources only. If you use the Apache HTTP server or the IIS as the reverse proxy, deploy the chat UI to the reverse proxy directly and expose it as /chatui.

1.3.8.7. Task 5: Configure reverse proxy for the chat components

This task is performed on proxyserver.yourdomain.net and currently only the following three types of reverse proxies are certified. This task applies to both Classic Service Manager Collaboration and SMA-SM Collaboration. When performing

this task, you need to replace the parameters with the actual values. Refer to the below table for different values for each parameter.

Parameter	Value	
	Classic SM Collaboration	SMA-SM Collaboration
CHATUI_SERVER	HTTP: chatuiserver.yourdomain.net:8080 HTTPS: chatuiserver.yourdomain.net:8443	Unnecessary to set reverse proxy for Chat UI
CHATUI_RESOURCE	chatui	Unnecessary to set reverse proxy for Chat UI
CHATUI_HOST	chatuiserver.yourdomain.net	Unnecessary to set reverse proxy for Chat UI
CHATSERVER_HOST	chatserver.yourdomain.net	<EXTERNAL_ACCESS_HOST>:443
HTTPBIND_SERVER	HTTP: chatserver.yourdomain.net:7070 HTTPS: chatserver.yourdomain.net:7443	<EXTERNAL_ACCESS_HOST>:443
HTTPBIND_RESOURCE	http-bind	of-http-bind
PLUGINS_SERVER	HTTP: chatserver.yourdomain.net:9090 HTTPS: chatserver.yourdomain.net:9091	<EXTERNAL_ACCESS_HOST>:443
PLUGINS_RESOURCE	plugins	of-plugins
CHATSERVICE_SERVER	HTTP: chatserver.yourdomain.net:8088 HTTPS: chatserver.yourdomain.net:8448	<EXTERNAL_ACCESS_HOST>:443
CHATSERVICE_RESOURCE		/chatservice



Note

SMA-SM Collaboration: Use HTTPS because the chat service in the suite requires SSL.

Important

For security reasons, we recommend adding two response headers to the Chat UI:

- Content-Security-Policy: frame-ancestors 'self'
- Cross-Origin-Embedder-Policy: require-corp

For detailed instructions about how to add a response header, refer to the Apache, IIS, or F5 documentation.

For example, in Apache, you can use the Header directive as below to set the headers to the /chatui location.

```
<Location "/chatui">
    Header set Content-Security-Policy "frame-ancestors 'self'"
    Header set Cross-Origin-Embedder-Policy "require-corp"
</Location>
```

Apache

Windows

1. (HTTP Only) Open the %APACHE_HOME%\conf\extra\httpd-vhosts.conf file with a text editor. Insert the following codes below the `ProxyPass /webtier` line in the default `<VirtualHost>` definition:

   ```
   ProxyPass /chatui ajp://%CHATUI_HOST%:8009/%CHATUI_RESOURCE% secret=
   ProxyPass /of-http-bind http://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE% timeout=120
   ProxyPassReverse /of-http-bind http://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE%
   ProxyPass /of-plugins http://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
   ProxyPassReverse /of-plugins http://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
   ProxyPass /chatservice http://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
   ProxyPassReverse /chatservice http://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
   ```

   Note that the `secret` value must be the same as `<tomcat_secret>` you have defined in the <tomcat_home>\server.xml file.
2. (HTTPS Only) Open the %APACHE_HOME%\conf\extra\httpd-ahssl.conf file with a text editor. Insert the following codes below the `ProxyPass /webtier` line in the default `<VirtualHost>` definition:

   ```
   ProxyPass /chatui ajp://%CHATUI_HOST%:8009/%CHATUI_RESOURCE% secret=
   ProxyPass /of-http-bind https://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE% timeout=120
   ProxyPassReverse /of-http-bind https://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE%
   ProxyPass /of-plugins https://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
   ProxyPassReverse /of-plugins https://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
   ProxyPass /chatservice https://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
   ProxyPassReverse /chatservice https://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
   ```

   Note that the `secret` value must be the same as `<tomcat_secret>` you have defined in the <tomcat_home>\server.xml file.
3. Go to Windows Services, and restart the **Apache2.4** service.

Linux

1. (HTTP Only) Open the file %APACHE_HOME%/conf.modules.d/00-proxy.conf in a text editor.

Insert the following codes below the `ProxyPass /webtier` line in the default `<VirtualHost>` definition:

```
ProxyPass /chatui ajp://%CHATUI_HOST%:8009/%CHATUI_RESOURCE% secret=
ProxyPass /of-http-bind http://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE% timeout=120
ProxyPassReverse /of-http-bind http://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE%
ProxyPass /of-plugins http://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
ProxyPassReverse /of-plugins http://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
ProxyPass /chatservice http://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
ProxyPassReverse /chatservice http://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
```

Note that the `secret` value must be the same as `<tomcat_secret>` you have defined in the <tomcat_home>\server.xml file.

2. (HTTPS Only) Open the file %APACHE_HOME%/conf.d/ssl.conf in a text editor.
   Insert the following codes below the `ProxyPass /webtier` line in the default `<VirtualHost>` definition:

```
ProxyPass /chatui ajp://%CHATUI_HOST%:8009/%CHATUI_RESOURCE% secret=
ProxyPass /of-http-bind https://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE% timeout=120
ProxyPassReverse /of-http-bind https://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE%
ProxyPass /of-plugins https://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
ProxyPassReverse /of-plugins https://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%
ProxyPass /chatservice https://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
ProxyPassReverse /chatservice https://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%
```

   Note that the `secret` value must be the same as `<tomcat_secret>` you have defined in the <tomcat_home>\server.xml file.
3. Run the **# systemctl restart httpd** command to restart Apache.

Apache HTTPS configuration examples

Classic Service Manager Collaboration:

```
ProxyPass /chatui ajp://chatuiserver.yourdomain.net:8009/chatui secret=MyT0mcAt
ProxyPass /of-http-bind https://chatserver.yourdomain.net:7443/http-bind  timeout=120
ProxyPassReverse /of-http-bind https://chatserver.yourdomain.net:7443/http-bind
ProxyPass /of-plugins https://chatserver.yourdomain.net:9091/plugins
ProxyPassReverse /of-plugins https://chatserver.yourdomain.net:9091/plugins
ProxyPass /chatservice https://chatserver.yourdomain.net:8448
ProxyPassReverse /chatservice https://chatserver.yourdomain.net:8448
```

SMA-SM Collaboration:

```
ProxyPass /of-http-bind https://<EXTERNAL_ACCESS_HOST>:443/of-http-bind  timeout=120
ProxyPassReverse /of-http-bind https://<EXTERNAL_ACCESS_HOST>:443/of-http-bind
ProxyPass /of-plugins https://<EXTERNAL_ACCESS_HOST>:443/of-plugins
ProxyPassReverse /of-plugins https://<EXTERNAL_ACCESS_HOST>:443/of-plugins
ProxyPass /chatservice https://<EXTERNAL_ACCESS_HOST>:443/chatservice
ProxyPassReverse /chatservice https://<EXTERNAL_ACCESS_HOST>:443/chatservice
```

Internet Information Services (IIS)

1. (HTTPS Only) Import the CA certificate of the chat server (chatserver.yourdomain.net) into the Certificates folder of the **Trusted Root Certification Authorities**.
   You must perform this step if the certificate of the IIS and the chat server don't share the same CA.

2. Add four rules for URL Rewrite.
    1. Select the server node.
    2. Double-click the **URL Rewrite** feature in the middle pane to open the feature. Alternatively, you can select it and then click **Open feature** in the Actions panel on the right.

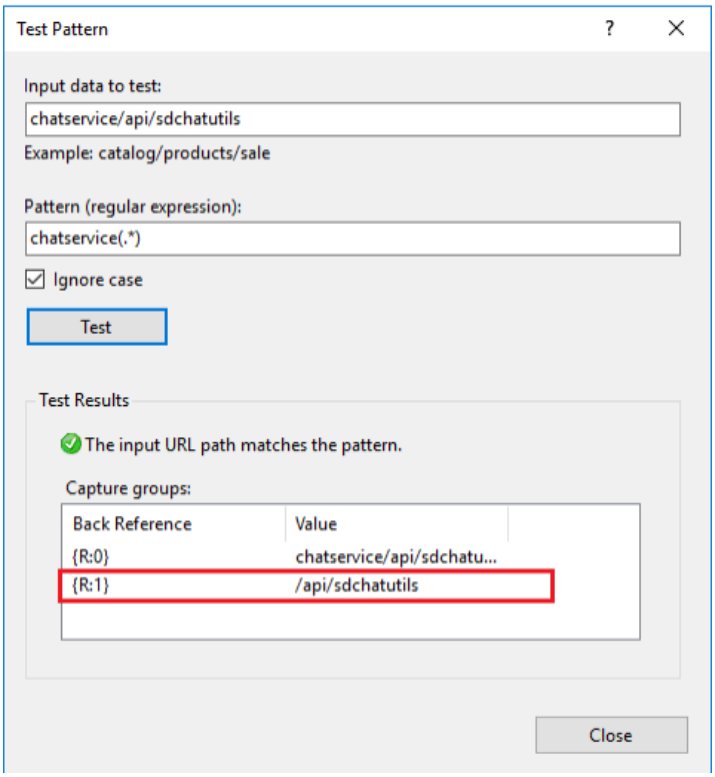
Use value in the following table to add four rules:

	Name	Pattern	Rewrite URL
1	httpbind	^of-http-bind	http(s)://%HTTPBIND_SERVER%/%HTTPBIND_RESOURCE%/ **Note** Do not miss the ending slash (/). HTTPS URL examples: Classic SM Collaboration: https://chatserver.yourdomain.net:7443/http-bind/ SMA-SM Collaboration: https://<EXTERNAL_ACCESS_HOST>:443/of-http-bind/
2	plugins	of-plugins(.*)	http(s)://%PLUGINS_SERVER%/%PLUGINS_RESOURCE%{R:1} HTTPS URL examples: Classic SM Collaboration: https://chatserver.yourdomain.net:9091/plugins{R:1} SMA-SM Collaboration: https://<EXTERNAL_ACCESS_HOST>:443/of-plugins{R:1}
3	chatservice	chatservice(.*)	http(s)://%CHATSERVICE_SERVER%%CHATSERVICE_RESOURCE%{R:1} HTTPS URL examples: Classic SM Collaboration: https://chatserver.yourdomain.net:8448{R:1} SMA-SM Collaboration: https://<EXTERNAL_ACCESS_HOST>:443/chatservice{R:1}

4	chatui	chatui.*	http(s)://%CHATUI_SERVER%/{R:0} HTTPS URL examples: https://chatuiserver.yourdomain.net:8443/{R:0}

3. In the **URL Rewrite** feature page, click the **Add Rule(s)...** link in the Actions panel on the right.
4. In the **Inbound rules** section, choose the **Blank rule** and click **OK**.
5. In the **Edit Inbound Rule** page, fill the Name, Pattern and Rewrite URL fields and use default value for other fields.
6. Click **Apply** in the Actions panel on the right to save the change.

You can click the **Test Pattern...** button to open a dialog and test your pattern. For example, here is the test for the chatservice rule:



F5

Follow these steps:

1. Create four pools and set properties and members as listed in the following table.

Pool name	Active health monitor	Member	Description
smc-http-bind	tcp	%HTTPBIND_SERVER%	For Http-bind of the chat server
smc-plugins	tcp	%PLUGINS_SERVER%	For plugins of the chat server
smc-chatservice	tcp	%CHATSERVICE_SERVER%	For chat service
smc-chatui	tcp	%CHATUI_SERVER%	For chatui

2. Open the **SMC** iRule, and then locate the following section:

```
when HTTP_REQUEST {
if { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/webtier" } {
  pool smc-webtier
  return
}
}
```

   Add the following additional iRules above the last `}`:

```
elseif { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/of-http-bind" } {
  set uri [string map {"/of-http-bind/" "/%HTTPBIND_RESOURCE%/"} [HTTP::uri]]
  HTTP::uri $uri
  pool smc-http-bind
  return
}
elseif { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/of-plugins" } {
  set uri [string map {"/of-plugins/" "/%PLUGINS_RESOURCE%/"} [HTTP::uri]]
  HTTP::uri $uri
  pool smc-plugins
  return
}
elseif { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/chatservice" } {
  set uri [string map {"/chatservice/" "%CHATSERVICE_RESOURCE%/"} [HTTP::uri]]
  HTTP::uri $uri
  pool smc-chatservice
  return
}
elseif { [HTTP::host] contains "<YOUR_F5_FQDN>" and [HTTP::uri] starts_with "/chatui" } {
  pool smc-chatui
  return
}
```

3. Save your changes.

1.3.8.8. Task 6: Select a portal for End User Chat

This task is performed on smwebtier.yourdomain.net and any computer that has a supported browser.

Service Manager supports the use of one of the following portals for End User Chat:

- SMA Service Portal (default): used for SMA Service Portal users
- ESS: used for Service Manager Employee Self-Service (ESS) client users

You need to select the right portal depending on which portal is being used for end users in your organization. To do this, follow these steps:

1. Use a browser to Access your web tier and log on to Service Manager as a system administrator.
2. Go to **System Administration** > **Base System Configuration** > **Miscellaneous** > **System Information Record**.
3. Select the **Active Integrations** tab.
4. In the SM Portal field, select the right portal from the drop-down list.
5. Specify the portal URL for your portal, as described in the following table.

Portal	steps
ESS	In the ESS URL field, type the fully qualified ess.do URL to your web tier. For example: http(s)://smwebtier.yourdomain.net/webtier/ess.do
Service Portal	In the Service Portal URL field, enter the following value: `https://<EXTERNAL_ACCESS_HOST>/saw/ess/smRequestTracking?TENANTID=<xxxxxxxx>`. You can find the tenant ID in the Service Manager tenant record. If you use a short URL, use the ess/smRequestTracking keyword to identify the Service Portal URL.

6. Click **Save**.
7. Restart the chat service.
8. (ESS ONLY) Update smc.ess.chat.url for the webtier. On smwebtier.yourdomain.net, follow these steps:
    1. Navigate to the %TOMCAT_HOME%\webtier\WEB-INF directory, and then open webtier.properties with a text editor.
    2. Locate the `SMC ESS Chat configuration` section, and then update the parameter of smc.ess.chat.url. You may also update other parameters. See the following table for detailed description:

Parameter	Description
smc.ess.chat.url	Specify the absolute URL to the End User Chat UI application. This chat URL is the real URL which you can access through the reverse proxy. For example, http(s)://proxyserver.yourdomain.net/chatui.
smc.ess.chat.style.width	Specify the width of the iFrame which contains the End User Chat window. The unit is pixel.
smc.ess.chat.style.height	Specify the height of the iFrame which contains the End User Chat window. The unit is pixel.
smc.ess.chat.style.position	Specify the position of the End User Chat UI. The available values are `bl` (bottom-left) or `br` (bottom-right).

    3. Save your changes and close this file.
    4. Restart Tomcat.

1.3.8.9. Task 7: Enable Service Manager Collaboration

This task is performed on any computer that has a supported browser.

By default, the Collaboration feature is disabled. In this task, you will sign in to Service Manager and set up the Collaboration Configuration.

Follow these steps:

1. Access http(s)://proxyserver.yourdomain.net/webtier/index.do in your web browser, and then sign in to Service Manager as a system administrator.
2. Click **System Administration** > **Ongoing Maintenance** > **Collaboration** > **Configuration** to open the Collaboration Settings form.
3. Select the **Enable Collaboration** checkbox to enable Service Manager Collaboration.

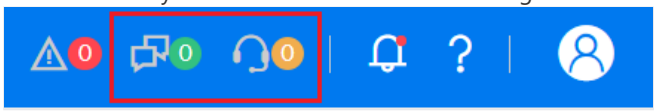
4. Select whether to enable the End User Chat virtual agent or not.

> **Important**
> Important
> To enable the End User Chat virtual agent, your enterprise must enable Smart Analytics first.

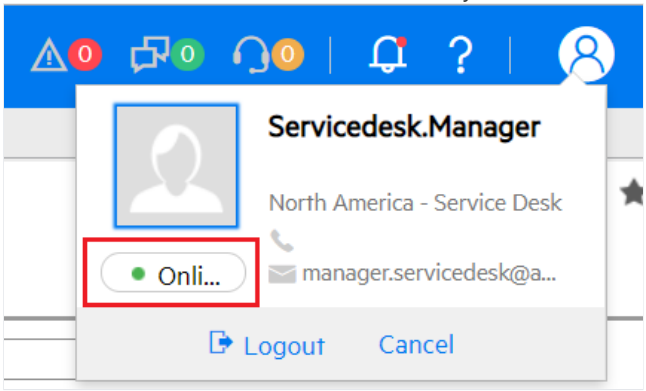
5. Specify field values as described in the following table.

Field	Value in this task	Description
Maximum Participants Per Conversation	200	The maximum number of participants in a conversation. The default value is 200.
Notification Delay Time (Seconds)	30	The maximum time that an online participant has to wait to receive the live conversation notifications. The default value is 30. Notification delay is disabled if this value is set to 0 or minus.
Chat Service Path	/chatservice/	The chat service path. The value is same as that in the reverse proxy parameter for chatservice. The default value is /chatservice/.

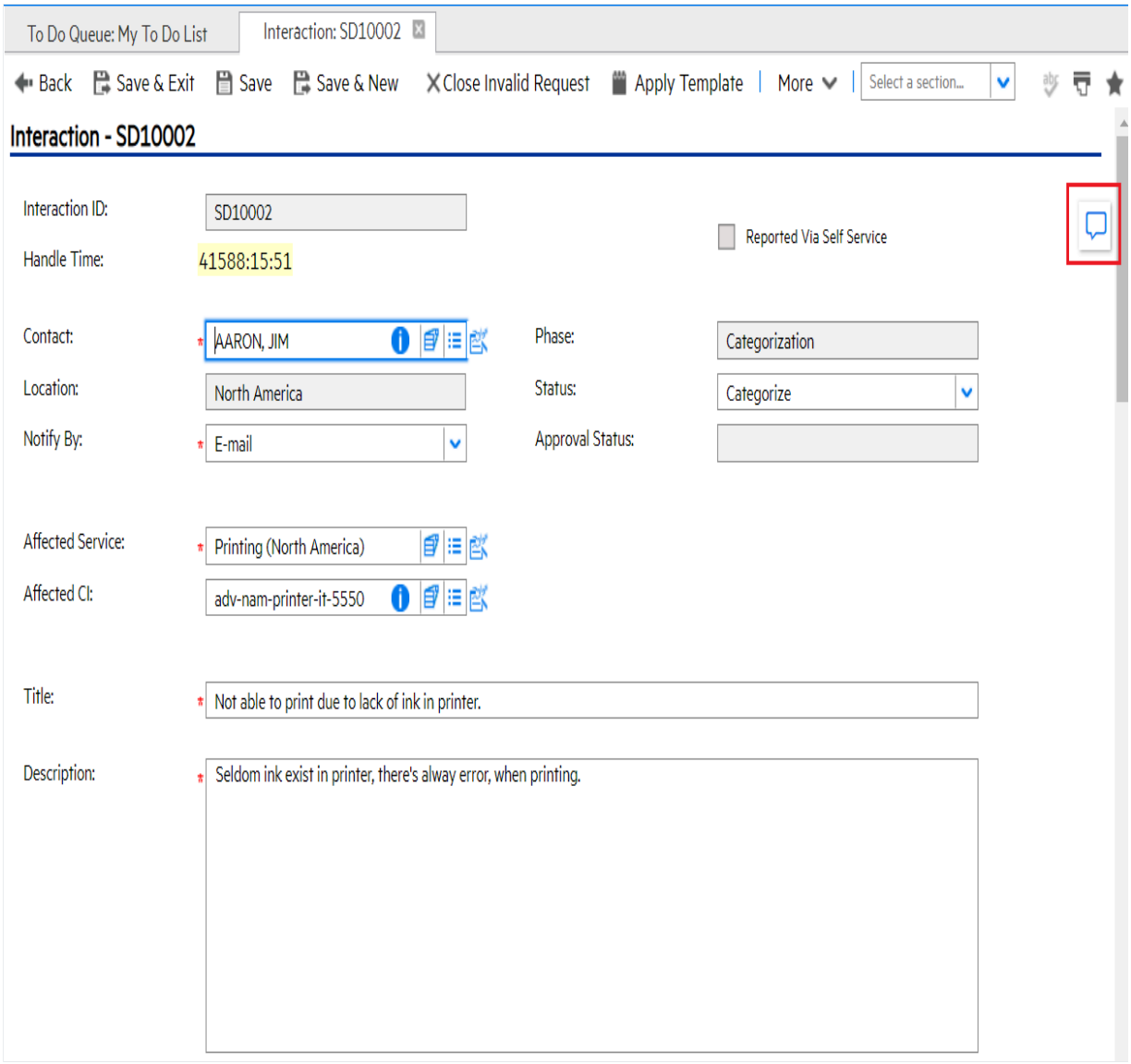
6. Click **Save** and **OK**. It may take a while for the configurations to take effect.
7. Sign out of the web client, and then sign in as the Service Desk manager.
8. Verify that a **Chat Notification** button and an **End user chat request** button are displayed at the top-right corner of the screen. They are clickable after the number is greater than 0.



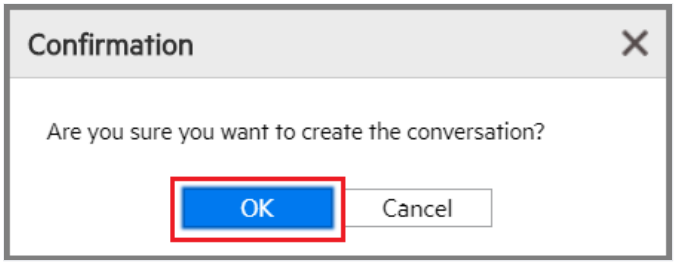
9. Click the **User Information** button to show your User Basic Information Card. Your presence status is now **Online**.



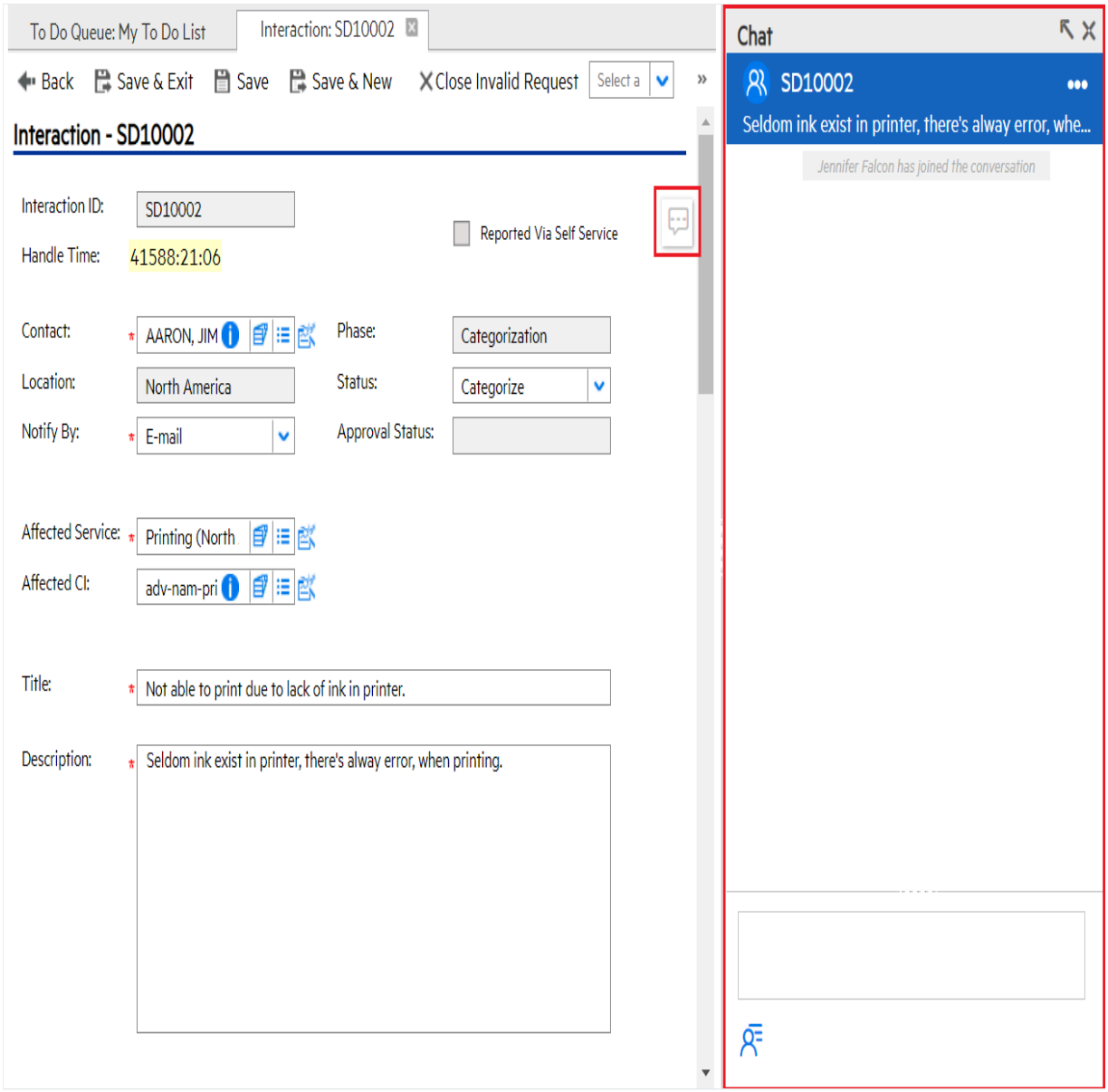
10. Open an interaction record. The **Start Conversation** button floats on the upper-right corner of the detailed view of this record.



11. Click the **Start Conversation** floating button, and then click **OK** in the confirmation dialog.



12. The **Start Conversation** floating button is disabled and the conversation window opens on the right of the window.



Congratulations! You have successfully deployed Service Manager Collaboration!

1.3.9. Configure a web server to redirect requests to the web tier

You can install and configure a web server to redirect web-client specific URLs to the Service Manager Web tier. The following sample configurations assume that the Service Manager web tier is deployed on a Tomcat web application server.

Apache HTTP server

In this example, the Apache module mod_proxy_ajp is used to connect Apache HTTP server with Tomcat.

Note

Note

If you are using HTTPS and have previously configured and used HTTP related proxy configuration file, such as httpd-vhosts.conf in Windows or 00-proxy.conf in Linux, you need to comment out all the configured proxies in this file so that the HTTPS settings aren't overwritten by the HTTP settings.

Windows

1. Follow the Using Apache HTTP Server on Microsoft Windows guide to install the latest Apache HTTP Server 2.4 on a Windows server. The install directory is referred to as <APACHE24_HOME> in the following steps.

   Note
   Note
   If you rename the directory to any other name instead of the default Apache24, you should update the *SRVROOT* parameter in the configuration file conf/httpd.conf accordingly. For example, `Define SRVROOT "c:/Apache2.4.58"`.

2. Navigate to the <APACHE24_HOME>\conf directory.
3. Open the httpd.conf file with a text editor (You can choose to make a backup of it before making any changes).
4. Find and uncomment the below lines to load required modules, and then save the change.

```
...
LoadModule headers_module modules/mod_headers.so
...
LoadModule lbmethod_bybusyness_module modules/mod_lbmethod_bybusyness.so
...
LoadModule proxy_module modules/mod_proxy.so
LoadModule proxy_ajp_module modules/mod_proxy_ajp.so
LoadModule proxy_balancer_module modules/mod_proxy_balancer.so
...
LoadModule proxy_http_module modules/mod_proxy_http.so
...
LoadModule slotmem_shm_module modules/mod_slotmem_shm.so
...
LoadModule rewrite_module modules/mod_rewrite.so
...
LoadModule xml2enc_module modules/mod_xml2enc.so
...
```

5. (HTTP Only) Find and uncomment the following line in the httpd.conf file to include the httpd-vhosts.conf file, and then save the change.

```
        ...
        # Virtual hosts
        Include conf/extra/httpd-vhosts.conf
        ...
```

   Open the <APACHE24_HOME>\conf\extra\httpd-vhosts.conf file with a text editor, edit the default virtual host as below, remove redundant virtual host definitions, and then save the change. Note that the `secret` value must be the same as the one you have defined for the AJP connector in the Tomcat configuration.

```
<VirtualHost _default_:80>
DocumentRoot "${SRVROOT}/htdocs"
#ServerName apache.yourdomain.net:80
ProxyPass /webtier ajp://smwebtier.yourdomain.net:8009/webtier secret=MyT0mcAt
</VirtualHost>
```

6. (HTTPS Only) (Recommended) Configure the <APACHE24_HOME>\conf\extra\httpd-ssl.conf file according to the Configure SSL for Apache HTTP Server doc.
7. Navigate to the <APACHE24_HOME>\bin directory, and then run the **httpd -k install** command to install the Windows service.

8. Go to Windows Services, and start the **Apache2.4** service.

Linux

1. Follow the Apache installation guide to install Apache on Linux.
   For example:

   ```
   # yum install httpd
   # yum install mod_ssl
   # systemctl enable httpd
   ```

   The Apache is usually installed in /etc/httpd. Refer to it as <APACHE24_HOME>.
2. (HTTP Only) Appends the virtual host as shown below to the end of the <APACHE24_HOME>/conf/httpd.conf file.

   ```
   <VirtualHost _default_:80>
     DocumentRoot "/var/www/html"
     ServerName apache.yourdomain.net:80
     ProxyPass /webtier ajp://smwebtier.yourdomain.net:8009/webtier secret=MyT0mcAt
   </VirtualHost>
   ```

   Note that the `secret` value must be the same as the one you have defined for the AJP connector in the Tomcat configuration.
3. (HTTPS Only)(Recommended) Configure the <APACHE24_HOME>/conf.d/ssl.conf file according to the Configure SSL for Apache HTTP Server doc.
4. Run the following command to start Apache.

   ```
   # systemctl start httpd
   ```

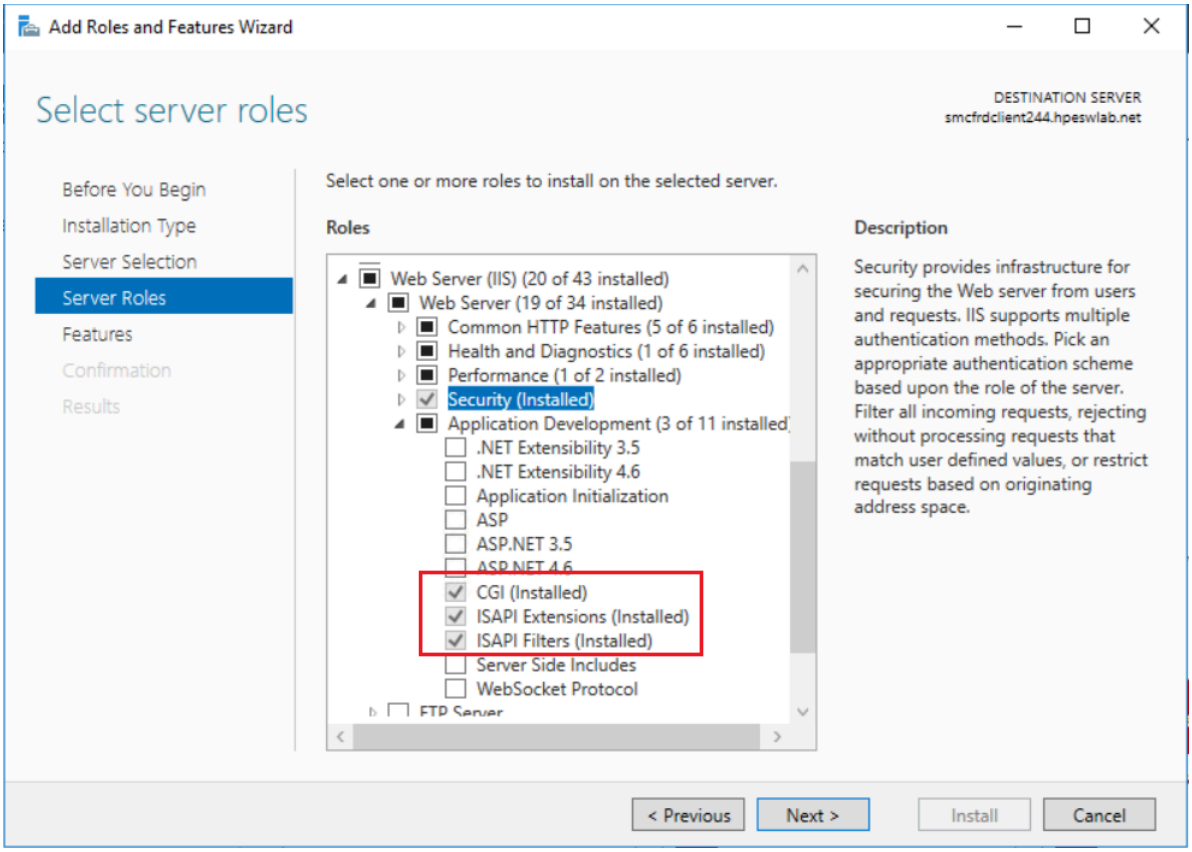
Windows Internet Information Services (IIS)

First, install Windows Internet Information Services (IIS). Next, depending on the protocol you want to use between IIS and Tomcat, you can use one of the following configuration examples: **Connect to Tomcat by using the BonCode AJP 1.3 connector**, **Connect to Tomcat by using the Apache JK connector**, or **Connect to Tomcat by using proxy**.

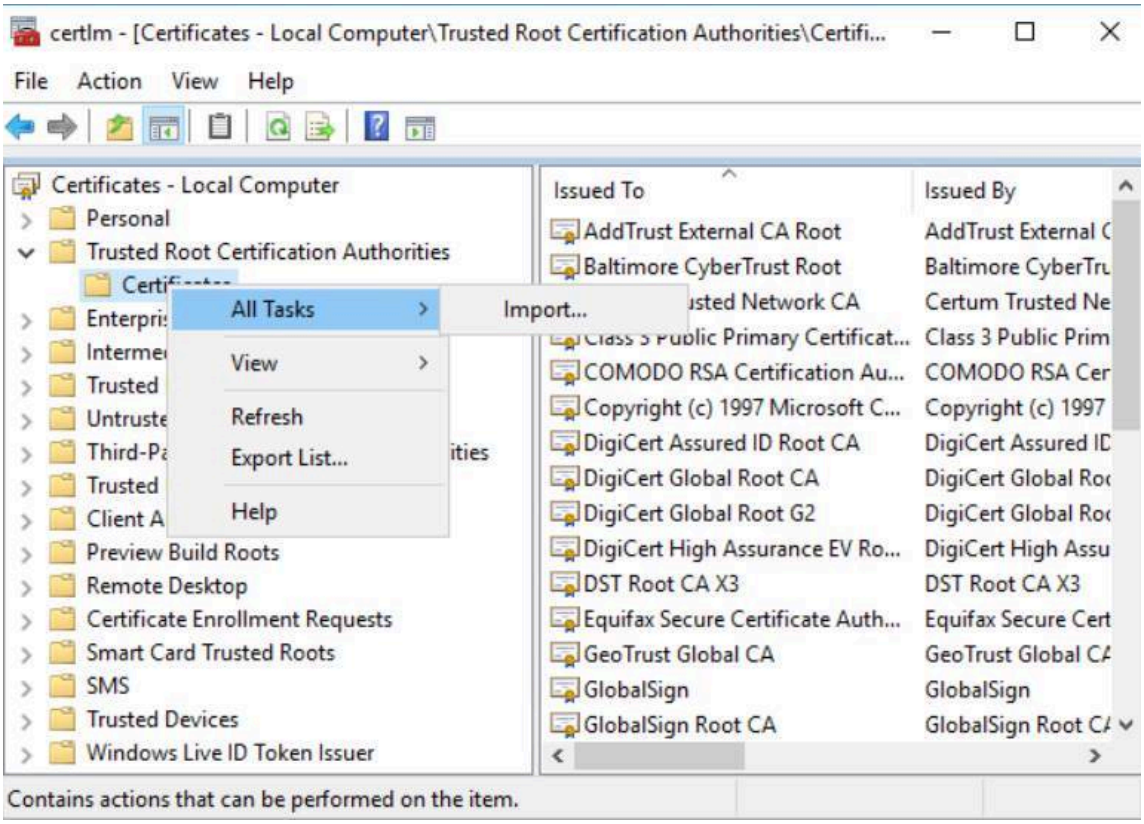
Install IIS

Follow these steps:

1. As shown in the following screenshot, install IIS and add **CGI** and **ISAPI** related features.



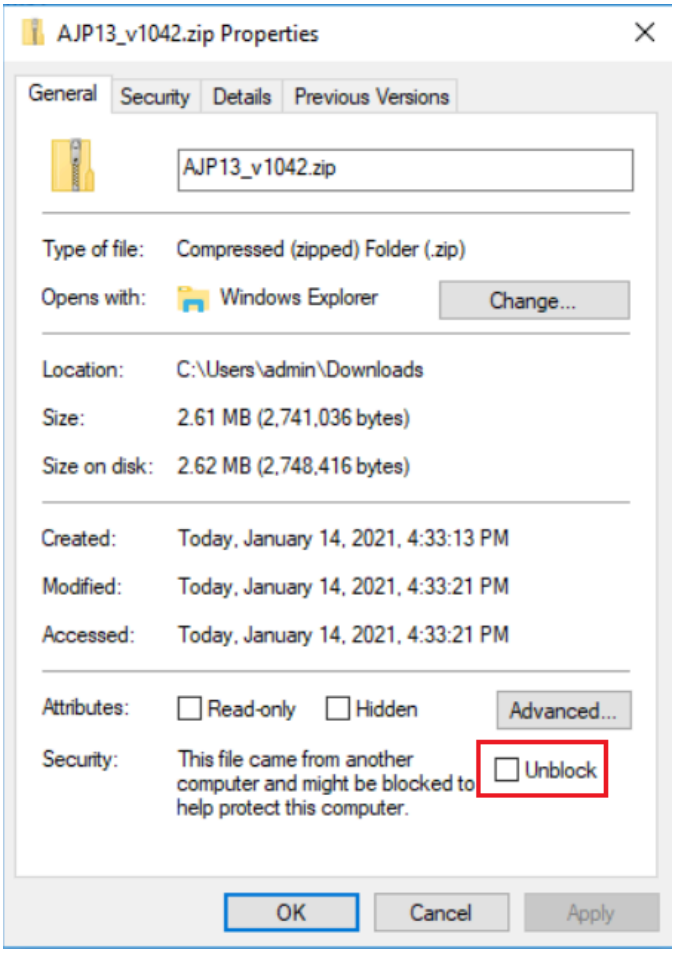
2. Download and install Application Request Routing as an extension for IIS.
3. Download and install URL Rewrite as an extension for IIS.
4. (HTTPS Only) Configure SSL on IIS. You may refer to https://techexpert.tips/iis/enable-https-iis/.
5. (HTTPS Only) Import the CA certificate of the web tier server (smwebtier.yourdomain.net) into the Certificates folder under **Trusted Roolt Certification Authorities**.
   Note: failure to do this step may result in a 502 error when you access the web tier server later.



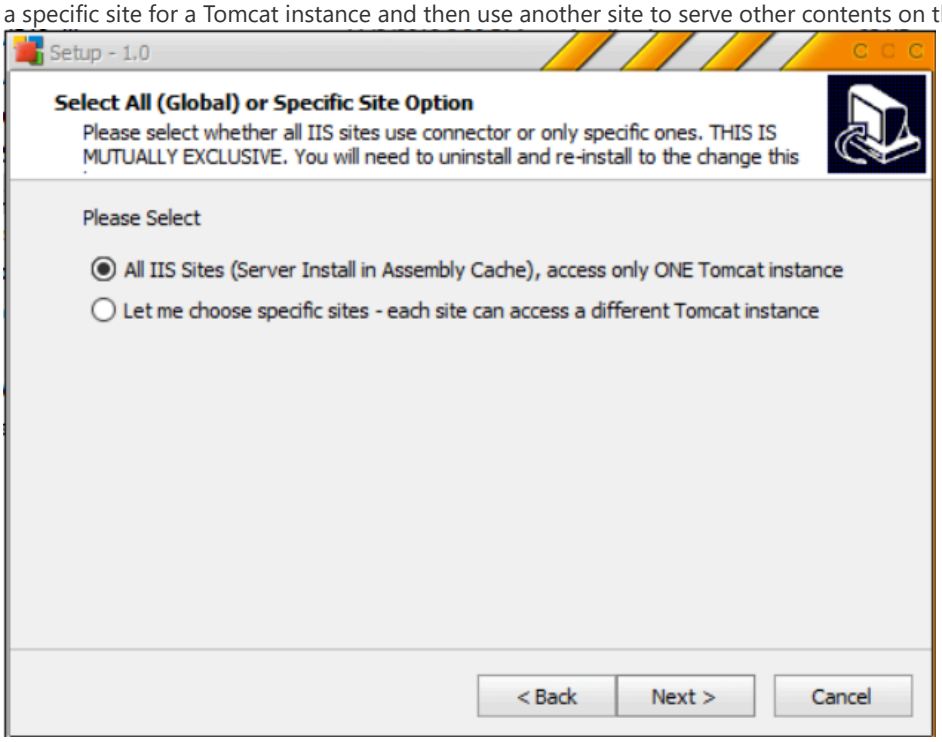
Connect to Tomcat by using the BonCode AJP 1.3 connector

Follow these steps:

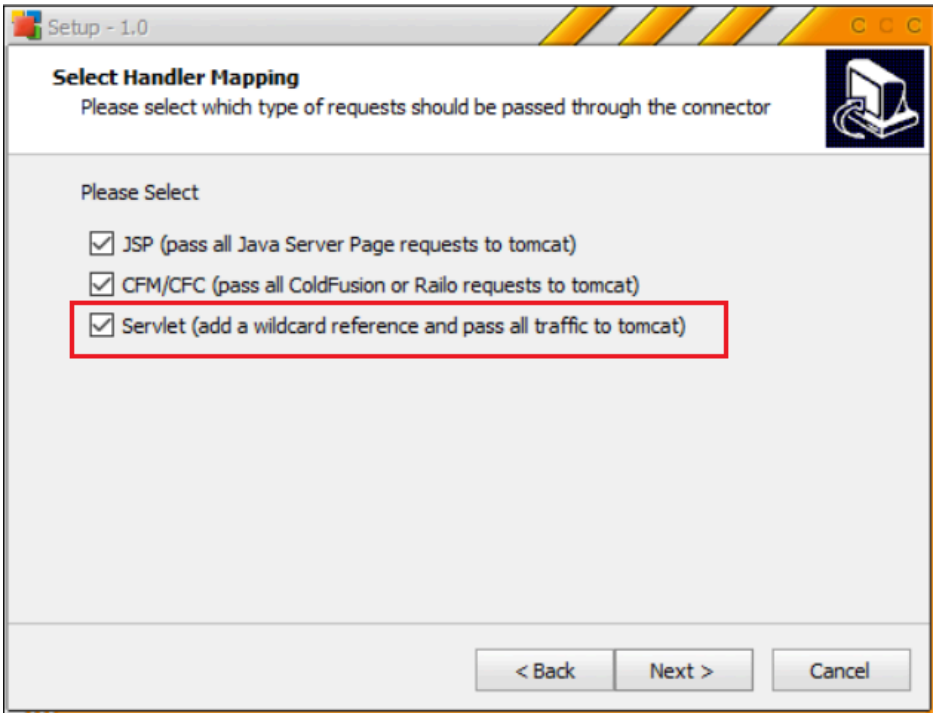
1. Download the BonCode AJP 1.3 connector from http://www.boncode.net/boncode-connector.
2. As illustrated below, unblock the zip file before you unzip it.



3. Unzip the zip file.
4. Run Connector_Setup.exe and follow the wizard to perform an automatic installation.
   In the site option step, choose the second option if you have other contents served on the IIS server. In this case, use a specific site for a Tomcat instance and then use another site to serve other contents on the IIS.



   In the handler mapping step, select the **Servlet** checkbox.



Wait till the setup process completes.

5. Copy BonCodeAJP13.settings from the unzipped BonCode AJP 1.3 connector folder to the BIN folder of the physical path of the specific site. For example, C:\inetpub\jsproot\BIN.
6. Open BIN/BonCodeAJP13.settings with a text editor.
7. Add the following child elements inside the root `<Settings>` element, and then save the change.

```
<RequestSecret>MyT0mcAt</RequestSecret>
<PacketSize>65536</PacketSize>
```

Connect to Tomcat by using the Apache JK connector

Refer to the Apache official documentation for detailed instructions. Meanwhile, you need to edit the workers.properties file in IIS to include the following five parameters:

- [uri:/webtier-x.xx/servlet/*]
  info=Prefix mapping
- [uri:/webtier-x.xx/*.jsp]
  info=Extension mapping
- [uri:/webtier-x.xx/*.do]
  info=Extension mapping
- [uri:/webtier-x.xx/attachments/*]
  info=Extension mapping
- [uri:/webtier-x.xx/cwc/nav.menu]
  info=Extension mapping

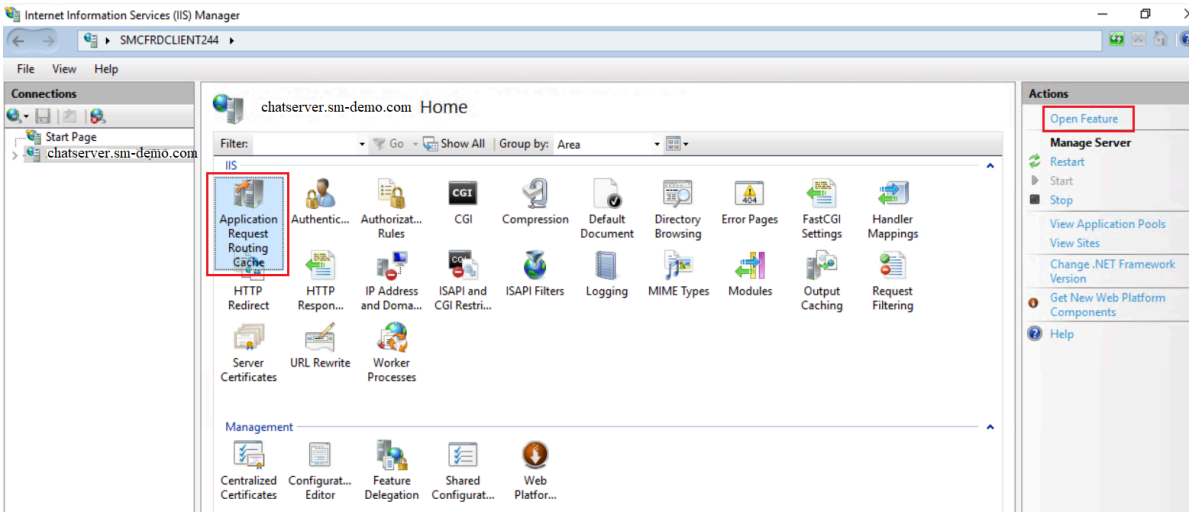
Note: If you change the default application name from webtier-x.xx to some other name such as "sm", you will need to change the URI mappings to match your web tier's application name.

Connect to Tomcat by using proxy

Follow these steps:

1. Open the **Internet Information Services (IIS) Manager**.
2. Enable proxy for the Application Request Routing.
    1. Select the server node.

2. Double-click the **Application Request Routing Cache** feature in the middle pane to open the feature. Alternatively, you can select it and then click **Open feature** in the Actions panel on the right.
3. In the **Application Request Routing Cache** feature page, click **Server Proxy Settings** in the Actions panel on the right to open the settings page.
4. In the Proxy Settings page, select the **Enable proxy** checkbox and set other fields by following your company policies.
5. Click **Apply** in the Actions panel on the right to save the change.



3. (Non-TSO Only) Add two server variables and one rule for URL Rewrite.
    1. Select the server node.
    2. Double-click the **URL Rewrite** feature in the middle pane to open the feature. Alternatively, you can select it and then click **Open feature** in the Actions panel on the right.
    3. Add the following two server variables:
        1. In the **URL Rewrite** feature page, click the **View Server Variables...** link in the Actions panel on the right.
        2. In the **Allowed Server Variables** page, click the **Add...** link in the Actions panel on the right to add two variables: `HTTP_X_FORWARDED_HOST` and `HTTP_X_FORWARDED_PROTO`.
    4. Add the following rule:
        1. In the **URL Rewrite** feature page, click the **Add Rule(s)...** link in the Actions panel on the right.
        2. In the popup of the **Inbound rules** section, choose the **Blank rule** and click **OK**.
        3. In the **Edit Inbound Rule** page, fill the Name, Pattern and Rewrite URL fields and use default value for other fields.

	Name	Pattern	Rewrite URL
1	webtier	webtier.*	http://smwebtier.yourdomain.net:8080/{R:0} or https://smwebtier.yourdomain.net:8443/{R:0}

        4. Expand the **Server Variables** panel and click **Add...** to add two variables as listed in below table.

	Name	Value	Replace
1	HTTP_X_FORWARDED_HOST	<the host of the webtier URL>:<the port of the webtier URL>	True
2	HTTP_X_FORWARDED_PROTO	<the scheme of the webtier URL>	True

You can skip `<the port of the webtier URL>` when working with the default HTTP port 80 or HTTPS port 443. For example:

	Name	Value	Replace
1	HTTP_X_FORWARDED_HOST	apache.yourdomain.net:8443	True
2	HTTP_X_FORWARDED_PROTO	https	True

5. Click **Apply** in the Actions panel on the right to save the change.

1.3.10. Install Identity Manager (IdM)

Service Manager (SM) leverages Identity Manager (IdM) to support Single Sign-On (SSO) by using SAML 2.0 or OpenID Connect (OIDC). You need to install the IdM service first, and then configure IdM authentication in Service Manager.

To set up SAML-based SSO for Service Manager, you need to create a trust relationship with a third-party identity provider (IdP, such as Microsoft ADFS). In the SAML-based SSO process, the IdM service acts as a service provider (SP) to the IdP.

To set up OIDC-based SSO for Service Manager, you need to register a client in the OpenID provider (OP, such as Keycloak). In the OIDC-based SSO process, the IdM service acts as a relying party to the OpenID provider.

The IdM service is released with Service Manager in Standalone mode (as a standalone WAR archive). This topic uses Tomcat as an example for the web application server.

Typically, you will deploy a reverse proxy (Apache HTTP Server or IIS) or a hardware load balancer on top of the IdM service in the production environment. This topic uses the Apache HTTP Server as an example for the reverse proxy and assumes the reverse proxy URL is https://apache.yourdomain.net. The later tasks use this sample reverse proxy URL and you need to replace it with the actual URL in your environment.



Note

Note

In SMA-SM, you don't need to install the standalone IdM service. Instead, leverage the IdM service deployed in the SMA suite and then integrate Service Manager with the IdM service.



Note

Tip

To enable FIPS mode for the IdM service, you need to perform some additional configurations during your installation. For more information, see Configure FIPS mode in the IdM Service.

To install the IdM service together with the IdM admin console, complete the following tasks.

Prerequisite

Make sure that you have completed the following procedure.

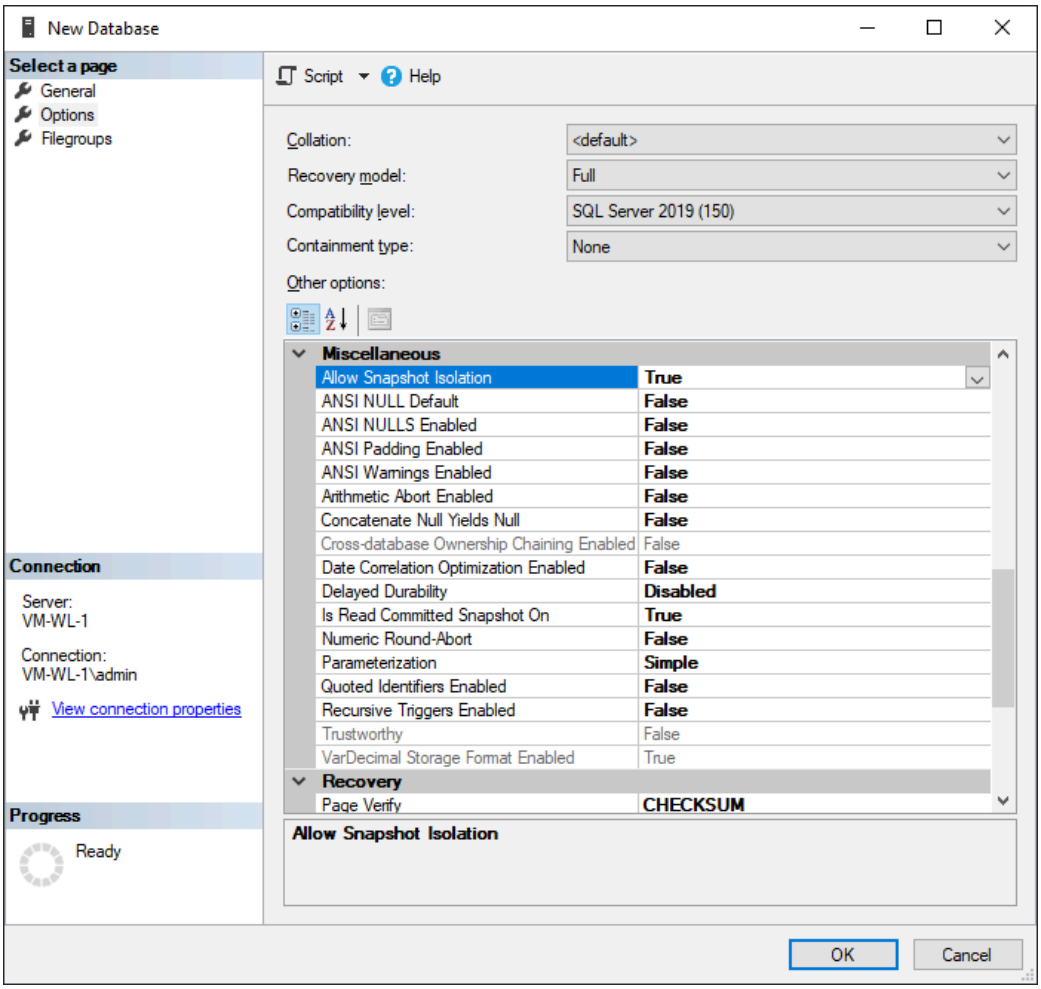
1. Install Oracle JDK 17 or Zulu OpenJDK 17 and set the `JAVA_HOME` environment variable properly.
2. Download and install Tomcat 10.
3. Generate a server certificate for the IdM service and configure SSL in Tomcat. It's mandatory to use the HTTPS port to access the IdM service. Otherwise, you will see 401 errors during the login process. For detailed instructions, see
4. Work with the certificate generation toolkit and IdM service.

Task 1: Create a database

You need to create a new database for the IdM service first. This database can locate on the same database server as Service Manager. Currently, the supported database types include SQL Server, Oracle, and PostgreSQL. For information about their supported versions, see the Service Manager Support Matrix.

Note: If your DBMS is MS SQLServer, you need to turn on the snapshot isolation when you create the database.

- **Allow Snapshot Isolation:** True
- **Is Read Commited Snapshot On:** True



Task 2: Install the IdM service

To deploy and set up idm-service, follow these steps:

1. Stop Tomcat.
2. Copy `idm-service-1.40.1-xxx.war` to the `<TOMCAT_HOME>/webapps` folder, rename it to `idm-service.war`, then extract it to the `<TOMCAT_HOME>/webapps/idm-service` folder.

3. Check that the `WEB-INF/classes/security/key-configuration.properties` file is using the `strong256bit` encryption profile. For example:

```
#####Strong128
#enckeytype.local.profile=strong128bit
#enckeytype.local.jce=SunJCE
#enckeytype.local.keyfile=key-local-strong128.bin
#enckeytype.local.passphrase.name=local strong128 passphrase
#enckeytype.local.passphrase.provider=file
#enckeytype.local.passphrase.file=key-local-strong128-passphrase.txt
#enckeytype.local.passphrase.deleteFile=false

#enckeytype.network.profile=strong128bit
#enckeytype.network.jce=SunJCE
#enckeytype.network.keyfile=key-network-strong128.bin
#enckeytype.network.passphrase.name=network strong128 passphrase
#enckeytype.network.passphrase.provider=file
#enckeytype.network.passphrase.file=key-network-strong128-passphrase.txt
#enckeytype.network.passphrase.deleteFile=false

#####Strong256
enckeytype.local.profile=strong256bit
enckeytype.local.jce=SunJCE
enckeytype.local.keyfile=key-local-strong256.bin
enckeytype.local.passphrase.name=local strong256 passphrase
enckeytype.local.passphrase.provider=file
enckeytype.local.passphrase.file=key-local-strong256-passphrase.txt
enckeytype.local.passphrase.deleteFile=false

enckeytype.network.profile=strong256bit
enckeytype.network.jce=SunJCE
enckeytype.network.keyfile=key-network-strong256.bin
enckeytype.network.passphrase.name=network strong256 passphrase
enckeytype.network.passphrase.provider=file
enckeytype.network.passphrase.file=key-network-strong256-passphrase.txt
enckeytype.network.passphrase.deleteFile=false
```

   Edit the following passphrase files in the WEB-INF/classes/security folder to provide your own passphrase. We recommend that the passphrase is 32 characters in length and consists of lower-case, upper-case, and at least 1 numerical digit.

   - key-local-strong256-passphrase.txt
   - key-network-strong256-passphrase.txt

   Remove the following out-of-the-box key files from the WEB-INF/classes/security folder.

   - key-local-strong256.bin
   - key-network-strong256.bin

4. Edit the `WEB-INF/spring/applicationContext.properties` file to configure the database connection. See the following sample configuration for the PostgresSQL database:

```
####PostgresSQL
idm.persistence.flyway.type = postgresql
# SQL dialect Change to commented out version for Hibernate 4.3.4.Final
# hibernate.dialect=org.hibernate.dialect.PostgreSQL9Dialect
idm.persistence.dialect = org.hibernate.dialect.PostgreSQL9Dialect
idm.persistence.connection.driver.class = org.postgresql.Driver
# Update this url to have your correct information
idm.persistence.connection.url = jdbc:postgresql://db.yourdomain.net:5432/idm
# Specify which database schema will be used. if not configured, or value is empty, it will be
ignored.
idm.persistence.connection.default_schema =
# Database connection user settings
idm.persistence.connection.username = postgres
idm.persistence.connection.password = <your encrypted password of the postgres user>
```

5. Create the **com.microfocus.sm__1.1.0.1__Add_UPDATE_Organization_Provider.json** file in the WEB-INF/classes/seeded folder to create the Provider tenant, the admin user, and the idmTransportUser user. Its content may resemble the following:

```
[
  {
    "operation":"ADD_OR_UPDATE",
    "type":"organization",
    "attributes":{
      "name":"Provider",
      "displayName":"Provider",
      "type":"PROVIDER",
      "description":"The admin tenant of the IdM service"
    }
  }, {
    "operation":"ADD_OR_UPDATE",
    "type":"databaseUser",
    "names":{
      "organizationName":"Provider"
    },
    "attributes":{
      "name":"admin",
      "password":"<your password for admin>",
      "email":"admin@yourdomain.net"
    },
    "associations":[{
      "name":"SUPER_IDM_ADMIN",
      "type":"permission"
    }]
  }, {
    "operation":"ADD_OR_UPDATE",
    "type":"databaseUser",
    "names":{
      "organizationName":"Provider"
    },
    "attributes":{
      "name":"idmTransportUser",
      "password":"<your password for idmTransportUser>",
      "type":"INTEGRATION_USER"
    }
  }
]
```



Note

Note

- The password must be at least 8 characters and consists of lower-case, upper-case, number, and special characters.
- For the `ADD_OR_UPDATE` operation, the **name** and **displayName** attributes are mandatory.
- You must include `idmTransportUser` in the seeded file because it is used for the Idm-client integration.

6. Create the **system_resource_config__1.0.0.1__update_signingKey_and_initString.json** file in the WEB-INF/classes/seeded folder to update signingKey and LW-SSO settings. Its content may resemble the following:

```
[
  {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"idm.encryptedSigningKey"
    },
    "attributes": {
      "value": "<your IdM signing key, 32 characters>"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.domain"
    },
    "attributes": {
      "value": "yourdomain.net"
    }
  }, {
    "operation": "UPDATE",
    "type": "systemResourceConfig",
    "names":{
      "systemResourceConfigName":"lwssoConfig.initString"
    },
    "attributes": {
      "value": "<your LWSSO init string, 32 characters>"
    }
  }
]
```



Note

Note

- The idm.encryptedSigningKey and the lwssoConfig.initString must be 32 characters.
- Update the lwssoConfig.domain value according to your environment.

7. Start Tomcat.
8. Verify the installation by accessing the https://idmservice.yourdomain.net:8443/idm-service/api/version-info URL. If the IdM service works fine, remove the following seeded files from the WEB-INF/classes/seeded folder. You should remember the idm.encryptedSigningKey and the password of the idmTransportUser for later use.
   - com.microfocus.sm__1.1.0.1__Add_UPDATE_Organization_Provider.json
   - system_resource_config__1.0.0.1__update_signingKey_and_initString.json

If you want to deploy the IdM service behind the Apache HTTP server, add ProxyPass directive to Apache configuration according to Apache HTTP Server and then restart the Apache HTTP server. If you use a hardware load balancer instead, you need to configure the load balancer to forward requests to the IdM service.

In a production environment, you must encrypt your passwords for security reasons. For detailed instructions about how to encrypt your password mentioned in *step 4*, *step 5*, and *step 6* above, see *Task 5: Encrypt the IdM service password*.

Task 3: Install the IdM admin console

To deploy and set up the idm-admin-console, follow these steps:

1. Stop Tomcat.

2. Copy the `idm-admin-console-1.40.1-xxx.war` to the `<TOMCAT_HOME>/webapps` folder, rename it to `idm-admin.war`, then extract it to the `<TOMCAT_HOME>/webapps/idm-admin` folder.
3. Edit the WEB-INF\classes\IdmConfig.properties file. The signingkey is same as the one for idm-service.
   Example:

   ```
   tenant=Provider
   integrationAcctUserName=idmTransportUser
   integrationAcctPassword=<your encrypted password for idmTransportUser of the Provider tenant>
   idm.serverUrl=https://apache.yourdomain.net/idm-service
   idm.serverUrlInternal=
   signingKey=<your encrypted IdM signing key>
   ```

   For detailed instructions about how to encrypt the password of `idmTransportUser` and the signing key, see *Task 6: Encrypt the IdM admin console password*.
   This example assumes that you have deployed the IdM service behind the Apache HTTP server. If there is no reverse proxy or load balancer exists on top of the IdM service, you need to use the URL exposed by Tomcat. For example, https://idmservice.yourdomain.net:8443/idm-service.
4. Copy the IdM client truststore to the `WEB-INF\classes\conf` folder. Edit the `WEB-INF\classes\conf\IdmConfig.xml` file, set the `trustAll` attribute to `false`, and then configure the client truststore. For example:

   ```
   <client trustAll="false" ...>
     ...
     <trustStoreFile>WEB-INF/classes/conf/smrootca.p12</trustStoreFile>
     <trustStoreType>PKCS12</trustStoreType>
     <trustStorePass>your encrypted password for the client truststore</trustStorePass>
     ...
   </client>
   ```

   See *Task 6: Encrypt the IdM admin console password* for detailed instructions about how to encrypt the `trustStorePass`.
   For the IdM client truststore, you need to import your corporate root CA that issues the server certificate for the reverse proxy or the load balancer to a truststore by using the certificate generation toolkit (such as `corprootca.p12`), and then use this truststore as the IdM client truststore.

   ```
   import-cert.bat --alias corprootca --file <your corporate root CA cert file> --keystore
   certs/corprootca.p12 --storepass <your password for corprootca.p12>
   ```

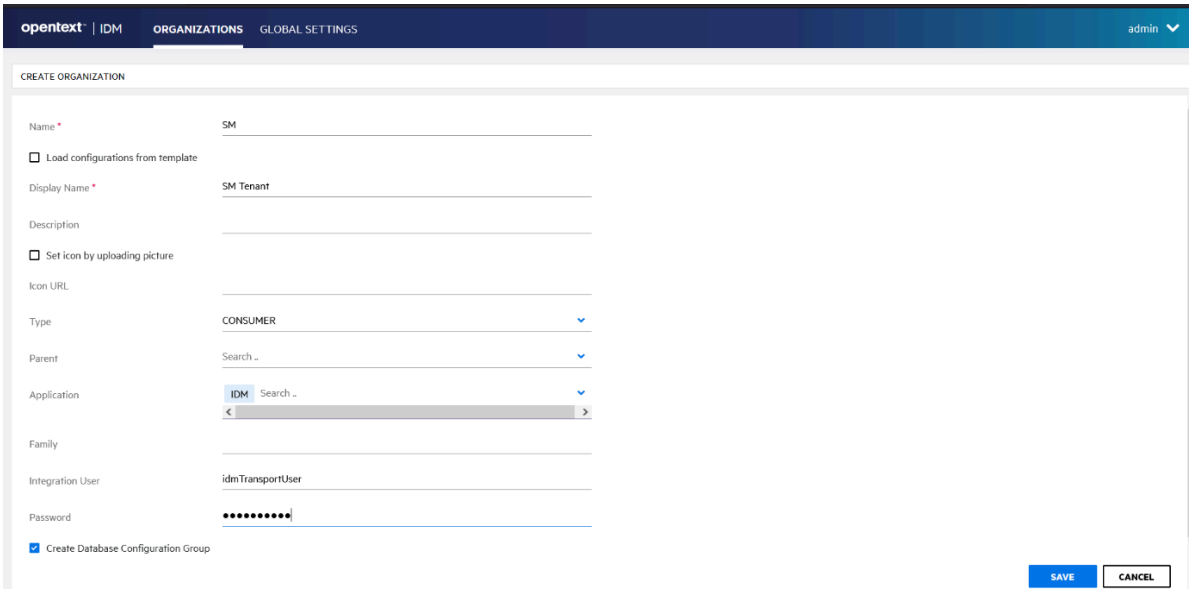
   If you use the SM private root CA to sign the server certificate for the reverse proxy in the development environment, or there is no reverse proxy or load balancer exists on top of the IdM service, use the `smrootca.p12` as the IdM client truststore.
5. Start Tomcat.

If you want to deploy the IdM admin console behind the Apache HTTP server, add ProxyPass directive to Apache configuration according to Apache HTTP Server and then restart the Apache HTTP server. If you use a hardware load balancer instead, you need to configure the load balancer to forward requests to the IdM service.

Task 4: Create the SM Tenant

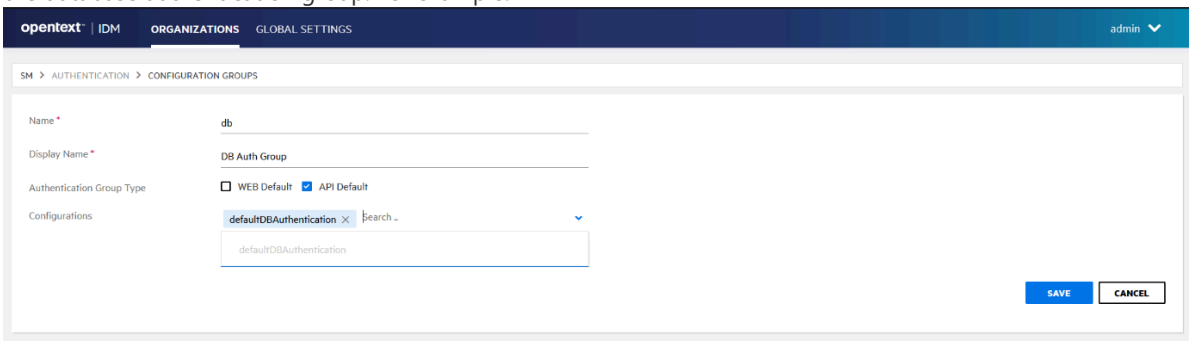
A tenant is a data space in IdM. IdM user authentication is based on a tenant. You need to create a tenant of CONSUMER type for Service Manager. Follow these steps:

1. Login to the IdM service through the admin console (https://apache.yourdomain.net/idm-admin) using the admin user you created in Task 2 > *step 5*.
2. Click the **Organization** tab, and then click **Add (+)** to create an organization. Enter `SM` in the **Name** field, enter **SM Tenant** in the **Display Name** field, enter **idmTransportUser** in the **Integration User** field, and then enter your password in the **Password** field. Click **Save**.

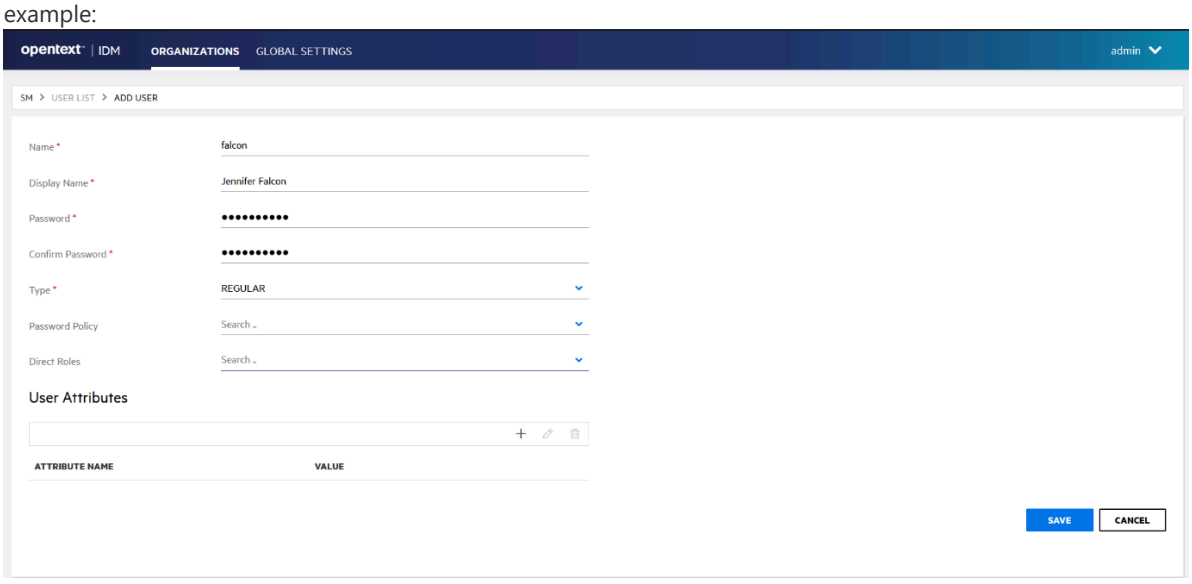


The Integration User and Password will be used when integrating SM with IdM.

3. Click **Save** to save the SM organization, and then click the SM organization to edit it.
4. Go to the **Authentication** page in the IdM admin console. Click **CONFIGURATION GROUPS** > **Add (+)** to create the database authentication group. For example:



5. Click **Save**.
6. (Optional) If you want to use some Service Manager out-of-the-box operators' accounts (such as `falcon`) but these operators don't exist in the external IdP (such as ADFS), click **USERS > Add (+)** to create database users in IdM. For example:



Click **Save**. You can then log in to the web tier as `falcon` later by using database authentication. For example, https://sm.yourdomain.net/webtier/index.do?auth=db.

Task 5: Encrypt the IdM service password

In this task, you will run the IdM encryption tool to encrypt the IdM service password.

Follow these steps:

1. Copy the dataprotection-passwordtool-1.0.5-standalone.jar to a folder. For example, `C:/passwordtool`.
2. Start the command prompt from the `C:/passwordtool` folder, and then run the following command to encrypt your password.

   ```
   java -jar dataprotection-passwordtool-1.0.5-standalone.jar --encrypt --conf "
   <TOMCAT_HOME>\webapps\idm-service\WEB-INF\classes\security" local <your password>
   ```

   For example, the value " `1Qaz2wsx` " is encrypted as " `ENC(kqtd/mNZ2Va81qs/c1ncIm7Te7qk5xIubOZTsHox0hmF6IH2iOW1ZhspmPgvqAAd9Mk4AVIMAJ1shRP5DjAm1Q==)`".

   ```
   C:\passwordtool>java -jar dataprotection-passwordtool-1.0.5-standalone.jar --encrypt --conf
   "C:\apache-tomcat-9.0.41\webapps\idm-service\WEB-INF\classes\security" local 1Qaz2wsx
   reading configuration: C:\apache-tomcat-9.0.41\webapps\idm-service\WEB-INF\classes\security
   encrypted:
   ENC(kqtd/mNZ2Va81qs/c1ncIm7Te7qk5xIubOZTsHox0hmF6IH2iOW1ZhspmPgvqAAd9Mk4AVIMAJ1shRP5DjAm1Q==)
   ```

Task 6: Encrypt the IdM admin console password

In this task, you will run the IdM encryption tool to encrypt the IdM console password.

Follow these steps:

1. Generate a key file first if you haven't done so. Start the command prompt from the `<TOMCAT_HOME>\webapps\idm-admin\WEB-INF\lib` folder, run the following command to generate a key file and save it to the `WEB-INF\classes` folder.

   ```
   java -jar idm-client-core-1.15.0-3.jar crypto -g -a AES -ep <your passphrase> -f
   ..\classes\conf\IdmConfig.key
   ```

   For example:

   ```
   C:\apache-tomcat-9.0.41\webapps\idm-admin\WEB-INF\lib>java -jar idm-client-core-1.15.0-3.jar
   crypto -g -a AES -ep wenmxjsk@#sF4545wejlkvhaslweyuvc -f ..\classes\conf\IdmConfig.key
   SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
   SLF4J: Defaulting to no-operation (NOP) logger implementation
   SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
   ====Output are below=======
   Key file has been generated successfully:
       ..\classes\conf\IdmConfig.key
   ```

   Then, update the `WEB-INF\classes\conf\IdmConfig.xml` file so that the `idm-client-core` uses the generated `IdmConfig.key`. For example:

   ```
   <client ...>
     ...
     <keyFile>WEB-INF/classes/conf/IdmConfig.key</keyFile>
     <encAlgorithm>AES</encAlgorithm>
     ...
   </client>
   ```

2. Start the command prompt from the `<TOMCAT_HOME>\webapps\idm-admin\WEB-INF\lib` directory, and then run the following command to encrypt the password.

```
java -jar idm-client-core-1.15.0-3.jar crypto -e -a AES -m <your password> -f
..\classes\conf\IdmConfig.key
```

For example, the value `"your_IdM_singing_key_with_32_len"` is encrypted to `"45zeg2ESNveBdFywBFdEDfWFib3wqfhb+E8OtYR1v372A99jCT1wh4P+lWqRwjWu"`, and you should use the enclosed form (such as `"ENC(45zeg2ESNveBdFywBFdEDfWFib3wqfhb+E8OtYR1v372A99jCT1wh4P+lWqRwjWu)"`) in the configuration files.

```
C:\apache-tomcat-9.0.41\webapps\idm-admin\WEB-INF\lib>java -jar idm-client-core-1.15.0-3.jar
crypto -e -a AES -m your_IdM_singing_key_with_32_len -f ..\classes\conf\IdmConfig.key
SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.
====Output are below=======
Encrypted Text: 45zeg2ESNveBdFywBFdEDfWFib3wqfhb+E8OtYR1v372A99jCT1wh4P+lWqRwjWu
```

1.4. Post-installation tasks

This section provides information about post-installation tasks.

This section includes the following:

Set up a replicated reporting database

Set up High Availability environment

1.4.1. Set up a replicated reporting database

The Service Manager Reporting module provides reports and dashboards with active operational data to achieve faster analysis and improved time to resolution. Since Reporting consumes additional system resources (memory and CPU), Service Manager enables you to optionally use a replicated database for reporting, which helps preserve the performance of your production database.

If you want to use a replicated database for the Service Manager Reports functionality, perform the following tasks.

Task 1: Prepare a replicated database

1. Prepare a database with the same database type and version as your production database. Follow the same instructions for your database type in Prepare your RDBMS.

   Note

   Note The two databases can reside on the same server or on different servers.

2. Perform periodic database synchronization between your production database and replicated database by using the standard database synchronization mechanism. For detailed instructions, refer to your specific database documentation.

Task 2: Configure a database connection between the **Service Manager** server and replicated database

Service Manager connects to the database through your RDBMS client (for example, Windows ODBC DSN defined for SQL Server). To set up the connection between your Service Manager application server and replicated database, know the name of the database and the login ID and password required to connect to the database server that you created above.

Configure the connection on the Service Manager Server host. In a horizontally scaled environment, you must do so on each of the Service Manager Server hosts. For example, if you use SQL Server, in addition to the ODBC Driver data source you configured for the production database, configure another ODBC Driver data source as a System DSN for the replicated database.

Task 3: Update the Service Manager server configuration file (sm.ini)

To do this, follow these steps:

1. In the sm.ini file, add the following parameters (the parameter values are for demonstration purposes only):

   Caution

   Caution The last four lines must be inserted as a group. Do not insert any other lines between them.

   For SQL Server:

   ```
   [sqlserver]
   sqldb:940pd
   sqllogin:pdadmin/passw0rd
   [sqlserver_replicate]
   sqldb:940rp
   sqllogin:rpadmin/passw0rd
   dashboardonreplicatedb
   ```

   For Oracle:

   ```
   [oracle]
   sqldb:940pd
   sqllogin:pdadmin/passw0rd
   [oracle_replicate]
   sqldb:940rp
   sqllogin:rpadmin/passw0rd
   dashboardonreplicatedb
   ```

Parameter	Description
[sqlserver]	This parameter creates a section header in the initialization file for information about an Microsoft™ SQL Server database. You only need to provide this parameter if you have set sqldictionary and are using a SQL Server database.
[sqlserver_replicate]	This parameter creates a section header in the initialization file for information about a replicated Microsoft™ SQL Server database. You only need to provide this parameter if you have created a replicated SQL Server reporting database.
[oracle]	This parameter creates a section header in the initialization file for information about an Oracle™ database. You only need to provide this parameter if you have set sqldictionary and are using an Oracle database.
[oracle_replicate]	This parameter creates a section header in the initialization file for information about a replicated Oracle™ database. You only need to provide this parameter if you have created a replicated Oracle reporting database.
sqllogin	This parameter defines the user name and password that Service Manager uses to authenticate connections to the RDBMS. You must use a slash character to separate the user name and password. If you omit this parameter, then the server attempts to authenticate the connection using the user name and password of the user who started the Service Manager Server, however this feature requires the Service Manager Server and the RDBMS server to use the same operating system. If the Service Manager Server and the RDBMS server use different operating systems, then you must specify a sqllogin value.
dashboardonreplicatedb	This parameter creates a section footer in the initialization file for information about a replicated reporting database.

2. (Optional) Add the following parameters to the sm.ini file of each Service Manager server:

```
dashboardquerycache_enable:1
dashboardquerycache_dbtime:100
dashboardquerycache_expire:10
```

Parameter	Description
dashboardquerycache_enable	Enables Service Manager to cache report query results in the database.
dashboardquerycache_dbtime	Defines a threshold for report query execution time that will trigger query caching (default: 100 milliseconds). Only when a query's execution time reaches or exceeds this threshold, the query results are cached in the database.
dashboardquerycache_expire	Defines how long cached report query results will be expired in the database (default: 10 minutes). When the specified time is reached, the cached query results are expired and will be refreshed at a later time.



Note

Note The query cache resides on the production database. It stores the results of queries against both the production database and replicated database.

3. Save the configuration file, and restart the Service Manager server.

In a horizontally scaled system, you need to repeat this task on each of the Service Manager server.

Task 4: Configure the Reporting module to use the replicated database

To do this, follow these steps:

1. Sign in to Service Manager as a system administrator.
2. Go to **Reporting** > **Administration** > **Report Settings**.
3. Select the **Use Replicated Database by default when users create reports** option.
   If this option is selected, the **Use Replicated Database** setting is enabled by default in each new report that users create. However, users can disable the **Use Replicated Database** setting in an individual report.
4. Select the **Use Production Database when Replicated Database is disabled** option.
   Select this option so that Service Manager can still use the production database to generate reports when the replicated database is disabled or the database connection parameters aren't defined correctly in the server configuration file (sm.ini).
5. Click **Save**.

1.4.2. Set up high availability environment

With an increased demand for reliable and performant infrastructures designed to serve critical systems, the term of high availability becomes more and more popular. High availability is a mode of running Service Manager on a cluster of two or more instances to enable load balancing and to ensure system availability.

Load balancing

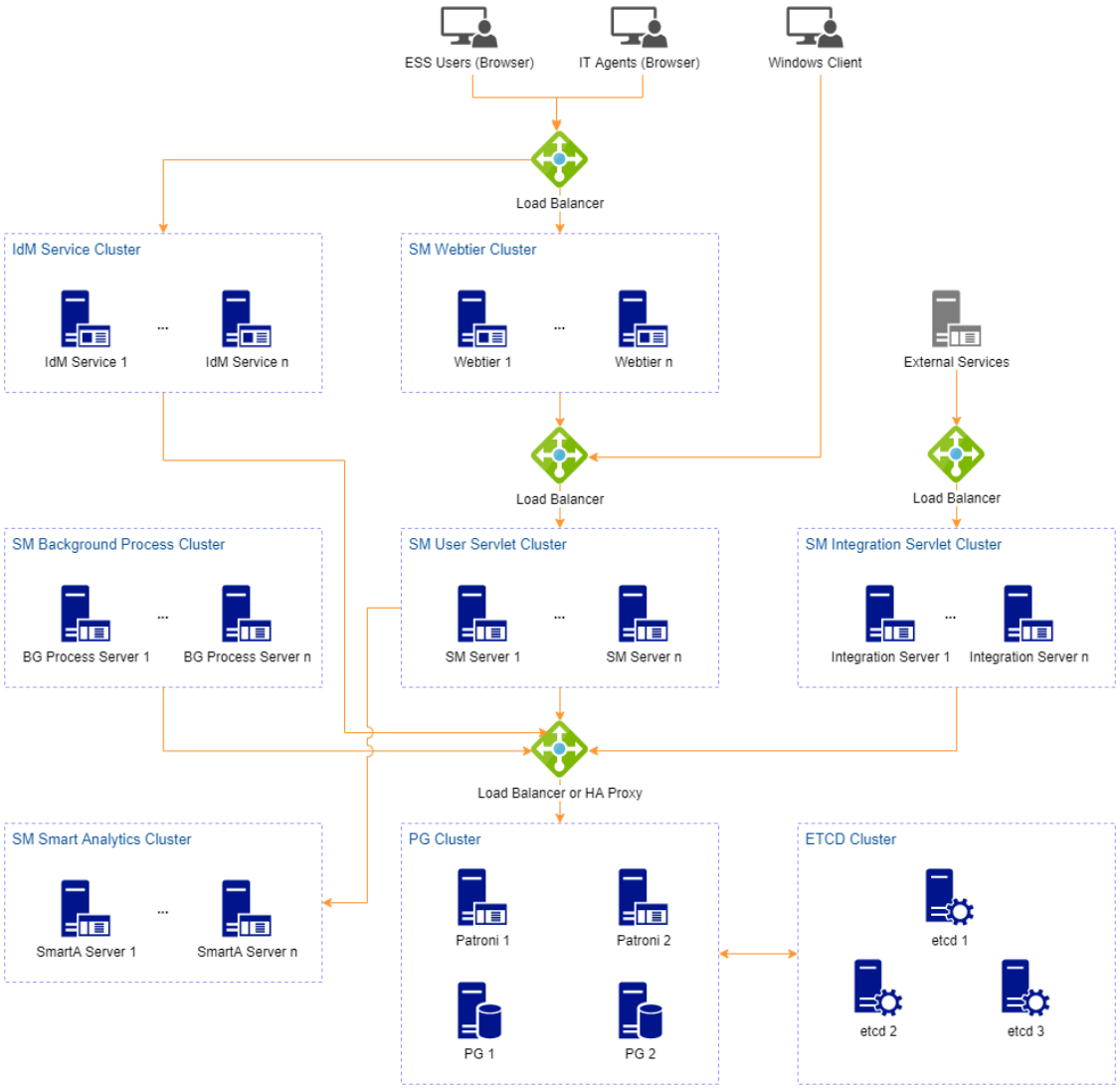
A typical configuration for a high availability environment is two or more Service Manager servers connect to the same database server. You can configure the servers to work behind a load balancer, that is, the load balancer serves as the entry point to the Service Manager servers. All the Service Manager servers are active at any given time and can handle incoming requests. The load balancer distributes requests to the servers in the cluster.

System availability

All Service Manager servers are active at any given time, and they all communicate with the database. If one of the servers becomes unavailable, the others can continue working as usual. Only the load will be redistributed.

Set up a high availability environment

The following diagram displays a configuration of Service Manager in high availability environment:



Note

Note

- In the lab environment, we have validated the configurations and deployments with the F5 hardware load balancer only.
- We have validated the single datacenter high availability only.

To do so, follow these steps:

1. **Set up the database**
   Service Manager supports Oracle, MS SQL Server, and PostgreSQL databases. The database vendors provide different approaches for high availability.
   - For the Oracle database, Service Manager supports Oracle real application clusters for high availability.
   - For the Microsoft SQL Server database, Service Manager supports SQL Server Alwayson for high availability. For detailed instructions, see Support of SQL Server AlwaysOn.
   - For the PostgreSQL database, Service Manager supports Patroni-based high availability. For detailed instructions, see High availability PostgreSQL with Patroni.

2. **Set up a Service Manager server cluster**
   If you have deployed the Service Manager server to serve incoming real user requests, you can leverage the horizontal scaling for high availability. For detailed instructions, see Horizontal scaling implementation.
   You can either use the Service Manager software load balancer for the user servlet cluster or deploy the F5 hardware load balancer on top of the user servlet cluster. For detailed instructions, see Parameter: loadBalancer or Configure an F5 hardware load balancer.
3. **Set up a Service Manager web tier cluster**
   The Service Manager web tier supports the Apache Tomcat web application server.
   Set up a web application server cluster and deploy the Service Manager web tier on top of it for high availability. For detailed instructions, see Install the Service Manager web tier.
   Service Manager supports the F5 hardware load balancer to forward the web client-specific HTTP requests to the Service Manager web tier cluster. The Service Manager web tier is stateful and you must configure session persistence (also known as sticky sessions) in the F5 hardware load balancer for it.
4. **(Optional) Set up an IdM service cluster**
   If you want to support SAML-based SSO or OIDC-based SSO for Service Manager, you need to install and integrate an IdM service with the Service Manager. The IdM service supports the Apache Tomcat web application server.
   Set up a web application server cluster and deploy the IdM service on top of it for high availability. For detailed instructions, see Install Identity Manager (IdM).
   The IdM service supports the F5 hardware load balancer to forward the IdM-specific HTTP requests to the IdM service cluster. The IdM service is stateful and you must configure session persistence (also known as sticky sessions) in the F5 hardware load balancer for it.
5. **Set up background processes high availability**
   Service Manager background processes support high availability. Multiple Service Manager threads for a scheduled class can work on multiple servers in parallel. With paralleled background processes, the Service Manager production system processes the tasks in the scheduled queue more efficiently. In the event of server restart or server failure, the other servers continue to work on the same class of tasks. For more information, see Set up background processes high availability.
   For detailed instructions about how to restart stopped background processes automatically, see Restart stopped background processes automatically.
6. **Set up an integration servlets cluster**
   If you have deployed the Service Manager server to serve the inbound integration requests, you can set up an integration servlets cluster for high availability.
   - Deploy two or more servlets for the inbound integration purpose.
   - Set up an F5 hardware load balancer in front of the servlets to redirect requests and load balancing.

   For detailed instructions, see Configure an F5 hardware load balancer.
7. **Smart Analytics high availability**
   To reduce potential downtime, we recommend configuring Smart Analytics for high availability. For detailed instructions, see Configure Smart Analytics for high availability.

1.5. Uninstall

This section provides information about how to uninstall different Service Manager components.

Uninstall the Service Manager Server

Follow these instructions if you need to uninstall the Service Manager Server.

Uninstall the Windows Server

You can uninstall the server from Add/Remove Programs or using the Service Manager uninstaller. Click a task below to view instructions for uninstalling the server using that method.

Uninstall the server using Add/Remove Programs

1. Sign in to the Windows server as a user with local administrator privileges.
2. Stop the Service Manager service.

3. From the Windows **Start** menu, click **Settings** > **Control Panel** > **Add/ Remove Programs**. The Add/Remove Programs dialog box opens.
4. Scroll to the Service Manager server program and click **Remove**. A message prompts you to verify that you want to remove the program.
5. Click **Yes**. The process takes several minutes. Additional messages indicate the progress of the uninstall. When you complete the uninstall, you return to the Add/Remove Programs dialog box.
6. Click **Close**.

Uninstall the server using the uninstaller

1. Sign in to the Windows server as a user with local administrator privileges.
2. Run the following commands:
   cd <install path>/Service Manager 9.80/Server
   unins000.exe

Uninstall the Linux Server

1. Stop the Service Manager server.
2. Run the following commands to uninstall the server:
   cd <*install path*>/_uninstall
   ./uninstall-ServiceManager9.80Server.pl
3. Delete all of the Service Manager server directories and subdirectories.

Uninstall the Windows Client or its components

You can uninstall the Windows Client by either using the Windows Control Panel or performing silent uninstallation.

Use Control Panel to uninstall the Windows Client

Follow these steps:

1. From the Windows **Start** menu, click **Settings** > **Control Panel** > **Add/ Remove Programs**. The Add/Remove Programs window opens.
2. Scroll to Service Manager Client and click **Remove**. A message prompts you to verify that you want to remove the program.
3. Click **Yes**. The uninstall process takes several minutes. Additional messages indicate the progress of the uninstall. When you complete the uninstall, click **Close** to close the Add/Remove Programs dialog box.

Perform silent uninstallation

Follow these steps:

1. Open a Windows command prompt and then navigate to the **Installation_uninstall** directory.
2. Run the following command to perform silent uninstallation:
   uninstaller.exe -i silent

> **Note**
> Note To keep the console waiting till the silent uninstallation is completed, you can save the above command in a batch file such as uninstall.bat, and then run the **call uninstall.bat** command.

The client uninstallation process intentionally preserves your client configuration settings and any other files that have changed since the initial installation. You must manually remove these files if you want to completely uninstall Service Manager from your system. We recommend that you delete the entire client installation folder and the local writeable workspace and configuration folder if you don't want to preserve any existing client settings.

Uninstall Smart Analytics

Follow these instructions if you need to uninstall Smart Analytics.

Note

Note

- Before you uninstall Smart Analytics, we recommend that you back up your index and category data if you want to restore it in the future. For backup and restore instructions, see *step 2* and *step 7* respectively in Transfer Smart Analytics intelligence.
- Restart the system after you uninstall Smart Analytics. Otherwise, the services and files can't be totally removed. Besides, the data and configurations remain after uninstall, and you must delete them manually for safety concerns.

Windows

To uninstall Service Manager Smart Analytics from Windows, follow these steps:

1. Go to **Control Panel** > **Programs** > **Uninstall a program**.
2. Select **SM x.xx Smart Analytics**, and then click **Uninstall/Change**. The Uninstall Smart Analytics wizard is displayed.
3. Click **Next**.
4. If you want to completely remove Smart Analytics, select **Complete Uninstall**. If you want to uninstall specific Smart Analytics features, select **Uninstall Specific Features**.

   Note

   Note If your Smart Analytics is upgraded from a former version, and there is a media server installed, you have to delete the files and service manually when you are uninstalling.

5. Click **Next**, and then follow the on-screen instructions to uninstall Smart Analytics.

Linux

To uninstall Service Manager Smart Analytics from Linux, follow these steps:

1. Go to the _uninstall folder under the Smart Analytics installation directory.
2. Type ./uninstaller -? from the command line interface to view the uninstallation options and instructions.

   Note

   Note As the Maintenance Mode is disabled in the SM Smart Analytics uninstaller, the parameters under the Maintenance Mode aren't applicable.

3. Use the available commands, and then follow the on-screen instructions to uninstall Smart Analytics.
   For example, you can use uninstaller -i console to launch a command line based interactive uninstall process. If you log on to the system through X-Window, you can launch the graphical uninstaller by using the uninstaller -i swing command.
   You can specify the features that you want to uninstall, or you can uninstall Service Manager Smart Analytics completely.

2. Install on AWS

This section walks you through on deploying Service Manager (SM) on AWS. You can deploy Service Manager in the classic mode (without integrating with SMA Service Portal), or in the SMA-SM mixed mode that integrates with SMA Service Portal (SMA SP). You can install all or part of the following SM components on AWS as needed:

Component	Classic	Mixed Mode (SMA-SM)
SM Server	Required	Required
SM Web tier	Required	Required
SM Windows Client	Optional	Optional
SM Mobility	Optional	Optional
IdM	Optional	N/A
SM Collaboration	Optional	N/A
SM Smart Analytics	Optional	Required

If you choose the mixed mode (SMA-SM), first install SMA SP, then install SM, set up the SMA-SM integration finally. Note that you must install the SMA SP and Service Manager in the same VPC, and use the same parent domain (for example, *yourdomain.net*).

Unless otherwise noted in the step heading, a step is obligatory for both the classic mode and the mixed mode. If a step is obligatory in one mode only, the step heading is with the (Classic Only) or (Mixed Mode Only) prefix.

In this section, EC2 instances are launched from the Windows Server 2019 image. Instructions and settings given in this guide are specific to the AWS console at the time of this writing. As the AWS console evolves, you may need to adapt these instructions and settings according to the latest AWS console and documentation when deploying SM on AWS.

You may tailor steps below depending on your own demand. For example, put RDS on separate private subnets.



Note

Text in the italic font means it's subject to change according to your deployment, you need to adapt it appropriately. For example:

- ***10.0.0.0/16*** is the CIDR block of your VPC. If you choose another CIDR block (for example, ***10.1.0.0/16***) for your VPC, you need to replace it with your CIDR block.
- ***us-west-2a*** means the availability zone A of the `US West (Oregon)` region. You need to replace it with an appropriate value according to the region you selected.
- ***9.81*** means the Service Manager version. You need to replace it with the actual version you installed.

2.1. Plan

When you deploy Service Manager on AWS, you can consider using one of the following two architectures based on your business needs:

- Public-facing architecture
- Hybrid architecture

Public-facing architecture

In this architecture, users access Service Manager via the internet. The VPC is composed of public subnets and private subnets, and all Service Manager components are put on private subnets for security consideration. See figure 1-1 as an example. This install guide is specific to the public-facing architecture.

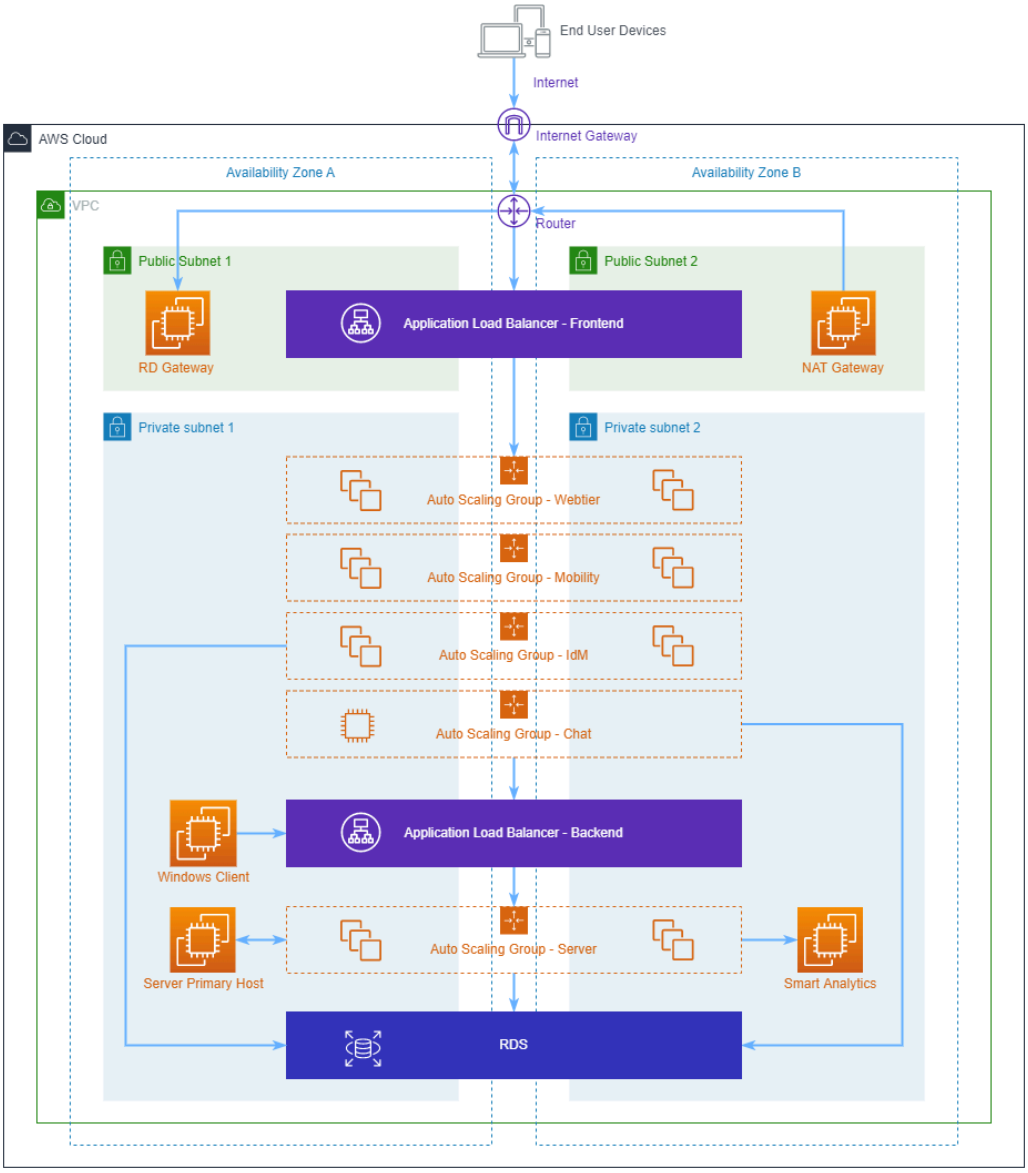


Figure 1-1 Public-facing Architecture

In this diagram:

- An AWS Application Load Balancer is created in the public subnets for the frontend applications (web tier, mobility, IdM, and chat).
- An RD Gateway is created in the public subnets as a bastion host.
- A NAT Gateway is created in the public subnets to allow internet access for EC2 instances in the private subnets.
- The AWS Auto Scaling Group (ASG) is used to scale out and scale in the web tier and mobility on demand.
- Chat is also put in an ASG, but with only one instance.
- An AWS Application Load Balancer is created in the private subnets for SM server instances.
- The SM server is horizontally scaled. The SM primary host is a standalone EC2 instance, while the SM server secondary hosts are managed by an ASG.
- Smart Analytics is a standalone EC2 instance.
- The SM server persists data to an RDS DB.

If you choose the mixed mode (SMA-SM), the Service Manager and the SMA Service Portal are deployed in the same VPC. Some resources are shared by Service Manager and the SMA SP to reduce cost. See figure 1-2 as an example. Note that public subnets and private subnets aren't illustrated in the diagram, resources that don't directly participate in the SMA-SM integration are omitted in the diagram either.

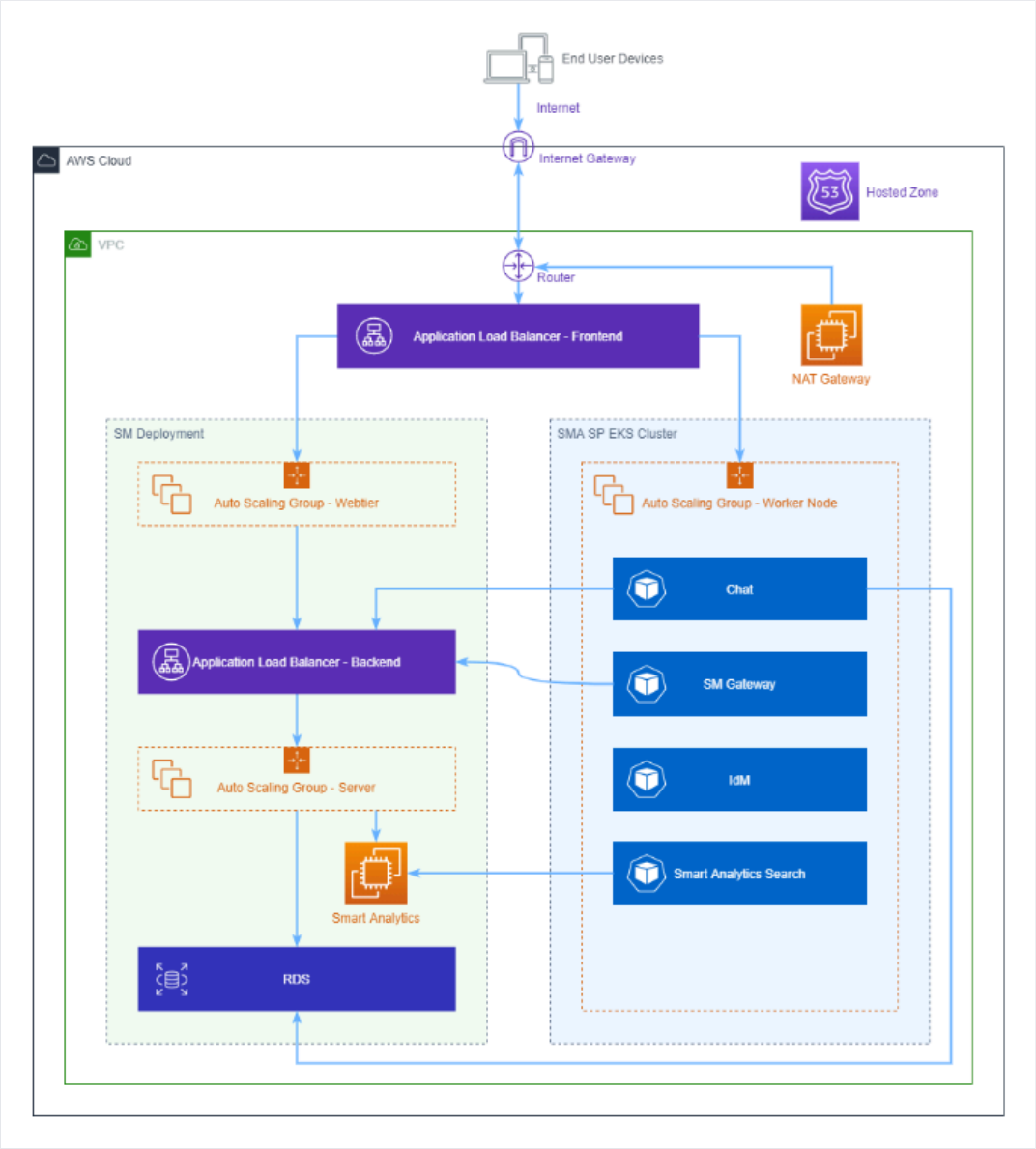


Figure 1-2 Public-facing Mixed Mode Architecture

In this diagram:

- The public hosted zone, the VPC, public subnets, the NAT gateway, and the frontend ALB are shared by Service Manager and the SMA SP.
- The frontend ALB and the NAT gateway are deployed in public subnets.
- SM and SMA SP are deployed in separate private subnets.
- The SM web tier is integrated with the IdM service in the SMA suite. Same for Service Manager mobility if you choose to install it.
- The SM gateway service and the Chat service (the chat-service container) in the SMA suite access the SM server through the backend ALB.
- The Chat service (the Openfire container) in the SMA suite accesses the SM database directly.
- The Smart Analytics Search service in the SMA suite delegates IDOL search to the SM Smart Analytics.

Hybrid architecture

In this architecture, you can extend your network to the cloud and users can access Service Manager through a VPN connection or AWS Direct Connect, just like Service Manager is deployed on the local network. See figure 1-3 as an example. The installation of the Service Manager components is similar to the public-facing architecture.

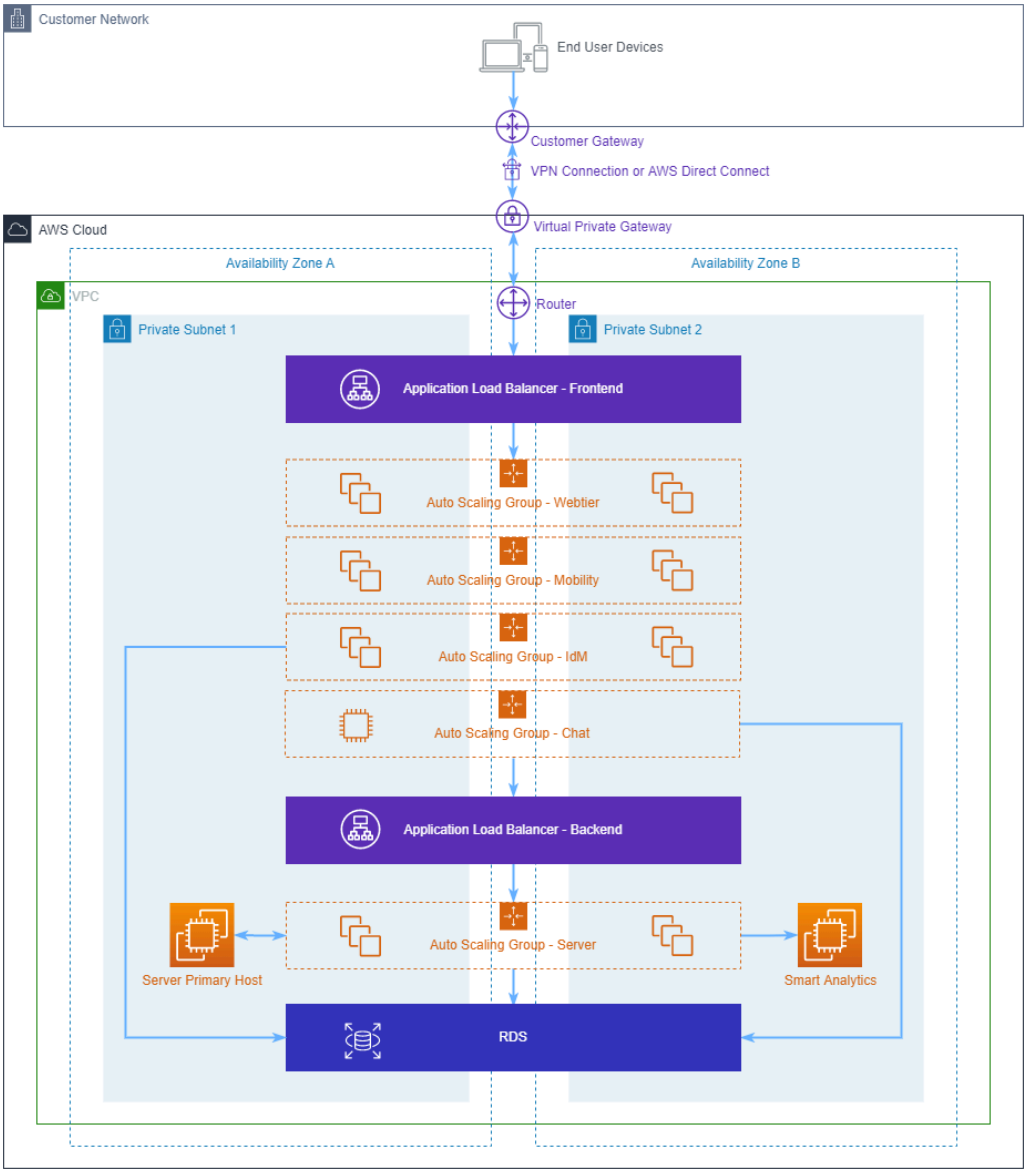


Figure 1-3 Hybrid Architecture

In this diagram:

- No public subnets and no bastion host.
- An AWS Application Load Balancer is created in the private subnets for the frontend applications (web tier, mobility, IdM, and chat).
- The AWS Auto Scaling Group (ASG) is used to scale out and scale in the web tier and mobility on demand.
- Chat is also put in an ASG, but with only one instance.
- An AWS Application Load Balancer is created in the private subnets for Service Manager server instances.
- The SM server is horizontally scaled. The SM primary host is a standalone EC2 instance, while the SM server secondary hosts are managed by an ASG.
- Smart Analytics is a standalone EC2 instance.
- The Service Manager server persists data to an RDS database.

2.2. Prepare

This section provides information required to prepare your Service Manager deployment on AWS. This section includes the following topics:

Topic	Description
(Mixed Mode Only) Install SMA Service Portal	If you choose the mixed mode (SMA-SM), you must install the SMA SP first. See (Mixed Mode Only) Install SMA Service Portal.
Create network facilities	Provides a detailed list of packages shipped with the release. See Create network facilities.
Create a database	Provides details of how to prepare the databases. See Create a database.

2.2.1. (Mixed Mode Only) Install SMA Service Portal

If you choose the mixed mode (SMA-SM), you must follow the instructions in the Install SMA Service Portal on AWS(EKS) topic and install the SMA SP first. We recommend that you download and install the latest version of SMA SP that supports your version of Service Manager. For compatibility information and installation documentation of SMA SP, see https://portal.microfocus.com/s/article/KM000001627.

After you install the SMA SP, you need to create an SM tenant and configure LW-SSO settings by completing step 4 to step 6 as described in the Set up SMA-SM topic. The SM tenant and the LW-SSO settings will be used in the Service Manager deployment later.

- Step 4: Configure a mail server
- Step 5: Create a Service Manager tenant
    - For the task Configure user authentication, you can use the database user authentication for demo purpose only. The DB user authentication requires you to manually create users with the same login name in the SMA suite and in Service Manager. For production usage, you are recommended to use your existing directory service, or create an AWS directory service according to the AWS Managed Microsoft AD AWS doc, and then configure the LDAP authentication for the account. You also need to configure the LDAP settings for the SM server later when you deploy Service Manager. You may need to configure the SAML authentication if you want to integrate with an existing ADFS. The IdM-related settings also need to be configured later when you deploy Service Manager.
    - For the task Create and deploy a Service Manager tenant, you need to skip step 7 to step 14 for now. You will finish these steps after Service Manager is deployed later.
- Step 6: Configure LW-SSO in Suite Administration

The following resources that are created during the SMA SP installation need to be shared with the SM deployment later:

- The public hosted zone
- The VPC
- Public subnets
- The frontend application load balancer
- The NAT gateway

You should skip certain steps in the following sections that create the above resources.

This guide refers to the frontend application load balancer as `sm-frontend-alb`. In the following sections, configuring the `sm-frontend-alb` refers to configure the ALB you created during the SMA SP installation.

Similarly, the VPC is referred to as `sm-vpc`, public subnets are referred to as *`sm-subnet-public-a`* and *`sm-subnet-public-b`*, the public hosted zone is referred to as *`yourdomain.net`*. You should replace these names in the following sections with real names you set during the SMA SP installation.

2.2.2. Create network facilities

Do the following to create network facilities:

- Create key pair
- Create VPC
- Create Split-view DNS
- Create a remote desktop gateway as the bastion host

Create key pair

A key pair is needed when you create EC2 instances later.

In the Amazon EC2 console, click **Navigation pane** > **Key Pairs** > **Create Key Pair** to create the key pair.

For example, create a key pair as follows:

- **Key pair name:** `sm-keypair`

Store the private key `sm-keypair.pem` in a safe place.

Create VPC



Note

Note

- If you have an existing VPC and you want to deploy the Service Manager on that VPC, you can skip the first step for creating VPC and directly create public subnets and private subnets in the existing VPC.
- If you choose the mixed mode (SMA-SM), you have already created VPC, public subnets and the NAT gateway when you deployed the SMA SP. To re-use these resources, skip the first step and then create private subnets for Service Manager.

Before you create the VPC, plan the network CIDR carefully, especially for the hybrid architecture, and make sure that the IP address range doesn't overlap with your corporate network.

You can create a VPC in the Amazon VPC console. To do this, follow these steps:

1. Create a VPC. In the VPC dashboard, click **Create VPC**. For example, create a VPC as follows:
    - **Resources to create:** VPC and more
    - **Auto-generate:** sm
    - **IPv4 CIDR block:** *10.0.0.0/16*
    - **IPv6 CIDR block:** No IPv6 CIDR Block
    - **Tenancy:** Default
    - **Number of Availability Zones (AZs):** 2
    - **Customize AZs:**
        - **First availability zone:** *us-west-2a*
        - **Second availability zone:** *us-west-2b*
    - **Number of public subnets:** 2
    - **Number of private subnets:** 2
    - **Customize subnets CIDR blocks**
        - **Public subnet CIDR block in us-west-2a:** *10.0.0.0/24*
        - **Public subnet CIDR block in us-west-2b:** *10.0.2.0/24*
        - **Private subnet CIDR block in us-west-2a:** *10.0.1.0/24*
        - **Private subnet CIDR block in us-west-2b:** *10.0.3.0/24*
    - **NAT gateways ($):** In 1 AZ
    - **VPC endpoints:** None
    - **Enable DNS hostnames:** Yes
    - **Enable DNS resolution:** Yes

2. Click **Navigation pane > Subnets** to edit the route table association, and then change the route table of public subnets to the one that forwards the traffic of the 0.0.0.0/0 destination to the internet gateway (igw-xxxx).
   The route table of public subnets should look like the following:

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	*igw-xxxx*

   The route table of private subnets should look like the following:

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	*nat-xxxx*

   Note that the target should look like `eni-xxxx` instead of `nat-xxxx` if you choose the NAT instance instead of the NAT gateway.

Create Split-view DNS

You need to bind a DNS name to the remote desktop gateway and the frontend application load balancer (ALB) that will be created later, so you need to register a public domain or use your existing public domain.

If you choose the mixed mode (SMA-SM), or if you want to install Service Manager Collaboration for the classic mode, you need to configure the LW-SSO in Service Manager. As LW-SSO requires you to bind a DNS name with the same parent domain name as the DNS name of the frontend ALB to the backend application load balancer (ALB) that forward requests to SM server nodes, you need to create a split-view DNS that could resolve the public DNS name for the frontend ALB and the private DNS name for the backend ALB.

1. Register your domain.

   Note
   Note
   - If you use an existing domain hosted by an external DNS provider other than AWS Route 53, skip this step. You should create alias records in your existing domain later for public resources (such as the remote desktop gateway and the frontend application load balancer).
   - If you use an existing VPC and you have already registered a domain, skip this step.

   To register your domain, go to the Amazon Route 53 console, and then click **Navigation pane** > **Registered domains** > **Register Domain**. For example, register a domain as follows:
   - Domain Name: *yourdomain.net*

   See the AWS doc Registering a New Domain for more details.
   After you finish the domain registration, a public hosted zone is created by AWS Route 53, you can create alias records later in this host zone for public resources.
2. Create a Private Hosted Zone.
   Now you need to create a private hosted zone with the same domain name to form a split-view DNS.
   Go to the Amazon Route 53 console, click **Navigation pane > Hosted zones > Create Hosted Zone** to create a private hosted zone. For example, create a private hosted zone as below:
   - **Domain Name:** *yourdomain.net*
   - **Type:** Private Hosted Zone for Amazon VPC
   - **VPC ID:** sm-vpc

   You can create alias records later in this hosted zone for internal resources (for example, the backend ALB).

> **Note**
> Note
> If you want to deploy Service Manager on an existing VPC and there is already an RD Gateway exists, skip this step

You can create the RD gateway in the Amazon EC2 console.

1. Create RD gateway instance
   Click **Navigation pane** > **Instances** > **Launch Instances** to create a EC2 instance. For example, create an EC2 instance as follows:
   - **Name:** `sm-rd-gateway`
   - **Number of instances:** 1
   - **Application and OS Images (Amazon Machine Image):** Microsoft Windows Server 2019 Base
   - **Instance type:** *t3.medium*
   - **Key pair name:** `sm-keypair`
   - **Network settings:**
     - **vpc:** `sm-vpc`
     - **Subnet:** `sm-subnet-public1-us-west-2a`
     - **Auto-assign public IP:** Disable
     - **Firewall(security groups):** Create security group
       - **Security group name:** `sm-rd-gateway-sg`
       - **Description:** The security group for RD Gateway
       - **Inbound security groups rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	0.0.0.0/0
HTTPS	TCP	443	0.0.0.0/0

> **Note**
> Note
> You can limit the source to your own IP for more strict access control.

     - **Advanced network configuration:**
       - **Network interface 1:** Default
   - **Configure Storage:** 50G / General Purpose SSD
   - **Advanced details:**
     - **Domain join directory:** No directory
     - **Shutdown behavior:** Stop
     - **Tenancy:** Default
2. Allocate an EIP and attach it to the RD gateway
   Click **Navigation pane** > **Elastic IPs** > **Allocate Elastic IP address** to allocate an EIP. For example, allocate an EIP as follows:
   - **Network Border Group:** *us_west_2*
   - **Public IPv4 address pool:** Amazon's pool of IPv4 addresses

   After creation, give it a meaningful name (for example, `sm-rd-gateway-eip`) by adding a `Name` tag, and then click **Actions** > **Associate address** to attach the EIP to the RD gateway instance. For example, fill in the fields as follows and then click the **Associate** button.

- **Resource type:** Instance
- **Instance:** sm-rd-gateway
- **Private IP:** *select the private IP of the* `sm-rd-gateway` *instance*

3. (Optional) Create a friendly DNS name.
   The auto-generated DNS name of the `sm-rd-gateway` ALB is very long (for example ec2-*public-ip.region*.compute.amazonaws.com). You can create a friendly DNS name for the `sm-rd-gateway` so that you can easily remember it.
   You can create a DNS name in AWS Route 53 by creating a record set in the public hosted zone.
   Go to the Amazon Route 53 console, click **Navigation pane** > **Hosted zones**, select the `yourdomain.net` of type `Public`, and then click **Create Record Set** to create a record set for the `sm-rd-gateway`. For example, create a record set as follows:
   - **Name**: `sm-rd-gateway.yourdomain.net`
   - **Type**: A - IPv4 address
   - **Alias**: No
   - **Value**: <*The elastic IP of the* `sm-rd-gateway-eip` >
   - **Routing Policy**: Simple
4. Configure the RD gateway
   You need to establish the Remote Desktop connection to the RD gateway instance, and then configure the RD gateway. See the AWS document Remote Desktop Gateway on AWS Partner Solution Deployment Guide for detailed instructions on manually setting up the RD gateway.
   1. Install the RD gateway by running the following command from a PowerShell instance started with administrative privileges:

      ```
      Install-WindowsFeature RDS-Gateway -IncludeManagementTools
      ```

   2. Implementing a self-signed certificate:
      1. Click **Windows Administrative Tools** > **Remote Desktop Gateway Manager** to launch **RD Gateway Manager**.
      2. Click the local server name in the tree pane, and then select the **Properties** link in the **Actions** pane.
      3. On the **SSL Certificate** tab, click the **Create and Import a Certificate** button to create a self-signed certificate. For example, create a self-signed certificate as follows:
         - **Certificate name:** The public DNS name of the `sm-rd-gateway`. For example, `sm-rd-gateway.yourdomain.net`
         - **Store the root certificate:** C:\Users\Administrator\Documents\sm-rd-gateway.cer

      Note
      Note
      - The self-signed certificate will expire in 6 months, and you need to do this step again when it is expired.
      - In the production environment, you are encouraged to import a certificate issued from your organization's certificate authority, instead of creating the self-signed certificate.

   3. Configure authorization policies.
      Click the **Policies** node in the tree pane, and then select **Create New Authorization Policies** in the **Actions** pane to launch the **Create New Authorization Policies** wizard. For example, create policies as follows:
      - **Select types of authorization policies to create:** Create a RD CAP and a RD RAP (recommended)
      - **Type a name for the RD CAP:** SM-RD-GATEWAY-CAP
        - **Select at least one supported Windows authentication method:** Password
        - **User group membership:** BUILTIN\Administrators

        - **Device redirection:** Enable device redirection for all client devices
    - **Type a name for the RD RAP:** SM-RD-GATEWAY-RAP
        - **User group membership:** BUILTIN\Administrators
        - **Network resource:** Allow users to connect to any network resource (computer)
        - **Allowed ports:** Allow connections to only port 3389
4. Restart the **Remote Desktop Gateway** service by running the following command in a PowerShell instance:

```
Start-Service TSGateway
```

5. Install the self-signed certificate to your local machine.
   Copy the self-signed certificate from the RD gateway (*C:-rd-gateway.cer*) to your local machine, and then double-click the certificate to install it.
    - **Store Location:** Local Machine
    - **Certificate Store:** Trusted Root Certification Authorities

   Now you can establish a remote desktop connection to EC2 Windows instances in private subnets through the `sm-rd-gateway` RD gateway. For example, you can connect to an EC2 instance with the following settings:
    - **Computer**: *<The private IP of the EC2 instance>*
    - **Advanced** > **Settings** > **Connection Settings** > **Use these RD Gateway server settings**
        - **Server name**: <The public DNS name of the `sm-rd-gateway`. For example, *sm-rd-gateway.yourdomain.net*.
        - **Bypass RD Gateway server for local addresses**: Yes

> **Note**
> Note
> When the RD client prompts for the RD Gateway Server Credentials, type the fully qualified user name (for example, *<RDGW hostname>\Administrator* or *.\Administrator*) in the **User name** field.

2.2.3. Create a database

The Service Manager Server supports PostgreSQL, Oracle, and SQL Server. See the Service Manager Support matrix for more details. In this example, the PostgreSQL engine is selected to create the RDS database.

You can create an RDS PostgreSQL database in the Amazon Database console.

Create DB Subnet Group

Click **Navigation pane** > **Subnet groups** > **Create DB Subnet Group** to create a subnet group. For example, create a subnet group as follows:

- **Name:** sm-rds-pg-subnets
- **Description:** The subnet group for the sm-rds-pg DB
- **VPC:** sm-vpc
- **Add subnets:** <add private subnets to this subnet group, you could identify private subnets by the CIDR block>

Create Parameter Group

Click **Navigation pane > Parameter groups > Create parameter group** to create a DB parameter group. For example, create a DB parameter group as follows:

- **Parameter group family:** postgres14
- **Type:** DB Parameter Group
- **Group name:** sm-rds-pg-params
- **Description:** The parameter group for the sm-rds-pg DB

After creation, click the newly created `sm-rds-pg-params` parameter group, edit the `rds.force_ssl` parameter, and then set it to `1`.

Create RDS PostgreSQL DB

Click **Navigation pane > Databases > Create database** to create a database. For example, create a database as follows:

- **Engine Options**
  - **Engine type:** PostgresQL
  - **Version:** PostgreSQL 14.4-R1
- **Templates:** Production
- **Availability and duralibility**
  - **Multi-AZ deployment:** Multi-AZ DB instance
- **Settings**
  - **DB cluster identifier:** sm-rds-pg
  - **Master username:** postgres
  - **Master password:** *<your password>*
- **DB Instance class:** Memory Optimized classes (*db.r5.xlarge*)
- **Storage**
  - **Storage type**: Provisioned IOPS (SSD)
  - **Allocated storage**: 100G
  - **Provisioned IOPS:** 1000
  - **Storage autoscaling**
    - **Enable storage autoscaling**: Yes
    - **Maximum storage threshold**: 500G
- **Connectivity**
  - **Compute resource:** Don't connect to an EC2 compute resource
  - **VPC:** sm-vpc
  - **DB Subnet group:** sm-rds-pg-subnets
  - **Publicly accessible:** No
  - **VPC security group:** Create new
  - **New VPC security group name:** sm-rds-pg-sg
  - **Database port:** 5432
- **Additional configuration**
  - **Database options**
    - **Initial database name:** sm
    - **DB parameter group:** sm-rds-pg-params
  - **Backup**
    - **Enable automatic backups**: Yes
    - **Backup retention period**: 14 days
    - **Backup window**: <Select a time window when the Service Manager is not busy>
  - **Log exports**: Postgresql log
  - **Maintenance**
    - **Enable auto minor version upgrade:** Yes
    - **Maintenance window:** *<Select a time window that the Service Manager is not busy>*
  - **Deletion protection:** Enable deletion protection



Note

Note
Make sure that you choose an appropriate instance size and storage size based on your workload.

After creation, take note of the endpoints of the `sm-rds-pg` DB, it looks like *`sm-rds-pg.xxxyyyzzz.us-west-2.rds.amazonaws.com`*. You also need to take note of the certificate authority of the `sm-rds-pg` DB, it looks like *`rds-`*

`ca-2019`, you need to download this CA certificate later when you configure the Service Manager server and Service Manager Collaboration.
Note that if you create the database for development or test purpose, choose the `Dev/Test` template to reduce the cost, for example:

- **Templates:** Dev/Test
- **DB Instance class:** Burstable classes (*db.t3.medium*)
- **Storage**
  - **Storage type**: General Purpose (SSD)
  - **Allocated storage**: 50G
- **Availability and durability**
  - **Multi-AZ deployment:** Single DB instance

Optionally, you can create a read replica of the `sm-rds-pg` DB for the Service Manager reporting module, so queries from the reporting module could be bypassed to the read replica. Click **Navigation pane** > **Databases**, select the `sm-rds-pg` DB, click **Actions** > **Create read replica** to create a read replica. For example, create a read replica as follows:

- **Settings**
  - **Read replica source**: `sm-rds-pg`
  - **DB instance identifier**: `sm-rds-pg-replica`
- **Instance configuration**
  - **DB instance class**: *db.r5.large*
- **Destination region**: *US West (Oregon)*
- **Storage**
  - **Storage type:** Provisioned IOPS (SSD)
  - **Storage autoscaling:** Enable
  - **Maximum storage threshold:** 1000GiB
- **Availability & durability**
  - Multi-AZ deployment: Do not create a standby instance
- **Connectivity**
  - **Network type:** IPv4
  - **DB subnet group:** `sm-rds-pg-subnets`
  - **Publicly accessible:** Not publicly accessible
  - **Existing VPC security groups:** `sm-rds-pg-sg`
  - **Availability zone:** No preference
  - **Database port:** *5432*
- **Monitoring:** Disable enhanced monitoring
- **Maintenance**
  - **Enable auto minor version upgrade:** Yes

To allow traffics from EC2 instances in the `sm-vpc` VPC to the DB, go to the Amazon EC2 console, click **Navigation pane > Security Groups**, select the `sm-rds-pg-sg` security group, and then click **Inbound rules > Edit inbound rules** to edit the source of the inbound rule to allow TCP traffic from the CIDR block of the `sm-vpc` to the port `5432`. The entire inbound rules look like the following:

Type	Protocol	Port Range	Source
PostgresQL	TCP	5432	*10.0.0.0/16*

2.3. Deploy

This section provides detailed instructions about how to install Service Manager on AWS. Topics in this section include:

Install Service Manager server

Install Service Manager web tier

Install Service Manager Windows Client

Install Service Manager Mobility

Install Identity Manager (IdM)

(Classic Only) Install Service Manager Collaboration

Install Service Manager Smart Analytics

(Mixed Mode Only) Setup SMA-SM Integration

References

2.3.1. Install Service Manager server

You can create resources below in the Amazon EC2 console.

Create AWS Application Load Balancer for SM Server

Create Application Load Balancer

Click **Navigation pane > Load Balancers > Create Load Balancer** to create ALB. For example, an ALB is created as follows:

- **Select load balancer type:** Application Load Balancer
- **Basic configuration**
  - **Load balancer name:** sm-backend-alb
  - **Scheme:** internal
  - **IP address type:** ipv4
  - **Listeners:** HTTP/13080
  - **VPC:** sm-vpc
  - **Availability Zones:**
    - ***us-west-2a*** sm-subnet-private-a
    - ***us-west-2b*** sm-subnet-private-b
- **Security Groups:** Create a new security group
  - **Security group name:** sm-backend-alb-sg
  - **Description:** The security group for the sm-backend-alb
  - **Inbound rules:**

Type	Protocol	Port Range	Source
Custom TCP Rule	TCP	13080	*10.0.0.0/16*

- **Listeners and routing**
  - **Listener:** HTTP:13080
    - **Default action:** Create target group
      - **Choose a target type:** Instances
      - **Target group name:** sm-server-secondary-tg
      - **Protocol:** HTTP
      - **Port:** 13080
      - **VPC:** sm-vpc
      - **Protocol version:** HTTP1
      - **Health checks**
        - **Health check protocol:** HTTP
        - **Path:** /mbeanclient-9.80/Action?getattribute=AcceptNewConnection

- **Port:** traffic port
- **Healthy threshold:** 3
- **Unhealthy threshold:** 2
- **Timeout:** 5
- **Interval:** 30
- **Success codes:** 200

After creation, click **sm-server-secondary-tg > Attributes > Edit** to turn on stickiness.

- **Stickiness:** on
- **Stickiness type:** Load balancer generated cookie
- **Stickiness duration:** 1 day

After creation, click **sm-backend-alb > Attributes > Edit**, change the **Idle timeout** to `200` seconds.

Create Private DNS Name

You can create a private DNS name in AWS Route 53 by creating a record set in the private hosted zone.

Go to the Amazon Route 53 console, click **Navigation pane > Hosted zones**, select the *yourdomain.net* of type `Private`, click **Create Record Set** to create a record set for the `sm-backend-alb`. For example, create a record set as below:

- **Name:** *sm-server.yourdomain.net*
- **Type:** A - IPv4 address
- **Alias:** Yes
- **Alias Target:** *<The auto-generated DNS name of the `sm-backend-alb` ALB>*
- **Routing Policy:** Simple

You can now use *sm-server.yourdomain.net* to refer to the backend ALB.

Create Custom AMI for SM Server

1. Create an EC2 instance.
   Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
   - **Name:** `sm-server-template`
   - **Number of instances:** 1
   - **Application and OS Images (Amazon Machine Image):** Microsoft Windows Server 2019 Base
   - **Instance type:** `r5dn.xlarge`
   - **Key pair name:** `sm-keypair`
   - **Network settings:**
     - **Network:** sm-vpc
     - **Subnet:** `sm-subnet-private1-us-west-2a`
     - **Auto-assign Public IP:** Disable
     - **Firewall (security groups):** Create security group
       - **Security group name:** `sm-server-sg`
       - **Description:** The security group for RD Gateway
       - **Inbound security groups rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32
Custom TCP Rule	TCP	13080	*10.0.0.0/16*

   - **Advanced network configuration:**

        - **Network interface 1:** Default
    - **Storage (volumes):** 50G / General Purpose SSD
    - **Configure Instance Details:**
        - **Domain join directory:** No directory
        - **IAM role:** None
        - **Shutdown behavior:** Stop
        - **Tenancy:** Default

To allow communications (for example, JGroups messages) between SM server instances, click **Navigation pane > Security Groups**, select the `sm-server-sg` security group, and then click **Inbound > Edit** to add an inbound rule that allows all traffic from the `sm-server-sg` itself. The entire inbound rules look like follows:

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32
Custom TCP Rule	TCP	13080	*10.0.0.0/16*
All traffic	All	All	The ID of the `sm-server-sg` security group

2. Install the Service Manager server.



Note

Note
When you obtain license keys, you can use the private IP (for example, *10.0.3.5*) that you planned for the primary host as the group license IP.

Establish a remote desktop connection to the `sm-server-template` instance through the `sm-rd-gateway` RD gateway. Download the Service Manager server installation packages to your computer, and then install the Service Manager server to a folder (referred to as `%SM_SERVER_HOME%` later) by following the instructions in the Install the Service Manager Server topic.

After installation, download the RDS root certificate (for example, download from https://s3.amazonaws.com/rds-downloads/rds-ca-2019-root.pem and refer to the AWS doc Using SSL/TLS to Encrypt a Connection to a DB Instance for more details) that matches the certificate authority of the `sm-rds-pg` DB, put it to the `%SM_SERVER_HOME%\RUN` folder, then edit the `%SM_SERVER_HOME%\sm.ini` as below:

```
shared_memory:192000000
log:../logs/sm.log
system:13080
httpPort:13080
httpsPort:13443
sslConnector:0
queryhashcode:sm-server.yourdomain.net:13080
plugin0:kmplugin.dll

sqldictionary:postgresql
[postgresql]
sqldb:host=sm-rds-pg.xxxyyyzzz.us-west-2.rds.amazonaws.com port=5432 dbname=sm sslmode=verify-full
sslrootcert=rds-ca-2019-root.pem sslcert=''
sqllogin:postgres/your_db_password
pgschema:public

dashboardonreplicatedb
[postgresql_replicate]
sqldb:host=sm-rds-pg-replica.xxxyyyzzz.us-west-2.rds.amazonaws.com port=5432 dbname=sm
sslmode=verify-full sslrootcert=rds-ca-2019-root.pem sslcert=''
sqllogin:postgres/your_db_password

grouplicenseip:10.0.1.5
groupname:mygroup1
jgroupstcp:1
gossiprouterhosts:10.0.1.5[12001]
external_lb

numberoflogfiles:10
threadsperprocess:80
JVMOption999:../../lib/mbeanclient-9.80.war
```

Note

Note

- Change the value of the `host` connection parameter to the actual value of your DB cluster.
- Adapt the value of the `sslrootcert` connection parameter to the RDS root certificate file name.
- Do not add the `dashboardonreplicatedb` parameter and the `postgresql_replicate` section if you don't create a DB replica.
- Set the `queryhashcode` parameter to *<the DNS of the sm-backend-alb ALB>:13080*.
- Set the `grouplicenseip` parameter to *the private IP you planned for the primary host*.
- Set the `gossiprouterhosts` parameter to *<the private IP of the sm-server-template instance>*:[12001].

If you deploy the Service Manager for the SMA-SM mixed mode, you need to configure the IdM settings. To do this, add the following properties to `%SM_SERVER_HOME%\sm.ini`:

```
idmssoenabled:1
idmsigningkey:<The key that's used to sign the IDM token>
```

Note

Note

The `idmsigningkey` must be same as the one configured in the SMA suite. You can get the signing key from Suite Administration, go to **CONFIGURATIONS > Security**, and then check the value of the `Signing key` field under **IDM configuration**.

If you want to use HTML email notification, you need to configure the SMTP settings. To do this, add the following properties to `%SM_SERVER_HOME%\sm.ini`:

```
smtphost:<The DNS name of your SMTP server>
smtpport:<The TCP port of your SMTP server>
smtpTLS:1
smtpusername:<The user name to access your SMTP server>
smtppassword:<The password of the SMTP user>
mailFrom:<The email address of the sender. For example, SM Notification <sm-
notification@yourdomain.net> >
```

Note

Note

- You can also set other JavaMail connection parameters. For more details, see Startup parameter: emailout.
- Your SMTP server must be accessible from the `sm-server-template` instance. If not, you can use the Amazon Simple Email Service to send emails.
- If the certificate of your SMTP server is issued by your private root CA or is self-signed, you need to import your private root CA or the self-signed certificate to the Java truststore `%SM_SERVER_HOME%\jre\lib\security\cacerts`. Similar policy should be applied to the certificate of your IMAP/POP3 mail server if you want to configure the Smart Email later.

Edit `%SM_SERVER_HOME%\sm.cfg` as follows:

```
# start Gossip Router
sm -gossiprouter -gossiprouterport:12001

# start background schedulers
sm system.start
```

Append the following lines to `%SM_SERVER_HOME%\sm.cfg` if you want to use the HTML email notification feature:

```
# start the emailout process
sm -emailout
```

Copy `%SM_SERVER_HOME%\sm.cfg` as `%SM_SERVER_HOME%\sm-secondary.cfg`, and edit it as follows:

```
# start the servlet
sm
# start the sync scheduler
sm -sync
```

If you want to deploy SM Chat later or if you deploy the Service Manager for the SMA-SM mixed mode, you need to configure the LWSSO settings. Edit the `%SM_SERVER_HOME%\lwssofmconf.xml`, set the `domain` and the `initString` to appropriate values. The settings look like the following:

```
<!-- ... omitted -->
<lwssoValidation>
  <domain>yourdomain.net</domain>
  <crypto ... initString="<your shared secret passphrase>" />
</lwssoValidation>
```

Note

Note

In the SMA-SM mixed mode, the `domain` and the `initString` value must be the same as the one configured in the SMA suite. You can get the `initString` value from Suite Administration, go to **CONFIGURATIONS > Security**, and then check the value of the `Encryption key` field under **Lightweight SSO configuration**.

If you want to load the OOB data, open the command prompt under the `%SM_SERVER_HOME%\RUN` folder, and then run the following command.

`sm -system_load -system_directory:../data`

Open the **Services** desktop app, change the startup type of the Service Manager Server service to `Automatic (Delayed)`. Verify that the service can be started successfully, and then check the Service Manager logs to make sure that no error occurs.

Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the `sm.exe` for the Private and Public network types.

Add a system environment variable, `SM_SERVER_HOME` = *`C:\Program Files\OpenText\Service Manager 9.80\Server`*.

Create a Powershell script `ChangeServerSetting.ps1` under the `C:\` folder, and edit it as follows:

```
param(
  [ValidateSet("Primary", "Secondary")]
  [string]$NodeType = "Primary",

  [string]$PrimaryIP,

  [hashtable]$Params
)

$SMRunDir = "$env:SM_SERVER_HOME\RUN"
$SMIniFile = "$SMRunDir\sm.ini"
if (!$PrimaryIP) {
  $PrimaryIP = (Test-Connection -ComputerName (hostname) -Count 1).IPV4Address.IPAddressToString
}
if ($NodeType -eq "Secondary") {
  Copy-Item "$SMRunDir\sm-secondary.cfg" -Destination "$SMRunDir\sm.cfg"
}

$Content = Get-Content -Encoding UTF8 -Raw $SMIniFile

$Content = $Content -replace '(.*):\d+\.\d+\.\d+\.\d+(.*)', ('$1:' + $PrimaryIP + '$2')

if ($Params) {
  foreach ($Name in $Params.keys) {
    $Value = $Params[$Name]
    $Param = $Name
    if ($Value) {
      $Param = $Param + ':' + $Value
    }

    $Pattern = '(?m)^#?' + [regex]::Escape($Name) + '\s*($|:).*'
    if ($Content -match $Pattern) {
      $Content = $Content -replace $Pattern, $Param
    } else {
      $Content = $Content + "`r`n$Param"
    }
  }
}

Set-Content -Encoding UTF8 $SMIniFile $Content
```

The script is to change the settings for the Service Manager server when you launch a new instance from the AMI. For example:

```
# Specify the instance as the primary host
C:\ChangeServerSetting.ps1 -NodeType Primary

# Specify the instance as the primary host with other settings
C:\ChangeServerSetting.ps1 -NodeType Primary -Params @{threadsperprocess = 100; shared_memory =
192000000; JVMOption0 = "512M"}

# Specify the instance as the secondary host with the specified primary IP
C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5

# Specify the instance as the secondary host with the specified primary IP and other settings
C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5 -Params @{threadsperprocess =
100; shared_memory = 192000000; JVMOption0 = "512M"}
```

3. Create AMI.
   You first need to prepare the `sm-server-template` instance for next boot before you create an AMI from it.
   1. Open the **Services** desktop app, stop the Service Manager Server service.
   2. Open the command prompt under the `%SM_SERVER_HOME%\RUN` folder, and then run the command to unlock the Service Manager database: `sm -unlockdatabase`
   3. Remove all log files under the `%SM_SERVER_HOME%\logs` folder.
   4. Configure EC2Launch by following the instructions in Configure EC2Launch for next boot.
   5. Shut down the `sm-server-template` instance.

   After the `sm-server-template` instance is stopped, you can now create the custom AMI from it. Click **Navigation pane > Instances**, select the `sm-server-template` instance, click **Actions > Image and templates > Create Image** to create custom AMI. For example, create a custom AMI as follows:
   - **Image name**: `sm-server-9.80-<timestamp>-ami`
   - **Image description:** Service Manager Server 9.80 on Windows Server 2019

Create SM Server Primary Host

Create EC2 instance. Click **navigation > AMIs**, select the `sm-server-9.80-<timestamp>-ami` AMI, and then click **Launch** to launch an EC2 instance. For example, create an EC2 instance as follows:

- **Name:** `sm-server-primary`
- **Number of instances:** 1
- **Application and OS Images (Amazon Machine Image):** Microsoft Windows Server 2019 Base
- **Instance type:** t3.large
- **Key pair name:** `sm-keypair`
- **Network settings:**
  - **vpc:** `sm-vpc`
  - **Subnet:** `sm-subnet-private1-us-west-2a`
  - **Auto-assign public IP:** Disable
  - **Firewall(security groups):** Select the existing security group `sm-server-sg`
  - **Common security groups:** `sm-server-sg`
  - **Advanced network configuration:**
    - **Network interface 1:** Default
- **Configure Storage:** 50G / General Purpose SSD
- **Advanced details:**
  - **Domain join directory:** No directory
  - **Shutdown behavior:** Stop
  - **Tenancy:** Default
  - **User data:** As text

```
<powershell>
C:\ChangeServerSetting.ps1 -NodeType Primary
</powershell>
```

Create an Auto Scaling Group for SM Server Secondary Hosts

1. Create launch template.
   Click **Navigation pane > Instances > Launch Template > Create launch template** to create a launch template. For example, create a launch template as below:
   - **Launch template name:** `sm-server-secondary-9.80-<timestamp>-lt`
   - **Application and OS Images (Amazon Machine Image)**
     - **My AMIs:** `sm-server-9.80-<timestamp>-ami`
   - **Instance Type:** *t3.medium*

- **Key pair(login)**
  - **Key pair options:** Choose an existing key pair
  - **Existing key pair:** `sm-keypair`
- **Security Groups:**
  - **Assign a security group:** Select an existing security group `sm-server-sg`
- **Storage (volumns):** 50G / General Purpose SSD
- **Advanced Details:**
  - **User data:** As text

```
<powershell>
C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5
</powershell>
```

2. Create an auto scaling group.
   Click **navigation > Auto Scaling Groups > Create an Auto Scaling group** to create an auto scaling group. For example, create an auto scaling group as below:
   - **Choose launch template or configuration:**
     - **Group name:** `sm-server-secondary-asg`
     - **Launch template:** Launch template `sm-server-secondary-9.80-<timestamp>-lt`
   - **Choose instance launch options:**
     - **Network:** `sm-vpc`
     - **Subnet:** `sm-subnet-private-a` `sm-subnet-private-b`
   - **Configure advanced options:**
     - **Load Balancing:** Attach to an existing load
     - **Attach to an existing load balancer:** Choose from your load balancer target groups `sm-backend-tg`
     - **Health Check Type:** EC2
     - **Health Check Grace Period:** 300 seconds
   - **Configure group size and scaling:** Keep this group at its initial size
   - **Add notifications:** Click **Add notification > create topic**
     - **Send a notification to:** *sm-notification*
     - **With these recipients:** *<your email address>*
     - **Whenever instances:**
       - launch
       - terminate
       - fail to launch
       - fail to terminate
     - **Add tags**: Click **Add tag**
       - **Key**: InstanceType
       - * **Value**: sm-server
       - **Tag New Instances**: true

Note

Note

- You should set appropriate **Desired Capacity**, **Min**, and **Max** values for the sm-server-secondary-asg ASG according to your load estimation. For more sizing information, see Sizing.
- The Service Manager server has a limit on connection count per instance, it's not suitable to be scaled in/out based on the **Average CPU Utilization** metric. You can add **Scheduled Actions** to scale out or scale in SM server secondary hosts at specific time based on the peak time and idle time of your Service Manager load.

2.3.2. Install Service Manager web tier

You can create the following resources in the Amazon EC2 console.

(Classic Only) Create AWS Application Load Balancer for SM web tier

Create Application Load Balancer

In the Amazon EC2 console, click **Navigation pane > Load Balancers > Create Load Balancer** to create ALB. For example, an ALB is created as follows.

- **Select load balancer type:** Application Load Balancer
- **Basic configuration**
  - **Load Balancer Name:** sm-frontend-alb
  - **Scheme:** internet-facing
  - **IP address type:** ipv4
- **Network mapping**
  - **VPC:** sm-vpc
  - **Mapping:**
    - ***us-west-2a*** `sm-subnet-public1-us-west-2a`
    - ***us-west-2b*** `sm-subnet-public2-us-west-2b`
- **Security Groups:** Create new security group
  - **Security group name:** sm-frontend-alb-sg
  - **Description:** The security group for the sm-frontend-alb
  - **Inbound rules:**

Type	Protocol	Port Range	Source
HTTP	TCP	80	*0.0.0.0/0*
HTTPS	TCP	443	0.0.0.0/0

- **Listeners and routing**
  - **Listener:** HTTP:80
    - **Default action:** Create target group
      - **Choose a target type:** Instances
      - **Target group name:** sm-webtier-tg
      - **Protocol:** HTTP
      - **Port:** 8080
      - **VPC:** sm-vpc
      - **Protocol version:** HTTP 1
      - **Health checks**
        - **Protocol:** HTTP
        - **Path:** */webtier*/api/healthz

- **Port:** traffic port
- **Healthy threshold:** 3
- **Unhealthy threshold:** 2
- **Timeout:** 5
- **Interval:** 30
- **Success codes:** 200

After creation, click **sm-webtier-tg > Attributes > Edit**, and then turn on stickiness.

- **Stickiness:** on
- **Stickiness type:** Load balancer generated cookie
- **Stickiness duration:** 1 day

Note that the context path of the web tier is assumed to be `/webtier`, you may change it according to your deployment.

After creation, click `sm-frontend-alb` **> Attributes > Edit**, and then update the following attributes.

- **Idle timeout:** 200 seconds
- **X-Forwarded-For header:** Append
- **Preserve host header:** on

Create a public DNS name

You can create a public DNS name in AWS Route 53 by creating a record set in the public hosted zone.

Go to the Amazon Route 53 console, click **Navigation pane > Hosted zones**. Select the `yourdomain.net` of type `Public`, and then click **Create Record Set** to create a record set for the `sm-frontend-alb`. For example, create a record set as below:

- **Name:** *sm.yourdomain.net*
- **Type:** A - Routes traffic to an IPv4 address and some AWS resources
- **Alias:** Yes
- **Route traffic to:** <*The auto-generated DNS name of the* `sm-frontend-alb` *ALB*>
- **Routing Policy:** Simple routing
- **Evaluate target health:** Yes

Enable HTTPS

1. Request a certificate.

   > **Note**
   > You can import a certificate issued by your organization's CA to the ACM instead of requesting a certificate in the ACM.

   Go to the AWS Certificate Manager console, and then click **Request a Certificate** to request a certificate in the ACM. For example, request a certificate as follows:

   - **Certificate type**: Request a public certificate
   - **Fully qualified domain name**: *sm.yourdomain.net*
   - **Select validation method**: DNS validation

   After you click **Confirm and Request**, expand the domain section, and then click **Create record in Route 53** to create the CNAME record that's required by the DNS validation.

2. Edit listeners.
   Go to the Amazon EC2 console, and then click **Navigation pane** > **Load Balancers**. Select the `sm-frontend-alb`, and then click the **Listeners** tab page. Select the `HTTP:80` listener, and then click **Edit** to edit the listener. For example, edit the listener as follows:
   - **Protocol:** HTTPS
   - **Port:** 443
   - **Security policy:** *ELBSecurityPolicy-TLS-1-2-2017-01*
   - **Default SSL certificate:** <From ACM *select the certificate you just requested above*>

Configure Application Load Balancer for SM Web tier

For the mixed mode (SMA-SM), you should re-use the frontend ALB you created during the SMA SP installation. `sm-frontend-alb` refers to the existing frontend ALB in the following sections. Note that the context path of the web tier is assumed to be `/webtier` in the following settings, you may change it according to your deployment.

1. Create a target group
   Click **Navigation pane > Target Groups > Create target group** to create a target group. For example, create a target group as below:
   - **Choose a target type:** Instances
   - **Target group name:** `sm-webtier-tg`
   - **Protocol:** HTTP
   - **Port:** 8080
   - **VPC:** `sm-vpc`
   - **Protocol version:** HTTP 1
   - **Health checks**
     - **Protocol:** HTTP
     - **Path:** */webtier*/api/healthz
     - **Port:** traffic port
     - **Healthy threshold:** 3
     - **Unhealthy threshold:** 2
     - **Timeout:** 5
     - **Interval:** 30
     - **Success codes:** 200

   After creation, click **sm-webtier-tg > Attributes > Edit**, and then turn on stickiness.
   - **Stickiness:** on
   - **Stickiness type:** Load balancer generated cookie
   - **Stickiness duration:** 1 day
2. Edit listener rules
   Click **Navigation pane -> Load Balancers**, select the `sm-frontend-alb` load balancer, click **Listeners** tab page, click the **View/edit rules** link, add a rule (#1) to the **HTTPS : 443** listener as below:

ORDER	IF	THEN
1	*Path is* `/webtier` ***OR*** `/webtier/*`	***Forward to*** `sm-webtier-tg`
last	Requests otherwise not routed	**Forward to** `example-sma-443-tg`

> **Note**
> You should keep the last rule unchanged, the last rule should forward other requests to the SMA 443 target group (for example, `example-sma-443-tg`).

Create custom AMI for SM web tier

You can create the custom AMI in the Amazon EC2 console.

If you want to deploy the classic SM Chat later, you need to configure the LW-SSO settings. For detailed information, see Configure LW-SSO in the Service Manager Web tier.

> **Note**
> If you choose the mixed mode (SMA-SM), you MUST NOT configure LW-SSO in the web tier as IdM will generate and renew the LW-SSO token.

Open the **Services** desktop app, and change the startup type of the **Tomcat10** service to `Automatic (Delayed)`. Verify if the service can be started successfully, and then access the `http://localhost:8080/webtier` URL in a web browser to check if the web tier works fine.

Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the `%TOMCAT_HOME%\bin\tomcat10.exe` for the Private and Public network types.

Create a PowerShell script `SetJVMRoute.ps1` in the `C:\` folder, and then edit it as follows:

```
$ConfigFile = "$env:TOMCAT_HOME\conf\server.xml"
# Get local IP
$IP = (Test-Connection -ComputerName (hostname) -Count 1).IPV4Address.IPAddressToString
# Use last two numbers as the jvmRoute
$JVMRoute = $IP -replace '\d+\.\d+\.(\d+\.\d+)', '$1'
# Add the jvmRoute attribute to the <Engine> element
(Get-Content -Encoding UTF8 $ConfigFile) -replace '(<Engine.*"localhost")>', ('$1 jvmRoute="' + $JVMRoute + '">') | Out-File -Encoding UTF8 $ConfigFile
```

The script is to add the jvmRoute attribute to the tomcat configuration file to specify which instance a user's requests are routed to, which can facilitate troubleshooting. For example, the suffix `0.100` in the JSESSIONID `BE4EB1B933698EBE53C6A62DD6BCC56F.0.100` indicates that all requests of the session are routed to the Tomcat instance running on the EC2 instance `10.0.0.100`.

1. Create an EC2 instance.
   Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
   - **Name:** `sm-webtier-template`
   - **Number of instances:** 1
   - **Application and OS Images (Amazon Machine Image):** Microsoft Windows Server 2019 Base
   - **Instance type:** *t3.large*
   - **Key pair name:** `sm-keypair`
   - **Network settings:**
     - **VPC:** `sm-vpc`
     - **Subnet:** `sm-subnet-private1-us-west-2a`
     - **Auto-assign Public IP:** Disable
     - **Firewall (security groups):** Create security group
       - **Security group name:** `sm-frontend-app-sg`
       - **Description:** The security group for SM frontend applications (such as SM web tier and mobility)
       - **Inbound security groups rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32
Custom TCP Rule	TCP	8080	*10.0.0.0/16*

     - **Advanced network configuration:**
       - **Network interface 1:** Default
     - **Configure Storage: 50G / General Purpose SSD**
     - **Advanced details:**
       - **Domain join directory:** No directory
       - **Shutdown behavior:** Stop

- **Tenancy:** Default

2. Install the Service Manager web tier
   Establish a remote desktop connection to the `sm-webtier-template` instance through the `sm-rd-gateway` to connect to EC2 instances in private subnets.
   Download the latest OpenJDK 17 and install it. Add a system environment variable, `JAVA_HOME` =<*the install dir of OpenJDK*>.
   Download the latest Tomcat 10 and install it. Add a system environment variable, `TOMCAT_HOME` =<*the install dir of Tomcat*>. Open the command prompt under the `%TOMCAT_HOME%\bin` folder, run below command to install the Tomcat service:

```
set JvmMs=1536
set JvmMx=1536
service.bat install
```



Note

If you choose a more powerful EC2 instance type (For example, *t3.large* with 8 GiB memory) for the web tier, you can set a large maximum heap size for the JVM. For example, you can use the following commands instead to install the Tomcat service:

```
set JvmMs=1536
set JvmMx=4096
service.bat install
```

   Download the Service Manager web tier installation packages to your computer, and then install the Service Manager server by following the instructions in the Install the Service Manager web tier topic. Note that you can change the context path to `/webtier` by renaming `webtier-x.xx.war` to `webtier.war` after you copy the war file to the `%TOMCAT_HOME%\webapps` folder. Extract `webtier.war` to the `webtier` sub-folder, and then remove the `webtier.war` file.
   After installation, edit `%TOMCAT_HOME%\webapps\webtier\WEB-INF\webtier.properties` by adding the following parameters:

```
cookie.secure=true
serverHost=sm-server.yourdomain.net
externalLB=true
secureLogin=false
```

   If you deploy the Service Manager for the SMA-SM mixed mode, you need to configure the IdM settings. To do this, set the properties as follows in `%TOMCAT_HOME%\webapps\webtier\WEB-INF\webtier.properties`:

```
idm.sso.enabled=true
idm.tenant=<ID of the SM tenant you created in the SMA suite>
idm.token.signingKey=<The key that's used to sign the IdM token>
idm.endpoint=<The absolute URL of the IdM service in the SMA suite. For example, https://itsma.yourdomain.net/idm-service>
idm.http.user=idmTransportUser
idm.http.password=<The password of the IdM transport user>
idm.ssl.trustAll=false
idm.truststore=<The trust store of the IdM service. For example, /WEB-INF/idm.truststore>
idm.keystore.password=<The password of the IdM trust store>
idm.returnUri=https://sm.yourdomain.net/webtier
```

Note

- The `idm.tenant` is the ID of the Service Manager tenant you created in the **(Mixed Mode Only) Install SMA Service Portal** section.
- The `idm.token.signingKey` must be the same as what configured in the SMA suite. You can get the signing key from Suite Administration, go to **CONFIGURATIONS -> Security**, and then check the value of the `Signing key` field under **IDM configuration**.
- The `idm.http.password` must be same with what configured in the SMA suite. You can get it by following the steps below.
  1. Sign in to the SMA bastion node.
  2. Identify the pod name for idm:

     ```
     kubectl get pod -n <namespace> | grep ^idm
     ```

  3. Go into the idm pod:

     ```
     kubectl exec -it <pod-name> -n <namespace> -c idm -- /bin/sh
     ```

  4. Execute the following command to get the password of the transport user:

     ```
     get_secret idm_transport_user_password_secret_key
     ```

  5. Exit the pod.

3. The Idm trust store must contain the SMA suite CA certificate. You can get the trust store by following steps below:
   - Get the SMA suite CA certificate (`<SMA cert file>`) according to the *Get the suite CA certificate* topic in the SMA-SM Help Center.
   - Import the SMA suite CA certificate to idm.truststore:

     ```
     keytool -importcert -alias suite -file <SMA cert file> -keystore idm.truststore -storepass <your password for idm.keystore>
     ```

   - Copy the idm.truststore file to the `%TOMCAT_HOME%\webapps\webtier\WEB-INF\` folder.
4. Create AMI.
   You first need to prepare the `sm-webtier-template` instance for next boot before you create an AMI from it. After the `sm-webtier-template` instance is stopped, you can now create the custom AMI from it. Click **Navigation pane > Instances**, select the `sm-webtier-template` instance, click **Actions > Image and templates > Create Image** to create custom AMI. For example, create a custom AMI as follows:
   1. Open the **Services** desktop app, stop the **Tomcat10** service.
   2. Remove all log files under the Service Manager server `%TOMCAT_HOME%\logs` folder, and remove the `sm.log` under the `%TOMCAT_HOME%` folder.
   3. Configure EC2Launch by following the instructions in Configure EC2Launch for next boot.
   4. Shut down the `sm-webtier-template` instance.
   - **Image name:** `sm-webtier-9.81-<timestamp>-ami`
   - **Image description:** Service Manager Web tier 9.81 on Windows Server 2019

Create an Auto Scaling Group for SM Web tier

1. Create launch template.
   Click **Navigation pane > Instances > Launch Template > Create launch template** to create a launch template. For example, create a launch template as below:
   - **Launch template name:** `sm-webtier-9.81-<timestamp>-lt`

- **Application and OS Images (Amazon Machine Image)**
  - **My AMIs:** `sm-webtier-9.81-<timestamp>-ami`
- **Instance Type:** *t3.medium*
- **Key pair(login)**
  - **Key pair options:** Choose an existing key pair
  - **Existing key pair:** `sm-keypair`
- **Security Groups:**
  - **Assign a security group:** Select an existing security group `sm-frontend-app-sg`
- **Storage (volumns):** 50G / General Purpose SSD
- **Advanced Details:**
  - **User data:** As text

```
<powershell>
C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5
</powershell>
```

2. Create an auto-scaling group.
   Click **navigation > Auto Scaling Groups > Create an Auto Scaling group** to create an auto-scaling group. For example, create an auto-scaling group as below:
   - **Choose launch template or configuration:**
     - **Group name:** `sm-webtier-asg`
     - **Launch template:** Launch template `sm-webtier-9.81-<timestamp>-lt`
   - **Choose instance launch options:**
     - **Network:** `sm-vpc`
     - **Subnet:** `sm-subnet-private1-us-west-2a` `sm-subnet-private2-us-west-2b`
   - **Configure advanced options:**
     - **Load Balancing:** Attach to an existing load
     - **Attach to an existing load balancer:** Choose from your load balancer target groups `sm-webtier-tg`
     - **Health Check Type:** EC2
     - **Health Check Grace Period:** 300 seconds
   - **Configure group size and scaling:** Keep this group at its initial size
   - **Add notifications:** Click **Add notification** > **create topic**
     - **Send a notification to**: *sm-notification*
     - **With these recipients:** *<your email address>*
     - **Whenever instances:**
       - launch
       - terminate
       - fail to launch
       - fail to terminate
   - **Add tags:** Click **Add tag**
     - **Key**: InstanceType
     - **Value**: sm-webtier
     - **Tag New Instances**: true

After creating the scaling group, you need to adjust the capacity of this group. Scale between 1 and 5 instances in the Details tab, and configure scaling policy in the Automatic scaling tab, then click **Create dynamic scaling policy** and configure scale policies as follows.

- **Increase Group Size:**
  - **Policy type:** Simple scaling

    - **Scaling policy name:** Scale Out
    - **CloudWatch alarm:** Create a CloudWatch alarm
        - **Metric name:** CPU Utilization
        - **AutoScalingGroupName:** `sm-webtier-asg`
        - **Statistic:** `Average`
        - **Period:** `1 minutes`
        - **Conditions:** `Whenever CPU Utilization is Greater/Equal than 70 for 5 datapoints within 5 minutes`
        - **Alarm name:** `awsec2-sm-webtier-asg-High-CPU-Utilization`
    - **Take the action:** `Add 1 capacity units`
- **Decrease Group Size:**
    - **Policy type:** Simple scaling
    - **Scaling policy name:** Scale In
    - **CloudWatch alarm:** Create a CloudWatch alarm
        - **Metric name:** CPU Utilization
        - **AutoScalingGroupName:** `sm-webtier-asg`
        - **Statistic:** `Average`
        - **Period:** `1 minutes`
        - **Conditions:** `Whenever CPU Utilization is Lower/Equal than 20 for 5 datapoints within 5 minutes`
        - **Alarm name:** `awsec2-sm-webtier-asg-Low-CPU-Utilization`
    - **Take the action:** `Remove 1 capacity units`



Note

- You need to set appropriate **Min** and **Max** values for the sm-webtier-asg ASG according to your load estimation. For more sizing information, see the Service Manager 9.7x Deployment Sizing Guide.
- The scaling policies above are for illustration purposes. You need to adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **Scheduled Actions** instead to scale out or scale in SM web tier instances at a specific time based on the peak time and idle time of your Service Manager web tier load.

After the `sm-webtier-asg` ASG launches desired EC2 instances, you can access the web tier by `https://sm.yourdomain.net/webtier`.

If you have created a read replica for the reporting module, sign in to the web tier, and then configure the replicated reporting database by following the instructions provided in task 4 of the Set up a replicated reporting database topic.

2.3.3. Install Service Manager Windows Client

This task is optional. To use some tailoring features such as Forms Designer that are only available in the Service Manager Windows Client, you can install the Windows Client on an EC2 instance.

To install Service Manager Windows Client, follow these steps:

1. Create an EC2 instance.
   You can create the EC2 instance in the Amazon EC2 console. Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
    - **Name:** `sm-windows-client`
    - **Number of instances:** 1
    - **Instance type:** *t3.medium*
    - **Key pair name:** `sm-keypair`
    - **Network settings:**
        - **vpc:** `sm-vpc`
        - **Subnet:** `sm-subnet-private1-us-west-2a`

        - **Auto-assign public IP:** Enable
        - **Firewall(security groups):** Create security group
            - **Security group name: sm-windows-client-sg**
            - **Description:** The security group for Windows Client
            - **Inbound security groups rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32

        - **Advanced network configuration:**
            - **Network interface 1:** Default
    - **Configure Storage:** 30 G / General Purpose SSD
    - **Advanced details:**
        - **Domain join directory:** No directory
        - **Shutdown behavior:** Stop
        - **Tenancy:** Default

> **Note**
> Note
> You can limit the source to your own IP for more strict access control.

2. Install the Service Manager Windows Client.
   Establish a remote desktop connection to the `sm-windows-client` instance through the `sm-rd-gateway` RD gateway. Download the Service Manager Windows Client installation package to the computer. Next, install the Service Manager Windows Client by following the instructions provided in the Install the Service Manager Windows Client topic.
   After installation, connect the Windows Client to the Service Manager server by setting the following parameters:
    - **Connection**
        - **Server host name**: *sm-server.yourdomain.net*
        - **Server port number**: 13080
    - **Advanced**
        - **Connect to External Load Balancer**: true

> **Tip**
> Tip
> You can stop the `sm-windows-client` instance when you don't use it to reduce cost.

2.3.4. Install Service Manager Mobility

You can create the following resources in the Amazon EC2 console.

Create custom AMI for SM Mobility

You can create the custom AMI in the Amazon EC2 console.

1. Create an EC2 instance.
   Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
    - **Name:** `sm-mobility-template`
    - **Number of instances:** 1
    - **Application and OS Images (Amazon Machine Image):** My AMIs / `sm-webtier-9.80-<timestamp>-ami`

- **Instance type:** t3.medium
- **Key pair name:** sm-keypair
- **Network settings:**
  - **Network:** sm-vpc
  - **Subnet:** sm-subnet-private1-us-west-2a
  - **Auto-assign Public IP:** Disable
  - **Firewall(security groups):** Select existing security group sm-fronted-app-sg
  - **Advanced network configuration:**
    - **Network interface 1:** Default
- **Storage (volumes):** 50G / General Purpose SSD
- **Advanced details:**
  - **Domain join directory:** No directory
  - **Shutdown behavior:** Stop
  - **Tenancy:** Default



Note

Note

The EC2 instance is launched from the sm-webtier-image-9.80-<timestamp> image in which the JDK and the Tomcat have already been installed and configured. You only need to install the mobility on the EC2 instance and remove the web tier from the EC2 instance.

2. Install Service Manager Mobility.
   Establish a remote desktop connection to the sm-mobility-template instance through the sm-rd-gateway RD gateway.
   Open the Services desktop app, and then stop the **Tomcat9** service.
   Remove the webtier sub-folder from the %TOMCAT_HOME%\webapps folder to uninstall the web tier application.
   Download the Service Manager Mobility installation packages to your computer, and then install the Service Manager Mobility by following the instructions provided in the Install the Mobile Applications client topic. Note that you can change the context path to /mobility by renaming webapp-x.xx.war to mobility.war after you copy the war file to the %TOMCAT_HOME%\webapps folder, and then expand the mobility.war .
   After installation, edit the %TOMCAT_HOME%\webapps\mobility\WEB-INF\web.properties , modify the parameters below.

   ```
   endpoint=http://sm-server.yourdomain.net:13080/SM/ui
   externalLB=true
   ```

   If you choose the mixed mode (SMA-SM), you need to configure IdM settings in the %TOMCAT_HOME%\webapps\mobility\WEB-INF\web.properties , silimar to what is configured in the web tier.
   Verify if the **Tomcat9** service can be started successfully, and then access the *http://localhost:8080/mobility/std/login* URL in a web browser to check if the login page can be displayed.
   Note that if you want to verify mobility functions, you need to install Chrome and access the mobility in Chrome by turning on the device simulator.
3. Create AMI.
   You first need to prepare the sm-mobility-template instance for next boot before you create an AMI from it.
   1. Open the **Services** desktop app, stop the **Tomcat9** service.
   2. Remove all log files under the %TOMCAT_HOME%\logs folder, and then remove the mobility.log , the mobility-perf.log and the sm.log under the %TOMCAT_HOME% folder.
   3. Configure EC2Launch by following the instructions in Configure EC2Launch for next boot.
   4. Shut down the sm-mobility-template instance.

After the `sm-mobility-template` instance is stopped, you can now create the custom AMI from it. Click **Navigation pane > Instances**, select the `sm-mobility-template` instance, and then click **Actions > Image and templates > Create Image** to create your custom AMI. For example, create a custom AMI as follows:

- **Image name:** `sm-mobility-9.80-<timestamp>-ami`
- **Image description:** Service Manager Mobility 9.80 on Windows Server 2019

Configure Frontend ALB for SM Mobility

1. Create a target group. Click **Navigation pane > Target Groups > Create target group** to create a target group. For example, create a target group as follows:
   - **Name**: `sm-mobility-tg`
   - **Target type**: Instances
   - **Protocol**: HTTP
   - **Port**: 8080
   - **VPC**: `sm-vpc`
   - **Protocol version:** HTTP 1
   - **Health checks**
     - **Protocol:** HTTP
     - **Path:** */mobility*
     - **Port:** traffic port
     - **Healthy threshold:** 3
     - **Unhealthy threshold:** 2
     - **Timeout**: 5
     - **Interval**: 30
     - **Success codes**: 200-399

   After creation, click **sm-mobility-tg > Attributes > Edit**, and then turn on stickiness.
   - **Stickiness:** on
   - **Stickiness type:** Load balancer generated cookie
   - **Stickiness duration:** 1 day
2. Edit listener rules.
   Click **Navigation pane > Load Balancers**, select the `sm-frontend-alb` load balancer, click **Listeners** tab page, click the View/edit rules link, and then add two rules (#1 and #2) to the **HTTPS : 443** listener as follows:

ORDER	IF	THEN
1	*Path is* `/webtier` *OR* `/webtier/*`	*Forward to* `sm-webtier-tg`
2	*Path is* `/mobility` *OR* `/mobility/*`	*Forward to* `sm-mobility-tg`
last	Requests otherwise not routed	Forward to `sm-webtier-tg`

> **Note**
> Note
> If you choose the mixed mode (SMA-SM), you only need to add the second rule as above, keep the last rule unchanged, the last rule should forward other requests to the SMA 443 target group (for example: `example-sma-443-tg`).

Create an Auto Scaling Group for SM Mobility

1. Create launch template.
   Click **Navigation pane > Instances > Launch Template > Create launch template** to create a launch template. For example, create a launch template as below:
   - **Launch template name:** `sm-mobility-9.80-<timestamp>-lt`

- **Application and OS Images (Amazon Machine Image)**
  - **My AMIs:** `sm-mobility-9.80-<timestamp>-ami`
- **Instance Type:** *t3.medium*
- **Key pair(login)**
  - **Key pair options:** Choose an existing key pair
  - **Existing key pair:** `sm-keypair`
- **Security Groups:**
  - **Assign a security group:** Select an existing security group `sm-server-sg`
- **Storage (volumns):** 50G / General Purpose SSD
- **Advanced Details:**
  - **User data:** As text

```
<powershell>
C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5
</powershell>
```

2. Create an auto scaling group.
   Click **navigation > Auto Scaling Groups > Create an Auto Scaling group** to create an auto scaling group. For example, create an auto scaling group as follows:
   - **Choose launch template or configuration:**
     - **Group name:** `sm-mobility-asg`
     - **Launch template:** Launch template `sm-mobility-9.80-<timestamp>-lt`
   - **Choose instance launch options:**
     - **Network:** `sm-vpc`
     - **Subnet:** `sm-subnet-private1-us-west-2a sm-subnet-private2-us-west-2b`
   - **Configure advanced options:**
     - **Load Balancing:** Attach to an existing load
     - **Attach to an existing load balancer:** Choose from your load balancer target groups `sm-mobility-tg`
     - **Health Check Type:** EC2
     - **Health Check Grace Period:** 300 seconds
   - **Configure group size and scaling:** Keep this group at its initial size
   - **Add notifications:** Click **Add notification > create topic**
     - **Send a notification to:** *sm-notification*
     - **With these recipients:** *<your email address>*
     - **Whenever instances:**
       - launch
       - terminate
       - fail to launch
       - fail to terminate
   - **Add tags:** Click **Add tag**
     - **Key:** InstanceType
     - *** Value:** sm-mobility
     - **Tag New Instances:** true

   After creating the scaling group, you need to adjust the capacity of this group. Scale between 1 and 5 instances in the Details tab, and configure scaling policy in the Automatic scaling tab, then click **Create dynamic scaling policy** and configure scale policies as follows:
   - **Increase Group Size**
     - **Policy type:** Simple scaling
     - **Scaling policy name:** Scale Out

        - **CloudWatch alarm:** Create a CloudWatch alarm
            - **Metric name:** CPU Utilization
            - **AutoScalingGroupName:** `sm-mobility-asg`
            - **Statistic:** `Average`
            - **Period:** `1 minutes`
            - **Conditions:** `Whenever CPU Utilization is Greater/Equal than 70 for 5 datapoints within 5 minutes`
            - **Alarm name:** `awsec2-sm-mobility-asg-High-CPU-Utilization`
        - **Take the action:** `Add 1 capacity units`
    - **Decrease Group Size:**
        - **Policy type:** Simple scaling
        - **Scaling policy name:** Scale In
        - **CloudWatch alarm:** Create a CloudWatch alarm
            - **Metric name:** CPU Utilization
            - **AutoScalingGroupName:** `sm-mobility-asg`
            - **Statistic:** `Average`
            - **Period:** `1 minutes`
            - **Conditions:** `Whenever CPU Utilization is Lower/Equal than 20 for 5 datapoints within 5 minutes`
            - **Alarm name:** `awsec2-sm-mobility-asg-Low-CPU-Utilization`
        - **Take the action:** `Remove 1 capacity units`



Note

Note

- You need to set appropriate **Min** and **Max** values for the `sm-mobility-asg` ASG according to your load estimation. For more sizing information, see Sizing.
- The scaling policies above are for illustration purposes. You need to adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **Scheduled Actions** instead to scale out or scale in SM mobility instances at a specific time based on the peak time and idle time of your Service Manager mobility load.

After the `sm-mobility-asg` ASG launches the EC2 instances, you can access the mobility by `https://sm.yourdomain.net/mobility/std/login`.

2.3.5. Install Identity Manager (IdM)

You can create the following resources in the Amazon EC2 console.

Create custom AMI for IdM

You can create the custom AMI in the Amazon EC2 console.

1. Create an EC2 instance.
   Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
    - **Name:** `idm-template`
    - **Number of instances:** 1
    - **Application and OS Images (Amazon Machine Image):** My AMIs / `sm-webtier-9.80-<timestamp>-ami`
    - **Instance type:** `t3.medium`
    - **Key pair name:** `sm-keypair`
    - **Network settings:**
        - **Network:** `sm-vpc`
        - **Subnet:** `sm-subnet-private1-us-west-2a`

        - **Auto-assign Public IP:** Disable
        - **Firewall(security groups):** Select existing security group `sm-fronted-app-sg`
        - **Advanced network configuration:**
            - **Network interface 1:** Default
    - **Storage (volumes):** 50G / General Purpose SSD
    - **Advanced details:**
        - **Domain join directory:** No directory
        - **Shutdown behavior:** Stop
        - **Tenancy:** Default



Note

Note

The EC2 instance is launched from the `sm-webtier-image-9.80-<timestamp>` image in which the JDK and the Tomcat have already been installed and configured. You only need to install IdM on the EC2 instance and remove the web tier from the EC2 instance.

2. Install IdM.

   Establish a remote desktop connection to the `idm-template` instance through the `sm-rd-gateway` RD gateway.
   Open the Services desktop app, and then stop the **Tomcat9** service.
   Remove the `webtier` sub-folder from the `%TOMCAT_HOME%\webapps` folder to uninstall the web tier application.
   Download the Identity Manager (IdM) installation packages to your computer, and then install IdM by following the instructions provided in the Install Identity Manager (IdM) topic. Note that you can change the context path to `/idm-service` and `/idm-admin` by renaming `idm-service-x.xx.war` and `idm-admin-console-x.xx.war` to `idm-service.war` and `idm-admin.war` after you copy the war file to the `%TOMCAT_HOME%\webapps` folder.
   Download the pgAdmin installer to the computer and install it. Run pgAdmin, and then connect to the `sm-db-pg` DB server with the following settings:
    - **Name:** `sm-rds-pg`
    - **Host name:** `sm-rds-pg.xxxyyyzzz.us-west-2.rds.amazonaws.com`
    - **Port:** 5432
    - **Maintenance database:** postgres
    - **Username:** *postgres*
    - **Password:** *<your DB password>*
    - **SSL mode:** Require

   After connection, create a new database named `idm` and follow the instructions provided in Install Identity Manager (IdM) to Install the IdM service. Note that the value of the idm.persistence.connection parameter in the WEB-INF/spring/applicationContext.properties file is as follows.

```
idm.persistence.connection.url = jdbc:postgresql://sm-rds-pg.xxxyyyzzz.us-west-
2.rds.amazonaws.com:5432/idm
```

   Next, follow the instructions provided in Install Identity Manager (IdM) to Install the IdM admin console. Note that the value of the idm.serverUrl parameter in the WEB-INF\classes\IdmConfig.properties file is as follows.

```
idm.serverUrl=https://sm.yourdomain.net/idm-service
```

   Edit idm-admin\WEB-INF\classes\conf\IdmConfig.xml to configure `returnUri` before `<logoutUrls>` as follows.

```
<!-- URL Where IdM will redirect to after authenticated successfully. if not specified, it will
redirect the URL accessed firstly  -->
<returnUri base="true" value="https://sm.yourdomain.net/idm-admin "></returnUri>
```

Verify if the **Tomcat9** service can be started successfully, and then access the *http://localhost:8080/idm-service/api/version-info* URL in a web browser to check if the version can be displayed.

3. Create AMI.
   You first need to prepare the idm-template instance for the next boot before creating an AMI from it.
   1. Open the **Services** desktop app, stop the **Tomcat9** service.
   2. Remove all log files under the %TOMCAT_HOME%\logs folder.
   3. Configure EC2Launch by following the instructions in Configure EC2Launch for next boot.
   4. Shut down the idm-template instance.

   After the idm-template instance is stopped, you can now create the custom AMI from it. Click **Navigation pane > Instances**, select the idm-template instance, and then click **Actions > Image and templates > Create Image** to create your custom AMI. For example, create a custom AMI as follows:
   - **Image name:** idm-9.80-<timestamp>-ami
   - **Image description:** IdM on Windows Server 2019

Configure Frontend ALB for IdM

1. Go to the Amazon Route 53 console, click **Navigation pane** > **Hosted zones**. Select *yourdomain.net* of type Private, and then click **Create Record Set** to create a record set for *sm.yourdomain.net*. For example, create a record set as follows.
   - **Record name:** *sm.yourdomain.net*
   - **Record type:** A - IPv4 address
   - **Alias:** Yes
   - **Route traffic to:** <The DNS name of the sm-frontend-alb Load Balancer >
   - **Routing Policy:** Simple
2. Create a target group. Click **Navigation pane > Target Groups > Create target group** to create a target group. For example, create a target group as follows.
   - **Name**: idm-tg
   - **Target type**: Instances
   - **Protocol**: HTTP
   - **Port**: 8080
   - **VPC**: sm-vpc
   - **Protocol version:** HTTP 1
   - **Health checks**
     - **Protocol:** HTTP
     - **Path:** */idm-service/api/version-info*
     - **Port:** traffic port
     - **Healthy threshold:** 3
     - **Unhealthy threshold:** 2
     - **Timeout**: 5
     - **Interval**: 30
     - **Success codes**: 200-399

   After creation, click **idm-tg > Attributes > Edit**, and then turn on stickiness.
   - **Stickiness:** on
   - **Stickiness type:** Load balancer generated cookie
   - **Stickiness duration:** 1 day
3. Edit listener rules.

Click **Navigation pane > Load Balancers**, select the `sm-frontend-alb` load balancer. Click the **Listeners** tab page, click the **View/edit rules** link, and then add a rules (#1) to the **HTTPS : 443** listener as follows:

ORDER	IF	THEN
1	*Path is* `/idm-service` *OR* `/idm-service/*` *OR* `/idm-admin` *OR* `/idm-admin/*`	*Forward to* `idm-tg`

Create an Auto Scaling Group for IdM

1. Create launch template.
   Click **Navigation pane > Instances > Launch Template > Create launch template** to create a launch template. For example, create a launch template as below:
   - **Launch template name:** `idm-9.80-<timestamp>-lt`
   - **Application and OS Images (Amazon Machine Image)**
     - **My AMIs:** `idm-9.80-<timestamp>-ami`
   - **Instance Type:** *t3.medium*
   - **Key pair(login)**
     - **Key pair options:** Choose an existing key pair
     - **Existing key pair:** `sm-keypair`
   - **Security Groups:**
     - **Assign a security group:** Select an existing security group `sm-server-sg`
   - **Storage (volumns):** 50G / General Purpose SSD
   - **Advanced Details:**
     - **User data:** As text

     ```
     <powershell>
     C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5
     </powershell>
     ```

2. Create an auto scaling group.
   Click **navigation > Auto Scaling Groups > Create an Auto Scaling group** to create an auto scaling group. For example, create an auto scaling group as follows:
   - **Choose launch template or configuration:**
     - **Group name:** `idm-asg`
     - **Launch template:** `idm-9.80-<timestamp>-lt`
   - **Choose instance launch options:**
     - **Network:** `sm-vpc`
     - **Subnet:** `sm-subnet-private1-us-west-2a sm-subnet-private2-us-west-2b`
   - **Configure advanced options:**
     - **Load Balancing:** Attach to an existing load balancer
     - **Attach to an existing load balancer:** Choose from your load balancer target groups `idm-tg`
     - **Health Check Type:** EC2
     - **Health Check Grace Period:** 300 seconds
   - **Configure group size and scaling:** Keep this group at its initial size
   - **Add notifications:** Click **Add notification > create topic**
     - **Send a notification to:** *sm-notification*
     - **With these recipients:** *<your email address>*
     - **Whenever instances:**
       - launch

        - terminate
        - fail to launch
        - fail to terminate
    - **Add tags:** Click **Add tag**
        - **Key:** InstanceType
        - *** Value:** idm
        - **Tag New Instances:** true

After creating the scaling group, you need to adjust the capacity of this group. Scale between 1 and 5 instances in the Details tab, and configure scaling policy in the Automatic scaling tab, then click **Create dynamic scaling policy** and configure scale policies as follows:

- **Increase Group Size**
    - **Policy type:** Simple scaling
    - **Scaling policy name:** Scale Out
    - **CloudWatch alarm:** Create a CloudWatch alarm
        - **Metric name:** CPU Utilization
        - **AutoScalingGroupName:** `idm-asg`
        - **Statistic:** `Average`
        - **Period:** `1 minutes`
        - **Conditions:** `Whenever CPU Utilization is Greater/Equal than 70 for 5 datapoints within 5 minutes`
        - **Alarm name:** `awsec2-idm-asg-High-CPU-Utilization`
    - **Take the action:** `Add 1 capacity units`
- **Decrease Group Size:**
    - **Policy type:** Simple scaling
    - **Scaling policy name:** Scale In
    - **CloudWatch alarm:** Create a CloudWatch alarm
        - **Metric name:** CPU Utilization
        - **AutoScalingGroupName:** `idm-asg`
        - **Statistic:** `Average`
        - **Period:** `1 minutes`
        - **Conditions:** `Whenever CPU Utilization is Lower/Equal than 20 for 5 datapoints within 5 minutes`
        - **Alarm name:** `awsec2-idm-asg-Low-CPU-Utilization`
    - **Take the action:** `Remove 1 capacity units`



Note

Note

- You need to set appropriate **Min** and **Max** values for the `idm-asg` ASG according to your load estimation. For more sizing information, see Sizing.
- The scaling policies above are for illustration purposes. You need to adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **Scheduled Actions** instead to scale out or scale in IdM instances at a specific time based on the peak time and idle time of your IdM load.

After the `idm-asg` ASG launches the EC2 instances, you can access IdM by *`https://sm.yourdomain.net/idm-admin`*.

2.3.6. (Classic Only) Install Service Manager Collaboration

You can create the following resources in the Amazon EC2 console.

Create custom AMI for SM Collaboration

1. Create an EC2 instance.

Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:

- **Name:** `sm-chat-template`
- **Number of instances:** 1
- **Application and OS Images (Amazon Machine Image):** Microsoft Windows Server 2019 Base
- **Instance type:** *t3.large*
- **Key pair name:** `sm-keypair`
- **Network settings:**
  - **Network:** sm-vpc
  - **Subnet:** `sm-subnet-private1-us-west-2a`
  - **Auto-assign Public IP:** Disable
  - **Firewall(security groups):** Create security group
    - **Security group name:** `sm-chat-sg`
    - **Description:** The security group for SM Collaboration
    - **Inbound security groups rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32
Custom TCP Rule	TCP	80	*10.0.0.0/16*

  - **Advanced network configuration:**
    - **Network interface 1:** Default
- **Storage (volumes):** 50G / General Purpose SSD
- **Configure Instance Details:**
  - **Domain join directory:** No directory
  - **IAM role:** None
  - **Shutdown behavior:** Stop
  - **Tenancy:** Default

2. Install Service Manager Collaboration components.
   Establish a remote desktop connection to the `sm-chat-template` instance through the `sm-rd-gateway` RD gateway.
   Download the latest Apache httpd 2.4.x and install it to a folder (referred to as `%APACHE_HTTPD_HOME%` later).
   Open the command prompt with administrative privileges under the `%APACHE_HTTPD_HOME%\bin` folder, and then run the command below to install the apache service:

   ```
   httpd.exe -k install
   ```

   Download the latest OpenJDK 11 and install it. Add a system environment variable, `JAVA_HOME` =*<the install dir of the OpenJDK>*.
   Download the Service Manager Chat Server installation package to the computer, and extract it to a folder (referred to as `%CHAT_SERVER_HOME%` later).
   Download RDS root certificate (e.g. https://s3.amazonaws.com/rds-downloads/rds-ca-2019-root.pem, see the AWS doc Using SSL/TLS to Encrypt a Connection to a DB Instance for more details) that matches the certificate authority of the `sm-rds-pg` database, put it to the `%CHAT_SERVER_HOME%\bin` folder. Then configure the Chat Server by following the instructions provided in the Task 2: Deploy and configure the chat server topic. You can open the configuration wizard by the URL http://localhost:9090/setup/index.jsp in a web browser. Note that you need to configure the following settings:
   - **Server Settings**:

        - **Domain**: *a unique string to identify the XMPP domain, for example:* `sm-chat`
    - **Database Settings**:
        - **Database Driver Presets**: PostgreSQL
        - **JDBC Driver Class**: org.postgresql.Driver
        - **Database URL**: jdbc:postgresql://*sm-rds-pg-xxxyyyzzz.us-west-2.rds.amazonaws.com/sm*?sslmode=verify-full&sslcert=&sslrootcert=rds-ca-2019-root.pem
        - **Username**: *postgres*
        - **Password**: *<your_db_password>*

    Download the Service Manager Chat Service installation packages to your computer, and then install it to a folder (referred to as `%CHAT_SERVICE_HOME%` later) by following the instructions provided in the Task 3: Deploy and configure the chat service topic. Note that you need to configure the following properties in the `%CHAT_SERVICE_HOME%\conf\app.properties`:

    - **daoFactory.serviceEndPoint**: `smserver.yourdomain.net:13080`
    - **chatServerConfig.domain**: *same value with the XMPP Domain parameter above*
    - **chatServerConfig.host**: localhost

    Download the Service Manager Chat UI installation package to the computer, extract the chatui.war from the zip file, unzip the war file, edit the `chatui\conf\env.js` file, and set the value of the `chatServiceUrl` parameter to `https://sm.yourdomain.net/chatservice`. Then, copy the `chatui` folder to the `%APACHE_HTTPD_HOME%\htdocs\` folder. Edit the `%APACHE_HTTPD_HOME%\conf\httpd.conf` file. Make sure `proxy_module` is loaded, and then append the following lines to the end of the file:

    ```
    ProxyPass /of-http-bind http://localhost:7070/http-bind timeout=120
    ProxyPass /of-plugins http://localhost:9090/plugins
    ProxyPass /chatservice http://localhost:8088
    ProxyPass /healthCheck http://localhost:8089/healthcheck
    ```

    Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the `%TOMCAT_HOME%\bin\httpd.exe` for the Private and Public network types.
    Verify that the Apache2.4 service, the Service Manager Chat Server service, and the Service Manager Chat Service service can be started successfully.
3. Create an AMI.
    You first need to prepare the `sm-chat-template` instance for next boot before you create an AMI from it.
    1. Open the **Services** desktop app, stop the **Apache2.4** service, the **Service Manager Chat Server** service, and the **Service Manager Chat Service** service.
    2. Remove all log files under the `%APACHE_HTTPD_HOME%\logs`, `%CHAT_SERVER_HOME%\logs`, and `%CHAT_SERVICE_HOME\log` folders.
    3. Configure EC2Launch by following the instructions in Configure EC2Launch for next boot.
    4. Shut down the `sm-chat-template` instance.

    After the `sm-chat-template` instance is stopped, you can now create the custom AMI from it. Click **Navigation pane > Instances**, select the `sm-chat-template` instance, and then click **Actions > Image and templates > Create Image** to create your custom AMI. For example, create a custom AMI as follows:

    - **Image name:** `sm-chat-9.80-<timestamp>-ami`
    - **Image description:** Service Manager Chat 9.80 on Windows Server 2019

Configure Frontend ALB for SM Collaboration

After creation, click **sm-chat-tg > Attributes > Edit**, and then turn on stickiness.

1. Create a target group. Click **Navigation pane > Target Groups > Create target group** to create a target group. For example, create a target group as follows:
    - **Choose a target type**: Instances
    - **Target group name**: `sm-chat-tg`
    - **Protocol**: HTTP

- **Port**: 80
- **VPC**: `sm-vpc`
- **Protocol version:** HTTP 1
- **Health checks**
  - **Protocol:** HTTP
  - **Path:** */healthCheck/*
  - **Port:** traffic port
  - **Healthy threshold:** 3
  - **Unhealthy threshold:** 2
  - **Timeout**: 5
  - **Interval**: 30
  - **Success codes**: 200-399

  After creation, click **sm-chat-tg > Attributes > Edit**, and then turn on stickiness.
  - **Stickiness:** on
  - **Stickiness type:** Load balancer generated cookie
  - **Stickiness duration:** 1 day

2. Edit listener rules.
   Click **Navigation pane > Load Balancers**, select the `sm-frontend-alb` load balancer, click **Listeners** tab page, click Listeners tab page, click the View/edit rules link, and then change the target group of the last rule of the **HTTPS : 443** listener to `sm-chat-tg` , as follows:

ORDER	IF	THEN
1	*Path is* `/webtier` *OR* `/webtier/*`	*Forward to* `sm-webtier-tg`
2	*Path is* `/mobility` *OR* `/mobility/*`	*Forward to* `sm-mobility-tg`
last	Requests otherwise not routed	Forward to `sm-chat-tg`

Create an Auto Scaling Group for SM Collaboration

1. Create launch template.
   Click **Navigation pane > Instances > Launch Template > Create launch template** to create a launch template. For example, create a launch template as below:
   - **Launch template name:** `sm-chat-9.80-<timestamp>-lt`
   - **Application and OS Images (Amazon Machine Image)**
     - **My AMIs:** `sm-chat-9.80-<timestamp>-ami`
   - **Instance Type:** *t3.medium*
   - **Key pair(login)**
     - **Key pair options:** Choose an existing key pair
     - **Existing key pair:** `sm-keypair`
   - **Security Groups:**
     - **Assign a security group:** Select an existing security group `sm-server-sg`
   - **Storage (volumns):** 50G / General Purpose SSD
   - **Advanced Details:**
     - **User data:** As text

       ```
       <powershell>
       C:\ChangeServerSetting.ps1 -NodeType Secondary -PrimaryIP 10.0.1.5
       </powershell>
       ```

2. Create an auto scaling group.
   Click **navigation > Auto Scaling Groups > Create an Auto Scaling group** to create an auto scaling group. For example, create an auto scaling group as follows:
   - **Choose launch template or configuration:**
     - **Group name:** `sm-chat-asg`
     - **Launch template:** Launch template `sm-chat-9.80-<timestamp>-lt`
   - **Choose instance launch options:**
     - **Network:** `sm-vpc`
     - **Subnet:** `sm-subnet-private1-us-west-2a sm-subnet-private2-us-west-2b`
   - **Configure advanced options:**
     - **Load Balancing:** Attach to an existing load
     - **Attach to an existing load balancer:** Choose from your load balancer target groups sm-chat-tg
     - **Health Check Type:** EC2
     - **Health Check Grace Period:** 300 seconds
   - **Configure group size and scaling:** Keep this group at its initial size
   - **Add notifications:** Click **Add notification > create topic**
     - **Send a notification to:** *sm-notification*
     - **With these recipients:** *<your email address>*
     - **Whenever instances:**
       - launch
       - terminate
       - fail to launch
       - fail to terminate
   - **Add tags:** Click **Add tag**
     - **Key:** InstanceType
     - *** Value:** sm-chat
     - **Tag New Instances:** true

   Note

   Note
   Service Manager Collaboration doesn't support being deployed as a cluster. You must set the **Desired Capacity**, **Min,** and **Max** value for the `sm-chat-asg` ASG to `1`. The only purpose of the `sm-chat-asg` ASG is to ensure that there is one and only one SM Collaboration instance is running.

   After the `sm-chat-asg` ASG launches the Service Manager Collaboration instance, complete the following tasks:
   1. Select the ESS as the portal by following the instructions provided in Task 6: Select a portal for End User Chat.
   2. Enable Service Manager Collaboration by following the instructions provided in Task 7: Enable Service Manager Collaboration. Specify the Chat Service URL as *https://sm.yourdomain.net/chatservice*.

2.3.7. Install Service Manager Smart Analytics

To install Service Manager Smart Analytics, follow these steps:

1. Create an EC2 instance.
   You can create the EC2 instance in the Amazon EC2 console. Click **Navigation pane > Instances > Launch Instances** to create an EC2 instance. For example, create an EC2 instance as follows:
   - **Choose an Amazon Machine Image (AMI):** Microsoft Windows Server 2019 Base
   - **Choose an Instance Type:** *t3.xlarge*
   - **Configure Instance Details:**
     - **Number of instances:** 1
     - **Network:** sm-vpc

        - **Subnet:** sm-subnet-private-b
        - **Auto-assign Public IP:** Disable
        - **Domain join directory:** No directory
        - **IAM role:** None
        - **Shutdown behavior:** Stop
        - **Tenancy:** Default
        - **T2/T3 Unlimited:** Enable
        - **Network interfaces:** eth0 - New network interface
    - **Add Storage:** 500G / General Purpose SSD
    - **Configure Security Group:**
        - **Assign a security group:** Create a new security group
        - **Security group name:** `sm-smarta-sg`
        - **Description:** The security group for Service Manager Smart Analytics
        - **Inbound rules:**

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the sm-rd-gateway instance>*/32
HTTP	TCP	80	*10.0.0.0/16*
All traffic	All	All	The ID of the `sm-server-sg` security group



Note

Note
The inbound rules above allow all traffic from the Service Manager Server instances.

    - **Choose an existing key:** sm-keypair

After creation, you can give a meaningful name (for example, `sm-smarta`) to the newly created instance.
To allow communications between SM Smart Analytics components, click **Navigation pane > Security Groups**, select the `sm-smarta-sg` security group, click **Inbound > Edit** to add an inbound rule that allows all traffic from the `sm-smarta-sg` itself. If you choose the mixed mode (SMA-SM), you should also add an inbound rule that allows all traffic from the security group of worker nodes of the SMA EKS cluster (for example: *`eks-cluster-sg-xxxx-eks-cluster-yyyy`*). The entire inbound rules look like the following:

Type	Protocol	Port Range	Source
RDP	TCP	3389	*<the private IP of the* `sm-rd-gateway` *instance>*/32
All traffic	All	All	The ID of the `sm-server-sg` security group
All traffic	All	All	The ID of the `sm-smarta-sg` security group
All traffic	All	All	*`eks-cluster-sg-xxxx-eks-cluster-yyyy`*

2. Create private DNS name
   You can create a private DNS name in AWS Route 53 by creating a record set in the private hosted zone.
   Go to the Amazon Route 53 console, click **Navigation pane > Hosted zones**, select the *yourdomain.net* of type Private, click **Create Record Set** to create a record set for the sm-backend-alb. For example, create a record set as below:
   - **Name:** *sm-smarta.yourdomain.net*
   - **Type:** A - IPv4 address
   - **Alias:** No
   - **Value:** *<The private IP of the sm-smarta EC2 instance>*
   - **Routing Policy:** Simple
3. Install SM Smart Analytics.
   Establish a remote desktop connection to the sm-smarta instance through the sm-rd-gateway RD gateway.
   Download the Service Manager Windows Client installation package to the computer, then install the Service Manager Windows Client by following the instructions provided in the Install the Service Manager Windows Client topic.
   Download the Service Manager Smart Analytics installer for Windows to the computer, and then install it to a folder (referred to as %SMART_ANALYTICS_HOME% later) by following the instructions provided in Install Smart Analytics.

   Note
   Note
   You need to specify * as the IP address of your Service Manager server during the installation because the IPs of SM Server secondary hosts are dynamic. The network access policy will be enforced by the sm-smarta-sg security group.

   Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the Smart Analytics programs as follows for the Private and Public network types:
   - %SMART_ANALYTICS_HOME%\IDOL\IDOLServer.exe
   - %SMART_ANALYTICS_HOME%\IDOL\agentstore\agentstore.exe
   - %SMART_ANALYTICS_HOME%\IDOL\category\category.exe
   - %SMART_ANALYTICS_HOME%\IDOL\community\community.exe
   - %SMART_ANALYTICS_HOME%\IDOL\dah\dah.exe
   - %SMART_ANALYTICS_HOME%\IDOL\dih\dih.exe
   - %SMART_ANALYTICS_HOME%\IDOL\view\view.exe
   - %SMART_ANALYTICS_HOME%\level2proxy\IDOLServer.exe
   - %SMART_ANALYTICS_HOME%\level2proxy\dah\dah.exe
   - %SMART_ANALYTICS_HOME%\level2proxy\dih\dih.exe
   - %SMART_ANALYTICS_HOME%\CFS\CFS.exe
   - %SMART_ANALYTICS_HOME%\Content1\Content1.exe (If you have Content2, Content3, and so on, add them also.)
   - %SMART_ANALYTICS_HOME%\ImageServer1\ImageServer1.exe (If you have ImageServer2, ImageServer3, and so on, add them also.)
4. Other considerations
   The installation illustrated above is to install all Smart Analytics components on a machine. To achieve better availability, consider deploying Smart Analytics on multiple servers and configuring Smart Analytics for high availability. For detailed instructions, see Example: Deploying Smart Analytics on multiple servers and Configure Smart Analytics for high availability.

2.3.8. (Mixed Mode Only) Setup SMA-SM Integration

After you install all Service Manager components, you can now begin to set up the SMA-SM integration according to the Set up SMA-SM topic, some steps should be adapted for cloud deployment as below.

- Step 1: Configure Service Manager

    - Skip the task (Optional) Configure one-way SSL between SMA and Service Manager as we don't configure SSL in the Service Manager server in this guide.
    - Skip the task Configure LW-SSO in Service Manager as you have already configured the LW-SSO settings in Service Manager.
- Step 3: Configure Smart Analytics
    - Skip this step as you have configured Smart Analytics in the **Install Service Manager Smart Analytics** topic.
- Step 4: Configure a mail server
    - Skip this step as you have finished it in the **(Mixed Mode Only) Install SMA Service Portal** topic.
- Step 5: Create a Service Manager tenant
    - You have finished most of the tasks in the **(Mixed Mode Only) Install SMA Service Portal** topic. You only need to follow step 7 to step 14 of the **Create and deploy a Service Manager tenant** task. Some settings need to be configured as below:
        - **Service Manager URL:** `http://sm-server.yourdomain.net:13080`
        - **Smart Analytics server URL:** `http://sm-smarta.yourdomain.net:9000`
        - **SM database type:** PostgreSQL
        - **SM database host/IP:** sm-rds-pg.*xxxyyyzzz.us-west-2*.rds.amazonaws.com
        - **SM database port:** 5432
        - **SM database name:** `sm`
        - **SM database account:** *postgres*
        - **SM database password:** *<your_db_password>*
        - **Enable SSL:** On
        - **SSL mode:** verify-full

      Note

      Note

      You need to make a copy of the RDS root certificate you downloaded when installing the Service Manager server, rename it to `smpgroot.crt`, then upload it to the `/mnt/efs/var/vols/itom/itsma/global-volume/certificate/smpgcert/` directory on the SMA bastion node.
- Step 6: Configure LW-SSO in Suite Administration
    - Skip this step as you have finished it in the **(Mixed Mode Only) Install SMA Service Portal** topic.

2.3.9. References

Configure EC2Launch for next boot

Follow these steps to configure EC2Launch for the initial boot of the new EC2 instance:

1. Edit the `C:\ProgramData\Amazon\EC2-Windows\Launch\Config\LaunchConfig.json` file as follows:

```
{
  "setComputerName": true,
  "setWallpaper": true,
  "addDnsSuffixList": true,
  "extendBootVolumeSize": true,
  "handleUserData": true,
  "adminPasswordType": "Specify",
  "adminPassword":  "password that adheres to your security policy"
}
```

   Or

```
{
  "setComputerName": true,
  "setWallpaper": true,
  "addDnsSuffixList": true,
  "extendBootVolumeSize": true,
  "handleUserData": true,
  "adminPasswordType": "Random",
  "adminPassword":  ""
}
```



Note

Note
If you specify the `adminPasswordType` as `Random`, you need to get the password by using your private key after you launch a new EC2 instance from the AMI.

2. In Windows PowerShell, run the following command to schedule the script to run as a Windows Scheduled Task. The script runs one time during the next boot and then disables these tasks from running again.

```
C:\ProgramData\Amazon\EC2-Windows\Launch\Scripts\InitializeInstance.ps1 -Schedule
```

See the AWS documentation Configuring a Windows Instance Using EC2Launch for more details.

Maintenance

You should install Windows updates to EC2 instances on a regular basis, and you can upgrade SM components to a newer version to get bug fixes or new features on your demand.

For standalone EC2 instances, you can install the latest Windows updates manually, or you can use AWS Systems Manager Patch Manager to automatically install Windows updates on your specific maintenance window. You can also upgrade SM components manually on your demand.

For EC2 instances managed by the ASG, you should create a new custom AMI and apply it to the ASG. For example, you can upgrade the custom AMI for the `sm-webtier-asg` ASG by following these steps:

1. Start the template EC2 instance (for example: `sm-webtier-template`).
2. Establish a Remote Desktop Connection to the EC2 instance through the `sm-rd-gateway` RD gateway.
3. Install the latest Windows updates manually, or use AWS Systems Manager Patch Manager to install Windows updates.
4. Upgrade SM components (for example: `webtier`) if you want to apply bug fixes or new features.
5. Create a new custom AMI (for example: *`sm-webtier-9.80p1-<newtimestamp>-ami`*) from the EC2 instance by following the *Create AMI* step in the **Create custom AMI for SM Web tier** section.
6. Copy a launch configuration (for example: `sm-webtier-9.80p1-<newtimestamp>-lc`) from the original launch configuration (for example: `sm-webtier-9.80-<timestamp>-lc`), change the AMI to the new custom AMI (for example: *`sm-webtier-9.80p1-<newtimestamp>-ami`*). Review the new launch configuration carefully and make sure that other settings aren't lost.
7. Update the ASG (for example: `sm-webtier-asg`) to use the new launch configuration (for example: `sm-webtier-9.80p1-<newtimestamp>-lc`).

For SM server primary host, you can update it as a standalone EC2 instance. Or you can delete it first, then create a new EC2 instance from the new custom AMI (for example: *`sm-server-9.802p1-<newtimestamp>-ami`*) by following the **Create EC2 instance** step in the **Create SM Server Primary Host** section.

3. Install on Azure

This section walks you through on deploying Service Manager (SM) on Azure. You can deploy Service Manager in the classic mode (without integrating with SMA Service Portal), or in the SMA-SM mixed mode that integrates with SMA Service Portal (SMA SP). You can install the following SM components on Azure:

Component	**Classic**	Mixed Mode (SMA-SM)
SM Server	Required	Required
SM Web tier	Required	Required
SM Windows Client	Optional	Optional
SM Mobility	Optional	Optional
IdM	Optional	N/A
SM Collaboration	Optional	N/A
SM Smart Analytics	Optional	Required

If you choose the mixed mode (SMA-SM), first install SMA SP, then install SM, setup the SMA-SM integration finally. Note that you must install the SMA SP and Service Manager in the same VNet, and use the same parent domain (for example, *yourdomain.net*).

Unless otherwise noted in the step heading, a step is obligatory for both the classic mode and the mixed mode. If a step is obligatory in one mode only, the step heading is with the (Classic Only) or (Mixed Mode Only) prefix.

In this guide, virtual machines are created from the Windows Server 2025 Datacenter image. You may tailor the steps below depending on your own demand.

Resources in this guide are created through the Azure portal. Instructions and settings given in this guide are specific to the Azure portal at the time of this writing. As the Azure portal evolves, you may need to adapt these instructions and settings according to the latest Azure portal and documentation when deploying SM on Azure.



Note

Text in the italic font means it's subject to change according to your deployment, you need to adapt it appropriately. For example:

- ***10.0.0.0/16*** is the CIDR block of your VNet. If you choose another CIDR block (for example, ***10.1.0.0/16***) for your VNet, you need to replace it with your CIDR block.
- ***(US) West US 3*** means the region in which resources are created. You can choose a region close to your location.
- ***9.81*** means the Service Manager version. You need to replace it with the actual version you installed.

3.1. Plan

Public-facing architecture

In this architecture, users access Service Manager via the internet. The VNet is composed of public subnets and private subnets, and all Service Manager components are put on private subnets for security consideration. See figure 1-1 as an example. This deployment procedure in this document is specific to the public-facing architecture.

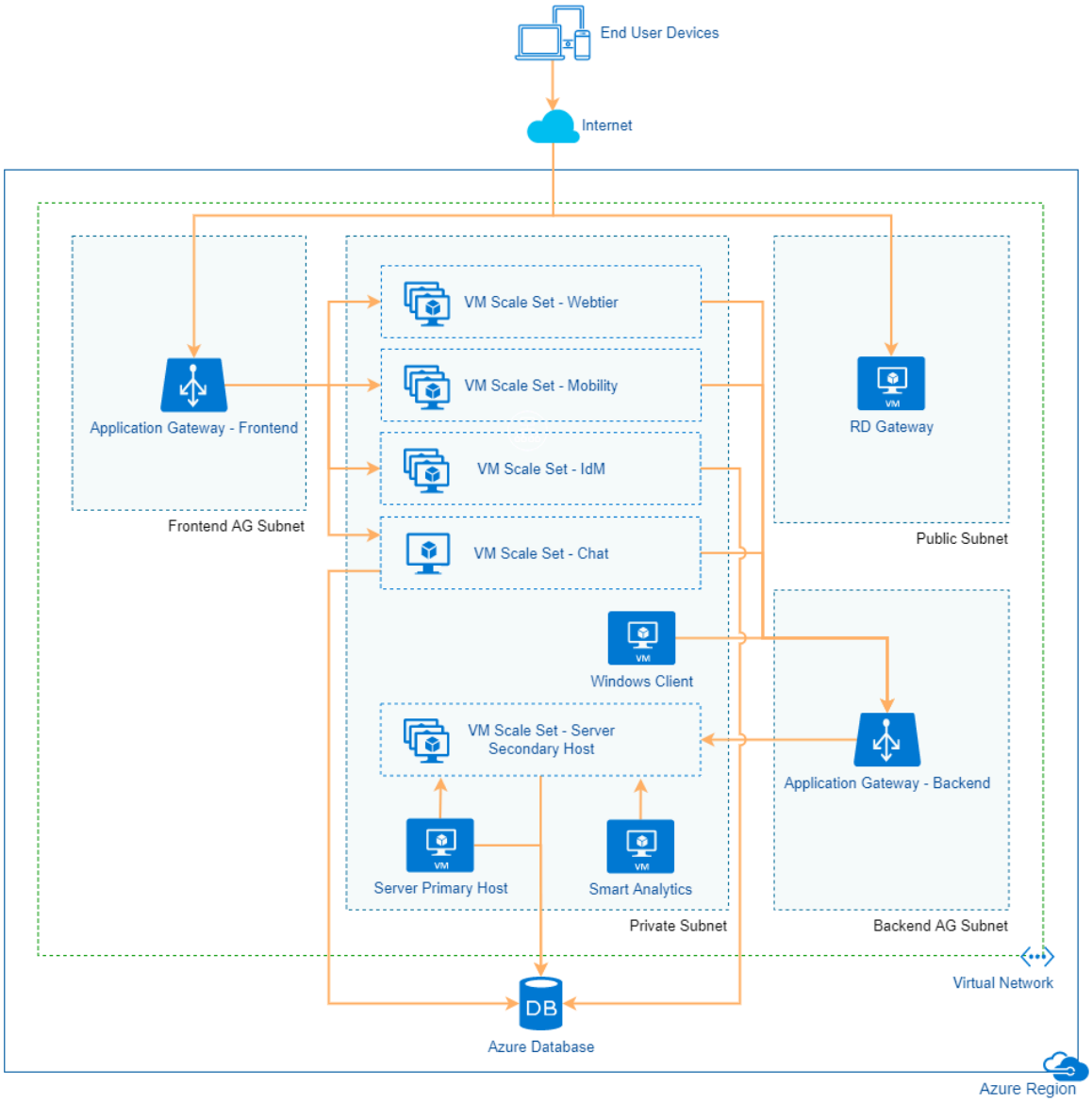


Figure 1-1 Public-facing Architecture

In this diagram:

- An Application Gateway is created in the frontend AG subnet for the frontend applications (web tier, mobility, IdM, and chat).
- An RD Gateway is created in the public subnet as a bastion host.
- The Virtual Machine Scale Set (VMSS) is used to scale out and scale in the web tier and mobility on demand.
- The chat is also put in a VMSS, but with only one instance.
- An Application Gateway is created in the backend AG subnet for SM server instances.
- The SM server is horizontally scaled by a VMSS. The SM primary host is a standalone VM instance, while the SM server secondary hosts are managed by a VMSS.
- Smart Analytics is a standalone VM instance.
- The SM server persists data to an Azure database.

If you choose the mixed mode (SMA-SM), Serivce Manager and the SMA Service Portal are deployed in the same VNet. Some resources are shared by the SM and the SMA SP to reduce cost. See figure 1-2 as an example. Public subnets and private subnets aren't illustrated in the diagram, resources that don't directly participate in the SMA-SM integration are omitted in the diagram either.

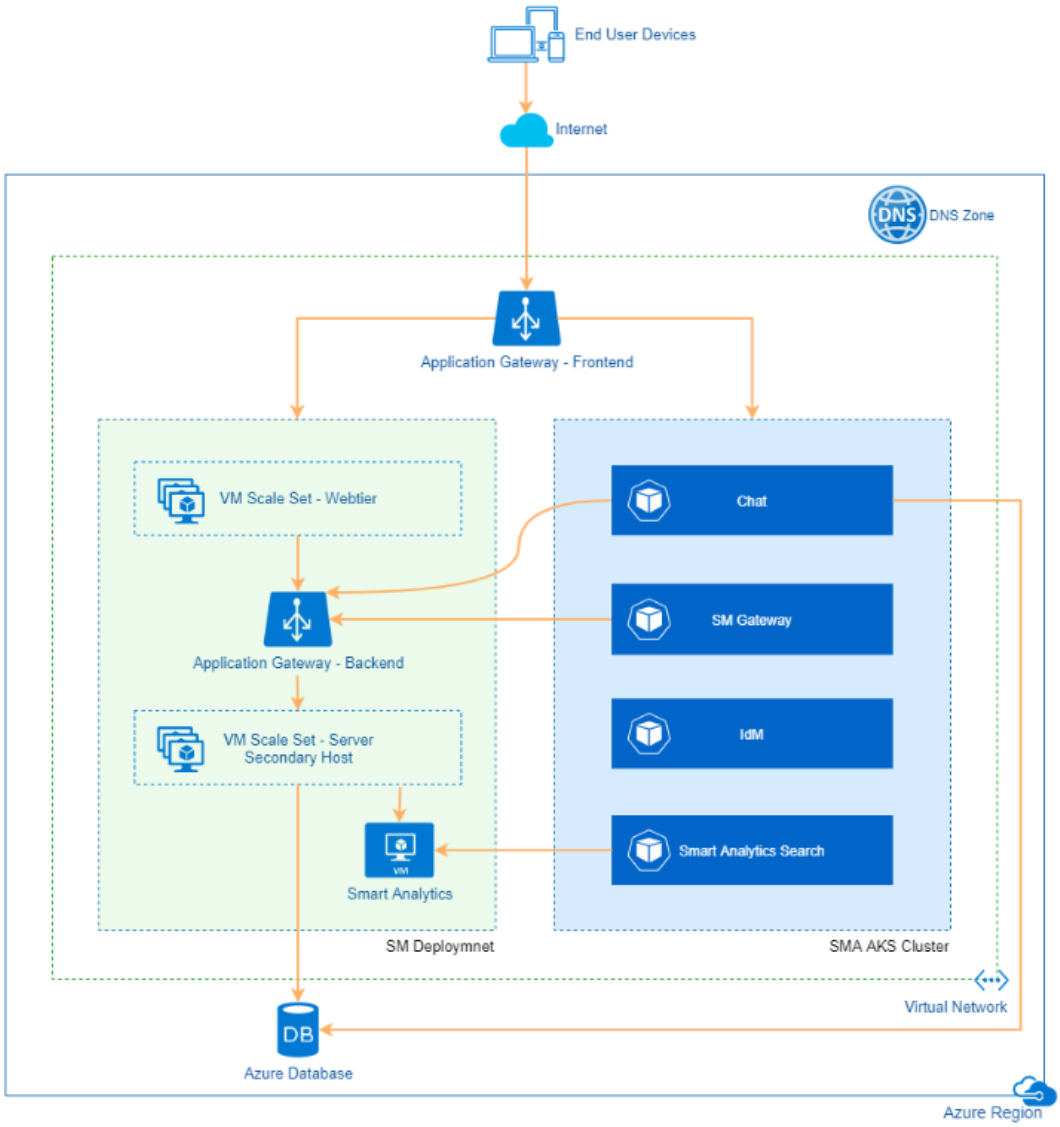


Figure 1-2 Public-facing Mixed Mode Architecture

In this diagram:

- The DNS zone, the VNet, and the frontend AG are shared by the SM and the SMA SP.
- SM and SMA SP are deployed in separate private subnets.
- The SM web tier is integrated with the IdM service in the SMA suite. Same for the SM mobility if you choose to install it.
- The SM gateway service and the Chat service (the `chat-service` container) in the SMA suite access the SM server through the backend AG.
- The Chat service (the `openfire` container) in the SMA suite accesses the SM DB directly.
- The Smart Analytics Search service in the SMA suite delegates IDOL search to the SM Smart Analytics.

Hybrid architecture

In this architecture, customers extend their network to the cloud and users access Service Manager via VPN connection or Azure ExpressRoute just like Service Manager is deployed on the local network. See figure 1-3 as an example. The installation of the Service Manager components is similar to the public-facing architecture.

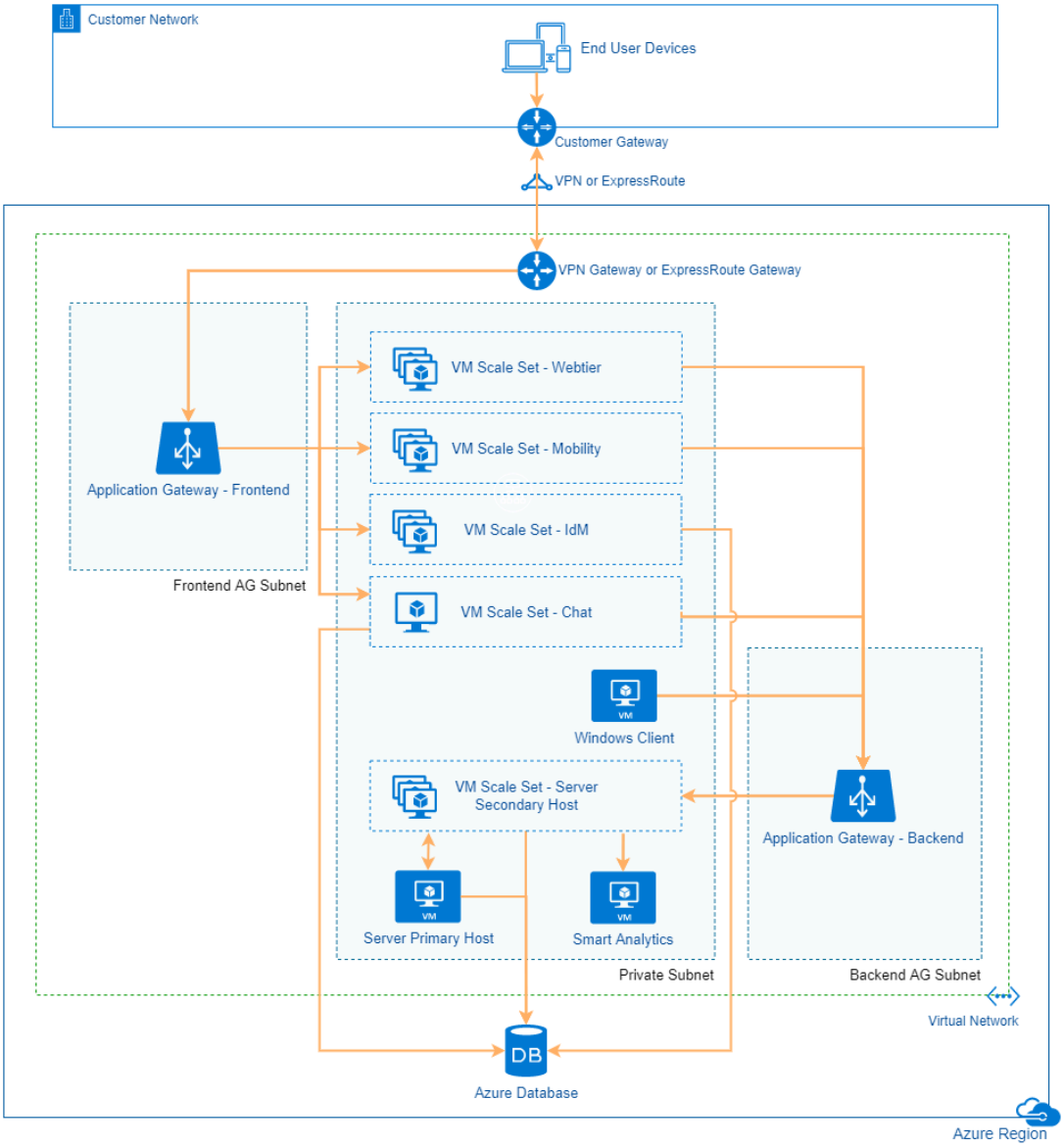


Figure 1-3 Hybrid Architecture

In this diagram:

- No public subnets and no bastion host.
- An Application Gateway is created in the frontend AG subnet for the frontend applications (web tier, mobility, IdM, and chat).
- The Virtual Machine Scale Set (VMSS) is used to scale out and scale in the web tier and mobility on demand.
- Collaboration is also put in a VMSS, but with only one instance.
- An Application Gateway is created in the backend AG subnet for SM server instances.
- The SM server is horizontally scaled by a VMSS. The SM primary host is a standalone VM instance, while the SM server secondary hosts are managed by a VMSS.
- Smart Analytics is a standalone VM instance.
- The SM server persists data to an Azure database.

3.2. Prepare

This section provides information required to prepare your Service Manager deployment on Azure. This section includes the following topics:

Topic	Description
(Mixed Mode Only) Install SMA Service Portal	If you choose the mixed mode (SMA-SM), you must install the SMA SP first. See (Mixed Mode Only) Install SMA Service Portal.
Create network facilities	Provides a detailed list of packages shipped with the release. See Create network facilities.
Create a database	Provides details of how to prepare the databases. See Create a database.

3.2.1. (Mixed Mode Only) Install SMA Service Portal

If you choose the mixed mode (SMA-SM), you must follow the instructions in the Install SMA Service Portal on Azure(AKS) and install the SMA SP first. We recommend that you download and install the latest version of SMA SP that supports your version of Service Manager. For compatibility information and installation documentation of SMA SP, see https://portal.microfocus.com/s/article/KM000001627.

After you install the SMA SP, you need to create an SM tenant and configure LW-SSO settings by completing step 4 to step 6 as described in the Set up SMA-SM topic. The SM tenant and the LW-SSO settings will be used in the SM deployment later.

- Step 4: Configure a mail server
- Step 5: Create a Service Manager tenant
    - For the task Configure user authentication, you cand use the database user authentication only for demo purpose. The DB user authentication requires you to manually create users with the same login name in the SMA suite and in the SM. For production usage, you are recommended to use your existing directory service, or use the Azure AD and the Azure AD DS, and then configure the LDAP authentication for the account. You also need to configure the LDAP settings for the SM server later when you deploy Service Manager. You may need to configure the SAML authentication if you use the Azure AD. The IdM-related settings also need to be configured later when you deploy Service Manager.
    - For the task Create and deploy a Service Manager tenant, you need to skip step 7 to step 14 for now. You will finish these steps after Service Manager is deployed later.
- Step 6: Configure LW-SSO in Suite Administration

Resources below you created during the SMA SP installation need to be shared with the SM deployment later:

- The DNS zone
- The VNet
- The frontend application gateway

You should skip certain steps in the following sections that create resources above.

This section refers the frontend application gateway as `sm-frontend-ag`. In the following sections, configuring the `sm-frontend-ag` AG refers to configure the AG you created during the SMA SP installation.

Similarly, the VNet is referred to as `sm-vnet`, the public hosted zone is referred to as *`yourdomain.net`*. You should replace these names in the following sections with real names you set during the SMA SP installation.

3.2.2. Create network facilities

Create Network Security Group

Click **Navigation pane > All services > Networking > Network security groups > Create** to create a network security group for private subnets that restricts access from the internet. For example, create a network security group as below:

- **Subscription**: *<Your subscription>*

- **Resource group**: Create new
  - **Name**: `sm-prod-rg`
- **Name:** `sm-private-subnet-nsg`
- **Location**: *(US) West US 3*

Create Application Security Group

Click **Navigation pane > All services > Networking > Application security groups > Create** to create an application security group to logically group VMs that need to access the SM Smart Analytics. For example, create an application security group as below:

- **Subscription:** *<Your subscription>*
- **Resource group:** `sm-prod-rg`
- **Name:** `sm-smarta-client-asg`
- **Region:** *(US) West US 3*

Create Virtual Network

Before you create the VNet, plan the network CIDR carefully, especially for the hybrid architecture. Make sure that the IP address range doesn't overlap with your corporate network.



Note

Note

- If you have an existing VNet and you want to deploy the Service Manager on that VNet, skip the first step to create a VNet. You can create public subnets and private subnets in the existing VNet.
- If you choose the mixed mode (SMA-SM), you have created VNet and public subnets when you deployed the SMA SP. To re-use these resources, skip the first step and then create private subnets for Service Manager.

Follow these steps:

1. Click the **Navigation pane** > **All services** > **Networking** > **Virtual networks** > **Create** to create a VNet. For example, create a VNet as follows:
   - **Subscription:** *<Your subscription>*
   - **Resource group:** `sm-prod-rg`
   - **Name:** `sm-vnet`
   - **Region:** *(US) West US 3*
   - **IPv4 address space:** *10.0.0.0/16*
   - **Subnet:** Change the name `default` to `sm-public-subnet`
   - **Address range:** *10.0.0.0/24*
   - **DDos Protection Standard:** Disable
   - **BastionHost:** Disable
   - **Firewall:** Disabled
2. Click the newly created `sm-vnet` VNet, click **Subnets** > **Subnet** to create additional subnets. For example, create additional subnets as follows:
   1. Subnet for the frontend application gateway
      - **Name**: `sm-frontend-ag-subnet`
      - **Address range**: *10.0.1.0/24*
   2. Subnet for the backend application gateway
      - **Name**: `sm-backend-ag-subnet`
      - **Address range**: *10.0.2.0/24*
   3. Private subnet for SM components
      - **Name**: `sm-private-subnet`

- **Address range**: *10.0.3.0/24*
- **Network security group**: sm-private-subnet-nsg
- **Service endpoints**: Microsoft.Sql

4. Private subnet for SM DB
   - **Name:** sm-private-db-subnet
   - **Address range:** *10.0.4.0/24*
   - **Network security group:** sm-private-subnet-nsg
   - **Service endpoints:** Microsoft.Sql
   - **Delegate subnet to a service:** Microsoft.DBforPostgresSQL/flexibleServers

Create Split-view DNS

You need to bind a DNS name to the remote desktop gateway and the frontend application gateway (AG) that will be created later, so you need to create a DNS zone or use your existing DNS zone.

If you want to install the SM chat, you need to configure LW-SSO in Service Manager. LW-SSO requires you to bind a DNS name with the same parent domain name as the DNS name of the frontend AG to the backend application gateway (AG) that forward requests to Service Manager server nodes, so you need to create a split-view DNS that could resolve the public DNS name for the frontend AG and the private DNS name for the backend AG.

1. Create DNS zone

   Note
   Note
   - If you have your own DNS server, you need to create alias records for public resources in you own DNS server, please skip this step.
   - If you choose an existing VNet and you have already created a DNS zone, please skip this step.
   - You must buy your domain (for example, *yourdomain.net*) from a domain name registrar first.

   Click **Navigation pane > All services > Networking > DNS zones > Create** to create a DNS zone. For example, create a DNS zone as below:
   - **Subscription:** *<Your subscription>*
   - **Resource group:** sm-prod-rg
   - **Name:** *yourdomain.net*

   After creation, click **Go to resource** to view the newly created *yourdomain.net* DNS zone, copy name servers, update them to the NS records in your registrar's DNS management page.
   Refer to the Azure doc Tutorial: Host your domain in Azure DNS for more details.
   You can create alias records later in this DNS zone for public resources.
2. Create a private DNS zone
   Click **Navigation pane > All services > Networking > Private DNS zones > Create** to create a private DNS zone. For example, create a private DNS zone as below:
   - **Subscription:** *<Your subscription>*
   - **Resource group:** sm-prod-rg
   - **Name:** *yourdomain.net*

   After creation, click **Go to resource** to view the newly created private DNS zone *yourdomain.net* , click **Virtual network links -> Add** to link the sm-vnet to the private DNS zone. For example, create a virtual network link as below:
   - **Link name:** sm-vnet-link
   - **Subscription:** *<Your subscription>*
   - **Virtual network:** sm-vnet
   - **Enable auto registration:** false

   You can create record sets later in this private DNS zone for internal resources (for example, the backend AG).

Create a Remote Desktop Gateway as the Bastion Host

Note

Note

- If you want to deploy the Service Manager on an existing VNet, and there is already a Remote Desktop (RD) Gateway exists, skip this task.
- Alternatively, you can leverage the Azure Bastion service to connect directly to VMs in private subnets through the Azure portal, please refer to the Azure doc What's Azure Bastion for more details. If you choose the Azure Bastion service, skip this task.

Follow these steps:

1. Create the RD gateway VM. Click **Navigation pane** > **All services** > **Compute** > **Virtual machines** > **Create** to create a VM. For example, create a VM as follows:
    - **Project details**
        - **Subscription**: *<Your subscription>*
        - **Resource group**: `sm-prod-rg`
    - **Instance details**
        - **Virtual machine name**: `sm-rd-gateway`
        - **Region**: *(US) West US 3*
        - **Availability options**: No infrastructure redundancy required
        - **Security type**: Standard
        - **Image**: Windows Server 2019 Datacenter - Gen1
        - **Run with Azure Spot discount**: no
        - **Size**: *Standard B2s (2 vcpus, 4 GiB memory)*
    - **Administrator account**
        - **Username**: *<Your username>*
        - **Password**: *<Your password that adheres to your security policy>*
    - **INBOUND PORT RULES**
        - **Public inbound ports**: Allow selected ports
        - **Select inbound ports**: `HTTPS` , `RDP`
    - **Disk options**
        - **OS disk type**: Standard SSD
    - **Network interface**
        - **Virtual network**: `sm-vnet`
        - **Subnet**: `sm-public-subnet`
        - **Public IP**: Create new
            - **Name**: `sm-rd-gateway-pip`
            - **SKU**: Basic
            - **Assignment**: Static
        - **NIC network security group**: Basic
    - **Monitoring**
        - **Boot diagnostics**: Off
        - **OS guest diagnostics**: Off
2. Create a friendly DNS name.

Note

Note

If you have your own DNS server, you can create an alias record for the `sm-rd-gateway-pip` public IP in your DNS server instead. After creation, click **Go to resource** to view the newly created `sm-rd-gateway` virtual machine. In the **Overview** page, click the public IP address. In the **Configuration** page, click **Create alias record** to create an alias record for the public IP. For example, create an alias record as follows:

- **Subscription**: *<Your subscription>*
- **DNS zone:** *yourdomain.net*
- **Name**: `sm-rd-gateway` .yourdomain.net

3. Configure the RD gateway. You need to establish the Remote Desktop connection to the RD gateway VM, and then configure the RD gateway.
    1. Install the RD gateway by running the following command from a PowerShell instance started with administrative privileges: `Install-WindowsFeature RDS-Gateway -IncludeManagementTools`
    2. Implement a self-signed certificate.
        1. Click **Windows Administrative Tools** > **Remote Desktop Gateway Manager** to launch the **RD Gateway Manager**.
        2. Click the local server name in the tree pane, and then choose **Properties** link in the **Actions** pane.
        3. On the **SSL Certificate** tab, click the **Create and Import a Certificate** button to create a self-signed certificate. For example, create a self-signed certificate as follows:
            - **Certificate name**: The public DNS name of the `sm-rd-gateway` . For example: *sm-rd-gateway.yourdomain.net*
            - **Store the root certificate**: C:\Users\Administrator\Documents\sm-rd-gateway.cer
              Notes:
            - The self-signed certificate will expire in 6 months, you need to do this step again when it is expired.
            - In the production environment, you are encouraged to import a certificate issued from your orgnization's certificate authority, instead of creating the self-signed certificate.
    3. Configure authorization policies. Click the **Policies** node in the tree pane, and then choose **Create New Authorization Policies** in the **Actions** pane to launch the **Create New Authorization Policies** wizard. For example, create policies as follows:
        - **Select types of authorization policies to create**: Create an RD CAP and an RD RAP (recommended)
        - **Type a name for the RD CAP**: SM-RD-GATEWAY-CAP
            - **Select at least one supported Windows authentication method**: Password
            - **User group membership**: BUILTIN\Administrators
            - **Device redirection**: Enable device redirection for all client devices
        - **Type a name for the RD RAP**: SM-RD-GATEWAY-RAP
            - **User group membership**: BUILTIN\Administrators
            - **Network resource**: Allow users to connect to any network resource (computer)
            - **Allowed ports**: Allow connections to only port 3389
    4. Start the **Remote Desktop Gateway** service by running following command in a PowerShell instance: `Start-Service TSGateway`
4. Install the self-signed certificate to your local machine. Copy the self-signed certificate from the `sm-rd-gateway` instance (*C:\Users\Administrator\Documents\sm-rd-gateway.cer*) to your local machine, and then double-click the certificate to install it.
    - **Store Location**: Local Machine
    - **Certificate Store**: Trusted Root Certification Authorities Now you can establish a remote desktop connection to VM instances in private subnets through the `sm-rd-gateway` RD gateway.

For example, you can connect to a VM instance by following settings:

- **Computer**: *<The private IP of the VM instance>*
- **Advanced** > **Settings** > **Connection Settings** > **Use these RD Gateway server settings**
  - **Server name:** <The public DNS name of the `sm-rd-gateway`. For example, *sm-rd-gateway.yourdomain.net* >
  - **Bypass RD Gateway server for local addresses**: Yes

> **Note**
> Note
> When the RD client prompts for the RD Gateway Server Credentials, you need to type the fully qualified user name (for example: *sm-rd-gateway\<Your username>* or `.\<Your username>`) in the **User name** field.

3.2.3. Create database

The Service Manager Server supports PostgreSQL, Oracle, and SQL Server. See the Support matrix for more details. You can create an Azure database of one of these types. In this example, the Azure Database for PostgreSQL server is used.

Create PostgreSQL DB

Click **Navigation pane** > **All services** > **Databases** > **Azure Database for PostgreSQL servers** > **Create** to create a database. For example, create a **flexible server** database as follows:

- **Project details**
  - **Subscription**: *<Your subscription>*
  - **Resource group**: `sm-prod-rg`
- **Server details**
  - **Server name**: `sm-db-pg`
  - **Admin username**: *postgres*
  - **Password**: *<Your password that adheres to your security policy>*
  - **Region**: *(US) West US 3*
  - **Workload type**: Production (Small / Medium-size)
  - **PostgresSQL Version**: 14
  - **Compute + storage**: *General purpose*
    - **vCore**: 2
    - **Storage**: 128GiB
    - **Backup retention period**: 7 days
- **Networking**
  - **Connectivity method**: Private access (Vnet Integration)
  - **Subscription**: *<Your subscription>*
  - **Virtual network**: `sm-vnet`
  - **Subnet**: `sm-private-db-subnet`
  - **Private DNS zone**: sm-db-pg.postgres.database.azure.com

> **Note**
> Note
> You need to choose appropriate compute and storage size based on your workload.

After creation, click **Go to resource** to view the DB details, and then take note of the server name of the `sm-db-pg` DB, which looks like `sm-db-pg.postgres.database.azure.com`. Note that the user referred in the connection string is `postgres`.

3.3. Deploy

This section provides detailed instructions about how to install Service Manager on Azure. Topics in this section include:

Install Service Manager server

Install Service Manager web tier

Install Service Manager Windows Client

Install Service Manager Mobility

Install Identity Manager (IdM)

(Classic Only) Install Service Manager Collaboration

Install Service Manager Smart Analytics

(Mixed Mode Only) Setup SMA-SM Integration

References

3.3.1. Install Service Manager server

Create Application Gateway for SM Server

Create Application Gateway

Click **Navigation pane** > **All services** > **Networking** > **Application gateways** > **Create** to create an application gateway. For example, create an application gateway as follows:

- **Project details**
  - **Subscription**: *<Your subscription>*
  - **Resource group**: `sm-prod-rg`
- **Instance details**
  - **Application gateway name**: `sm-backend-ag`
  - **Region**: *(US) West US 3*
  - **Tier**: Standard
  - **Instance count**: 1
  - **SKU**: *Medium*
  - **HTTP/2**: Disabled
- **Configure virtual network**
  - **Virtual network**: `sm-vnet`
  - **Subnet**: `sm-backend-ag-subnet`
- **Frontends**
  - **Frontend IP address type**: Private
  - **Use a specific (static) IP address**: Yes
  - **Private IP address**: *<An IP in the address range of the* `sm-backend-ag-subnet` *. For example:* `10.0.2.5` *>*
- **Backends**: Add a backend pool
  - **Name**: `sm-server-pool`
  - **Add backend pool without target**: Yes
- **Routing rules**: Add a rule
  - **Rule name**: `sm-server-rule`
  - **Priority**: 1
  - **Listener**
    - **Listener name**: `sm-server-listener`
    - **Frontend IP**: Private
    - **Protocol**: `HTTP`
    - **Port**: 13080
    - **Additional settings**
      - **Listen type**: Basic

        - **Error page url**: No
    - **Backend targets**
        - **Backend target**: `sm-server-pool`
        - **Backend setting**: Create new
            - **Backend setting name**: `sm-server-http`
            - **Backend protocol**: `HTTP`
            - **Backend port**: 13080
            - **Additional settings**
                - **Cookie-based affinity**: Enabled
                - **Affinity cookie name**: ROUTEID
                - **Connection draining**: Enabled
                - **Drain timeout (seconds)**: 300
                - **Request timeout (seconds)**: 300
            - **Host name**
                - **Override with new host name**: No

After creation, go to the newly created `sm-backend-ag` application gateway, click **Health probes** > **Add**, and then add a health probe as follows:

- **Name**: `sm-server-health-probe`
- **Protocol**: HTTP
- **Host**: `sm-server-health-probe`
- **Path**: /mbeanclient-*9.80*/Action?getattribute=AcceptNewConnection
- **Interval (seconds)**: 30
- **Timeout (seconds)**: 10
- **Unhealthy threshold**: 2

Click **Backend settings** > **`sm-server-http`**, and then change the settings as follows:

- **Use custom probe**: Yes
- **Custom probe**: `sm-server-health-probe`

Create Private DNS Name

You can create a private DNS name for the backend AG by creating a record set in the private DNS zone.

Click **Navigation pane > All services > Networking > Private DNS zones >** **`yourdomain.net`**, and then click **Overview > Record set** to add a record set. For example, add a record set as below:

- **Name:** `sm-server`
- **Type:** A
- **IP address:** <*The frontend private IP address of the* `sm-backend-ag.` *For example,* `10.0.2.5` >

You can now use the DNS name `sm-server.yourdomain.net` to refer to the backend AG.

Create SM server primary host

1. Create a VM.
   Click **Navigation pane** > **All services** > **Compute** > **Virtual machines** > **Create** to create a VM. For example, create a VM as follows:
    - **Project details**
        - **Subscription**: <*Your subscription*>
        - **Resource group**: `sm-prod-rg`
    - **Instance details**
        - **Virtual machine name**: `sm-server-pri`
        - **Region**: *(US) West US 3*
        - **Availability options**: No infrastructure redundancy required

        - **Security type**: Standard
        - **Image**: Windows Server 2019 Datacenter - Gen1
        - **Run with Azure Spot discount**: no
        - **Size**: *Standard B2ms (2 vcpus, 8 GiB memory)*
    - **Administrator account**
        - **Username**: *<Your username>*
        - **Password**: *<Your password that adheres to your security policy>*
    - **INBOUND PORT RULES**
        - **Public inbound ports**: None
    - **Disk options**
        - **OS disk type**: Standard SSD
    - **Network interface**
        - **Virtual network**: `sm-vnet`
        - **Subnet**: `sm-private-subnet`
        - **Public IP**: None
        - **NIC network security group**: None
    - **Monitoring**
        - **Boot diagnostics**: Off
        - **OS guest diagnostics**: Off

    After creation, click **Go to resource** to view the newly created `sm-server-pri` virtual machine. As the server primary host needs a static IP, you need to change the IP assignment to static. Click **Networking > Network Interface: sm-server-pri<*num*>**, click **IP configurations > ipconfig1**, change the **Assignment** to `Static`.
    Click **Networking > Application security groups > Configure the application security groups** to assign the application security group `sm-smarta-client-asg` to the VM.
2. Install SM server.

    Note

    When you obtain license keys for your product, you can use the private IP (for example, *10.0.3.4*) you assigned for the primary host above as the group license IP.
    Establish a remote desktop connection to the `sm-server-pri` VM instance through the `sm-rd-gateway` RD gateway.

    Download the pagAdmin installer to the computer and install it. Run pgAdmin, and then connect to the `sm-db-pg` DB server with the following settings:
    - **Name**: `sm-db-pg`
    - **Host name**: *sm-db-pg.postgres.database.azure.com*
    - **Port**: 5432
    - **Maintenance database**: postgres
    - **Username**: *postgres*
    - **Password**: *<your DB password>*
    - **SSL mode**: Require

    After connection, create a new database named as `sm`. Download the Service Manager server installation packages to the computer, and then install the Service Manager server to a folder (referred to as `%SM_SERVER_HOME%` later) by following the instructions in Install the Service Manager Server.

After installation, download the Azure database root CA certificate at https://dl.cacerts.digicert.com/DigiCertGlobalRootCA.crt.pem and rename it to `DigiCertGlobalRootCA.crt`, put it to the `%SM_SERVER_HOME%\RUN` folder, then edit the `%SM_SERVER_HOME%\sm.ini` as follows:

```
 shared_memory:192000000
 log:../logs/sm.log
 system:13080
 httpPort:13080
 httpsPort:13443
 sslConnector:0
 queryhashcode:sm-server.yourdomain.net:13080

 sqldictionary:postgresql
 [postgresql]
 sqldb:host=sm-db-pg.postgres.database.azure.com dbname=sm sslmode=verify-ca sslcert=''
sslrootcert=DigiCertGlobalRootCA.crt
 sqllogin:postgres/your_db_password
 pgschema:public

 grouplicenseip:10.0.3.4
 groupname:mygroup1
 jgroupstcp:1
 gossiprouterhosts:10.0.3.4[12001]
 external_lb

 numberoflogfiles:10
 threadsperprocess:100
 sessiontimeout:60
 JVMOption0:512M
 JVMOption999:../../lib/mbeanclient-9.80.war
 pgsslEnableCiphers: TLS_AES_256_GCM_SHA384
```

Note

- Change the value of the `host` connection parameter to the actual value of your DB cluster.
- SM doesn't support the `sslmode=verify-full` connection parameter so far so that you need to use `sslmode=verify-ca` instead.
- Set the `queryhashcode` parameter to *<the DNS of the `sm-backend-ag` AG>:13080*.
- Set the `grouplicenseip` parameter to the private IP that you planned for the primary host.
- Set the `gossiprouterhosts` parameter to *<the private IP of the `sm-server-template` instance>*:[12001].

If you deploy the Service Manager for the mixed mode (SMA-SM), you need to configure the IdM settings. Add properties below to the `%SM_SERVER_HOME%\sm.ini`:

```
idmssoenabled:1
idmsigningkey:<The key that's used to sign the IDM token>
```

Note

The `idmsigningkey` must be the same with what configured in the SMA suite. You can get the signing key from Suite Administration, go to **CONFIGURATIONS > Security**, and then check the value of the `Signing key` field under **IDM configuration**.

If you want to use the HTML email notification feature, you need to configure the SMTP settings. Add the following properties to `%SM_SERVER_HOME%\sm.ini`:

```
smtphost:<The DNS name of your SMTP server>
smtpport:<The TCP port of your SMTP server>
smtpTLS:1
smtpusername:<The user name to access your SMTP server>
smtppassword:<The password of the SMTP user>
mailFrom:<The email address of the sender. For example: SM Notification <sm-
notification@yourdomain.net> >
```



Note

- There are other JavaMail connection parameters that can be set. See Startup parameter: emailout for more details.
- Your SMTP server must be accessible from the `sm-server-template` instance. If not, you can use Microsoft Office 365 or other cloud email services (for example, SendGrid) to send emails.
- If the certificate of your SMTP server is issued by your private root CA or is self-signed, you need to import your private root CA or the self-signed certificate to the Java truststore `%SM_SERVER_HOME%\jre\lib\security\cacerts`. Similar policy should be applied to the certificate of your IMAP/POP3 mail server if you want to configure the Smart Email later.

Edit the `%SM_SERVER_HOME%\sm.cfg` file as follows:

```
# start Gossip Router
sm -gossiprouter -gossiprouterport:12001
# start background schedulers
sm system.start
```

Append the following lines to the `%SM_SERVER_HOME%\sm.cfg` follow if you want to use the HTML email notification feature:

```
# start the emailout process
sm -emailout
```

Copy the `%SM_SERVER_HOME%\sm.cfg` as `%SM_SERVER_HOME%\sm-secondary.cfg`, and edit it as follows:

```
# start the servlet
sm

# start the sync scheduler
sm -sync
```

If you want to install SM Collaboration later, or if you deploy Service Manager for the mixed mode (SMA-SM), you need to configure LW-SSO settings. Edit `%SM_SERVER_HOME%\lwssofmconf.xml`, and then set the `domain` and the `initStrin`g to appropriate values, which looks like the following:

```
<!-- ... omitted -->
<lwssoValidation>
  <domain>yourdomain.net</domain>
  <crypto ... initString="your shared secret passphrase" />
</lwssoValidation>
```

> **Note**
> In the mixed mode (SMA-SM), the `domain` and the `initString` must be the same as those configured in the SMA suite. You can get the `initString` from Suite Administration. Go to **CONFIGURATIONS > Security**, and then check the value of the `Encryption key` field under **Lightweight SSO configuration**.

If you want to load the OOB data, open the command prompt under the `%SM_SERVER_HOME%\RUN` folder, and then run the following command.

```
sm -system_load -system_directory:../data
```

Open the **Services** desktop app, and then change the startup type of the Service Manager Server service to `Automatic (Delayed)`. You need to verify the service can be started successfully, check the SM logs to make sure that no error occurs.
Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the `sm.exe` for the Private and Public network types.

Create a custom image for SM server secondary host

1. Clone a VM. In `sm-server-pri` VM page, click **Overview** > **Stop**to shutdown the `sm-server-pri` VM, and then click **Disks** > **sm-server-pri_OsDisks_1_<*uuid*>** to open the disk page, click **Overview** > **Create snapshot** to create a snapshot as follows:
   - **Name**: `sm-server-primary-host-disk-snapshot`
   - **Snapshot type**: Full
   - **Resource group**: `sm-prod-rg`
   - **Storage type**: Standard HDD
   - **Network access**: Disable public access and enable private access
   - **Disk access**: (Create new)
     - **Name**: sm-private-disk-access
     - **Private endpoint**: (Add)
       - **Subscription**: <Your subscription>
       - **Resource group**: sm-prod-rg
       - **Location**: West US 3
       - **Name**: sm-private-endpoint
       - **Target resource**: disks
       - **Virtual network**: sm-vnet
       - **Subnet**: sm-private-subnet
       - **Integrate with private DNS zone**: Yes
       - **Private DNS Zone**: (New)privatelink.blob.core.window.net

   After the snapshot is created, click **Navigation pane** > **All services** > **Compute** > **Disks** > **Create** to create a managed disk. For example, create a managed disk as follows:
   - **Project details**
     - **Subscription**: *<Your subscription>*
     - **Resource group**: `sm-prod-rg`
   - **Disk details**
     - **Disk name**: `sm-server-secondary-host-disk`
     - **Region**: *(US) West US 3*
     - **Availability zone**: None
     - **Source type**: Snapshot
     - **Source snapshot**: `sm-server-primary-host-disk-snapshot`
     - **Size**: *128 GiB, Standard SSD*

After creation, click **Go to resource** to view the newly created disk. Click **Overview** > **Create VM** to create a VM from the `sm-server-secondary-host-disk` disk. For example, create a VM as follows:

- **Project details**
  - **Subscription**: *<Your subscription>*
  - **Resource group**: `sm-prod-rg`
- **Instance details**
  - **Virtual machine name**: `sm-server-sec`
  - **Region**: *(US) West US 3*
  - **Availability options**: No infrastructure redundancy required
  - **Image**: `sm-server-secondary-host-disk`
  - **Size**: *Standard_E4ds_v4(4 vcpus, 32GiB memory)*
- **INBOUND PORT RULES**
  - **Public inbound ports**: None
- **Disk options**
  - **OS disk type: Standard SSD**
- **Network interface**
  - **Virtual network**: `sm-vnet`
  - **Subnet**: `sm-private-subnet`
  - **Public IP**: None
  - **NIC network security group**: None
- **Monitoring**
  - **Boot diagnostics**: Off
  - **OS guest diagnostics**: Off

Finally, click **Navigation pane** > **All services** > **Compute** > **Snapshots** > **`sm-server-primary-host-disk-snapshot`**, and then click **Overview** > **Delete** to delete the snapshot.

2. Create a custom image. Establish a remote desktop connection to the `sm-server-sec` instance through the `sm-rd-gateway` RD gateway. Edit the `%SM_SERVER_HOME%\sm.cfg` as follows.

```
# start the servlet
sm

# start the sync scheduler
sm -sync
```

Prepare the `sm-server-sec` VM before you capture image from it.

- Open the **Services** desktop app, stop the Service Manager Server service.
- Remove all log files under the `%SM_SERVER_HOME%\logs` folder.
- Generalize the VM by following the instructions in Generalize the Windows VM.

After the VM is generalized, go to `sm-server-sec` VM in the Azure portal, and then click **Overview** > **Capture** to create an image as follows:

- **Name**: `sm-server-secondary-host-image-9.80-<timestamp>`
- **Resource group**: `sm-prod-rg`
- **Automatically delete this virtual machine after creating the image**: Yes
- **Share image to Azure compute gallery**: No
- **Zone resiliency**: On

After the image is created, click **Navigation pane** > **All services** > **Other** > **Disks** > **`sm-server-secondary-host-disk`**, and then click **Overview** > **Delete** to delete the unattached managed disk.

Create a virtual machine scale set for SM server secondary hosts

Create a virtual machine scale set. Click **Navigation pane** > **All services** > **Compute** > **Virtual machine scale sets** > **Create** to create a virtual machine scale set. For example, create a virtual machine scale set as follows.

- **Basics**
  - **Project details**
    - **Subscription:** < *Your subscription*>
    - **Resource group:** `sm-prod-rg`
  - **Scale set details**
    - **Virtual machine scale set name:** `SmServerSecVmss`
    - **Region:** *(US) West US 3*
    - **Availability zone:** Zone 1, 2, 3
  - **Orchestration**
    - **Orchestration mode:** `Uniform`
  - **Instance details**
    - **Image:** See all images -> My Images -> `sm-server-secondary-host-image-9.80-<timestamp>`
    - **Azure Spot instance:** No
    - **Size:** *Standard_E4ds_v4(4 vcpus, 32GiB memory)*
  - **Administrator account**
    - **Username:** <*Your username*>
    - **Password:** <*Your password that adheres to your security policy*>
- **Disks**
  - **Disk options**
    - **OS disk type:** Standard SSD
    - **Encryption type:** Default
    - **Enable Ultra Disk compatibility:** No
  - **Advanced**
    - **Use managed disks:** Yes
- **Networking**
  - **Virtual network configuration**
    - **Virtual network:** `sm-vnet`
  - **Network interface** edit the network interface `sm-vnet-nic01` as below
    - **Name:** `sm-vnet-nic01`
    - **Subnet:** `sm-private-subnet`
    - **NIC network security group:** None
    - **Public IP address:** Disabled
  - **Load balancing**
    - **Use a load balancer:** Yes
  - **Load balancing settings**
    - **Load balancing options:** Application Gateway
    - **Select an application gateway:** `sm-backend-ag`
    - **Select a backend pool:** `sm-server-pool`
- **Scaling**
  - **Initial instance count:** 1
  - **Scaling policy:** Manual
  - **Scale-in policy:** Default
- **Management**
  - **Upgrade policy**

        - **Upgrade mode:** Manual
    - **Monitoring**
        - **Boot diagnostics:** Off
    - **Identity**
        - **System assigned managed identity:** Off
    - **Instance termination**
        - **Instance termination notification:** Off
- **Health**
    - **Monitor application health:** Disabled



Note

- You need to set appropriate **Instance count** for the `SmServerSecVmss` VMSS according to your load estimation. For more sizing information, see Sizing.
- The SM server has limit on connection count per instance, it is unsuitable to be scaled in/out based on the **Average Percentage CPU** metric. You can configure the **Custom autoscale** after creation to scale out or scale in SM server secondary hosts at specific time based on the peak time and idle time of your Service Manager load.

After creation, disable Windows Automatic Updates for the `SmServerSecVmss` VMSS by following the instructions in Disable Windows automatic updates for a VMSS.

VMs created by the VMSS `SmServerSecVmss` need to access the SM smart analytics, you need to assign the application security group `sm-smarta-client-asg` to the VMSS `SmServerSecVmss` by following instructions provided in Assign an Application Security Group to a VMSS.

3.3.2. Install Service Manager web tier

(Classic Only) Create an application gateway for SM Web tier

Create an application gateway



Note

You need to prepare a certificate for the frontend application gateway to configure the HTTPS listener. The certificate can be issued by your organization's certificate authority, or is bought from a trusted 3rd-party certificate authority. The certificate must be stored as the PFX format, and its **Comman Name (CN)** must match the DNS name that you planned for the frontend application gateway.

Click **Navigation pane** > **All services** > **Networking** > **Application gateways** > **Create** to create an application gateway. For example, an application gateway is created as follows:

- **Project details**
    - **Subscription**: *<Your subscription>*
    - **Resource group**: `sm-prod-rg`
- **Instance details**
    - **Application gateway name**: `sm-frontend-ag`
    - **Region**: *(US) West US 3*
    - **Tier**: Standard V2
    - **Enable autoscaling**: Yes
    - **Minimum instances**: 2
    - **Maximum instances**: 10
    - **Availability zone**: Zone 1, 2, 3
    - **HTTP/2**: Disabled

- **Configure virtual network**
  - **Virtual network**: `sm-vnet`
  - **Subnet**: `sm-frontend-ag-subnet`
- **Frontends**
  - **Frontend IP address type**: Public
  - **Public IP address**: Create new
    - **Name**: `sm-frontend-ag-pip`
- **Backends**: Add a backend pool
  - **Name**: `sm-webtier-pool`
  - **Add backend pool without target**: Yes
- **Routing rules**: Add a rule
  - **Rule name**: `sm-frontend-rule`
  - **Listener**
    - **Listener name**: `sm-frontend-public-listener`
    - **Frontend IP**: Public
    - **Protocol**: `HTTPS`
    - **Port**: 443
    - **HTTPS Certificate**
      - **PFX certificate file**: *<Select the certificate file from your local machine>*
      - **Certificate name**: *<The name of your certificate>*
      - **Password**: *<The password of your certificate>*
    - **Additional settings**
      - **Listen type**: Basic
      - **Error page url**: No
  - **Backend targets**
    - **Backend target**: `sm-webtier-pool`
    - **Backend setting**: Create new
      - **Backend setting name**: `sm-webtier-http`
      - **Backend protocol**: `HTTP`
      - **Backend port**: 8080
      - **Additional settings**
        - **Cookie-based affinity**: Enabled
        - **Affinity cookie name**: WEBTIER_ROUTE
        - **Connection draining**: Enabled
        - **Drain timeout (seconds)**: 300
        - **Request timeout (seconds)**: 300
      - **Host name**
        - **Override with new host name**: No
    - **Path-based routing**: Click **Add multiple targets to create a path-based rule** to add a path rule
      - **Target type:** Backend pool
      - **Path:** `/webtier,/webtier/*`
      - **Target name:** `sm-webtier`
      - **Backend setting**: `sm-webtier-http`
      - **Backend target**: `sm-webtier-pool`

After creation, go to the newly created `sm-frontend-ag` application gateway, click **Health probes** > **Add**, and then add a health probe as follows:

- **Name**: `sm-webtier-health-probe`

- **Protocol**: HTTP
- **Host**: sm-webtier-health-probe
- **Path**: */webtier/api/healthz*
- **Interval (seconds)**: 30
- **Timeout (seconds)**: 10
- **Unhealthy threshold**: 2

Click **Backend settings** > **sm-webtier-http**, and then change its settings as follows:

- **Use custom probe**: Yes
- **Custom probe**: sm-webtier-health-probe

Note that the context path of the web tier is assumed to be /webtier. You can change it according to your deployment.

Create Public DNS name

> **Note**
>
> If you have your own DNS server, you can create an alias record for the sm-frontend-ag-pip public IP in your DNS server instead.

In the **Overview** page, click the public IP address. In the **Configuration** page, click **Create alias record** to create an alias record for the public IP. For example, create an alias record as follows:

- **Subscription**: *<Your subscription>*
- **DNS zone**: *yourdomain.net*
- **Name**: sm.yourdomain.net

(Mixed Mode Only) Configure Application Gateway for SM Web tier

For the mixed mode (SMA-SM), you should re-use the frontend AG you created during the SMA SP installation. We still use the name sm-frontend-ag to refer to the existing frontend AG in the following sections.

1. Create backend pool
   Click **Navigation > All services > Networking > Application gateways >** **sm-frontend-ag**, then click **Backend pools > Add** to add a backend pool for the web tier. For example, create a backend pool as below:
   - **Name:** sm-webtier-pool
   - **Add backend pool without target:** Yes
2. Create health probe and HTTP settings
   Click **Health probes -> Add**, create a health probe as below:
   - **Name:** sm-webtier-health-probe
   - **Protocol:** HTTP
   - **Host:** sm-webtier-health-probe
   - **Path:** */webtier/api/healthz*
   - **Interval (seconds):** 30
   - **Timeout (seconds):** 10
   - **Unhealthy threshold:** 2

   Then click **Backend settings -> Add**, create an HTTP setting as below:
   - **Name:** sm-webtier-http
   - **Cookie-based affinity:** Enabled
   - **Affinity cookie name:** WEBTIER_ROUTE
   - **Connection draining:** Enabled
   - **Drain timeout (seconds):** 300
   - **Protocol:** HTTP
   - **Port:** 8080

- **Request timeout (seconds):** 300
- **Use custom probe:** Yes
- **Custom probe:** `sm-webtier-health-probe`

3. Edit routing rule
   The routing rule you created for the port 443 when you deployed the SMA SP is a `Basic` type rule, you need to change it to a `PathBasedRouting` type rule, and add a path-based rule to the routing rule for the web tier. At the time of writing, the Azure portal UI doesn't support to change the rule type, you need to run following PowerShell command in the Azure Cloud Shell to change the rule type and add a path-based rule for the web tier.

```
# The resource group name that the frontend AG belongs to
$rgName = "sma-rg"
# The name of the frontend AG you created when you deployed the SMA SP
$agName = "sm-frontend-ag"
# The listener name of the https port 443
$listenerName = "sma-listener-443"
# The backend pool name for the SMA port 443
$backendPoolName = "sma-pool-443"
# The HTTP settins name for the SMA port 443
$httpSettingName = "sma-http-443"
# The routing rule name for the SMA port 443
$ruleName = "sma-rule-443"

# Get the fontend AG and its listener
$ag = Get-AzApplicationGateway -ResourceGroupName $rgName -Name $agName
$listener = Get-AzApplicationGatewayHttpListener -ApplicationGateway $ag -Name $listenerName

# Get the backend pool and the HTTP settings you created for SMA port 443
$defaultPool = Get-AzApplicationGatewayBackendAddressPool -ApplicationGateway $ag -Name
$backendPoolName
$defaultHttp = Get-AzApplicationGatewayBackendHttpSetting -ApplicationGateway $ag -Name
$httpSettingName

# Get the backend pool and the HTTP settings you just created for the SM webtier
$pool = Get-AzApplicationGatewayBackendAddressPool -ApplicationGateway $ag -Name "sm-webtier-pool"
$http = Get-AzApplicationGatewayBackendHttpSetting -ApplicationGateway $ag -Name "sm-webtier-http"

# Add a URL path map that contains a path-based rule for the SM webtier
$pathRule = New-AzApplicationGatewayPathRuleConfig -Name "sm-webtier" -Paths
"/webtier","/webtier/*" -BackendAddressPool $pool -BackendHttpSettings $http
Add-AzApplicationGatewayUrlPathMapConfig -ApplicationGateway $ag -Name "sm-frontend-url-path-map"
-PathRules $pathRule -DefaultBackendAddressPool $defaultPool -DefaultBackendHttpSettings
$defaultHttp
$pathMap = Get-AzApplicationGatewayUrlPathMapConfig -ApplicationGateway $ag -Name "sm-frontend-
url-path-map"

# Update the routing rule and save the applicaton gateway
Set-AzApplicationGatewayRequestRoutingRule -ApplicationGateway $ag -Name $ruleName -RuleType
"PathBasedRouting" -HttpListener $listener -UrlPathMap $pathMap
Set-AzApplicationGateway -ApplicationGateway $ag
```

Note

- You need to adapt the value of various name variables to the real name of resources you created when you deployed the SMA SP.
- The context path of the web tier is assumed to be `/webtier`, you could change it according to your deployment.
- Please see the Azure doc Quickstart for PowerShell in Azure Cloud Shell for details on how to use the PowerShell in Cloud Shell in the Azure portal.

Create a custom image for SM Web tier

1. Create a VM.
   Click **Navigation pane > All services > Compute > Virtual machines > Create** to create a VM. For example, create a VM as follows:
   - **Project details**
     - **Subscription**: *<Your subscription>*
     - **Resource group**: `sm-prod-rg`
   - **Instance details**
     - **Virtual machine name**: `sm-webtier`
     - **Region**: *(US) West US 3*
     - **Availability options**: No infrastructure redundancy required
     - **Security type**: Standard
     - **Image**: Windows Server 2019 Datacenter - Gen1
     - **Run with Azure Spot discount**: no
     - **Size**: *Standard B2s (2 vcpus, 4 GiB memory)*
   - **Administrator account**
     - **Username**: *<Your username>*
     - **Password**: *<Your password that adheres to your security policy>*
   - **INBOUND PORT RULES**
     - **Public inbound ports**: None
   - **Disk options**
     - **OS disk type**: Standard SSD
   - **Network interface**
     - **Virtual network**: `sm-vnet`
     - **Subnet**: `sm-private-subnet`
     - **Public IP**: None
     - **NIC network security group**: None
   - **Monitoring**
     - **Boot diagnostics**: Off
     - **OS guest diagnostics**: Off
2. Install web tier.
   Establish a remote desktop connection to the `sm-webtier` VM through the `sm-rd-gateway` RD gateway.
   Download the latest OpenJDK 11 and install it. Add a system environment variable, `JAVA_HOME` =*<the install dir of OpenJDK>*.

Download the latest Tomcat 9 and install it. Add a system environment variable, `TOMCAT_HOME` =<*the install dir of Tomcat*>. Open the command prompt under the `%TOMCAT_HOME%\bin` folder, and then run the following command to install the Tomcat service:

```
set JvmMs=1536
set JvmMx=1536
service.bat install
```

Note

If you choose a more powerful VM instance size (For example, *Standard B2ms* with 8 GiB memory) for the web tier, you can set a large maximum heap size for the JVM. For example, you can use the following commands instead to install the Tomcat service:

```
set JvmMs=1536
set JvmMx=4096
service.bat install
```

Download the Service Manager Web tier installation package to the computer, and then install the Service Manager web tier by following the instructions in Install the Service Manager Web Tier. Note that you can change the context path to `/webtier` by renaming `webtier-x.xx.war` to `webtier.war` after you copy the war file to the `%TOMCAT_HOME%\webapps` folder. Extract `webtier.war` to the `webtier` sub-folder, and then remove the `webtier.war` file.
After installation, edit the `%TOMCAT_HOME%\webapps\webtier\WEB-INF\webtier.properties`, and then add the following parameters:

```
cookie.secure=true
serverHost=Private IP / FQDN of sm-backend-ag
externalLB=true
secureLogin=false
```

If you want to deploy the classic SM Collaboration later, you need to configure the LW-SSO settings. Edit the `%TOMCAT_HOME%\webapps\webtier\WEB-INF\webtier.properties`, set the `enableLWSSOFramework` to `true`, and then set the `domain` and the `initString` to appropriate values. See the following example:

```
<enableLWSSO
    enableLWSSOFramework="true" .../>
<webui>
    <validation>
        <in-ui-lwsso>
            <lwssoValidation ...>
                <domain>yourdomain.net</domain>
                <crypto ... initString="your shared secret passphrase"/>
<!-- ... omitted -->
```

If you want to deploy the classic SM Chat later, you need to configure the LW-SSO settings. For detailed information, see Configure LW-SSO in the Service Manager Web tier.

Note

If you choose the mixed mode (SMA-SM), you MUST NOT configure LW-SSO in the web tier as IdM will generate and renew the LW-SSO token.

You need to verify if the **Tomcat9** service can be started successfully, and access the URL *http://localhost:8080/webtier* in a web browser to check if the web tier works fine.
Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the %TOMCAT_HOME%\bin\tomcat9.exe for the Private and Public network types.
Create a PowerShell script named Startup.ps1 under the C:\ folder, and then edit it as follows:

```
param(
  [switch]$Schedule
)

if ($Schedule) {
  # Create a scheduled task that runs this script file at system startup
  $Trigger = New-ScheduledTaskTrigger -AtStartup
  $Action = New-ScheduledTaskAction -Execute "PowerShell.exe" -Argument "-File C:\Startup.ps1"
  Register-ScheduledTask -TaskName "Startup the Tomcat9 Service" -Action $Action -Trigger $Trigger
-User SYSTEM -RunLevel Highest -Force
  return
}

$ConfigFile = "$env:TOMCAT_HOME\conf\server.xml"
# Get local IP
$IP = (Test-Connection -ComputerName (hostname) -Count 1).IPV4Address.IPAddressToString
# Use last two numbers as the jvmRoute
$JVMRoute = $IP -replace '\d+\.\d+\.(\d+\.\d+)', '$1'
# Add the jvmRoute attribute to the <Engine> element
(Get-Content -Encoding UTF8 $ConfigFile) -replace '(<Engine.*"localhost").*>', ('$1 jvmRoute="' +
$JVMRoute + '">') | Out-File -Encoding UTF8 $ConfigFile

# Start the Tomcat9 service
Start-Service Tomcat9
```

Then, run the following command in a PowerShell instance to create a scheduled task that runs the script file at system startup:
powershell C:\Startup.ps1 -Schedule
The script is to add the jvmRoute attribute to the Tomcat config file and start the **Tomcat9** service. The jvmRoute attribute tells which instance a user's requests are routed to, which can facilitate troubleshooting in the future. For example, the suffix 0.100 in the JSESSIONID BE4EB1B933698EBE53C6A62DD6BCC56F.0.100 indicates that all requests of the session are routed to the Tomcat instance running on the VM instance 10.0.0.100.

3. Create a custom image.
   First, prepare the sm-webtier VM.
   - Open the **Services** desktop app, and then stop the **Tomcat9** service.
   - Remove all log files under the %TOMCAT_HOME%\logs folder, and then remove the sm.log under the %TOMCAT_HOME% folder.
   - Generalize the VM by following the instructions in Generalize the Windows VM.

   After the VM is generalized, go to the sm-webtier VM in the Azure portal, and then click **Overview > Capture** to create an image as follows:
   - **Name:** sm-webtier-image-9.80-<timestamp>
   - **Resource group:** sm-prod-rg
   - **Automatically delete this virtual machine after creating the image:** Yes
   - **Share image to Azure compute gallery**: No
   - **Zone resiliency:** On

   After the image is created, click **Navigation pane > All services > Other > Disks >** **sm-webtier_OsDisk_1_<uuid>**, and then click **Overview > Delete** to delete the unattached managed disk.

Create a virtual machine scale set for SM web tier

Click **Navigation pane > All services > Compute > Virtual machine scale sets > Create** to create a virtual machine scale set. For example, create a virtual machine scale set as follows:

- **Basics**
  - **Project details**
    - **Subscription:** *<Your subscription>*
    - **Resource group:** `sm-prod-rg`
  - **Orchestration**
    - **Orchestration mode:** `Uniform`
  - **Scale set details**
    - **Virtual machine scale set name:** `SmWebtierVmss`
    - **Region:** *(US) West US 3*
    - **Availability zone:** Zone 1, 2, 3
  - **Instance details**
    - **Image:** See all images -> My Images -> `sm-webtier-image-9.80-<timestamp>`
    - **Azure Spot instance:** No
    - **Size:** *Standard B2s (2 vcpus, 4 GiB memory)*
  - **Administrator account**
    - **Username:** *<Your username>*
    - **Password:** *<Your password that adheres to your security policy>*
- **Disks**
  - **Disk options**
    - **OS disk type:** Standard SSD
    - **Encryption type:** Default
    - **Enable Ultra Disk compatibility:** No
  - **Advanced**
    - **Use managed disks:** Yes
- **Networking**
  - **Virtual network configuration**
    - **Virtual network:** `sm-vnet`
  - **Network interface** edit the network interface `sm-vnet-nic01` as below
    - **Name:** `sm-vnet-nic01`
    - **Subnet:** `sm-private-subnet`
    - **NIC network security group:** None
    - **Public IP address:** Disabled
  - **Load balancing**
    - **Use a load balancer:** Yes
  - **Load balancing settings**
    - **Load balancing options:** Application Gateway
    - **Select an application gateway:** `sm-frontend-ag`
    - **Select a backend pool:** `sm-webtier-pool`
- **Scaling**
  - **Initial instance count:** 1
  - **Scaling**
    - **Scaling policy:** Custom
    - **Minimum number of VMs:** 1
    - **Maximum number of VMs:** 10

    - **Scale out**
        - **CPU threshold (%):** 60
        - **Duration in minutes:** 10
        - **Number of VM to increase by:** 1
    - **Scale in**
        - **CPU threshold (%):** 20
        - **Number of VM to decrease by:** 1
    - **Collect diagnostic logs from Autoscale:** Disabled
    - **Scale-In policy:** Default
- **Management**
    - **Upgrade policy**
        - **Upgrade mode:** Manual
    - **Monitoring**
        - **Boot diagnostics:** Off
    - **Identity**
        - **System assigned managed identity:** Off
    - **Instance termination**
        - **Instance termination notification:** Off
- **Health**
    - **Monitor application health:** Disabled



Note

- You need to set appropriate **Instance count**, **Minimum number of VMs**, and **Maximum number of VMs** for the `SmWebtierVmss` VMSS according to your load estimation. For more sizing information, see Sizing.
- The scaling policies above are for illustration purposes, you should adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **scale conditions** after creation to scale out or scale in SM web tier instances at a specific time based on the peak time and idle time of your Service Manager Web tier load.

After creation, disable Windows Automatic Updates for the `SmWebtierVmss` VMSS by following the instructions in Disable Windows automatic updates for a VMSS.

After the `SmWebtierVmss` VMSS launches the desired VM instances, you can access the web tier at *`https://sm.yourdomain.net/webtier`*.

If you have created a read replica for the reporting module, log in to the web tier, and then configure the replicated reporting database by following the instructions provided in task 4 of Set up a replicated reporting database.

3.3.3. Install Service Manager Windows Client

This task is optional. If you want to use some tailoring features such as Forms Designer that are only available in the Service Manager Windows Client, you can install the Windows Client on a VM.

Create a VM

Click **Navigation pane** > **All services** > **Compute** > **Virtual machines** > **Create** to create a VM. For example, create a VM as follows:

- **Project details**
    - **Subscription**: *<Your subscription>*
    - **Resource group**: `sm-prod-rg`
- **Instance details**

    - **Virtual machine name**: `sm-win-client`
    - **Region**: *(US) West US 3*
    - **Availability options**: No infrastructure redundancy required
    - **Security type**: Standard
    - **Image**: Windows 10 Pro, Version 21H2 - Gen2
    - **Run with Azure Spot discount**: no
    - **Size**: *Standard B2ms (2 vcpus, 8 GiB memory)*
- **Administrator account**
    - **Username**: *<Your username>*
    - **Password**: *<Your password that adheres to your security policy>*
- **INBOUND PORT RULES**
    - **Public inbound ports: None**
- **Disk options**
    - **OS disk type: Standard SSD**
- **Network interface**
    - **Virtual network**: `sm-vnet`
    - **Subnet**: `sm-private-subnet`
    - **Public IP**: None
    - **NIC network security group**: None
- **Monitoring**
    - **Boot diagnostics**: Off
    - **OS guest diagnostics**: Off

Install Service Manager Windows Client

Establish a remote desktop connection to the `sm-win-client` VM through the `sm-rd-gateway` RD gateway.

Download the Service Manager Windows Client installation package to the computer, and then install the Service Manager Windows Client by following the instructions in Install a Windows Client.

After installation, you can connect the Windows Client to the SM server with the following parameters:

- **Connection**
    - **Server host name**: sm-server.yourdomain.net
    - **Server port number**: 13080
- **Advanced**
    - **Connect to External Load Balancer**: true



Tip

Tip

You can stop the `sm-win-client` VM when you don't use it to save cost.

3.3.4. Install Service Manager Mobility

Create a custom image for SM Mobility

1. Create a VM.
   Click **Navigation pane > All services > Compute > Virtual machines > Create** to create a VM. For example, create a VM as follows:
    - **Project details**
        - **Subscription**: *<Your subscription>*
        - **Resource group**: `sm-prod-rg`
    - **Instance details**

        - **Virtual machine name**: `sm-mobility`
        - **Region**: *(US) West US 3*
        - **Availability options**: No infrastructure redundancy required
        - **Security type**: Standard
        - **Image**: See all images > My Images > `sm-webtier-image-9.80-<timestamp>`
        - **Run with Azure Spot discount**: no
        - **Size**: *Standard B2s (2 vcpus, 4 GiB memory)*
    - **Administrator account**
        - **Username**: *<Your username>*
        - **Password**: *<Your password that adheres to your security policy>*
    - **INBOUND PORT RULES**
        - **Public inbound ports**: None
    - **Disk options**
        - **OS disk type**: Standard SSD
    - **Network interface**
        - **Virtual network**: `sm-vnet`
        - **Subnet**: `sm-private-subnet`
        - **Public IP**: None
        - **NIC network security group**: None
    - **Monitoring**
        - **Boot diagnostics**: Off
        - **OS guest diagnostics**: Off

    Note

    Note
    The VM is created from the `sm-webtier-image-9.80-<timestamp>` image in which the JDK and the Tomcat have already been installed and configured. You only need to install the mobility on the VM and remove the web tier from the VM.

2. Install mobility.
   Establish a remote desktop connection to the `sm-mobility` instance through the `sm-rd-gateway` RD gateway.
   Open the **Services** desktop app, and then stop the **Tomcat9** service.
   Remove the `webtier` sub-folder from the `%TOMCAT_HOME%\webapps` folder to uninstall the web tier application.
   Download the Service Manager Mobility installation package to the computer, and then install the Service Manager Mobility by following the instructions in Install the Mobile Applications client. Note that you can change the context path to `/mobility` by renaming `webapp-x.xx.war` to `mobility.war` after you copy the war file to the `%TOMCAT_HOME%\webapps` folder, and then expand the `mobility.war`.
   After installation, edit the `%TOMCAT_HOME%\webapps\mobility\WEB-INF\web.properties`, and then modify the parameters as follows:

   ```
   endpoint=http://sm-server.yourdomain.net:13080/SM/ui
   externalLB=true
   ```

   Verify if the **Tomcat9** service can be started successfully and access the URL in a web browser to check if the login page can be displayed. Note that if you want to verify the mobility features, you need to install Chrome and access the mobility in Chrome by turning on the device simulator.
3. Create a custom image.
   First prepare the `sm-mobility` VM.
    - Open the **Services** desktop app, stop the **Tomcat9** service.

- Remove all log files under the `%TOMCAT_HOME%\logs` folder, and then remove `mobility.log`, `mobility-perf.log` and `sm.log` under the `%TOMCAT_HOME%` folder.
- Generalize the VM by following the instructions in Generalize the Windows VM.

After the VM is generalized, go to the `sm-mobility` VM in the Azure portal, and then click **Overview > Capture**, create an image as follows:

- **Name:** `sm-mobility-image-9.80-<timestamp>`
- **Resource group:** `sm-prod-rg`
- **Automatically delete this virtual machine after creating the image:** Yes
- **Zone resiliency:** On

After the image is created, click **Navigation pane > All services > Other > Disks >** **`sm-mobility_OsDisk_1_<uuid>`**, and then click **Overview > Delete** to delete the unattached managed disk.

Create a virtual machine scale set for SM Mobility

1. Create a virtual machine scale set.
   Click **Navigation pane > All services > Compute > Virtual machine scale sets > Create** to create a virtual machine scale set. For example, create a virtual machine scale set as follows:
   - **Basics**
     - **Project details**
       - **Subscription:** *<Your subscription>*
       - **Resource group:** `sm-prod-rg`
     - **Scale set details**
       - **Virtual machine scale set name:** `SmMobilityVmss`
       - **Region:** *(US) West US 3*
       - **Availability zone:** Zone 1, 2, 3
     - **Orchestration**
       - **Orchestration mode:** `Uniform`
     - **Instance details**
       - **Image:** See all images -> My Images -> `sm-mobility-image-9.80-<timestamp>`
       - **Azure Spot instance:** No
       - **Size:** *Standard B2s (2 vcpus, 4 GiB memory)*
     - **Administrator account**
       - **Username:** *<Your username>*
       - **Password:** *<Your password that adheres to your security policy>*
   - **Disks**
     - **Disk options**
       - **OS disk type:** Standard SSD
       - **Encryption type:** Default
       - **Enable Ultra Disk compatibility:** No
     - **Advanced**
       - **Use managed disks:** Yes
   - **Networking**
     - **Virtual network configuration**
       - **Virtual network:** `sm-vnet`
     - **Network interface** edit the network interface `sm-vnet-nic01` as below
       - **Name:** `sm-vnet-nic01`
       - **Subnet:** `sm-private-subnet`
       - **NIC network security group:** None
       - **Public IP address:** Disabled
     - **Load balancing**

        - **Use a load balancer:** Yes
    - **Load balancing settings**
        - **Load balancing options:** Application Gateway
        - **Select an application gateway:** `sm-frontend-ag`
        - **Select a backend pool:** Create new
            - **Name:** `sm-mobility-pool`
- **Scaling**
    - **Initial instance count:** 1
    - **Scaling**
        - **Scaling policy:** Custom
        - **Minimum number of VMs:** 1
        - **Maximum number of VMs:** 10
    - **Scale out**
        - **CPU threshold (%):** 60
        - **Duration in minutes:** 10
        - **Number of VM to increase by:** 1
    - **Scale in**
        - **CPU threshold (%):** 20
        - **Number of VM to decrease by:** 1
    - **Collect diagnostic logs from Autoscale:** Disabled
    - **Scale-In policy:** Default
- **Management**
    - **Upgrade policy**
        - **Upgrade mode:** Manual
    - **Monitoring**
        - **Boot diagnostics:** Off
    - **Identity**
        - **System assigned managed identity:** Off
    - **Instance termination**
        - **Instance termination notification:** Off
- **Health**
    - **Monitor application health:** Disabled



Note

Note

- You need to set appropriate **Instance count**, **Minimum number of VMs**, and **Maximum number of VMs** for the `SmMobilityVmss` VMSS according to your load estimation. For more sizing information, see Sizing.
- The scaling policies above are for illustration purpose, you should adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **scale conditions** after creation to scale out or scale in SM mobility instances at specific time based on the peak time and idle time of your Service Manager Mobility load.

After creation, disable Windows Automatic Updates for the `SmMobilityVmss` VMSS by following the instructions in Disable Windows automatic updates for a VMSS.

2. Configure the application gateway.

Click **Navigation > All services > Networking > Application gateways >** **sm-frontend-ag**, and then click **Health probes > Add** to create a health probe as follows:

- **Name:** sm-mobility-health-probe
- **Protocol:** HTTP
- **Host:** sm-mobility-health-probe
- **Path:** */mobility*
- **Interval (seconds):** 30
- **Timeout (seconds):** 10
- **Unhealthy threshold:** 2

Click **Backend settings > Add** to create a backend setting as follows:

- **Name:** sm-mobility-http
- **Cookie-based affinity:** Enabled
- **Affinity cookie name:** MOBILITY_ROUTE
- **Connection draining:** Enabled
- **Drain timeout (seconds):** 300
- **Protocol:** HTTP
- **Port:** 8080
- **Request timeout (seconds):** 300
- **Use custom probe:** Yes
- **Custom probe:** sm-mobility-health-probe

Click **Rules >** **sm-frontend-rule** **> Backend targets > Add multiple targets to create a path-based rule** to add a path rule as follows:

- **Target type:** Backend pool
- **Path:** */mobility,/mobility/**
- **Target name:** sm-mobility
- **Backend settings:** sm-mobility-http
- **Backend target:** sm-mobility-pool

After the SmMobilityVmss VMSS launches the desired VM instances, you can access the mobility at *https://sm.yourdomain.net/mobility/std/login*.

3.3.5. Install Identity Manager (IdM)

Create a custom image for IdM

1. Create a VM.
   Click **Navigation pane > All services > Compute > Virtual machines > Create** to create a VM. For example, create a VM as follows:
   - **Project details**
     - **Subscription**: <*Your subscription*>
     - **Resource group**: sm-prod-rg
   - **Instance details**
     - **Virtual machine name**: idm
     - **Region**: *(US) West US 3*
     - **Availability options**: No infrastructure redundancy required
     - **Security type**: Standard
     - **Image**: See all images > My Images > sm-webtier-image-9.80-<timestamp>
     - **Run with Azure Spot discount**: no
     - **Size**: *Standard B2s (2 vcpus, 4 GiB memory)*
   - **Administrator account**
     - **Username**: <*Your username*>

        - **Password**: *<Your password that adheres to your security policy>*
    - **INBOUND PORT RULES**
        - **Public inbound ports**: None
    - **Disk options**
        - **OS disk type**: Standard SSD
    - **Network interface**
        - **Virtual network**: `sm-vnet`
        - **Subnet**: `sm-private-subnet`
        - **Public IP**: None
        - **NIC network security group**: None
    - **Monitoring**
        - **Boot diagnostics**: Off
        - **OS guest diagnostics**: Off

    > **Note**
    > Note
    > The VM is created from the `sm-webtier-image-9.80-<timestamp>` image in which the JDK and the Tomcat have already been installed and configured. You only need to install IdM on the VM and remove the web tier from the VM.

2. Install IdM.
   Establish a remote desktop connection to the `idm-template` instance through the `sm-rd-gateway` RD gateway.
   Open the Services desktop app, and then stop the **Tomcat9** service.
   Remove the `webtier` sub-folder from the `%TOMCAT_HOME%\webapps` folder to uninstall the web tier application.
   Download the Identity Manager (IdM) installation packages to your computer, and then install IdM by following the instructions provided in the Install Identity Manager (IdM) topic. Note that you can change the context path to `/idm-service` and `/idm-admin` by renaming `idm-service-x.xx.war` and `idm-admin-console-x.xx.war` to `idm-service.war` and `idm-admin.war` after you copy the war file to the `%TOMCAT_HOME%\webapps` folder.
   Connect to the `sm-db-pg` DB and then create a new database named `idm` and follow the instructions provided in Install Identity Manager (IdM) to install the IdM service. Note that the value of the idm.persistence.connection parameter in the WEB-INF/spring/applicationContext.properties file is as follows.

   ```
   idm.persistence.connection.url = jdbc:postgresql://sm-rds-pg.xxxyyyzzz.us-west-
   2.rds.amazonaws.com:5432/idm
   ```

   Next, follow the instructions provided in Install Identity Manager (IdM) to Install the IdM admin console. Note that the value of the idm.serverUrl parameter in the WEB-INF\classes\IdmConfig.properties file is as follows.

   ```
   idm.serverUrl=https://sm.yourdomain.net/idm-service
   ```

   Edit idm-admin\WEB-INF\classes\conf\IdmConfig.xml to configure `returnUri` before `<logoutUrls>` as follows.

   ```
   <!-- URL Where IdM will redirect to after authenticated successfully. if not specified, it will
   redirect the URL accessed firstly  -->
   <returnUri base="true" value="https://sm.yourdomain.net/idm-admin "></returnUri>
   ```

   Verify if the **Tomcat9** service can be started successfully, and then access the `http://localhost:8080/idm-service/api/version-info` URL in a web browser to check if the version can be displayed.
3. Create a custom image.
   First prepare the `idm` VM.
    - Open the **Services** desktop app, stop the **Tomcat9** service.

- Remove all log files under the `%TOMCAT_HOME%\logs` folder.
- Generalize the VM by following the instructions in Generalize the Windows VM.

After the VM is generalized, go to the `idm` VM in the Azure portal, and then click **Overview > Capture**, create an image as follows:

- **Name:** idm `-image-9.80-<timestamp>`
- **Resource group:** `sm-prod-rg`
- **Automatically delete this virtual machine after creating the image:** Yes
- **Zone resiliency:** On

After the image is created, click **Navigation pane > All services > Other > Disks > idm** `_OsDisk_1_<uuid>`, and then click **Overview > Delete** to delete the unattached managed disk.

Create a virtual machine scale set for IdM

1. Create a virtual machine scale set.
   Click **Navigation pane > All services > Compute > Virtual machine scale sets > Create** to create a virtual machine scale set. For example, create a virtual machine scale set as follows.
   - **Basics**
     - **Project details**
       - **Subscription:** *<Your subscription>*
       - **Resource group:** `sm-prod-rg`
     - **Scale set details**
       - **Virtual machine scale set name:** `IdMVmss`
       - **Region:** *(US) West US 3*
       - **Availability zone:** Zone 1, 2, 3
     - **Orchestration**
       - **Orchestration mode:** `Uniform`
     - **Instance details**
       - **Image:** See all images -> My Images -> `idm-image-9.80-<timestamp>`
       - **Azure Spot instance:** No
       - **Size:** *Standard B2s (2 vcpus, 4 GiB memory)*
     - **Administrator account**
       - **Username:** *<Your username>*
       - **Password:** *<Your password that adheres to your security policy>*
   - **Disks**
     - **Disk options**
       - **OS disk type:** Standard SSD
       - **Encryption type:** Default
       - **Enable Ultra Disk compatibility:** No
     - **Advanced**
       - **Use managed disks:** Yes
   - **Networking**
     - **Virtual network configuration**
       - **Virtual network:** `sm-vnet`
     - **Network interface** edit the network interface `sm-vnet-nic01` as below
       - **Name:** `sm-vnet-nic01`
       - **Subnet:** `sm-private-subnet`
       - **NIC network security group:** None
       - **Public IP address:** Disabled
     - **Load balancing**

        - **Use a load balancer:** Yes
    - **Load balancing settings**
        - **Load balancing options:** Application Gateway
        - **Select an application gateway:** `sm-frontend-ag`
        - **Select a backend pool:** Create new
            - **Name:** `idm-pool`
- **Scaling**
    - **Initial instance count:** 1
    - **Scaling**
        - **Scaling policy:** Custom
        - **Minimum number of VMs:** 1
        - **Maximum number of VMs:** 10
    - **Scale out**
        - **CPU threshold (%):** 60
        - **Duration in minutes:** 10
        - **Number of VM to increase by:** 1
    - **Scale in**
        - **CPU threshold (%):** 20
        - **Number of VM to decrease by:** 1
    - **Collect diagnostic logs from Autoscale:** Disabled
    - **Scale-In policy:** Default
- **Management**
    - **Upgrade policy**
        - **Upgrade mode:** Manual
    - **Monitoring**
        - **Boot diagnostics:** Off
    - **Identity**
        - **System assigned managed identity:** Off
    - **Instance termination**
        - **Instance termination notification:** Off
- **Health**
    - **Monitor application health:** Disabled



Note

Note

- You need to set the appropriate **Instance count**, **Minimum number of VMs**, and **Maximum number of VMs** for the `IdMVmss` VMSS according to your load estimation. For more sizing information, see Sizing.
- The scaling policies above are for illustration purpose, you should adjust the high/low threshold appropriately according to your production environment. If the average CPU utilization is insensitive enough to your workload, you can add **scale conditions** after creation to scale out or scale in IdM instances at a specific time based on the peak time and idle time of your Service Manager IdM load.

After creation, disable Windows Automatic Updates for the `IdMVmss` VMSS by following the instructions in Disable Windows automatic updates for a VMSS.

2. Configure the application gateway.

Click **Navigation > All services > Networking > Application gateways >** `sm-frontend-ag`, and then click **Health probes > Add** to create a health probe as follows:

- **Name:** idm-health-probe
- **Protocol:** HTTP
- **Host:** idm-health-probe
- **Path:** */idm-service/api/version-info*
- **Interval (seconds):** 30
- **Timeout (seconds):** 10
- **Unhealthy threshold:** 2

Click **Backend settings > Add** to create a backend setting as follows:

- **Name:** idm-http
- **Cookie-based affinity:** Enabled
- **Affinity cookie name:** IDM_ROUTE
- **Connection draining:** Enabled
- **Drain timeout (seconds):** 300
- **Protocol:** HTTP
- **Port:** 8080
- **Request timeout (seconds):** 300
- **Use custom probe:** Yes
- **Custom probe:** idm-health-probe

Click **Rules >** `sm-frontend-rule` **> Backend targets > Add multiple targets to create a path-based rule** to add a path rule as follows:

- **Target type:** Backend pool
- **Path:** */idm-service,/idm-service/*,/idm-admin,/idm-admin/**
- **Target name:** idm
- **Backend settings:** idm-http
- **Backend target:** idm-pool

After the IdMVmss VMSS launches the desired VM instances, you can access IdM at *https://sm.yourdomain.net/idm-admin*.

3.3.6. (Classic Only) Install Service Manager Collaboration

Create a custom image for SM Collaboration

1. Create a VM.
   Click **Navigation pane > All services >Compute > Virtual machines > Create** to create a VM. For example, create a VM as follows:
   - **Project details**
     - **Subscription**: *<Your subscription>*
     - **Resource group**: sm-prod-rg
   - **Instance details**
     - **Virtual machine name**: sm-chat
     - **Region**: *(US) West US 2*
     - **Availability options**: No infrastructure redundancy required
     - **Security type**: Standard
     - **Image**: Windows Server 2019 Datacenter - Gen1
     - **Run with Azure Spot discount**: no
     - **Size**: *Standard B2ms (2 vcpus, 8 GiB memory)*
   - **Administrator account**
     - **Username**: *<Your username>*
     - **Password**: *<Your password that adheres to your security policy>*

- **INBOUND PORT RULES**
  - **Public inbound ports**: None
- **Disk options**
  - **OS disk type**: Standard SSD
- **Network interface**
  - **Virtual network**: `sm-vnet`
  - **Subnet**: `sm-private-subnet`
  - **Public IP**: None
  - **NIC network security group**: None
- **Monitoring**
  - **Boot diagnostics**: Off
  - **OS guest diagnostics**: Off

2. Install the SM Collaboration components.
   Install all SM Collaboration components on the same VM instance.
   Establish a remote desktop connection to the `sm-chat` VM instance through the `sm-rd-gateway` RD gateway.
   Download the latest Apache httpd 2.4.x and install it to a folder (referred to as `%APACHE_HTTPD_HOME%` later).
   Open the command prompt with administrative privileges under the `%APACHE_HTTPD_HOME%\bin` folder, and then run the following command to install the apache service:
   `httpd.exe -k install`
   Download the latest OpenJDK 8 and install it. Add a system environment variable, `JAVA_HOME` =*<the install dir of OpenJDK>*.
   Download the Service Manager Chat Server install package to the computer, extract it to a folder (referred to as `%CHAT_SERVER_HOME%` later). Download the Azure database root CA certificate at https://dl.cacerts.digicert.com/DigiCertGlobalRootCA.crt.pem, rename it to `DigiCertGlobalRootCA.crt`, and then save it to the `%CHAT_SERVER_HOME%\bin` folder. Configure the Chat Server by following the instructions provided in the Task 2: Deploy and configure the chat server topic. You can open the configuration wizard by visiting the URL http://localhost:9090/setup/index.jsp in a web browser. Note that you need to configure the settings as follows:
   - **Server Settings**:
     - **Domain**: *a unique string to identify the XMPP domain, such as* `sm-chat`
   - **Database Settings**:
     - **Database Driver Presets**: PostgreSQL
     - **JDBC Driver Class**: org.postgresql.Driver
     - **Database URL:** jdbc:postgresql://*sm-db-pg.postgres.database.azure.com/sm*?sslmode=verify-full&sslcert=&sslrootcert=DigiCertGlobalRootCA.crt
     - **Username**: *postgres*
     - **Password**: *your_db_password*

   Download the Service Manager Chat Service installation package to the computer, and then install it to a folder (referred to as `%CHAT_SERVICE_HOME%` later) by following the instructions in Task 3: Deploy and configure the chat service. Note that you need to configure some properties in `%CHAT_SERVICE_HOME%\conf\app.properties` as follows:
   - **daoFactory.serviceEndPoint:** *http://smserver.yourdomain.net:13080*
   - **chatServerConfig.domain:** *same value with the **XMPP Domain** parameter above*
   - **chatServerConfig.host:** localhost

   Download the Service Manager Chat UI installation package to the computer, extract the chatui.war from the zip file, unzip the war file, edit the `chatui\conf\env.js` file, and then set the value of the `chatServiceUrl` parameter to `https://sm.yourdomain.net/chatservice`. Then, copy the `chatui` folder to the `%APACHE_HTTPD_HOME%\htdocs\` folder.

Edit the `%APACHE_HTTPD_HOME%\conf\httpd.conf` file, make sure that the `proxy_module` is loaded, and then append the following lines to the end of the file:

```
ProxyPass /of-http-bind http://localhost:7070/http-bind timeout=120
ProxyPass /of-plugins http://localhost:9090/plugins
ProxyPass /chatservice http://localhost:8088
ProxyPass /healthCheck http://localhost:8089/healthcheck
```

Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the `%APACHE_HTTPD_HOME%\bin\httpd.exe` for the Private and Public network types.
Verify if the **Apache2.4** service, the **Service Manager Chat Server** service, and the **Service Manager Chat Service** service can be started successfully.

3. Create a custom image.
   First prepare the `sm-chat` VM before you capture an image from it.
   - Open the **Services** desktop app, and then stop the **Apache2.4** service, the **Service Manager Chat Server** service, and the **Service Manager Chat Service** service.
   - Remove all log files under the `%APACHE_HTTPD_HOME%\logs`, `%CHAT_SERVER_HOME%\logs` and `%CHAT_SERVICE_HOME\log` folders.
   - Generalize the VM by following the instructions in Generalize the Windows VM.

   After the VM is generalized, go to the `sm-chat` VM in the Azure portal, and then click **Overview > Capture** to create an image as follows:
   - **Name:** `sm-chat-image-9.70-<timestamp>`
   - **Resource group:** `sm-prod-rg`
   - **Automatically delete this virtual machine after creating the image:** Yes
   - **Share image to Azure compute gallery**: No
   - **Zone resiliency:** On

   After the image is created, click **Navigation pane > All services > Other > Disks >** **`sm-chat_OsDisk_1_<uuid>`**, and then click **Overview > Delete** to delete the unattached managed disk.

Create a virtual machine scale set for SM Collaboration

1. Create a virtual machine scale set.
   Click **Navigation pane > All services > Compute > Virtual machine scale sets > Create** to create a virtual machine scale set. For example, create a virtual machine scale set as follows:
   - **Basics**
     - **Project details**
       - **Subscription:** *<Your subscription>*
       - **Resource group:** `sm-prod-rg`
     - **Scale set details**
       - **Virtual machine scale set name:** `SmChatVmss`
       - **Region:** *(US) West US 3*
       - **Availability zone:** Zone 1, 2, 3
     - **Orchestration**
       - **Orchestration mode:** `Uniform`
     - **Instance details**
       - **Image:** See all images -> My Images -> `sm-chat-image-9.70-<timestamp>`
       - **Azure Spot instance:** No
       - **Size:** *Standard B2s (2 vcpus, 4 GiB memory)*
     - **Administrator account**
       - **Username:** *<Your username>*
       - **Password:** *<Your password that adheres to your security policy>*

- **Disks**
  - **Disk options**
    - **OS disk type:** Standard SSD
    - **Encryption type:** Default
    - **Enable Ultra Disk compatibility:** No
  - **Advanced**
    - **Use managed disks:** Yes
- **Networking**
  - **Virtual network configuration**
    - **Virtual network:** `sm-vnet`
  - **Network interface** edit the network interface `sm-vnet-nic01` as below
    - **Name:** `sm-vnet-nic01`
    - **Subnet:** `sm-private-subnet`
    - **NIC network security group:** None
    - **Public IP address:** Disabled
  - **Load balancing**
    - **Use a load balancer:** Yes
  - **Load balancing settings**
    - **Load balancing options:** Application Gateway
    - **Select an application gateway:** `sm-frontend-ag`
    - **Select a backend pool:** Create new
      - **Name:** `sm-chat-pool`
- **Scaling**
  - **Initial instance count:** 1
  - **Scaling policy:** Manual
  - **Scale-in policy:** Default
- **Management**
  - **Upgrade policy**
    - **Upgrade mode:** Manual
  - **Monitoring**
    - **Boot diagnostics:** Off
  - **Identity**
    - **System assigned managed identity:** Off
  - **Instance termination**
    - **Instance termination notification:** Off
- **Health**
  - **Monitor application health:** Disabled



Note

The SM Collaboration doesn't support to be deployed as a cluster. You must set the **Instance count** for the `SmChatVmss` VMSS to **`1`**. The only purpose of the `SmChatVmss` VMSS is to ensure that there is one and only one SM Collaboration instance is running.

After creation, disable Windows Automatic Updates for the `SmChatVmss` VMSS by following the instructions in Disable Windows automatic updates for a VMSS.

2. Configure the application gateway.

Click **Navigation > All services > Networking > Application gateways >** `sm-frontend-ag`, and then click **Health probes > Add** to create a health probe as follows:

- **Name:** `sm-chat-health-probe`
- **Protocol:** HTTP
- **Host:** `sm-chat-health-probe`
- **Path:** */healthCheck*
- **Interval (seconds):** 30
- **Timeout (seconds):** 10
- **Unhealthy threshold:** 2

Click **Backend settings > Add** to create a backend setting as follows:

- **Name:** `sm-chat-http`
- **Cookie-based affinity:** Disabled
- **Connection draining:** Enabled
- **Drain timeout (seconds):** 300
- **Protocol:** `HTTP`
- **Port:** 80
- **Request timeout (seconds):** 300
- **Use custom probe:** Yes
- **Custom probe:** `sm-chat-health-probe`

Click **Rules >** `sm-frontend-rule` **> Backend targets** to edit the rule as follows:

- **Default backend pool:** `sm-chat-pool`
- **Default backend settings:** `sm-chat-http`

After the `SmChatVmss` VMSS launches the SM Collaboration instance, you need to finish the following tasks:

- Select ESS as the portal by following the instructions in Task 6: Select a portal for End User Chat.
- Enable Service Manager Collaboration by following the instructions in Task 7: Enable Service Manager Collaboration. Note that you need to specify the **Chat Service URL** as *https://sm.yourdomain.net/chatservice*.

3.3.7. Install Service Manager Smart Analytics

Folloe these steps:

1. Create a VM. Click **Navigation pane > All services > Add > Virtual machines > Create** to create a VM. For example, create a VM as follows:
   - **Project details**
     - **Subscription:** *<Your subscription>*
     - **Resource group:** `sm-prod-rg`
   - **Instance details**
   - **Virtual machine name:** `sm-smarta`
     - **Region:** *(US) West US 3*
     - **Availability options:** No infrastructure redundancy required
     - **Security type**: Standard
     - **Image**: Windows Server 2019 Datacenter - Gen1
     - **Run with Azure Spot discount**: no
     - **Size:** *Standard B4ms (4 vcpus, 16 GiB memory)*
   - **Administrator account**
     - **Username:** *<Your username>*
     - **Password:** *<Your password that adheres to your security policy>*
   - **INBOUND PORT RULES**
     - **Public inbound ports:** None
   - **Disk options**

        - **OS disk type:** Standard SSD
    - **Network interface**
        - **Virtual network:** `sm-vnet`
        - **Subnet:** `sm-private-subnet`
        - **Public IP:** None
        - **NIC network security group:** Advanced
        - **Configure network security group:** (new) `sm-smarta-nsg`
    - **Monitoring**
        - **Boot diagnostics:** Off
        - **OS guest diagnostics:** Off

    After creation, click **Go to resource** to view the newly created `sm-smarta` virtual machine. Click **Networking > Network Interface: sm-smarta<*num*>**, click **IP configurations > ipconfig1**, change the **Assignment** to `Static`. Click **Networking > Inbound port rules > Add inbound port rule** to add an inbound port rule that allows traffic from the application security group `sm-smarta-client-asg`. For example, add an inbound port rule as below:

    - **Source:** Applicaton security group
    - **Source application security group:** `sm-smarta-client-asg`
    - **Source port ranges:** *
    - **Destination:** Any
    - **Destination port ranges:** *
    - **Protocol:** Any
    - **Action:** Allow
    - **Priority:** 1010
    - **Name:** `allow-smarta-client`

    Add an inbound port rule to deny all other inbound traffic. For example, add an inbound port rule as below:

    - **Source:** Any
    - **Source port ranges:** *
    - **Destination:** Any
    - **Destination port ranges:** *
    - **Protocol:** Any
    - **Action:** Deny
    - **Priority:** 1020
    - **Name:** `deny-others`

    Finally, click **Networking > Application security groups > Configure the application security groups** to assign the application security group `sm-smarta-client-asg` to the VM.

2. Create private DNS name.
   You can create a private DNS name for the `sm-smarta` VM by creating a record set in the private DNS zone. Click **Navigation pane > All services > Networking > Private DNS zones >** **`yourdomain.net`**, and then click **Overview > Record set** to add a record set. For example, add a record set as below:
    - **Name:** `sm-smarta`
    - **Type:** A
    - **IP address:** *<The private IP of the `sm-smarta` VM>*

   You can now use the DNS name *`sm-smarta.yourdomain.net`* to refer to the `sm-smarta` VM.
3. Install SM Smart Analytics.
   Establish a remote desktop connection to the `sm-smarta` VM instance through the `sm-rd-gateway` RD gateway. Download the Service Manager Smart Analytics installer for Windows to the computer, and then install it to a folder (referred to as `%SMART_ANALYTICS_HOME%` later) by following the instructions in Install Smart Analytics. Note that, you need to specify `*` as the IP address of your Service Manager Server during the installation because the IPs of SM Server secondary hosts are dynamic. The network access policy will be enforced by the `sm-smarta-sg` network security group.

Open the **Windows Firewall**, click **Allow an app or feature through Windows Firewall**, and then allow the following Smart Analytics programs for the Private and Public network types:

- `%SMART_ANALYTICS_HOME%\IDOL\IDOLServer.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\agentstore\agentstore.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\category\category.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\community\community.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\dah\dah.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\dih\dih.exe`
- `%SMART_ANALYTICS_HOME%\IDOL\view\view.exe`
- `%SMART_ANALYTICS_HOME%\level2proxy\IDOLServer.exe`
- `%SMART_ANALYTICS_HOME%\level2proxy\dah\dah.exe`
- `%SMART_ANALYTICS_HOME%\level2proxy\dih\dih.exe`
- `%SMART_ANALYTICS_HOME%\CFS\CFS.exe`
- `%SMART_ANALYTICS_HOME%\Content1\Content1.exe` (Note that if you have `Content2`, `Content3`, and so on, add them also.)
- `%SMART_ANALYTICS_HOME%\ImageServer1\ImageServer1.exe` (Note that if you have `ImageServer2`, `ImageServer3`, and so on, add them also.)

4. The installation illustrated above is to install all Smart Analytics components on a machine. To achieve better availability, you can consider deploying Smart Analytics on multiple servers and configuring Smart Analytics for high availability. For detailed instructions, see Example: Deploying Smart Analytics on multiple servers and Configure Smart Analytics for high availability.

3.3.8. (Mixed Mode Only) Set up SMA-SM Integration

After you install all Service Manager components, you need to finish some post-installation tasks.

First, you need to assign the application security group `sm-smarta-client-asg` to the VMSS (for example, `aks-agentpool-19318116-vmss`) of the node pool of the SMA AKS cluster by following the instructions provided in the *Assign an Application Security Group to a VMSS* section in References. This operation allows traffics from the SMA AKS cluster worker nodes to Service Manager Smart Analytics.

Then you need to set up the SMA-SM integration according to the Set up SMA-SM topic. Some steps should be adapted for cloud deployment as below.

- Step 1: Configure Service Manager
  - Skip the task (Optional) Configure one-way SSL between SMA and Service Manager as we don't configure SSL in the SM server in this guide.
  - Skip the task Configure LW-SSO in Service Manager as you have already configured the LW-SSO settings in Service Manager.
- Step 3: Configure Smart Analytics
  - Skip this step as you have configured Smart Analytics in the **Install Service Manager Smart Analytics** topic.
- Step 4: Configure a mail server
  - Skip this step as you have finished it in the **(Mixed Mode Only) Install SMA Service Portal** topic.
- Step 5: Create a Service Manager tenant
  - You have finished most of the tasks in the **(Mixed Mode Only) Install SMA Service Portal** topic. You only need to follow step 7 to step 14 of the **Create and deploy a Service Manager tenant** task. Some settings need to be configured as below:
    - **Service Manager URL:** *http://sm-server.yourdomain.net:13080*
    - **Smart Analytics server URL:** *http://sm-smarta.yourdomain.net:9000*
    - **SM database type:** PostgreSQL
    - **SM database host/IP:** *sm-db-pg.postgres.database.azure.com*
    - **SM database port:** 5432
    - **SM database name:** `sm`

- **SM database account:** *postgres@sm-db-pg*
- **SM database password:** *<your_db_password>*
- **Enable SSL:** On
- **SSL mode:** verify-full

> **Note**
> Note
> You need to make a copy of the Azure database root CA certificate you downloaded when installed the SM server, rename it to `smpgroot.crt`, then upload it to the `/mnt/efs/var/vols/itom/itsma/global-volume/certificate/smpgcert/` directory on the SMA bastion node.

- Step 6: Configure LW-SSO in Suite Administration
  - Please skip this step as you have finished it in the **(Mixed Mode Only) Install SMA Service Portal** topic.

3.3.9. References

Disable Windows automatic updates for a VMSS

Run the following PowerShell command in the Azure Cloud Shell to disable the Windows Automatic Updates for a VMSS (for example, `SmWebtierVmss`).

```
# Your resource group name
$rgName = "sm-prod-rg"
# Your VMSS name
$vmssName = "SmWebtierVmss"

# Disable Windows Automatic Updates for the VMSS
Update-AzVmss -ResourceGroupName $rgName -VMScaleSetName $vmssName -EnableAutomaticUpdate $false
# Upgrade existing instances to the latest VMSS model
Update-AzVmssInstance -ResourceGroupName $rgName -VMScaleSetName $vmssName -InstanceId <All Instance
IDs>
```

> **Note**
> Note
> - You can view all instance IDs of the VMSS in its **Instances** page, you should specify all instance IDs for the **InstanceId** parameter. For example, `-InstanceId "1","2","3"`.
> - Please see the Azure doc Quickstart for PowerShell in Azure Cloud Shell for details on how to use the PowerShell in Cloud Shell in the Azure portal.

Assign an Application Security Group to a VMSS

Run following PowerShell command in the Azure Cloud Shell to assign an application security group (for example, `sm-smarta-client-asg`) to a VMSS (for example, `SmServerSecVmss`).

```
# Your resource group name
$rgName = "sm-prod-rg"
# Your VMSS name
$vmssName = "SmServerSecVmss"
# Your application security group name
$asgName = "sm-smarta-client-asg"

# Get the VMSS model and the IP configurations of it
$vmss = Get-AzVmss  -ResourceGroupName $rgName -VMScaleSetName $vmssName
$ipConfs = $vmss.VirtualMachineProfile.NetworkProfile.NetworkInterfaceConfigurations[0].IpConfigurations
# Add the application security group to the VMSS
$asg = Get-AzApplicationSecurityGroup -ResourceGroupName $rgName -Name $asgName
$asgList = New-Object
System.Collections.Generic.List[Microsoft.Azure.Management.Compute.Models.SubResource]
$asgList.Add($asg.id)
foreach ($ipConf in $ipConfs) { $ipConf.applicationSecurityGroups = $asgList }
Update-AzVmss -ResourceGroupName $rgName -VMScaleSetName $vmssName -VirtualMachineScaleSet $vmss
# Upgrade existing instances to the latest VMSS model
Update-AzVmssInstance -ResourceGroupName $rgName -VMScaleSetName $vmssName -InstanceId <All Instance
IDs>
```

Note

Note

- If the VMSS is auto-generated by the AKS cluster, its resource group is auto-generated either (for example, *MC_sm-prod- rg_aksname_westus2*) and is different from the resource group (sm-prod-rg) of the application security group. In this case, you should set the $rgName variable to the resource group of the VMSS, and then use an explicit resource group when getting the application security group. For example:

  ```
  $rgName = "MC_sm-prod-rg_aksname_westus2"
  $vmssName = "aks-agentpool-19318116-vmss"
  ...
  $vmss = Get-AzVmss  -ResourceGroupName $rgName -VMScaleSetName $vmssName
  ...
  $asg = Get-AzApplicationSecurityGroup -ResourceGroupName "sm-prod-rg" -Name $asgName
  ...
  ```

- You can view all instance IDs of the VMSS in its **Instances** page, you should specify all instance IDs for the **InstanceId** parameter. For example, -InstanceId "1","2","3" .
- Please see the Azure doc Quickstart for PowerShell in Azure Cloud Shell for details on how to use the PowerShell in Cloud Shell in the Azure portal.

Generalize the Windows VM

Run the C:\Windows\System32\Sysprep\sysprep.exe program as an administrator. In the **System Preparation Tool** dialog box, change options as follows, and then select **OK**.

- **System Cleanup Action:** Enter System Out-of-Box Experience (OOBE)
  - **Generize:** Yes
- **Shutdown Options:** Shutdown

When Sysprep completes, it shuts down the VM. Do not restart the VM.

See the Azure document Capture VM to image for more details.

Maintenance

You need to install Windows updates to virtual machines on a regular basis. You can also upgrade SM components to a newer version to get bug fixes or new features based on your demands.

For standalone virtual machines, the Windows updates will be installed automatically. You can change the active hours setting in the Windows Updates Settings according to your origination's work hours policy. You can also upgrade SM components manually as needed.

For virtual machines managed by the VMSS, you need to create a new custom image and apply it to the VMSS. For example, you can upgrade the custom image for the `SmWebtierVmss` VMSS by following these steps:

1. Create a VM from the custom image (for example: *sm-webtier-9.80-<timestamp>*).
2. Establish a Remote Desktop Connection to the VM through the `sm-rd-gateway` RD gateway.
3. Install latest Windows updates.
4. Upgrade SM components (for example, web tier) if you want to apply bug fixes or new features.
5. Create a new custom image (for example: *sm-webtier-9.80p1-<newtimestamp>*) from the VM by following the **Create a custom image for SM Web tier** step in the Install Service Manager web tier topic.
6. Apply the new custom image to the VMSS (for example: `SmWebtierVmss`) by running the following PowerShell commands in the Cloud Shell:

```
# Your resource group name
$rgName = "sm-prod-rg"
# Your VMSS name
$vmssName = "SmWebtierVmss"
# New custom image name
$newImageName = "sm-webtier-image-9.80p1-<newtimestamp>"

# Get the image object
$image = Get-AzImage -ResourceGroupName $rgName -ImageName $newImageName
# Update the ImageReferenceId of your VMSS
Update-AzVmss -ResourceGroupName $rgName -VMScaleSetName $vmssName -ImageReferenceId $image.Id
# Update the specified VM instances to the latest image. You could update VM instances batchly to
simulate the rolling update
Update-AzVmssInstance -ResourceGroupName $rgName -VMScaleSetName $vmssName -InstanceId "1","2"
# Waiting for minitues to allow instances 1 and 2 being replaced, then update instaces 3 and 4
Update-AzVmssInstance -ResourceGroupName $rgName -VMScaleSetName $vmssName -InstanceId "3","4"
# And so on...
```



Note

Note

- You can view all instance IDs of the VMSS in its **Instances** page.
- See the Azure document Quickstart for PowerShell in Azure Cloud Shell for details on how to use the PowerShell in Cloud Shell in the Azure portal.

For SM server secondary hosts, you can create a new custom image by following the steps above. Or, you can update the SM server primary host first, and then follow the steps in Create a custom image for SM Server Secondary Host.

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